CHAPTER 1

CONIC SECTION

1 ELLIPSES

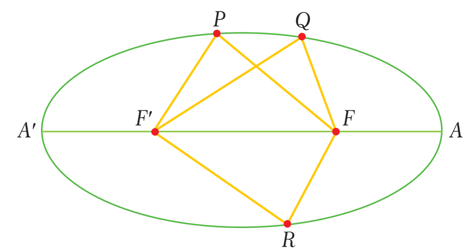
A. BASIC ELEMENTS OF AN ELLIPSE

1. Ellipse and Foci

Definition

ellipse, focus, foci

An **ellipse** is a set of points in a plane whose distances from two fixed points in the plane have a constant sum.



The fixed points are called **foci** (the plural form of **focus**).

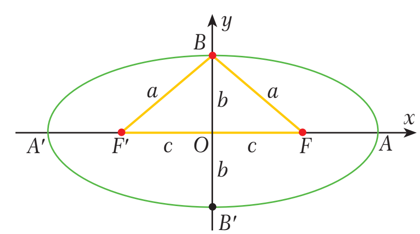
In the figure, P, Q and R are points on the ellipse and F and F' are the foci. By the definition of an ellipse, $PF + PF' = QF + QF' = RF + RF'$.

2. Axes and Vertices

How to draw an ellipse

1. Cut a piece of string of length x units.
2. Pin or nail the ends of the string to a board at a distance of y units, where $x > y$.
3. Use a pencil to stretch the string upwards then downwards to draw a figure. The figure is an ellipse and the two pins or nails are the foci.

Look at the figure. The two lines AA' and BB' are perpendicular and intersect at the center of the ellipse (O). A, A', B and B' mark the intersection of the lines and the ellipse. The longer line (AA') is called the **major axis** of the ellipse and the shorter line (BB') is called the **minor axis** of the ellipse. The points A, A', B and B' are called the **vertices** of the ellipse.



By the definition of an ellipse we have

$AF + AF' = 2a$ and $AF = A'F'$. So $AA' = AF' + A'F' = 2a$, i.e. the major axis has length $2a$.

Note 1

The longer axis of an ellipse is always the major axis. The shorter axis is always the minor axis. The foci always lie on the major axis. In the figure above, the major axis is AA'. It lies on the horizontal axis and has length $2a$. The major axis of an ellipse can also lie on the vertical axis, in which case the major axis is BB' and has length $2b$.

An ellipse is symmetric about its origin O. AA' and BB' are the axes of symmetry.

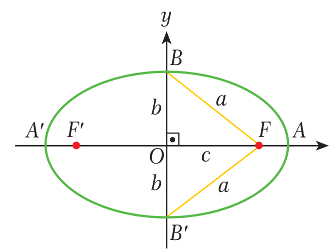
Note 2

In this book, unless stated otherwise we assume that the axes of an ellipse are the x-axis and y-axis of the coordinate plane and that the center of the ellipse is at the origin of the plane.

The coordinates of the vertices and foci of the ellipse are therefore

$A(a, 0)$, $A'(-a, 0)$, $B(0, b)$, $B'(0, -b)$, $F(c, 0)$ and $F'(-c, 0)$.

3. Lengths of the Axes of an Ellipse



Look at the figure. We know that $AA' = 2a$ and $BB' = 2b$. Since an ellipse is symmetric about its axes, we have $AO = A'O = a$ and $BO = B'O = b$.

Also, by the definition of ellipse we have $BF + BF' = 2a$ and $BF = BF'$.

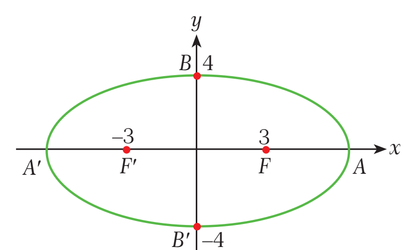
So $BF = BF' = a$.

Let $FF' = 2c$, then $FO = OF' = c$.

By the Pythagorean Theorem we can write $b^2 + c^2 = a^2$, i.e. $c^2 = a^2 - b^2$. For an ellipse whose major axis is the vertical axis, this equation becomes $a^2 + c^2 = b^2$.

EXAMPLE 1 The foci of an ellipse are $F(3, 0)$ and $F'(-3, 0)$ and the length of its minor axis is 8 cm. Find the coordinates of the vertices of the ellipse.

Solution The figure shows the ellipse.



By the properties of the axes of an ellipse,

$c = 3$ and $BB' = 8$ cm so $b = 4$.

$b^2 + c^2 = a^2$ so

$4^2 + 3^2 = a^2$, i.e. $a = 5$.

So the vertices are $A(5, 0)$, $A'(-5, 0)$, $B(0, 4)$ and $B'(0, -4)$.

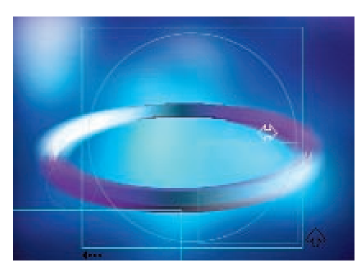
EXAMPLE 2 The lengths of the major axis and the minor axis of an ellipse are 26 cm and 10 cm respectively. What is the distance between the foci?

Solution $2a = 26$ cm so $a = 13$. Similarly, $2b = 10$ cm so $b = 5$.

$c^2 = a^2 - b^2 = 13^2 - 5^2 = 169 - 25 = 144$, i.e. $c = 12$.

So $FF' = 2c = 24$ cm.

EXAMPLE 3 One of the foci of an ellipse is $F(3, 0)$ and the length of the minor axis is 6 cm. Find the length of the major axis.



Solution $2b = 6$ cm so $b = 3$. We are also given $c = 3$. So

$b^2 + c^2 = a^2$ implies $a^2 = 3^2 + 3^2 = 18$, i.e. $a = 3\sqrt{2}$.

So the length of the major axis is $2a = 6\sqrt{2}$ cm.

4. Eccentricity

Definition

eccentricity of an ellipse

The ratio of the distance between the foci of an ellipse to the length of its major axis is called the **eccentricity** of the ellipse, denoted by e. If the major axis is the horizontal axis, we can write $e = \frac{2c}{2a}$, i.e. $e = \frac{c}{a}$.

We know that $c < a$ since $c^2 = a^2 - b^2$. So $e = \frac{c}{a} < 1$ and we can conclude that $\mathbf{0 \leq e < 1}$ for any ellipse.

The eccentricity of an ellipse defines its overall shape. An ellipse with eccentricity zero is a circle. As the eccentricity approaches 1, the shape of the ellipse becomes closer to a straight line.

Note

If the major axis of an ellipse is the vertical axis then $b > a$ and the equation for the eccentricity becomes $e = \frac{c}{b}$.

5. Circles of an Ellipse

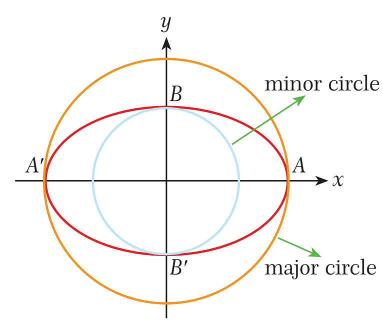
Definition

major and minor circles of an ellipse, circles of the directrix

Let E be an ellipse with a major axis of length $2a$ parallel to the x-axis and a minor axis of length $2b$.

The circle whose diameter is the major axis of E and whose center is at the center of E is called the **major circle** of the ellipse. It has the equation $x^2 + y^2 = a^2$.

The circle whose diameter is the minor axis of E and whose center is at the center of E is called the **minor circle** of the ellipse. It has the equation $x^2 + y^2 = b^2$.



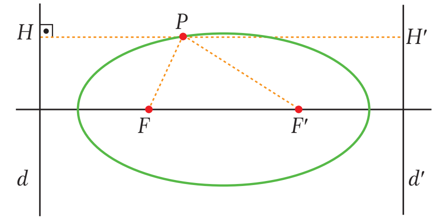
The two circles whose radii are $2a$ and whose centers are at the foci of the ellipse are called the **circles of the directrix**. Their equations are $(x \pm c)^2 + y^2 = 4a^2$.

Note

If the major axis of an ellipse is the vertical axis then the formulas for the major and minor circles are interchanged. In other words, if $b > a$ then the major circle is $x^2 + y^2 = b^2$ and the minor circle is $x^2 + y^2 = a^2$.

6. Directrices

At the beginning of this chapter we defined an ellipse as the set of points in a plane whose distances from two fixed points have a constant sum. We can also define an ellipse in a different way, as a set of points whose distances from a fixed point F and a fixed line d have a constant ratio, e.



The fixed line is called a **directrix** of the ellipse and the fixed point F is a focus of the ellipse. In fact, we can find two different parallel lines and two different fixed points which satisfy this definition for an ellipse, so each ellipse has two foci and two directrices.

The equations of the two directrices are usually written as one equation, called the **equation of the directrices**.

The ratio e is the same as the eccentricity of the ellipse. So in the figure above, $e = \dfrac{PF}{PH} = \dfrac{PF'}{PH'}$.

The lines d and d' are the directrices of the ellipse and the equations of the directrices are given by $x = \pm\dfrac{a^2}{c}$.

Note

If the major axis is BB' then the equation of the directrices becomes $y = \pm\dfrac{b^2}{c}$.

EXAMPLE 4 The vertices of an ellipse are $A(3, 0)$, $A'(-3, 0)$, $B(0, 2)$ and $B'(0, -2)$. Find the eccentricity of the ellipse and the coordinates of its foci.

Solution From the given coordinates we have $a = 3$ and $b = 2$. Since $b^2 + c^2 = a^2$, $2^2 + c^2 = 3^2$ and so $c = \sqrt{5}$. So the eccentricity is $e = \dfrac{c}{a} = \dfrac{\sqrt{5}}{3}$ and the foci are $F(\sqrt{5}, 0)$ and $F'(-\sqrt{5}, 0)$.

EXAMPLE 5 An ellipse has a focus at $F(6, 0)$ and the length of its major axis is 20 cm. Find the eccentricity, the vertices of the minor axis, and the equations of the major and minor circles and the directrices.

Solution If $F(6, 0)$ is a focus then $c = 6$.

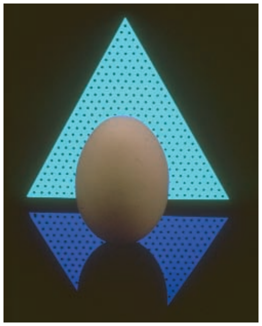
The length of the major axis is $2a = 20$ so $a = 10$ and $e = \dfrac{c}{a} = \dfrac{6}{10} = \dfrac{3}{5}$.

$b^2 + c^2 = a^2$ gives $b^2 = 10^2 - 6^2 = 64$. So $b = 8$ and the vertices of the minor axis are $B(0, 8)$ and $B'(0, -8)$.

equation of the major circle: $x^2 + y^2 = a^2$ so $x^2 + y^2 = 100$

equation of the minor circle: $x^2 + y^2 = b^2$ so $x^2 + y^2 = 64$

equation of the directrices: $x = \pm\dfrac{a^2}{c} = \pm\dfrac{10^2}{6} = \pm\dfrac{100}{6} = \pm\dfrac{50}{3}$



EXAMPLE 6 The eccentricity of an ellipse is $\frac{1}{2}$ and the length of its major axis is 12 cm. Find the length of the minor axis, the distance between the foci of the ellipse and the equations of its major and minor circles.

Solution $2a = 12$ so $a = 6$. Also, $e = \frac{c}{a} = \frac{1}{2}$ so $c = 3$. Since $b^2 + c^2 = a^2$ we have $b^2 + 3^2 = 6^2$ and $b = 3\sqrt{3}$.

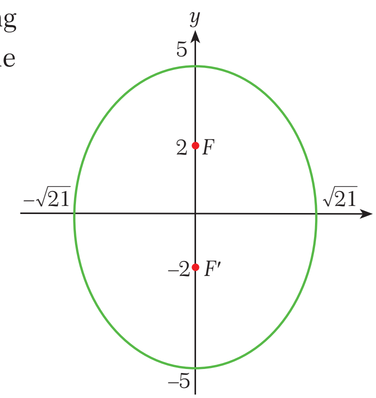
length of the minor axis: $2b = 2 \cdot 3\sqrt{3} = 6\sqrt{3}$ cm

distance between the foci: $2c = 2 \cdot 3 = 6$ cm

equation of the major circle: $x^2 + y^2 = a^2$, i.e. $x^2 + y^2 = 36$

equation of the minor circle: $x^2 + y^2 = b^2$, i.e. $x^2 + y^2 = 27$

EXAMPLE 7 The major axis of an ellipse measures 10 cm and lies along the y-axis. The eccentricity of the ellipse is 0.4. Find the length of the minor axis and the coordinates of the foci.



Solution Length of the major axis: $2b = 10$ so $b = 5$.

$e = 0.4 = \frac{2}{5} = \frac{c}{b} = \frac{c}{5}$ so $c = 2$.

$a^2 + c^2 = b^2$ gives us $a^2 + 2^2 = 5^2$, i.e. $a = \sqrt{21}$.

length of the minor axis: $2a = 2\sqrt{21}$ cm

coordinates of the foci: $F(0, 2)$ and $F'(0, -2)$

EXAMPLE 8 The foci of an ellipse are $F(0, 3)$ and $F'(0, -3)$ and the minor axis AA' measures 8 cm. Find the length of the major axis BB' and the eccentricity e.

Solution If the foci are $F(0, 3)$ and $F'(0, -3)$ then $c = 3$.

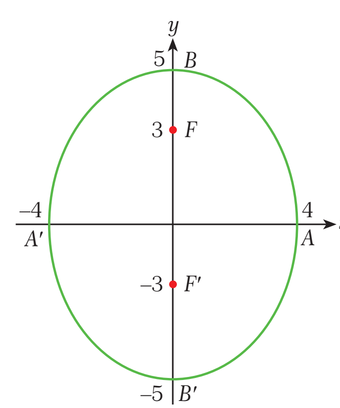
AA' is the minor axis and BB' is the major axis. So

$AA' = 2a = 8$ and $a = 4$.

Using $a^2 + c^2 = b^2$ we get $4^2 + 3^2 = b^2$, i.e. $b = 5$.

If $b = 5$ then the length of the major axis is $BB' = 2b = 10$ cm.

Finally, the eccentricity is $e = \frac{c}{b} = \frac{3}{5} = 0.6$.



Check Yourself 1

1. $F(2, 0)$ is a focus of an ellipse and the length of the major axis is 8 cm. Find the length of the minor axis.
2. The lengths of the major and minor axes of an ellipse are $AA' = 8$ cm and $BB' = 6$ cm respectively. Find the coordinates of the foci and the vertices of the ellipse.
3. E is an ellipse such that $c = 4$ and the length of the minor axis is 8 cm. Find the eccentricity and the length of the major axis.
4. The vertices of an ellipse are $A(3, 0)$, $A'(-3, 0)$, $B(0, 6)$ and $B'(0, -6)$. Find the eccentricity of the ellipse and the coordinates of the foci.
5. E is an ellipse with eccentricity $e = \frac{1}{3}$ and a major axis measuring 18 cm. Find the length of the minor axis and the equations of the major and minor circles of the ellipse.
6. The coordinates of the foci of an ellipse are $F(0, 5)$ and $F'(0, -5)$. Given that the length of the major axis is $BB' = 20$ cm, find the eccentricity and the coordinates of the vertices.

Answers

1. $4\sqrt{3}$ cm
2. foci: $F(\sqrt{7}, 0)$, $F'(-\sqrt{7}, 0)$; vertices: $A(4, 0)$, $A'(-4, 0)$, $B(0, 3)$, $B'(0,-3)$
3. $e = \frac{\sqrt{2}}{2}$, major axis $= 8\sqrt{2}$ cm
4. $e = \frac{\sqrt{3}}{2}$, foci: $F(0, 3\sqrt{3})$, $F'(0, -3\sqrt{3})$
5. minor axis $=12\sqrt{2}$ cm, major circle: $x^2 + y^2 = 81$, minor circle: $x^2 + y^2 = 72$
6. $e = \frac{1}{2}$, vertices: $A(5\sqrt{3}, 0)$, $A'(-5\sqrt{3}, 0)$, $B(0, 10)$, $B'(0, -10)$

Omar Khayyam (1048-1131)

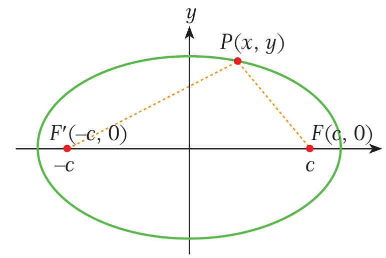
Omar Khayyam was a famous Muslim scientist and philosopher. He studied mathematics, physics, astronomy, poetry, medicine and music. His most famous work as a mathematician was a book called *Treatise on Demonstration of Problems of Algebra*. In part of this book, Khayyam showed how to solve cubic equations geometrically using conic sections. Khayyam classified cubic equations into thirteen different types, based on their complexity. This was the first classification of cubic equations in the history of mathematics. Omar Khayyam's other important contributions to mathematics were the discovery of binomial expansion and his works on geometry, which contributed to the development of non-Euclidean geometry.

Omar Khayyam's work as an astronomer led him to invent a new calendar, known as the Jalali calendar. This calendar loses one day in only 5000 years. The Gregorian calendar we use today loses one day every 3330 years. Khayyam also demonstrated that the universe was not moving around the Earth, as many people thought at that time. His demonstration used candles which lit star charts as they revolved on a platform around the room, showing the difference between night and day.

Omar Khayyam traveled to Samarkand, Bukhara and Isphahan during his life. He died in 1131 in Nishapur, Iran.

B. EQUATION OF AN ELLIPSE

1. Equation of an Ellipse Centered at the Origin



Let $P(x, y)$ be any point on the ellipse opposite. Then P satisfies the equation $PF + PF' = 2a$ where

$PF = \sqrt{(x-c)^2 + y^2}$ and $PF' = \sqrt{(x+c)^2 + y^2}$.

So $\sqrt{(x-c)^2 + y^2} + \sqrt{(x+c)^2 + y^2} = 2a$, i.e.

$(x-c)^2 + y^2 + 2\sqrt{(x-c)^2 + y^2} \cdot \sqrt{(x+c)^2 + y^2} + (x+c)^2 + y^2 = 4a^2$ (take the square of both sides)

$\sqrt{(x-c)^2 + y^2} \cdot \sqrt{(x+c)^2 + y^2} = 2a^2 - x^2 - y^2 - c^2$

$[(x-c)^2 + y^2] \cdot [(x+c)^2 + y^2] = (2a^2 - x^2 - y^2 - c^2)^2$ (take the square of both sides)

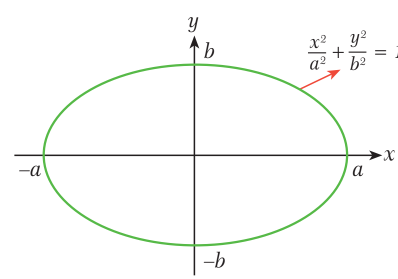
$(x^2 + y^2 + c^2)^2 - (2a^2 - x^2 - y^2 - c^2)^2 = 4c^2x^2$ (expand the parentheses and rearrange the terms)

$2 \cdot (x^2 + y^2 + c^2 - a^2) \cdot 2a^2 = 4x^2c^2$ ($b^2 + c^2 = a^2$, so $c^2 = a^2 - b^2$)

$4a^2(x^2 + y^2 + a^2 - b^2 - a^2) = 4x^2(a^2 - b^2)$

$a^2x^2 + a^2y^2 - a^2b^2 = a^2x^2 - b^2x^2$.

Rearranging this gives us $b^2x^2 + a^2y^2 = a^2b^2$, and dividing both sides of this by a^2b^2 gives $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$.



The equation $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ is called the **canonical equation** of the ellipse, which means the equation of an ellipse which is not rotated or translated and which has its center at the origin.

The y-intercepts of the ellipse are $(0, \pm b)$.

The x-intercepts are $(\pm a, 0)$.

EXAMPLE 9 The lengths of the major and minor axes of an ellipse are $AA' = 12$ cm and $BB' = 8$ cm respectively. Find the canonical equation of the ellipse.

Solution major axis = $2a$ = 12 cm, $a = 6$

minor axis = $2b$ = 8 cm, $b = 4$

Using $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ gives us $\frac{x^2}{6^2}+\frac{y^2}{4^2}=1$.

So the canonical equation is $\frac{x^2}{36}+\frac{y^2}{16}=1$, or $16x^2+36y^2=576$.

EXAMPLE 10 $\frac{x^2}{36}+\frac{y^2}{25}=1$ is the equation of an ellipse. Find

a. the length of the major axis.

b. the length of the minor axis.

c. the eccentricity.

d. the distance between the foci.

e. the coordinates of the vertices and the focii.

Solution $a^2 = 36$ so $a = 6$. $b^2 = 25$ so $b = 5$.

$b^2 + c^2 = a^2$ so $5^2 + c^2 = 6^2$, i.e. $c^2 = 11$ and $c = \sqrt{11}$.

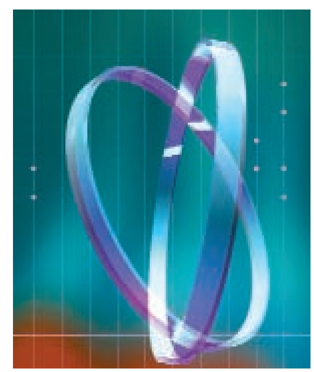
a. length of the major axis = $2a = 2 \cdot 6 = 12$

b. length of the minor axis = $2b = 2 \cdot 5 = 10$

c. eccentricity = $e = \frac{c}{a} = \frac{\sqrt{11}}{6}$

d. distance between the foci = $2c = 2\sqrt{11}$

e. vertices: $A(6, 0)$, $A'(-6, 0)$, $B(0, 5)$, $B'(0, -5)$; foci: $F(\sqrt{11}, 0)$, $F'(-\sqrt{11}, 0)$



EXAMPLE 11 $9x^2 + 16y^2 = 144$ is the equation of an ellipse. Find the lengths of the major and minor axes of the ellipse.

Solution $9x^2 + 16y^2 = 144$

$\frac{9x^2+16y^2}{144}=\frac{144}{144}$, $\frac{x^2}{16}+\frac{y^2}{9}=1$ (divide both sides by 144)

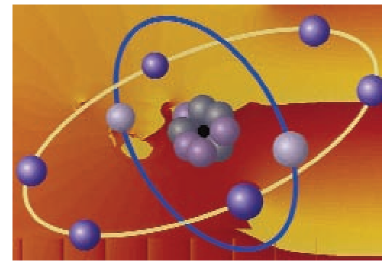
From this we get $a^2 = 16$, i.e. $a = 4$ and the length of the major axis is $2a = 2 \cdot 4 = 8$.

Similarly, $b^2 = 9$ so $b = 3$ and the length of the minor axis is $2b = 2 \cdot 3 = 6$.

EXAMPLE 12 The equation of an ellipse is $x^2 + \frac{y^2}{4} = 1$. Find

a. the lengths of the major and minor axes.

b. the eccentricity.



Solution $a^2 = 1$ so $a = 1$. $b^2 = 4$ so $b = 2$.

So $b > a$ and so the major axis is BB' and $a^2 + c^2 = b^2$, i.e.

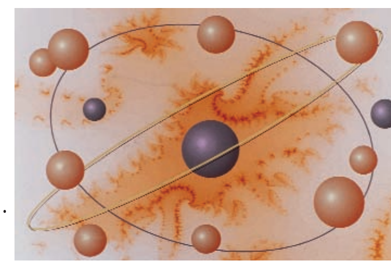
$1^2 + c^2 = 2^2$, $c^2 = 3$, $c = \sqrt{3}$.

a. length of the major axis: $2b = 2 \cdot 2 = 4$

length of the minor axis: $2a = 2 \cdot 1 = 2$

b. eccentricity $=$ $e = \frac{c}{b} = \frac{\sqrt{3}}{2}$

EXAMPLE 13 $9x^2 + 16y^2 = 1$ is an ellipse. Find



a. the lengths of the major and minor axes.

b. the eccentricity.

Solution $9x^2 + 16y^2 = 1$, i.e. $\frac{x^2}{\frac{1}{9}} + \frac{y^2}{\frac{1}{16}} = 1$. So $a^2 = \frac{1}{9}$ and $a = \frac{1}{3}$.

Similarly, $b^2 = \frac{1}{16}$ and $b = \frac{1}{4}$. $b^2 + c^2 = a^2$, $\frac{1}{16} + c^2 = \frac{1}{9}$, $c^2 = \frac{1}{9} - \frac{1}{16} = \frac{7}{144}$ and $c = \frac{\sqrt{7}}{12}$.

a. length of the major axis: $2a = \frac{2}{3}$, length of the minor axis: $2b = \frac{2}{4} = \frac{1}{2}$

b. eccentricity: $e = \frac{c}{a} = \frac{\frac{\sqrt{7}}{12}}{\frac{1}{3}} = \frac{\sqrt{7}}{12} \cdot \frac{3}{1} = \frac{\sqrt{7}}{4}$

EXAMPLE 14 The length of a line segment is 12 cm. Find the equation for the set of points whose distances from the endpoints of this line segment have a sum of 20 cm.

Solution This is the definition of an ellipse whose foci are the endpoints of the line.

So $FF' = 2c = 12$ cm and $c = 6$.

The sum of the distances from the foci is $2a = 20$ cm, so $a = 10$.

$b^2 + c^2 = a^2$ gives us $b^2 + 6^2 = 10^2$, $b^2 = 100 - 36 = 64$ and so $b = 8$.

The equation for the set of points is the same as the equation of the ellipse:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,\ \frac{x^2}{10^2} + \frac{y^2}{8^2} = 1 \text{ so } \frac{x^2}{100} + \frac{y^2}{64} = 1 \text{ or } 64x^2 + 100y^2 = 6400.$$

EXAMPLE 15 From each point on the circle $x^2 + y^2 = 100$ we draw a line perpendicular to the x-axis. What is the equation of the figure which is obtained by connecting the midpoints of all the perpendicular lines?

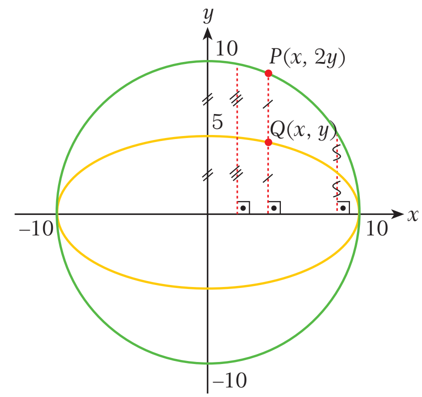
Solution Let $Q(x, y)$ be any point on the new figure. Then $P(x, 2y)$ is a point on the circle.

So $P(x, 2y)$ must satisfy the equation of the circle ($x^2 + y^2 = 100$), i.e. $x^2 + (2y)^2 = 100$.

From this equation we get the equation of the coordinates satisfying $Q(x, y)$: $x^2 + 4y^2 = 100$ or

$$\frac{x^2}{100} + \frac{4y^2}{100} = \frac{x^2}{100} + \frac{y^2}{25} = 1.$$

This is the canonical equation of an ellipse, so the new figure is an ellipse.



Check Yourself 2

1. The lengths of the major and minor axes of an ellipse are $AA' = 16$ cm and $BB' = 10$ cm respectively. What is the equation of the ellipse?
2. Find the lengths of the major and minor axes of the ellipse whose equation is $9x^2 + 25y^2 = 225$.
3. $\frac{x^2}{9} + \frac{y^2}{16} = 1$ is the equation of an ellipse. Find each component.
   a. the length of the major axis
   b. the length of the minor axis
   c. the eccentricity
   d. the distance between the foci
   e. the coordinates of the foci
   f. the equations of the circles of the ellipse
   g. the equation of the directrix
4. $9x^2 + 25y^2 = 1$ is the equation of an ellipse. Find
   a. the lengths of the major and minor axes of the ellipse.
   b. the eccentricity of the ellipse.
5. $9x^2 + 4y^2 = 36$ is the equation of an ellipse. Find
   a. the lengths of the major and minor axes of the ellipse.
   b. the eccentricity of the ellipse.

Answers

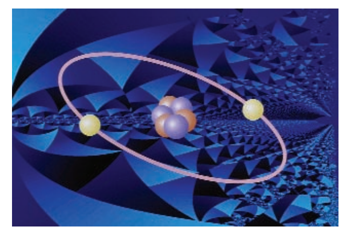
1. $\dfrac{x^2}{64}+\dfrac{y^2}{25}=1$ 2. major axis = 10, minor axis = 6

3. a. 8 b. 6 c. $\dfrac{\sqrt{7}}{4}$ d. $FF' = 2\sqrt{7}$ e. $F(0, \sqrt{7})$, $F'(0, -\sqrt{7})$

f. major circle: $x^2 + y^2 = 16$, minor circle: $x^2 + y^2 = 9$ g. $y=\pm\dfrac{16\sqrt{7}}{7}$

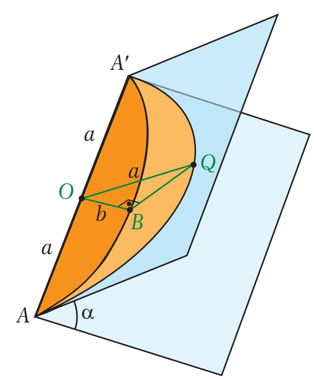
4. a. major axis = $\dfrac{2}{3}$, minor axis = $\dfrac{2}{5}$ b. $\dfrac{4}{5}$

5. a. major axis = 6, minor axis = 4 b. $\dfrac{\sqrt{5}}{3}$



C. AREA OF AN ELLIPSE

Recall the concept of orthogonal projection: to find the orthogonal projection of an object on a plane P we draw lines perpendicular to P from all the points on the object and trace their intersection with P.



Look at the figure opposite. We can say that the ellipse passing through A, B and A' is an orthogonal projection of a circle with diameter AA' whose radius is a and center is O.

The angle between two planes is called their **dihedral angle**.

Let us assume that the point Q is the midpoint of the semicircle and that the dihedral angle between the planes containing the circle and the ellipse is α. We can say that the projection of the point Q is B.

So $OQ = a$ and $OB = b$, and therefore $\cos\alpha = \dfrac{b}{a}$. By the rules of projection we can write

$\text{area}_{projection} = \text{area}_{figure} \cdot \cos\alpha$, i.e.

$\text{area}_{ellipse} = \text{area}_{circle} \cdot \cos\alpha$

$\text{area}_{ellipse} = \pi a^2 \cdot \dfrac{b}{a}$

$\boxed{\text{area}_{ellipse} = \pi ab}$. This is the formula for the area of an ellipse.

EXAMPLE 16 What is the area of the ellipse defined by $\frac{x^2}{36}+\frac{y^2}{16}=1$?

Solution From the given equation we get $a = 6$ and $b = 4$. So the area of the ellipse is

$\text{area}_{ellipse} = \pi ab = \pi \cdot 6 \cdot 4 = 24\pi.$

EXAMPLE 17 What is the area of the region R between the ellipse $\frac{x^2}{144}+\frac{y^2}{81}=1$ and its major circle?

Solution From the equation we get $a = 12$ and $b = 9$.

So the major axis of the ellipse is horizontal and the radius of the major circle is 12. The area between the ellipse and its major circle is the difference between the area of the major circle and the area of the ellipse. So

$\text{area}_R = \text{area}_{major\ circle} - \text{area}_{ellipse} = \pi a^2 - (\pi \cdot a \cdot b) = (\pi \cdot 12^2) - (\pi \cdot 12 \cdot 9) = 144\pi - 108\pi = 36\pi.$

Remember:
The major circle of an ellipse is a circle whose center is the center of the ellipse and whose diameter is the major axis of the ellipse. The equation of the major circle for an ellipse with a horizontal major axis is $x^2 + y^2 = a^2$.

Check Yourself 3

1. Find the area of the ellipse defined by $\frac{x^2}{121}+\frac{y^2}{49}=1$.
2. Find the area of the ellipse defined by $2x^2 + 5y^2 = 20$.
3. What is the area of the region between the ellipse $\frac{x^2}{9}+\frac{y^2}{16}=1$ and its major circle?
4. What is the area of the region between the ellipse $400x^2 + 9y^2 = 3600$ and its minor circle?
5. What is the area of the ellipse defined by $9x^2 + y^2 = 9$?

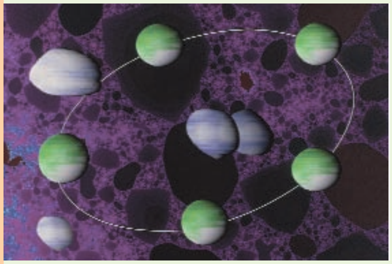
Answers

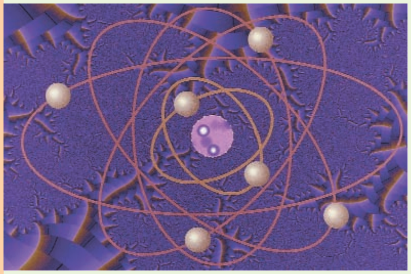
1. 77π 2. $2\sqrt{10} \cdot \pi$ 3. 4π 4. 51π 5. 3π

ELLIPSES AROUND US

Although the ellipse is not as simple as a circle, it is the most common geometric figure in nature. If you look at the circular objects and buildings around you, you will generally see an ellipse because of the effect of perspective.

Today we know that the planets in our solar system trace out elliptical paths (called *orbits*) as they move through space. This discovery was made by the mathematician and astronomer Johannes Kepler (1571-1630) in the 17th century. The ancient Greek astronomers, many centuries before Kepler, thought that the paths were circular, but Kepler proved that the Earth's orbit around the Sun is an ellipse and that the Sun is one of the foci of this ellipse. This, combined with the fact that the Earth rotates on an axis tilted at 23.5 degrees to its orbit, creates the different seasons we experience on Earth. On January 3 each year the Earth is at its closest point to the Sun, a distance of 149.5 million kilometers. On July 4 it reaches its farthest point from the Sun at 152.5 million kilometers. This means that in the equation for the ellipse of the Earth's orbit, $a - c = 149.5$ and $a + c = 152.5$, i.e. $a = 151$ and $c = 1.5$. So the eccentricity of this ellipse is $e = \frac{c}{a} = \frac{1.5}{151}$, which is approximately 0.01. We know that as the eccentricity of an ellipse approaches zero, the shape of the ellipse gets closer and closer to a circle. So the orbit of the Earth is an ellipse that is almost a circle. The moon and artifical satellites moving around the Earth also have elliptical orbits with the Earth at one focus. Comets have been proven to have orbits shaped like a conic section whose eccentricity increases as the speed of the comet increases.





Ellipses appear elsewhere in nature, too. The electrons in an atom move around the nucleus in an elliptical orbit. And ellipses can be used to model waves of sound and light. Special elliptical rooms and buildings called 'whispering galleries' show how the ellipse relates to sound waves. If two people stand at the two foci of a whispering gallery, they can hear each other whisper. However, if one person moves away from a focus, he or she can no longer hear the other person. The White House in Washington has a whispering gallery.

EXERCISES 1.1

A. Basic Elements of an Ellipse

1. An ellipse has foci $F(6, 0)$ and $F'(-6, 0)$ and two vertices at $A(10, 0)$ and $A'(-10, 0)$. Find the coordinates of the vertices B and B'.

2. An ellipse has foci $F(0, 5)$ and $F'(0, -5)$ and two vertices at $B(0, 13)$ and $B'(0, -13)$. Find the coordinates of the vertices A and A'.

3. The vertices of an ellipse are $A(25, 0)$, $A'(-25, 0)$, $B(0, 24)$ and $B'(0,-24)$. What are the coordinates of the foci F and F' of the ellipse?

4. The lengths of the major and minor axes of an ellipse are $AA' = 12$ cm and $BB' = 6$ cm respectively. What are the coordinates of the vertices?

5. The major axis BB' of an ellipse measures 30 cm and the distance between the foci of the ellipse is 24 cm. Find the length of the minor axis.

6. An ellipse has a focus at $F'(-8, 0)$ and the length of its major axis is 20 cm. Find the eccentricity of the ellipse, the vertices on its minor axis, and the equations of its major and minor circles and the directrices.

7. An ellipse has a focus at $F(4, 0)$. If the eccentricity of the ellipse is $\frac{4}{5}$, find the lengths of the major and minor axes.

8. An ellipse has an eccentricity of 0.8 and foci at $F(6, 0)$ and $F'(-6, 0)$. Find the coordinates of the vertices of the ellipse and the equation of its directrices.

9. The major axis of an ellipse lies along the y-axis and measures 12 cm. Given that the eccentricity is $\frac{5}{6}$, find the length of the minor axis and the coordinates of the foci.

10. An ellipse has foci $F(0, 5)$ and $F'(0, -5)$. Given that $AA' = 18$ cm, find BB' and the eccentricity e.

11. An ellipse has eccentricity $e = \frac{2}{5}$ and the length of its major axis is 20 cm. Find the length of the minor axis and the equations of the major and minor circles of this ellipse.

12. The eccentricity of an ellipse is $\frac{2}{3}$ and the equation of its major circle is $x^2 + y^2 = 81$. Find the length of the major axis BB' and the minor axis, and the equation of the directrices.

13. An ellipse has a focus $F(3, 0)$ and the equation of one directrix is $x = 4$. What is the eccentricity of this ellipse?

14. $P(2, 3)$ and $Q(10,9)$ are given. The set of points whose distances from P and Q have a sum of 26 cm form an ellipse. Find the coordinates of the vertices A, A', B and B' of this ellipse.

B. Equation of an Ellipse

15. The lengths of the major and minor axes of an ellipse are $AA' = 22$ cm and $BB' = 14$ cm respectively. Find the canonical equation of this ellipse.

16. The lengths of the major and minor axes of an ellipse are $BB' = 16$ cm and $AA' = 10$ cm respectively. Find the canonical equation of this ellipse.

17. Find the lengths of the major and minor axes of the ellipse defined by $100x^2 + 16y^2 = 1600$.

18. An ellipse is defined by $\dfrac{x^2}{100} + \dfrac{y^2}{64} = 1$. Find
    a. the length of the major axis.
    b. the length of the minor axis.
    c. the eccentricity.
    d. the distance between the foci.

19. The equation $25x^2 + 16y^2 = 1$ defines an ellipse. Find
    a. the lengths of the major and minor axes.
    b. the eccentricity.

20. A line segment PQ measures 8 cm. Find an equation for the set of points whose distances from the endpoints of this line segment have a sum of 14 cm.

21. From each point on the circle $x^2 + y^2 = 144$ we draw a line perpendicular to the y-axis. What is the equation of the figure which is obtained by connecting the midpoints of all the perpendicular lines?

C. Area of an Ellipse

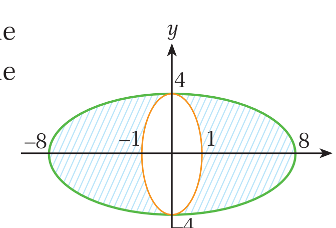
22. Find the area of the ellipse defined by $\dfrac{x^2}{81} + \dfrac{y^2}{36} = 1$.

23. Find the area of the ellipse defined by $3x^2 + 8y^2 = 48$.

24. What is the area of the region between the ellipse $\dfrac{x^2}{100} + \dfrac{y^2}{36} = 1$ and its major circle?

25. What is the area of the region between the ellipse $\dfrac{x^2}{5} + \dfrac{y^2}{16} = 1$ and its minor circle?

26. Find the area of the shaded region in the figure.



2 HYPERBOLAS

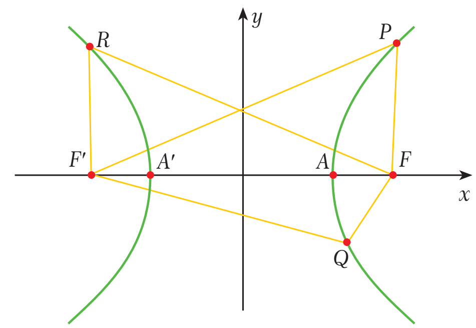
A. BASIC ELEMENTS OF A HYPERBOLA

1. Hyperbola and Foci

Definition

hyperbola, foci

A **hyperbola** is the set of all points in a plane whose distances from two fixed points in the plane have a constant difference. The fixed points are called the **foci** of the hyperbola.



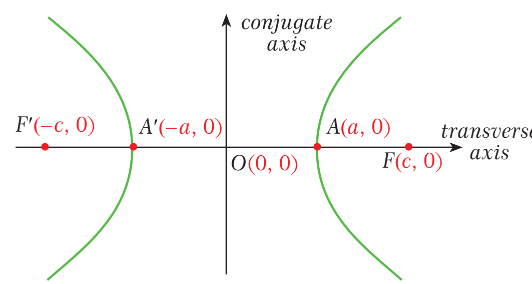
In the figure, P, Q and R are points on the hyperbola and F and F' are the foci. By the definition of a hyperbola,

$PF' - PF = QF' - QF = RF - RF'$.

2. Axes and Vertices

On an ellipse, the distances from the foci have a constant **sum**. On a hyperbola, the distances have a constant **difference**.

Look at the figure. It shows a hyperbola with two axes which intersect at O. The line $F'F$ is called the **transverse axis** or **major axis** of the hyperbola. The axis perpendicular to this is called the **conjugate axis**. The midpoint O of the line segment FF' is called the **center** of the hyperbola. The intersection points of the hyperbola with its transverse axis (A and A' in this diagram) are the **vertices** of the hyperbola. Since the conjugate axis does not contain any points on the hyperbola it is also sometimes called the **imaginary axis**.



Note 1

If we center the hyperbola in the coordinate plane then the coordinate axis which includes FF' is the transverse axis and the other axis is the conjugate axis. The vertices of the hyperbola are always on the transverse axis.

Note 2

A hyperbola is symmetric about its origin. The transverse axis and conjugate axis are the axes of symmetry.

Note 3

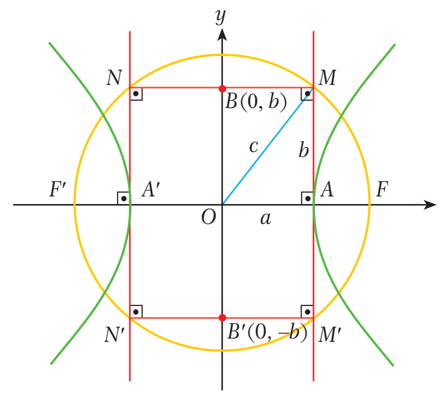
In this book, unless stated otherwise we will assume that the axes of a hyperbola are the x- and y-axes of the coordinate plane and that the center of the hyperbola is at the origin of the coordinate plane.

3. Lengths of the Axes of a Hyperbola

By the definition of a hyperbola, the difference of the distances from F and F' is constant for all points on the hyperbola. Let us say that this difference is $2a$, then we have $AF' - AF = 2a$ and $AF = A'F'$ due to symmetry. So $\boldsymbol{AA' = 2a}$, i.e. the length of the transverse axis of a hyperbola is $2a$.

The symmetric property of a hyperbola also gives us $OA = OA' = a$ and $OF = OF'$. If we write $OF = c$ then the distance between the foci is $FF' = 2c$.

Now look at the figure opposite. It shows a circle of radius c whose center coincides with the center of the hyperbola. Two perpendicular lines intersect the transverse axis of the hyperbola at A and A'.



Let M, M', N and N' be the points of intersection of these two lines and the circle.

So $MNN'M'$ is a rectangle.

Let the length $MM' = NN'$ be $2b$, then $OB = OB' = b$.

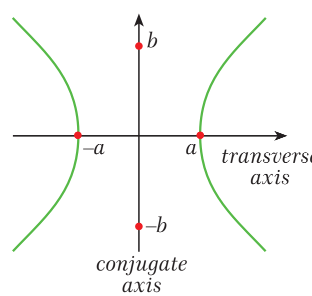
From ΔOAM we get $a^2 + b^2 = c^2$.

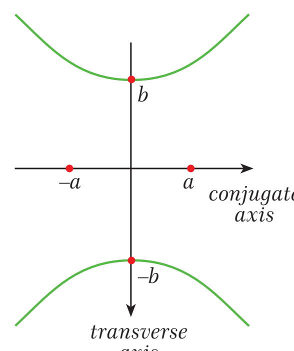
So the length of the conjugate axis of the hyperbola is $\boldsymbol{BB' = 2b}$.

Note

If the transverse axis of a hyperbola is horizontal in the coordinate plane, the transverse axis is $AA' = 2a$ and the conjugate axis is $BB' = 2b$.

If the transverse axis is vertical in the coordinate plane, the transverse axis is $BB' = 2b$ and the conjugate axis is $AA' = 2a$.





Remember:
The foci and vertices of a parabola always lie on its transverse axis.

EXAMPLE 18 The foci of a hyperbola are $F(5, 0)$ and $F'(-5, 0)$ and the length of the transverse axis is 6 cm. Find the length of the conjugate axis.

Solution $c = 5$ and $AA' = 6$ cm so $a = 3$.

$a^2 + b^2 = c^2$ gives us $3^2 + b^2 = 5^2$, i.e. $b = 4$.

So the length of the conjugate axis is $2b = 8$ cm.

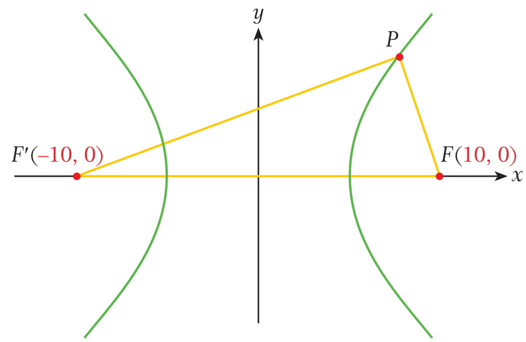
EXAMPLE 19 The lengths of the transverse and conjugate axes of a hyperbola are 24 cm and 10 cm respectively. Find the distance between the foci.

Solution $2a = 24$ cm so $a = 12$; $2b = 10$ cm so $b = 5$.

$c^2 = a^2 + b^2$ means $12^2 + 5^2 = 144 + 25 = 169$, i.e. $c = 13$.

So $FF' = 2c = 26$ cm.

EXAMPLE 20 The figure shows a hyperbola with foci $F(10, 0)$ and $F'(-10, 0)$. If $PF = 4$ cm and $PF' = 16$ cm then find the coordinates of the vertices and the lengths of the axes of the hyperbola.



Solution By the definition of a hyperbola we have $PF' - PF = 2a$. So $16 - 4 = 12 = 2a$, i.e. $a = 6$.

From the figure we get $c = 10$, so $c^2 = a^2 + b^2$ means $10^2 = 6^2 + b^2$ and so $b = 8$.

coordinates of the vertices: $A(6, 0)$ and $A'(-6, 0)$.

length of the transverse axis: $2a = 2 \cdot 6 = 12$ cm

length of the conjugate axis: $2b = 2 \cdot 8 = 16$ cm

EXAMPLE 21 $F(0, 15)$ is a focus of a hyperbola. Given that the length of the conjugate axis of the hyperbola is 18 cm, find the length of the transverse axis.

Solution Since the focus is on the y-axis, the transverse axis of the hyperbola lies on the y-axis. So the length of the conjugate axis is $2a = 18$ and $a = 9$. We also have $c = 15$.

$a^2 + b^2 = c^2$ means $9^2 + b^2 = 15^2$, i.e. $b = 12$.

So the length of the transverse axis is $2b = 2 \cdot 12 = 24$ cm.

4. Eccentricity

Definition

eccentricity of a hyperbola

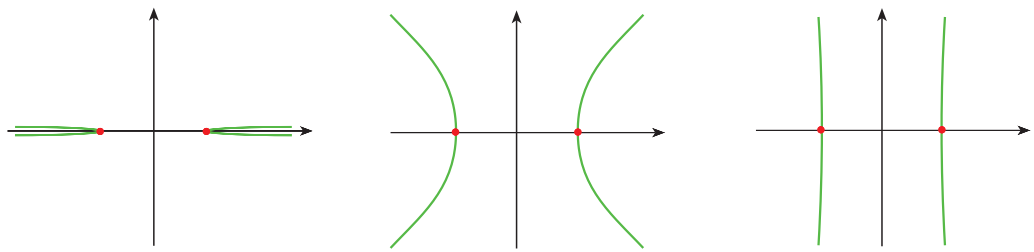
The ratio of the distance between the foci of a hyperbola to the length of its transverse axis is called the **eccentricity** of the hyperbola, denoted e. In other words, for a hyperbola whose transverse axis is parallel to the x-axis,

$$e = \frac{2c}{2a}, \text{ i.e. } \boxed{e = \frac{c}{a}}.$$

$0 < e < 1$ for an ellipse.
$e > 1$ for a hyperbola.

We know $c > a$ since $c^2 = a^2 + b^2$. So $e = \dfrac{c}{a} > 1$ and we can conclude that $\boldsymbol{e > 1}$ for any hyperbola.

The eccentricity of a hyperbola defines its overall shape, as shown in the figures.



$e \cong 1$ $\qquad$ $e > 1$ $\qquad$ $e \to \infty$

Note 1

The coordinates of the vertices of a hyperbola are $A(a, 0)$, $A'(-a, 0)$, $B(0, b)$ and $B'(0, -b)$. The foci are $F(c, 0)$ and $F'(-c, 0)$ $(c > a)$.

Note 2

If the transverse axis of a hyberbola is parallel to the y-axis then its eccentricity becomes $e = \dfrac{c}{b}$.

EXAMPLE 22 A hyperbola has vertices $A(3, 0)$ and $A'(-3, 0)$. Given that the length of its conjugate axis is 8 cm, find its eccentricity and the coordinates of the foci.

Solution If the vertices are $A(3, 0)$ and $A'(-3, 0)$ then $a = 3$.

The length of the conjugate axis is $2b = 8$ so $b = 4$.

$a^2 + b^2 = c^2$ means $3^2 + 4^2 = c^2$ so $c = 5$.

So the eccentricity is $e = \dfrac{c}{a} = \dfrac{5}{3}$ and the foci are $F(5, 0)$ and $F'(-5, 0)$.

EXAMPLE 23 A hyperbola has a focus at $F(8, 0)$ and an eccentricity of 2. Find the lengths of the transverse and conjugate axes of the hyperbola and the coordinates of its vertices.

Solution If $F(8, 0)$ is a focus then $c = 8$.

$e = \dfrac{c}{a}$ means $2 = \dfrac{8}{a}$ so $a = 4$.

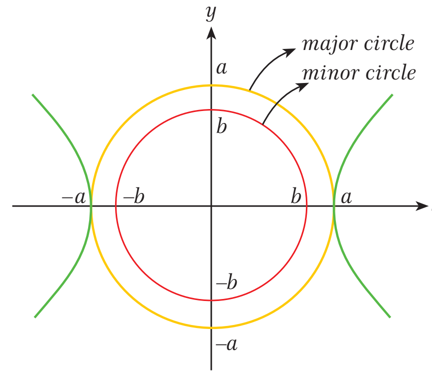
$a^2 + b^2 = c^2$ means $4^2 + b^2 = 8^2$ so $b = 4\sqrt{3}$.

length of the transverse axis: $2a = 8$

length of the conjugate axis: $2b = 8\sqrt{3}$

The vertices are $A(4, 0)$ and $A'(-4, 0)$.

5. Circles of a Hyperbola



Let H be a hyperbola with a transverse axis of length $2a$ and a conjugate axis of length $2b$.

The circle whose diameter is the transverse axis of H and whose center is at the center of H is called the **major circle** of the hyperbola. It has the equation $x^2 + y^2 = a^2$.

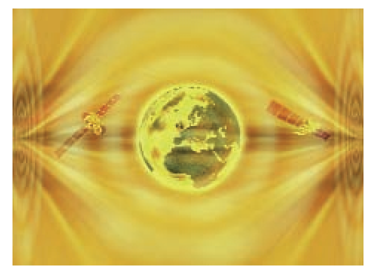
The circle whose diameter is the conjugate axis of H and whose center is at the center of H is called the **minor circle** of the hyperbola. It has the equation $x^2 + y^2 = b^2$.

The two circles with radius $2a$ whose centers are the foci of the hyperbola are called the **circles of the directrices**. They have the equations

$(x - c)^2 + y^2 = 4a^2$ and $(x + c)^2 + y^2 = 4a^2$.

Note

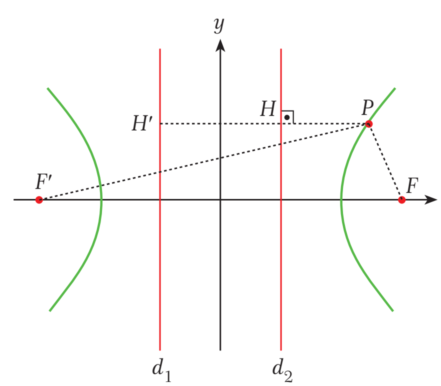
If the transverse axis of the hyperbola is parallel to the y-axis then the equation of the major circle is $x^2 + y^2 = b^2$ and the equation of the minor circle is $x^2 + y^2 = a^2$.



6. Directrices

Look at the figure opposite. $d_1 \parallel d_2$ and $PH \perp d_2$, $PH' \perp d_2$. Suppose that H and H' are such that $\frac{PF}{PH} = \frac{PF'}{PH'} = e$. Then the lines d_1 and d_2 are called the **directrices** of the hyperbola. The directrices are often described in a single equation called the equation of the directrices.

The eccentricity e of a parabola is the same as $\frac{PF}{PH}$ and $\frac{PF'}{PH'}$.



If the transverse axis of a hyperbola lies along the x-axis then the equation of the directrices is $x = \pm\frac{a^2}{c}$.

If the transverse axis lies along the y-axis then the equation of the directrices is $y = \pm\frac{b^2}{c}$.

EXAMPLE 24 The transverse axis of a hyperbola lies on the x-axis and has length 8 cm. Given that the length of the conjugate axis is 12 cm, find the equations of the circles and the directrices of this hyperbola.

Solution length of the transverse axis: $2a = 8$ so $a = 4$

length of the conjugate axis: $2b = 12$ so $b = 6$

$a^2 + b^2 = c^2$ means $4^2 + 6^2 = c^2$. So $c = 2\sqrt{13}$.

equation of the major circle: $x^2 + y^2 = a^2$, i.e. $x^2 + y^2 = 16$

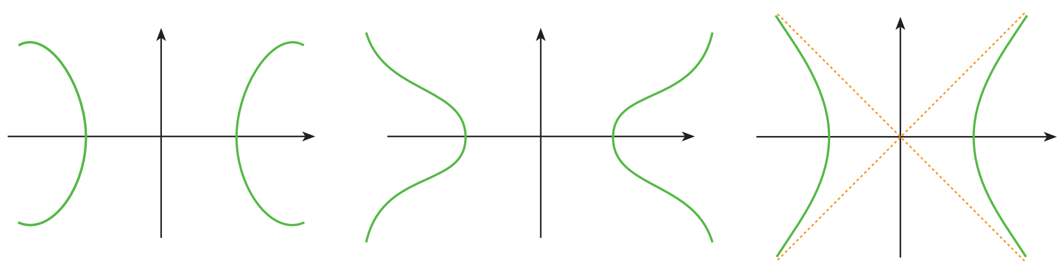
equation of the minor circle: $x^2 + y^2 = b^2$, i.e. $x^2 + y^2 = 36$

equations of the circles of the directrices: $(x - 2\sqrt{13})^2 + y^2 = 64$ and $(x + 2\sqrt{13})^2 + y^2 = 64$

equation of the directrices: $x = \pm\dfrac{a^2}{c}$, i.e. $x = \pm\dfrac{4^2}{2\sqrt{13}} = \pm\dfrac{8}{\sqrt{13}} = \pm\dfrac{8\sqrt{13}}{13}$

7. Asymptotes

To better understand the graph of a hyperbola we need to know the behavior of the graph at infinity. To define this behavior we use the term **asymptote**. An asymptote of a hyperbola is a line (or a curve) that the hyperbola approaches but never actually touches.



a. b. c.

If we do not know the behavior of the graph at infinity, we cannot decide which of the graphs a, b or c is the graph of a hyperbola. In fact, figure c shows a hyperbola and the shapes in figures a and b are not hyperbolas.

A hyperbola has two asymptotes. The equations of the asymptotes of a hyperbola are given by $y = \pm\dfrac{b}{a}x$. The yellow lines in figure c above are the asymptotes of the hyperbola.

Note

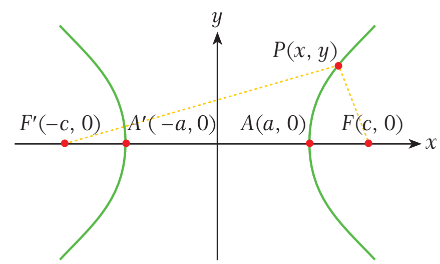
If the y-axis is the transverse axis of the hyperbola then the equation of the asymptotes does not change. It is still $y = \pm\dfrac{b}{a}x$.

B. EQUATION OF A HYPERBOLA

1. Equation of a Hyperbola Centered at the Origin

Distance formula
If $P(x_1, y_1)$ and $Q(x_2, y_2)$ are two points in a plane then the distance between P and Q is
$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Let $P(x, y)$ be a point on a hyperbola whose transverse axis is on the x-axis. Then P satisfies the equation $|PF' - PF| = 2a$, i.e. $PF' - PF = \pm 2a$ where $PF = \sqrt{(x-c)^2 + y^2}$ and $PF' = \sqrt{(x+c)^2 + y^2}$.



So $\sqrt{(x+c)^2 + y^2} - \sqrt{(x-c)^2 + y^2} = 2a$, i.e.

$\sqrt{(x+c)^2 + y^2} = 2a + \sqrt{(x-c)^2 + y^2}$

and taking the square of both sides gives us

$x^2 + 2cx + c^2 + y^2 =$

$$= 4a^2 + 4a\sqrt{(x-c)^2 + y^2} + x^2 - 2cx + c^2 + y^2.$$

If we cancel the common terms, take the square of both sides and move everything to the left-hand side we get

$xc - a^2 = a\sqrt{(x-c)^2 + y^2}$,

$x^2c^2 - 2xca^2 + a^4 - a^2x^2 + 2xca^2 - a^2c^2 - a^2y^2 = 0.$

Factoring gives

$$x^2(c^2 - a^2) - a^2(c^2 - a^2) - a^2y^2 = 0$$

$$b^2x^2 - a^2b^2 - a^2y^2 = 0 \qquad \text{(substitute } b^2 = c^2 - a^2\text{)}$$

$$b^2x^2 - a^2y^2 = a^2b^2.$$

Dividing both sides by a^2b^2 gives us $\boxed{\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1}$ or $\boxed{b^2x^2 - a^2y^2 = a^2b^2}$. This is the equation of a hyperbola whose transverse axis is on the x-axis.

Notice that when $y = 0$ we get $x_1 = -a$ and $x_2 = a$. These are the x-intercepts of the hyperbola. But when $x = 0$ the value of y is not defined. So we can say that the graph of a hyperbola with a transverse axis on the x-axis does not intercept the y-axis, but it intercepts the x-axis at the points $A(a, 0)$ and $A'(-a, 0)$.

Note

If the transverse axis of the hyperbola is on the y-axis then the equation of the hyperbola becomes $\boxed{\dfrac{y^2}{b^2} - \dfrac{x^2}{a^2} = 1}$ or $\boxed{a^2y^2 - b^2x^2 = a^2b^2}$. This hyperbola does not intercept the x-axis, and intercepts the y-axis at $B(0, b)$ and $B'(0, -b)$.

EXAMPLE 25 The lengths of the transverse and conjugate axes of a hyperbola are $AA' = 14$ cm and $BB' = 6$ cm respectively. Find the equation of the hyperbola.

Solution transverse axis: $2a = 14$ so $a = 7$

conjugate axis: $2b = 6$ so $b = 3$

So the equation of the hyperbola is

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, i.e. $\frac{x^2}{7^2} - \frac{y^2}{3^2} = 1$ which gives $\frac{x^2}{49} - \frac{y^2}{9} = 1$ or $9x^2 - 49y^2 = 441$.

EXAMPLE 26 Find the lengths of the transverse and conjugate axes of the hyperbola defined by $\frac{x^2}{121} - \frac{y^2}{36} = 1$.

Solution $a^2 = 121$, $a = 11$

$b^2 = 36$, $b = 6$

length of the transverse axis: $2a = 2 \cdot 11 = 22$

length of the conjugate axis: $2b = 2 \cdot 6 = 12$

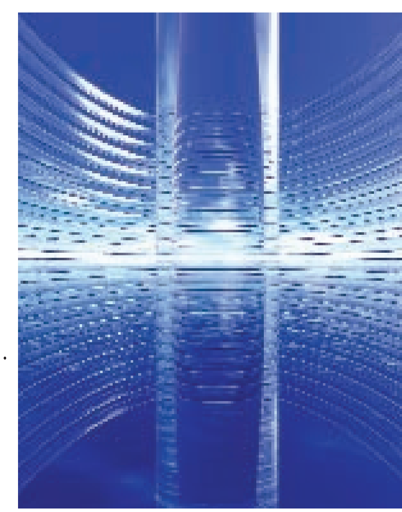
EXAMPLE 27 $\frac{x^2}{144} - \frac{y^2}{25} = 1$ is the equation of a hyperbola. Find

a. the lengths of the axes.

b. the eccentricity.

c. the coordinates of the vertices and the foci.

d. the equations of the directrices and the asymptotes.



Solution $a^2 = 144$ so $a = 12$; $b^2 = 25$ so $b = 5$.

$a^2 + b^2 = c^2$ means $12^2 + 5^2 = c^2$ so $c^2 = 169$ and $c = 13$.

a. length of the transverse axis: $2a = 2 \cdot 12 = 24$

length of the conjugate axis: $2b = 2 \cdot 5 = 10$

b. eccentricity: $e = \frac{c}{a} = \frac{13}{12}$

c. vertices: $A(12, 0)$, $A'(-12, 0)$; foci: $F(13, 0)$, $F'(-13, 0)$

d. directrices: $x = \pm\frac{a^2}{c} = \pm\frac{144}{13}$, asymptotes: $y = \pm\frac{b}{a}x = \pm\frac{5}{12}x$

EXAMPLE 28 The equation of a hyperbola is $\frac{y^2}{4} - x^2 = 1$. Find

a. the lengths of the axes.

b. the eccentricity.

Solution The transverse axis is the y-axis because x^2 has a negative coefficient in the equation.

$b^2 = 4$ so $b = 2$; $a^2 = 1$ so $a = 1$.

$a^2 + b^2 = c^2$ means $1^2 + 2^2 = c^2$ so $c^2 = 5$ and $c = \sqrt{5}$.

a. length of the transverse axis: $2b = 2 \cdot 2 = 4$

length of the conjugate axis: $2a = 2 \cdot 1 = 2$

b. eccentricity: $e = \frac{c}{b} = \frac{\sqrt{5}}{2}$

EXAMPLE 29 The center of a hyperbola is at the origin and its foci are on the x-axis. Given that the hyperbola passes through the points $M(3, 1)$ and $N(4, -2)$, find the equation of the hyperbola.

Solution The equation of the hyperbola will be in the form $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If the points M and N are on this hyperbola then they satisfy this equation.

For $M(3, 1)$ we have $\frac{9}{a^2} - \frac{1}{b^2} = 1$, and for $N(4, -2)$ we have $\frac{16}{a^2} - \frac{4}{b^2} = 1$.

Let us solve this system:

$$\begin{array}{lll} -4/ \; \frac{9}{a^2} - \frac{1}{b^2} = 1 & & -\frac{36}{a^2} + \frac{4}{b^2} = -4 \quad (1) \\ & \text{gives} & \\ \frac{16}{a^2} - \frac{4}{b^2} = 1 & & \frac{16}{a^2} - \frac{4}{b^2} = 1. \quad (2) \end{array}$$

Combining these gives us $a^2 = \frac{20}{3}$.

Substituting this in (1) gives us

$$\frac{9}{\frac{20}{3}} - \frac{1}{b^2} = 1, \text{ i.e. } \frac{1}{b^2} = \frac{27}{20} - 1 = \frac{7}{20} \text{ and } b^2 = \frac{20}{7}.$$

So the equation of the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, i.e. $\frac{x^2}{\frac{20}{3}} - \frac{y^2}{\frac{20}{7}} = 1$ or $\frac{3x^2}{20} - \frac{7y^2}{20} = 1$ or $3x^2 - 7y^2 = 20$.

EXAMPLE 30 The equation $x^2 - y^2 = 64$ defines an isosceles hyperbola. Find the distance between the foci of the hyperbola and the positive angles between the asymptotes of the hyperbola and the transverse axis.

Solution The equation of the hyperbola is $x^2 - y^2 = 64$ which means $\frac{x^2}{64} - \frac{y^2}{64} = 1$. So $a = b = 8$.

The equation $a^2 + b^2 = c^2$ gives us $c = 8\sqrt{2}$. So the distance between the foci is $16\sqrt{2}$.

The equations of the asymptotes are $y = \pm x$. So the slope of the line ($\tan \alpha$) is 1 or –1 and therefore the angles between the asymptotes and the transverse axis are 45° and 135°.

Check Yourself 4

1. The lengths of the transverse and conjugate axes of a hyperbola are $AA' = 12$ cm and $BB' = 8$ cm respectively. What is the equation of the hyperbola?
2. Find the lengths of the transverse and conjugate axes of the hyperbola defined by $64x^2 - 25y^2 = 1600$.
3. $\frac{x^2}{36} - \frac{y^2}{64} = 1$ is the equation of a hyperbola. Find
   a. the lengths of the axes. b. the eccentricity.
   c. the distance between the foci.
   d. the equations of the directrices and the asymptotes.
4. $4y^2 - 18x^2 = 1$ is the equation of a hyperbola. Find
   a. the lengths of the axes. b. the eccentricity.
5. A vertical line is drawn through one of the foci of the hyperbola defined by $\frac{x^2}{9} - 4y^2 = 1$.
   What is the distance between the intersection points of the line and the hyperbola?

Answers

1. $\frac{x^2}{36} - \frac{y^2}{16} = 1$
2. transverse axis = 10, conjugate axis = 16
3. a. transverse axis = 12, conjugate axis = 16 b. $e = \frac{5}{3}$ c. 20
   d. directrices: $x = \pm\frac{18}{5}$, asymptotes: $y = \pm\frac{4}{3}x$
4. a. transverse axis = 1, conjugate axis = $\frac{\sqrt{2}}{3}$ b. $e = \frac{\sqrt{11}}{3}$
5. $\frac{1}{6}$

MENAECHMUS (380-320 BC)

Menaechmus was a pupil of Eudoxus (408-355 BC) and studied with Plato.

He was famous for his discovery of conic sections and was the first to show that the ellipse, the parabola and the hyperbola can be obtained by intersecting a plane with a pair of circular cones placed vertex to vertex.

In Menaechmus' time there were three well-known problems for geometers: duplicating a cube, trisecting an angle and squaring a circle. Menaechmus also tried to solve these problems, and discovered conic sections while he was attempting to solve the problem of duplicating a cube.

EUCLID OF ALEXANDRIA (325-265 BC)

Euclid of Alexandria is one of the best-known mathematicians of the ancient world. His most famous work is his book *The Elements*, which contains almost all the mathematical knowledge of the time, organized in thirteen sections.

Euclid studied at Academia, a school in Athens. Then he became a teacher. He is known as the father of geometry.

Geometry had its beginnings in ancient Egypt. Long before Euclid's time, Egyptian land surveyors had to recalculate land boundaries every year when the Nile burst its banks. They used a kind of geometry, but in the beginning there were no theorems, proofs or common rules. Euclid collected the rules and theorems of geometry from Thales, Pythagoras and Eudoxus. Then he proved some of the theorems in his book.

Euclid organised his information into five postulates and five axioms. Today we still use these postulates and axioms.

He wrote a book about conics, but this book has been lost.

ARCHIMEDES (278-212 BC)

Archimedes has been accepted as one of the three greatest scientists of all time. He was a mathematician, a physicist and a philosopher and invented a lot of interesting devices. He used lenses to focus light on enemy ships to burn them. He also invented pulleys, the hydraulic screw and infinite screws.

In geometry, Archimedes calculated formulas for the area and the volume of a sphere. He calculated the number π to be between $3\frac{1}{7}$ and $3\frac{10}{71}$. While he was trying to square a parabola he used the concept of infinitely small parts, which formed the basis of differential integrals.

Archimedes studied equilibrium and said, 'If you can give me a suitable base then I can move the Earth.' He also discovered the rules of the lifting power of water.

In 212 BC The Romans conquered Syracus. At the time, Archimedes was working on a problem so he did not escape and a Roman soldier killed him.

EXERCISES 1.2

A. Basic Elements of a Hyperbola

1. The vertices of a hyperbola are $A(4, 0)$ and $A'(-4, 0)$. Given that the length of the conjugate axis is 6 cm, find the eccentricity and the coordinates of the foci.

2. The vertices of a hyperbola are $A(3, 0)$ and $A'(-3, 0)$. Given that one of the foci is $F(6, 0)$, find the length of the conjugate axis and the eccentricity.

3. The transverse axis of a hyperbola lies on the x–axis and has length 10 cm. If the length of conjugate axis is 6 cm, find the equations of the circles and the directrices of this hyperbola.

4. A hyperbola has a focus at $F(6, 0)$ and its eccentricity is 3. Find the lengths of the transverse and conjugate axes of the hyperbola and the coordinates of its vertices.

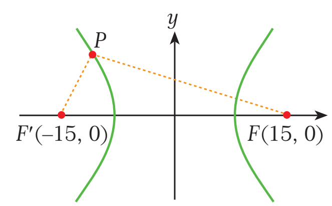
5. The eccentricity of a hyperbola is $\frac{6}{5}$ and the length of its transverse axis along the x-axis is 20 cm. Find the length of the conjugate axis, the distance between the foci and the equations of the directrices.

6. A hyperbola has a focus at $F(9, 0)$ and eccentricity $\frac{3}{2}$. Find the equations of the asymptotes and the directrices.

7. The transverse axis of a hyperbola lies along the x-axis. Given that the lengths of the transverse and conjugate axes are 4 cm and 6 cm respectively, find
   a. the eccentricity.
   b. the equations of the asymptotes.

8. The equations of the asymptotes of a hyperbola are $y = \pm\frac{12}{5}x$. If the distance between the foci is 78 cm, calculate the lengths of the transverse and conjugate axes of the hyperbola.

9. The figure shows a hyperbola with foci $F(15, 0)$ and $F'(-15, 0)$.



If $PF' = 3$ cm and $PF = 27$ cm, find the lengths of the axes of the hyperbola and its eccentricity.

10. One of the foci of a hyperbola is $F(0, 25)$. Given that the length of the conjugate axis is 14 cm, find the length of the transverse axis and the eccentricity.

11. The vertices of a hyperbola are $B(0, 12)$ and $B'(0, -12)$ and the length of the conjugate axis is 10 cm. Find
    a. the coordinates of the foci.
    b. the eccentricity.
    c. the equation of the directrices.

12. The transverse axis of a hyperbola lies along the y-axis and the lengths of the transverse and conjugate axes are 8 cm and 20 cm respectively. Find the eccentricity of the hyperbola and the equation of its directrices.

B. Equation of a Hyperbola

13. The lengths of the transverse and conjugate axes of a hyperbola are $AA' = 8$ cm and $BB' = 12$ cm respectively. Find the equation of this hyperbola.

14. Find the lengths of the transverse and conjugate axes of the hyperbola whose equation is $\frac{x^2}{144} - \frac{y^2}{64} = 1$.

15. $\frac{x^2}{64} - \frac{y^2}{16} = 1$ is the equation of a hyperbola. Find
    a. the lengths of the axes.
    b. the eccentricity.
    c. the equations of the directrices and the asymptotes.

16. $9x^2 - 49y^2 = 441$ is the equation of a hyperbola. Find the length of each axis of the hyperbola and the distance between its foci.

17. Find the lengths of the axes and the eccentricity of the hyperbola defined by $16x^2 - y^2 = 16$.

18. $81y^2 - 9x^2 = 1$ is the equation of a hyperbola. Find
    a. the lengths of the axes.
    b. the equation of the directrices.

19. The equation $x^2 - y^2 = 49$ defines an isosceles hyperbola. Find the distance between the foci of the hyperbola and the positive angle between the asymptotes of the hyperbola and the transverse axis.

20. The equation of a hyperbola is $\frac{y^2}{9} - \frac{x^2}{16} = 1$. Find
    a. the lengths of the axes.
    b. the eccentricity.

21. A hyperbola centered at the origin has its foci on the x-axis. Given that the hyperbola passes through the points $M(2\sqrt{3}, 3\sqrt{2})$ and $N(4, 3\sqrt{3})$, find the equation of the hyperbola.

22. A vertical line is drawn through one of the foci of the hyperbola $\frac{x^2}{16} - \frac{y^2}{36} = 1$. What is the distance between the two intersection points of the line and the hyperbola?

3 PARABOLAS

A. BASIC ELEMENTS OF A PARABOLA

1. Parabola and Focus

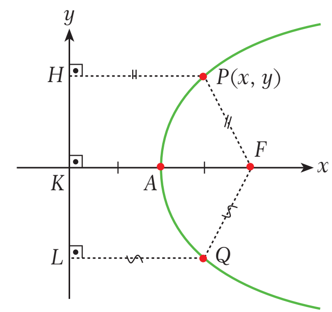
Definition

parabola, focus, directrix

A **parabola** is a set of points in a plane that are equidistant from a fixed point and a fixed line.

The fixed point is called the **focus** of the parabola and the fixed line is the **directrix** of the parabola.

In the figure opposite, P, Q and A are points on the parabola. F is the focus and HL lies on the directrix. By the definition of a parabola, $PF = PH$, $AF = AK$ and $QF = QL$.



2. Axis and Vertex

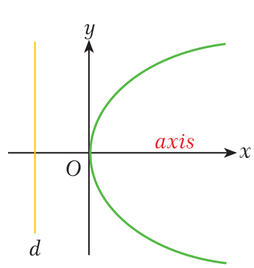
The point on a parabola which is closest to the directrix is called the **vertex** of the parabola. In the figure above, A is the vertex of the parabola.

The line perpendicular to the directrix which passes throught the focus is called the **axis** of the parabola. KF is the axis of the parabola above. Notice that we can also define the vertex as the intersection point of the parabola and its axis.

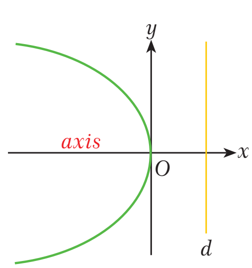
Since $AF = AK$ by the definition of a parabola, the vertex of a parabola lies at the midpoint of its axis. Any parabola is symmetric about its axis.

Note

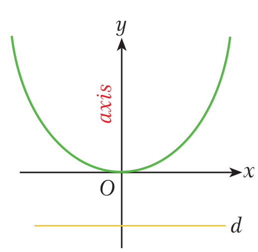
In this book we generally take the x-axis or y-axis as the axis of the parabola. There are four basic types of parabola:



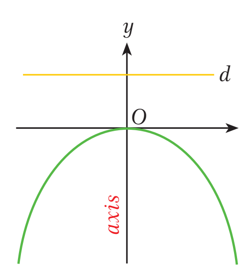
opens to the right



opens to the left



opens upward



opens downward

3. Parameter

The length of the line segment between the focus and the directrix of a parabola is called the **parameter** of the parabola, denoted by $\boldsymbol{p}$.

In the figure at the start of this section, $p = FK$ and $FA = AK = \dfrac{p}{2}$.

4. Eccentricity

ellipse: $0 < e < 1$
parabola: $e = 1$
hyperbola: $e > 1$

The eccentricity of a parabola is the ratio of the distance between a point on a parabola and its focus to the distance between the same point and the directrix.

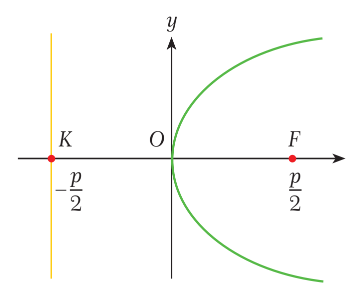
In the figure at the start of this section, the eccentricity is $\dfrac{PF}{PH}$. But we know from the definition of a parabola that $PF = PH$, so $e = 1$.

In fact, we can see that the eccentricity of any parabola is 1 by simply considering the definition of a parabola. So we can define a parabola in an alternative way as a conic section whose eccentricity is 1.

5. Directrix

When we define a parabola we look at its vertex and focus. To make our calculations easier, we often place the vertex of the parabola at the origin.

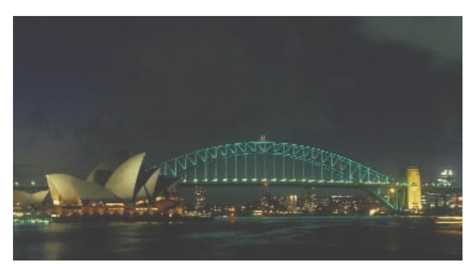
In the figure, $O(0, 0)$ is the vertex of the parabola.



We know that $FK = p$ so $KO = OF = \dfrac{p}{2}$. So the focus of the parabola is at $F(\dfrac{p}{2}, 0)$ and the equation of the directrix is $x = -\dfrac{p}{2}$.

Depending on the type of parabola (opening to the left, right, upward or downward), we can see that the four possible equations of the directrix are $x = \pm\dfrac{p}{2}$ and $y = \pm\dfrac{p}{2}$.

EXAMPLE 31 The focus of a parabola is $F(3, 0)$ and its vertex is at the origin. Find the equation of the directrix.



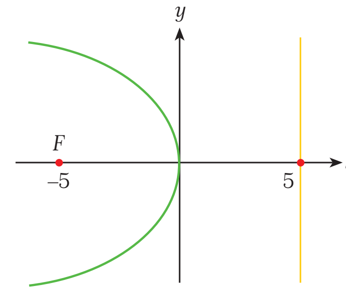
Solution The parabola opens to the right.

$F(\dfrac{p}{2}, 0) = F(3, 0)$ so $\dfrac{p}{2} = 3$.

So the equation of the directrix is $x = -3$.

EXAMPLE 32 Find the focus and eccentricity of the parabola whose vertex is at the origin and whose directrix is $x = 5$.

Solution By considering the position of the directrix, we can see that the parabola opens to the left.



The focus is $F(\frac{p}{2}, 0)$ and the directrix is $x = -\frac{p}{2}$ so $\frac{p}{2} = 5$.

So the focus is $F(–5, 0)$.

The eccentricity of a parabola is always 1.

Note

When solving problems like this, we look at the position of the focus and then write – or + because the parameter is always positive.

EXAMPLE 33 Find the parameter of the parabola with focus $F(0, –4)$ whose vertex is at the origin.

Solution The focus is $F(0, -\frac{p}{2})$ or $F(0, \frac{p}{2})$. We know p is the distance between the focus and the directrix and this distance cannot be negative, so we have $\frac{p}{2} = 4$ and so $p = 8$.

Check Yourself 5

1. $F(2, 0)$ is the focus of a parabola whose vertex is at the origin. Find the parameter p of the parabola and the equation of its directrix d.
2. The equation of the directrix of a parabola is $y = –7$. Find the focus F and parameter p of the parabola if its vertex is at the origin.
3. A parabola whose vertex is at the origin has parameter $p = 8$. Find the focus F and the equation of the directrix d if the axis of the parabola lies along the x-axis and the focus is on the positive side of this axis.

Answers

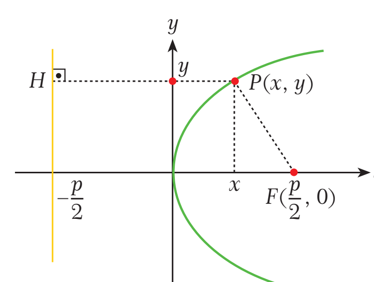
1. $p = 4$, d: $x = –2$ 2. $F(0, 7)$, $p = 14$ 3. $F(4, 0)$, d: $x = –4$

B. EQUATION OF A PARABOLA

1. Equation of a Parabola with Vertex at the Origin

Let $P(x, y)$ be a point on a parabola whose vertex is at the origin and which lies on the positive side of the x-axis. Then P satisfies the equation of the parabola.

Look at the figure. By the distance formula we get



$$PF = \sqrt{(x-\frac{p}{2})^2 + y^2} \text{ and } PH = x + \frac{p}{2}.$$

But $PF = PH$ (by the definition of a parabola), so

$$\sqrt{(x-\frac{p}{2})^2 + y^2} = x + \frac{p}{2}$$

$$(x-\frac{p}{2})^2 + y^2 = x^2 + px + \frac{p^2}{4} \quad \text{(take the square of both sides)}$$

$$x^2 - px + \frac{p^2}{4} + y^2 = x^2 + px + \frac{p^2}{4}. \quad \text{(expand)}$$

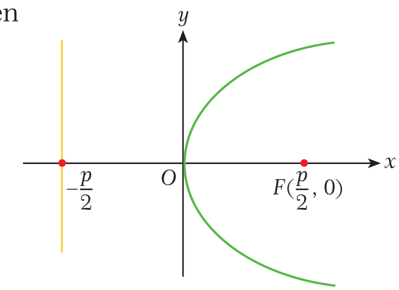
So $y^2 = 2px$ is the equation of the parabola in the figure. It is also the equation of any parabola which lies on the positive side of the x-axis. Let us now summarize the results for all four different possible positions of a parabola whose vertex is at the origin.

1. If the parabola lies on the positive side of the x-axis then we have

   focus: $F(\frac{p}{2}, 0)$

   directrix: $x = -\frac{p}{2}$

   equation: $y^2 = 2px$.

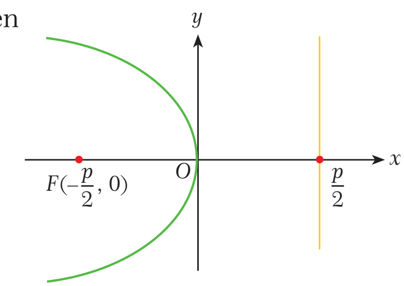


2. If the parabola lies on the negative side of the x-axis then we have

   focus: $F(-\frac{p}{2}, 0)$

   directrix: $x = \frac{p}{2}$

   equation: $y^2 = -2px$.

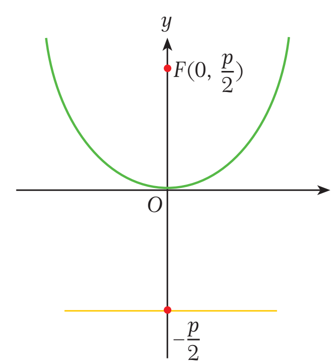


3. If the parabola lies on the positive side of the y-axis then we have

   focus: $F(0, \frac{p}{2})$

   directrix: $y = -\frac{p}{2}$

   equation: $x^2 = 2py$.

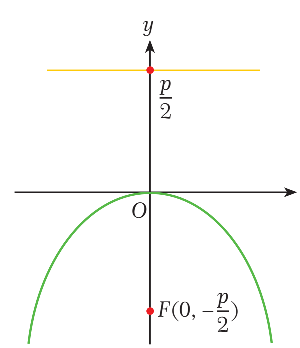


4. If the parabola lies on the negative side of the y-axis then we have

   focus: $F(0, -\frac{p}{2})$

   directrix: $y = \frac{p}{2}$

   equation: $x^2 = -2py$.



EXAMPLE 34 Find the equation of the parabola whose vertex is at the origin and focus is $F(5, 0)$.

Solution $\frac{p}{2} = 5$ so $p = 10$.

If the focus is $F(5, 0)$ then the parabola lies on the positive side of the x-axis and so its equation is of the form $y^2 = 2px$.

So the equation is $y^2 = 2 \cdot 10 \cdot x = 20x$.

EXAMPLE 35 Find the equation of the parabola which has its vertex at the origin and each additional component.

a. focus $F(4, 0)$

b. focus $F(0, -5)$

c. focus $F(-\frac{1}{2}, 0)$

d. directrix d: $x - 2 = 0$

e. directrix d: $3y + 1 = 0$

f. directrix d: $x = -\frac{1}{4}$

Solution We can examine the focus or the directrix to determine where the parabola lies in the plane.

a. The focus is of the form $F(\frac{p}{2}, 0)$ so the parabola lies on the right side of the y-axis and gives $\frac{p}{2} = 4$, $p = 8$. So the equation is $y^2 = 2px = 16x$.

b. The focus is of the form $F(0, -\frac{p}{2})$ so the parabola is below the x-axis and $\frac{p}{2} = 5$, $p = 10$. The equation is $x^2 = -2py = -20y$.

c. $F(-\frac{p}{2}, 0)$ is on the left of the y-axis and $\frac{p}{2} = \frac{1}{2}$, $p = 1$. The equation is $y^2 = -2px = -2x$.

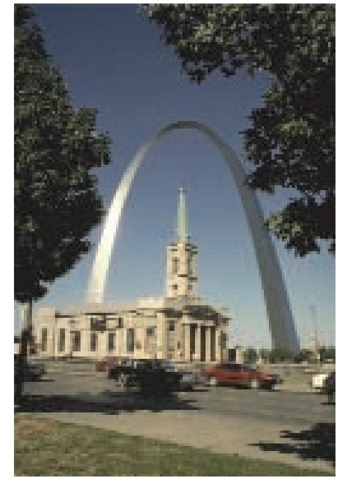
d. The directrix is $x = 2$ so the parabola is on the left of the y-axis, $\frac{p}{2} = 2$ and $p = 4$. equation: $y^2 = -2px = -8x$

e. directrix: $y = -\frac{1}{3}$ so $\frac{p}{2} = \frac{1}{3}$ and $p = \frac{2}{3}$; equation: $x^2 = 2py = \frac{4}{3}y$

f. directrix: $x = -\frac{1}{4}$ so $\frac{p}{2} = \frac{1}{4}$ and $p = \frac{1}{2}$; equation: $y^2 = 2px = x$

EXAMPLE 36 Find the foci and the directrix of each parabola given that its vertex is at the origin.

a. $x^2 = 4y$ b. $x^2 = -\frac{1}{6}y$ c. $y^2 = 6x$ d. $y^2 = -x$



Solution a. $x^2 = 4y$ means $p = 2$, $\frac{p}{2} = 1$ and the axis of the parabola is the y-axis.

focus: $F(0, \frac{p}{2}) = F(0, 1)$, directrix: $y = -\frac{p}{2} = -1$

b. $x^2 = -\frac{1}{6}y$ means $p = \frac{1}{12}$, $\frac{p}{2} = \frac{1}{24}$ and the axis of the parabola is the y-axis.

focus: $F(0, -\frac{p}{2}) = F(0, -\frac{1}{24})$, directrix: $y = \frac{p}{2} = \frac{1}{24}$

c. $y^2 = 6x$ means $p = 3$, $\frac{p}{2} = \frac{3}{2}$ and the axis of the parabola is the x-axis.

focus: $F(\frac{p}{2}, 0) = F(\frac{3}{2}, 0)$, directrix: $y = -\frac{p}{2} = -\frac{3}{2}$

d. $y^2 = -x$ means $p = \frac{1}{2}$, $\frac{p}{2} = \frac{1}{4}$ and the axis of the parabola is the x-axis.

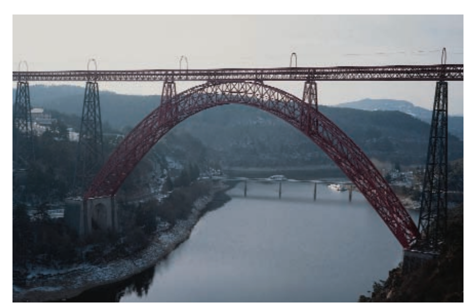
focus: $F(-\frac{p}{2}, 0) = F(-\frac{1}{4}, 0)$, directrix: $y = \frac{p}{2} = \frac{1}{4}$

EXAMPLE 37 $P(3, a)$ is a point on a parabola which has its vertex at the origin and directrix $y = -3$. Find a.

Solution The directrix is $y = -3$ so $\frac{p}{2} = 3$, $p = 6$ and the equation is $x^2 = 2py$, i.e. $x^2 = 12y$.

If $P(3, a)$ is on the parabola then it must satisfy the equation of the parabola. So $3^2 = 12 \cdot a$ and so $a = \frac{9}{12} = \frac{3}{4}$.

EXAMPLE 38 The vertex of a parabola is at the origin of the coordinate plane and $P(1, 2)$ is a point on the parabola. Find the equation and focus of this parabola if its focus is on the positive side of the x-axis.



Solution If the focus is on the positive side of the x-axis then the equation is of the form $y^2 = 2px$.

The point $P(1, 2)$ must satisfy this equation,

so $2^2 = 2 \cdot p \cdot 1$ and $p = 2$.

So the equation is $y^2 = 4x$ and the focus is $F(\frac{p}{2}, 0) = F(1, 0)$.

Check Yourself 6

1. Find the equation of the parabola which has its vertex at the origin and each additional component.

   a. focus $F(1, 0)$
   b. focus $F(0, -6)$
   c. focus $F(-2, 0)$
   d. focus $F(0, 4)$
   e. directrix d: $2x + 3 = 0$
   f. directrix d: $y = -1$
   g. directrix d: $4y - 5 = 0$
   h. directrix d: $x = \frac{2}{5}$

2. Find the focus F and the directrix d of each parabola if its vertex is at the origin.

   a. $x^2 = y$
   b. $x^2 = -\frac{2}{3}y$
   c. $y^2 = \frac{1}{2}x$
   d. $y^2 = -4x$
   e. $4x^2 - 5y = 0$

3. The vertex of a parabola is at the origin and $Q(3, -2)$ is a point on the parabola. Find the equation and focus of this parabola if the focus is on the negative side of the y-axis.

4. The equation of a parabola is $y^2 = -8x$. A line parallel to the y-axis is drawn through the focus of the parabola. Find the distance between the two points at which the parabola intersects this line.

Answers

1. a. $y^2 = 4x$ b. $x^2 = -24y$ c. $y^2 = -8x$ d. $x^2 = 16y$
   e. $y^2 = 6x$ f. $x^2 = 4y$ g. $x^2 = -5y$ h. $y^2 = -\frac{8}{5}x$

2. a. $F(0, \frac{1}{4})$, d: $y = -\frac{1}{4}$ b. $F(0, -\frac{1}{6})$, d: $y = \frac{1}{6}$ c. $F(\frac{1}{8}, 0)$, d: $x = -\frac{1}{8}$
   d. $F(-1, 0)$, d: $x = 1$ e. $F(0, \frac{5}{16})$, d: $y = -\frac{5}{16}$

3. equation: $x^2 = -\frac{9}{2}y$, focus: $F(0, -\frac{9}{8})$

4. 8

2. General Equation of a Parabola in the Plane (*OPTIONAL*)

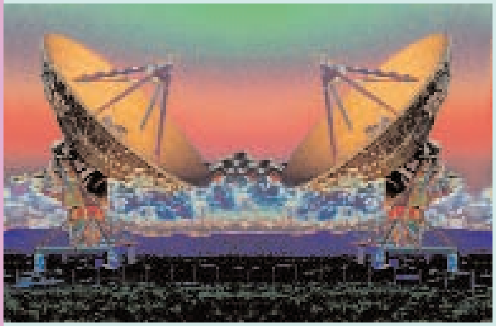
We have so far learned the possible equations of a parabola whose vertex is at the origin. How does the formula change if the vertex is not at the origin but at the point $A(h, k)$?

In this case the axis of the parabola will be parallel to the x-axis or the y-axis and the form of the equation of the parabola will change.

Let $A(h, k)$ be the vertex of a parabola and let us look at each possible case in turn.

PARABOLIC REFLECTORS

People all over the world today use digital satellite receivers and antennas in their houses. If you use a mobile phone, your service provider will also use receivers on its antenna masts. But do you know how these devices work?

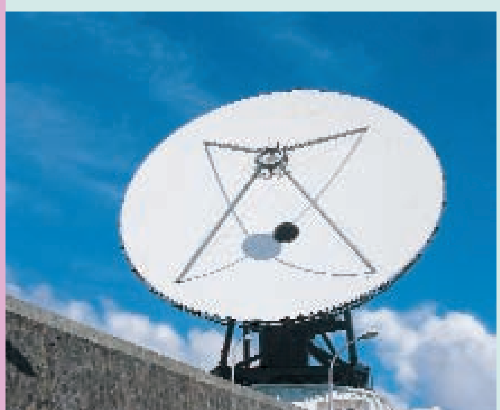


Parabolas have useful reflective properties. A *parabolic reflector* (also called a parabolic mirror) is a dish shaped like a parabola, with a receiver at its focus. Parabolic reflectors are used to collect and distribute radio, sound and light waves. They are designed so that any wave traveling towards the reflector which is parallel to the parabola's axis will be reflected to the parabola's focus. Similarly, any wave emitted from the parabola's focus will be reflected back parallel to the parabola's axis.

Satellite antennas use this principle to emit and gather radio waves and microwaves. They are often used for sending and receiving radio and telephone communications. These communications are transmitted at high frequencies. The satellite antennas have a special conversion device at their focus. This device receives the incoming high-frequency waves, converts them to a lower frequency and transmits them to an indoor receiver by cable.

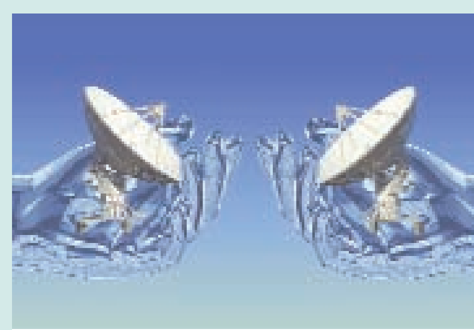
Parabolic reflectors are used in satellite links, radio astronomy, microwave links, radio relay towers and mobile phone antenna masts. These applications require high levels of gain to receive the incoming signals.

Some antennas use a different system, called a *Cassegrain reflector* system, to feed the antenna. In this system, the waves hit the center of a parabolic reflector from the center of the parabola and are then reflected back. Cassegrain reflectors make it easier to receive and send wawes efficiently.



Parabolic reflectors are also used to reflect light beams. If a light source is placed at the focus of a parabolic reflector, the light will be reflected parallel to the axis. In this way, we can get a straight beam of light. Spotlights and car headlights both use parabolic mirrors in this way.

In 1721, the astronomer John Hadley invented a telescope which used parabolic reflectors. Before Hadley, telescopes had used spherical mirrors, which do not always reflect rays to the same focus. Hadley's design made it easier to reflect rays to a single focus, and so allowed astronomers to observe light sources more clearly.



EXERCISES 1.3

A. Basic Elements of a Parabola

1. $F(5, 0)$ is the focus of a parabola whose vertex is at the origin. Find the equation of its directrix.

2. Find the focus of the parabola whose vertex is at the origin and whose directrix is $x = 3$.

3. Find the parameter of the parabola whose focus is $F(0, -6)$ and whose vertex is at the origin.

4. A parabola whose vertex is at the origin has parameter $p = 6$. Find the focus F and the equation of the directrix d, given that the axis of the parabola lies along the x-axis and the focus is on the positive side of this axis.

B. Equation of a Parabola

5. In each case, find the equation of the parabola with a vertex at the origin and focus F.

   a. $F(6, 0)$ b. $F(0, -2)$

   c. $F(-4, 0)$ d. $F(0, 8)$

6. In each case, find the equation of the parabola with a vertex at the origin and the given directrix.

   a. $2x - 3 = 0$ b. $5y - 2 = 0$

   c. $x = -\sqrt{3}$ d. $y = -\dfrac{3}{4}$

7. Find the focus F and the equation of the directrix d of each parabola if its vertex is at the origin.

   a. $y^2 = 6x$ b. $y^2 = -\dfrac{2}{3}x$

   c. $5y^2 - 3x = 0$ d. $x^2 = -36y$

   e. $x^2 = -y$ f. $2x^2 - 9y = 0$

8. A parabola whose vertex is at the origin has directrix $x = -2$ and passes through the point $P(4, a)$. Find a.

9. The vertex of a parabola is at the origin and $P(3, 4)$ is a point on the parabola. Find the equation and focus of this parabola if its focus is on the positive side of the y-axis.

10. The equation of a parabola is $x^2 = 12y$. A line parallel to the x-axis is drawn through the focus of this parabola. Find the distance between the two points at which the parabola intersects this line.

11. The equation of a parabola is $y^2 = 10x$. A line parallel to the y-axis is drawn through the focus of this parabola. Find the distance between the two points at which the parabola intersects this line.

CHAPTER REVIEW TEST 1A

1. An ellipse has foci $F(6, 0)$ and $F'(-6, 0)$ and its major axis measures 20 cm. Find the length of its minor axis.

A) 6 cm B) 8 cm C) 12 cm D) 16 cm E) 18 cm

2. The lengths of the major and minor axes of an ellipse are 13 cm and 5 cm respectively. Which point is a possible focus?

A) $F(3, 0)$ B) $F(5, 0)$ C) $F(6, 0)$ D) $F(12, 0)$ E) $F(0, 18)$

3. The vertices of an ellipse are $A(4, 0)$, $A'(-4, 0)$, $B(0, 3)$ and $B'(0, -3)$. What is its eccentricity?

A) $\frac{3}{4}$ B) $\frac{5}{4}$ C) $\frac{4}{5}$ D) $\frac{\sqrt{7}}{3}$ E) $\frac{\sqrt{7}}{4}$

4. An ellipse has eccentricity $e = \frac{3}{5}$ and its minor axis measures 24 cm. What is the length of its major axis?

A) 5 cm B) 10 cm C) 15 cm D) 24 cm E) 30 cm

5. The lengths of the major and minor axes of an ellipse are 14 cm and 10 cm respectively. What is its equation?

A) $25x^2 + 49y^2 = 1225$
B) $100x^2 + 196y^2 = 1$
C) $25x^2 + 49y^2 = 1$
D) $100x^2 + 196y^2 = 19600$
E) $5x^2 + 7y^2 = 35$

6. An ellipse is defined by the equation $\frac{x^2}{9} + \frac{y^2}{81} = 1$. Find the sum of the lengths of its major and minor axes.

A) 6 B) 12 C) 24 D) 90 E) 180

7. An ellipse is defined by $x^2 + 9y^2 = 9$. What is the eccentricity of this ellipse?

A) $\frac{\sqrt{3}}{2}$ B) $\frac{2\sqrt{2}}{3}$ C) $\frac{\sqrt{2}}{3}$ D) $\frac{1}{3}$ E) $\frac{\sqrt{10}}{3}$

8. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is the equation of an ellipse with $\frac{c}{a} = \frac{4}{5}$ and $a - b = 4$. What is the value of a?

A) 3 B) 4 C) 6 D) 10 E) 15

9. Which point is a focus of the ellipse defined by $9x^2 + 25y^2 = 1$?

A) $F(0, \frac{9}{25})$ B) $F(\frac{4}{5}, 0)$ C) $F(0, \frac{4}{5})$ D) $F(\frac{4}{15}, 0)$ E) $F(4, 0)$

10. Find the area between the major circle of the ellipse $\frac{x^2}{18} + \frac{y^2}{50} = 1$ and the ellipse itself.

A) 20π B) 32π C) 24π D) 30π E) 68π

11. The foci of a hyperbola are $F(6, 0)$ and $F'(-6, 0)$ and the length of its transverse axis is 8 cm. Find the length of its conjugate axis.

A) 12 cm B) $4\sqrt{7}$ cm C) $\sqrt{3}$ cm D) $2\sqrt{5}$ cm E) $4\sqrt{5}$ cm

12. The lengths of the transverse and conjugate axes of a hyperbola are 8 cm and 10 cm respectively. Find the distance between the foci.

A) 12 cm B) $2\sqrt{41}$ cm C) 6 cm D) $\sqrt{41}$ cm E) 18 cm

13. A hyperbola has vertices $B(0, 4)$ and $B'(0, -4)$. Given that the length of its conjugate axis is 12 cm, find the eccentricity of this hyperbola.

A) $\frac{\sqrt{13}}{2}$ B) $\sqrt{17}$ C) $\frac{2\sqrt{5}}{5}$ D) $\sqrt{5}$ E) $\sqrt{13}$

14. A hyperbola has eccentricity 3 and the length of its conjugate axis is 16 cm. Find the length of its transverse axis.

A) $6\sqrt{2}$ cm B) $2\sqrt{3}$ cm C) 6 cm D) $4\sqrt{2}$ cm E) $8\sqrt{2}$ cm

15. The lengths of the transverse and conjugate axes of a hyperbola are 16 cm and 20 cm respectively. What is a possible equation for this hyperbola?

A) $\frac{x^2}{16}-\frac{y^2}{20}=1$ B) $\frac{x^2}{64}-\frac{y^2}{100}=1$ C) $\frac{x^2}{8}-\frac{y^2}{10}=1$ D) $\frac{x^2}{196}-\frac{y^2}{400}=1$ E) $\frac{x^2}{16}-\frac{y^2}{36}=1$

16. A hyperbola centered at the origin has eccentricity $\frac{4}{3}$ and a focus at $F(4, 0)$. Find the equation of the hyperbola.

A) $\frac{x^2}{3}-\frac{y^2}{5}=1$ B) $\frac{x^2}{9}-\frac{y^2}{49}=1$ C) $\frac{x^2}{3}-\frac{y^2}{5}=1$ D) $\frac{x^2}{9}-\frac{y^2}{7}=1$ E) $\frac{x^2}{9}-\frac{y^2}{25}=1$

17. $\frac{y^2}{144}-\frac{x^2}{100}=1$ is the equation of a hyperbola. What is its eccentricity?

A) 2 B) $\frac{\sqrt{61}}{6}$ C) $\frac{6}{5}$ D) $\frac{36}{25}$ E) $\frac{12\sqrt{11}}{12}$

18. Which of the following is the equation of a hyperbola?

A) $3x^2 + 5y^2 = 1$ B) $4x^2 - 9y^2 = 0$ C) $16x^2 = 9y^2$ D) $x^2 = 36y$ E) $y^2 - 8x^2 = 1$

19. Which equation describes a directrix of the hyperbola $9x^2 - 4y^2 = 36$?

A) $x=\frac{4\sqrt{13}}{13}$ B) $y=\frac{\sqrt{13}}{13}$ C) $x=\frac{\sqrt{13}}{13}$ D) $x=\frac{2\sqrt{5}}{5}$ E) $x=\frac{9\sqrt{13}}{13}$

20. Which equation defines the asymptotes of the hyperbola $\frac{x^2}{100}-\frac{y^2}{36}=1$?

A) $y=\pm\frac{1}{2}x$ B) $y=\pm\frac{2}{3}x$ C) $y=\pm\frac{3}{4}x$ D) $y=\pm\frac{3}{5}x$ E) $y=\pm\frac{4}{5}x$

21. Which point is a focus of the hyperbola $9y^2 - 16x^2 = 1$?

A) $F(\frac{5}{12}, 0)$ B) $F(0, \frac{3}{5})$ C) $F(\frac{3}{5}, 0)$ D) $F(0, 25)$ E) $F(0, \frac{5}{12})$

22. $F(4, 0)$ is the focus of a parabola whose vertex is at the origin. Find the equation of its directrix.

A) $y = 4$ B) $y = 16$ C) $y = -8$ D) $x = -4$ E) $x = -2$

23. Find the focus of the parabola whose vertex is at the origin and whose directrix is $x = \frac{1}{3}$.

A) $F(\frac{1}{3}, 0)$ B) $F(-\frac{1}{3}, 0)$ C) $F(0, \frac{1}{3})$ D) $F(0, -\frac{1}{3})$ E) $F(\frac{2}{3}, 0)$

24. Find the parameter of the parabola whose focus is $F(0, -6)$ and whose vertex is at the origin.

A) 12 B) 24 C) 6 D) 3 E) –6

25. Find the equation of the parabola whose focus is $F(8, 0)$ and whose vertex is at the origin.

A) $y^2 = 32x$ B) $y^2 = 16x$ C) $y^2 = 8x$ D) $x^2 = 16y$ E) $x^2 = y + 2$

26. Find the equation of the parabola whose directrix is the line $3x + 4 = 0$ and whose vertex is at the origin.

A) $y^2 = \frac{8}{3}x$ B) $y^2 = \frac{8}{3}y$ C) $y^2 = \frac{32}{3}x$ D) $y^2 = \frac{16}{3}x$ E) $y^2 = \frac{16}{3}y$

27. Find the equation of the parabola whose vertex is at the origin and whose focus is $F(0, -5)$.

A) $x^2 = 10y$ B) $x^2 = -10y$ C) $y^2 = 10x$ D) $y^2 = -20x$ E) $x^2 = -20y$

28. Find the focus F and equation of the directrix d of the parabola $x^2 = 6y$ if its vertex is at the origin.

A) $F(3, 0)$, d: $y = -3$

B) $F(\frac{3}{2}, 0)$, d: $y = -\frac{3}{2}$

C) $F(0, \frac{3}{2})$, d: $y = -\frac{3}{2}$

D) $F(0, \frac{3}{2})$, d: $y = \frac{3}{2}$

E) $F(\frac{3}{2}, 0)$, d: $y = \frac{3}{2}$

29. A parabola with directrix $y = 5$ and vertex at the origin passes through the point $P(10, a)$. What is a?

A) 3 B) 5 C) –5 D) 10 E) 2

30. The vertex of a parabola is at the origin and $P(6, 9)$ is a point on this parabola. Find the equation of the parabola if its focus is on the positive side of the y-axis.

A) $y^2 = 4x$ B) $x^2 = 4y$ C) $x^2 = 2y$ D) $x^2 = 8y$ E) $y^2 = 8x$

31. The equation of a parabola is $y^2 = 12x$. A line parallel to the y-axis is drawn through the focus of this parabola. Find the distance between the two points at which the parabola intersects the line.

A) 3 B) 6 C) 8 D) 10 E) 12

CHAPTER 2

MATRICES AND DETERMINANTS

1 MATRICES

A. BASIC CONCEPTS

Three car dealers each sell three models of a car. The table below shows the number of cars each dealer sold in a given month.

	Dealer 1	Dealer 2	Dealer 3
Model A	5	7	4
Model B	3	4	9
Model C	2	0	1

We can organize data like this in a **matrix**. The matrix for our car dealer data is

$$A = \begin{array}{r} \\ \text{Model A} \\ \text{Model B} \\ \text{Model C} \end{array} \begin{array}{c} \begin{array}{ccc} D_1 & D_2 & D_3 \end{array} \\ \begin{bmatrix} 5 & 7 & 4 \\ 3 & 4 & 9 \\ 2 & 0 & 1 \end{bmatrix} \end{array}.$$

The plural form of matrix is **matrices**, pronounced 'may-trih-sees'.

Two large square brackets contain the numbers in the matrix. This matrix has three rows and three columns.

The column $\begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix}$ represents the cars sold by the first dealer. The row $[5 \quad 7 \quad 4]$ represents all the model A cars sold by the three dealers.

Matrices give us an effective way of organizing and manipulating data in different problems. We can begin our study of matrices with a more formal definition.

Definition

matrix

A **matrix** is a rectangular arrangement of numbers in rows and columns.

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} \cdots & a_{mn} \end{bmatrix}}_{n \text{ columns}} \Bigg\} \, m \text{ rows}$$

The horizontal lines of numbers in a matrix are called **rows** and the vertical lines are called **columns**. The number of rows and the number of columns determine the **dimensions** (also called the **order**) of the matrix. A matrix with m rows and n columns has dimensions $m \times n$ and is called an $\boldsymbol{m \times n}$ (read as 'm by n') **matrix**. Notice that the number of rows is always given first.

Each number in a matrix is called an **entry** of the matrix. $\boldsymbol{a_{ij}}$ means the entry in the ith row and jth column of the matrix A.

$$\underset{\text{rows}}{m} \times \underset{\text{columns}}{n}$$

3 columns

$$A = \begin{bmatrix} 2 & 5 & 4 \\ 6 & 3 & 0 \end{bmatrix} \quad \text{2 rows}$$

Matrix A is a 2 × 3 ('two by three') matrix.

a_{13}

1st row 3rd column

$$A = \begin{bmatrix} 2 & 5 & 4 \\ 6 & 3 & 0 \end{bmatrix}$$

a_{13} is the entry in the first row and the third column: $a_{13} = 4$.

EXAMPLE 1 Write the dimensions of each real matrix.

If each entry of a matrix is a real number then the matrix is called a **real matrix**.

a. $A = \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix}$ **b.** $B = \begin{bmatrix} \pi & -1 & \frac{1}{2} \end{bmatrix}$ **c.** $C = \begin{bmatrix} 0.5 \\ -2 \\ 7 \end{bmatrix}$ **d.** $D = \begin{bmatrix} 2 & 0 & 4 & 1 \\ 5 & 9 & 0 & 7 \\ 1 & 0 & 4 & 9 \end{bmatrix}$

Solution **a.** $[A]_{2\times 2}$ is a 2×2 matrix **b.** $[B]_{1\times 3}$ is a 1×3 matrix

c. $[C]_{3\times 1}$ is a 3×1 matrix **d.** $[D]_{3\times 4}$ is a 3×4 matrix

$[A]_{m\times n}$ means a matrix A with m rows and n columns.

EXAMPLE 2 $A = \begin{bmatrix} 2 & 0 & 4 & 1 \\ 5 & 9 & 0 & 7 \\ 1 & 0 & 4 & 9 \end{bmatrix}$ is given. Write each matrix entry.

a. a_{31} **b.** a_{23} **c.** a_{34} **d.** a_{22}

Solution **a.** 1 **b.** 0 **c.** 9 **d.** 9

EXAMPLE 3 Given $A = \begin{bmatrix} -3 & 0 & 5 \\ \sqrt{3} & 1 & \frac{3}{2} \end{bmatrix}$, find $2a_{13} - 3a_{21}^2 - 4a_{23}$.

Solution We have $a_{13} = 5$, $a_{21} = \sqrt{3}$ and $a_{23} = \frac{3}{2}$, so the expression becomes

$$(2\cdot 5) - 3(\sqrt{3})^2 - (4\cdot\frac{3}{2}) = 10 - 9 - 6 = -5.$$

B. TYPES OF MATRIX

1. Square Matrix

A **square matrix** is a matrix which has the same number of rows and columns.

$[2]$, $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 6 & 0 \end{bmatrix}$ are all square matrices. We say that a square matrix has **order n** if it has n rows and n columns.

A matrix with only one row, such as $[1 \quad 3 \quad 2]$, is called a **row matrix**.

Likewise, a **column matrix** such as $\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$ has only one column.

2. Zero Matrix

A **zero matrix** is a matrix whose entries are all zeros. We write **0** to mean a zero matrix.

$[0]$, $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ are all zero matrices.

3. Identity Matrix

A square matrix whose main diagonal elements (from top left to bottom right) are 1 and whose other entries are all zero is called an **identity matrix**. We write **I** to mean the identity matrix.

$[1]$, $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ are all identity matrices. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is not an identity matrix.

The **main diagonal** of a square matrix always runs from top left to bottom right.

$$\begin{bmatrix} a_{11} & \\ & a_{nn} \end{bmatrix}$$

main diagonal

4. Diagonal Matrix

A square matrix in which all the entries except the main diagonal entries are zero is called a **diagonal matrix**.

$\begin{bmatrix} -1 & 0 \\ 0 & 8 \end{bmatrix}$ and $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ are diagonal matrices.

5. Scalar Matrix

A square matrix whose main diagonal elements are all equal ($a_{11} = a_{22} = a_{33} = \ldots$) and whose other entries are all zero is called a **scalar matrix**.

$[8]$, $\begin{bmatrix} -5 & 0 \\ 0 & -5 \end{bmatrix}$ and $\begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}$ are all scalar matrices.

Notice that a scalar matrix is a type of diagonal matrix.

C. EQUAL MATRICES

Definition

equal matrices

Two matrices are called **equal matrices** if they have the same dimension and their corresponding entries are all equal. We write $\boldsymbol{A} = \boldsymbol{B}$ to mean that two matrices A and B are equal.

Notice that $[1 \quad 2 \quad 3] \neq \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, because these matrices do not have the same dimension: $(1\times3) \neq (3\times1)$.

EXAMPLE 4 Find a_{11}, a_{12}, a_{21} and a_{22} in the matrix equation.

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -3 & 0 \end{bmatrix}$$

Solution Since the two matrices are equal, their corresponding entries are equal, and so

$$a_{11} = 2, \qquad a_{12} = -1,$$
$$a_{21} = -3, \qquad a_{22} = 0.$$

EXAMPLE 5 $\begin{bmatrix} x+y & -3 \\ 3 & x-y \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ 3 & 2x+3y \end{bmatrix}$ is given. Find x, y and z.

Solution Since the two matrices are equal, their corresponding entries are equal. So

$$x+y=2 \quad (1)$$
$$x-y=2x+3y \Rightarrow x+4y=0. \quad (2)$$

Solving (1) and (2) for x and y gives us $x=\frac{8}{3}$, $y=-\frac{2}{3}$.

Check Yourself 1

Solve $\begin{bmatrix} 4x+5 & 9 & 15 \\ 7 & -2y+3 & -1 \end{bmatrix} = \begin{bmatrix} 21 & 9 & 15 \\ 7 & y-12 & -1 \end{bmatrix}$ for x and y.

Answers

$x = 4$, $y = 5$

D. OPERATIONS ON MATRICES

1. Matrix Addition

Adddition and subtraction are not defined for matrices with different dimensions.

Although we can always add two real numbers together, we cannot always add two matrices. In fact, we can only perform matrix addition on matrices with equal dimensions. To add or subtract two matrices A and B, we simply add or subtract corresponding entries.

$$\underset{A}{\begin{bmatrix} a & b \\ c & d \end{bmatrix}} + \underset{B}{\begin{bmatrix} e & f \\ g & h \end{bmatrix}} = \begin{bmatrix} (a+e) & (b+f) \\ (c+g) & (d+h) \end{bmatrix}$$

EXAMPLE 6 $A = \begin{bmatrix} 5 & 0 & 4 \\ 2 & 3 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 2 & 1 \\ 5 & 3 & 7 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 5 \\ 3 & 0 \end{bmatrix}$ are given. Write the matrices.

a. $A + B$ b. $A - B$ c. $A + C$

Solution a. Since the matrices have the same dimensions, we can add them.

$$\begin{bmatrix} 5 & 0 & 4 \\ 2 & 3 & -1 \end{bmatrix} + \begin{bmatrix} 2 & 2 & 1 \\ 5 & 3 & 7 \end{bmatrix} = \begin{bmatrix} 5+2 & 0+2 & 4+1 \\ 2+5 & 3+3 & -1+7 \end{bmatrix} = \begin{bmatrix} 7 & 2 & 5 \\ 7 & 6 & 6 \end{bmatrix}$$

$A - B \neq B - A$

b. Since the matrices have the same dimensions, we can subtract them.

$$\begin{bmatrix} 5 & 0 & 4 \\ 2 & 3 & -1 \end{bmatrix} - \begin{bmatrix} 2 & 2 & 1 \\ 5 & 3 & 7 \end{bmatrix} = \begin{bmatrix} 5-2 & 0-2 & 4-1 \\ 2-5 & 3-3 & -1-7 \end{bmatrix} = \begin{bmatrix} 3 & -2 & 3 \\ -3 & 0 & -8 \end{bmatrix}$$

Notice that $A - B \neq B - A$.

c. $A + C$ is undefined, since A and B have different dimensions.

EXAMPLE 7 Find each matrix sum or difference.

a. $\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ -4 & -3 \end{bmatrix}$ b. $\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ c. $\begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix} - \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$

Solution a. $\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ -4 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ b. $\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$

c. $\begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix} - \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ 2 & -8 \end{bmatrix}$

PROPERTIES OF MATRIX ADDITION

For any three real $m \times n$ matrices A, B and C, the following properties hold.

Closure property : $A + B$ is also an $m \times n$ matrix.
Commutative property : $A + B = B + A$
Associative property : $A + (B + C) = (A + B) + C$
Additive identity : $A + 0 = A$
Additive inverse : $A + (-A) = (-A) + A = 0$

Remember!
0 is the $m \times n$ matrix whose elements are all zero.

2. Multiplying a Matrix and a Scalar

In matrix algebra, a real number is often called a *scalar*. To multiply a matrix by a scalar, we multiply each entry in the matrix by the scalar. This operation is called **scalar multiplication**.

$$c\cdot\begin{bmatrix} a & b \\ b & d \end{bmatrix}=\begin{bmatrix} c\cdot a & c\cdot b \\ c\cdot b & c\cdot d \end{bmatrix}$$

EXAMPLE 8 The matrices $A=\begin{bmatrix} 1 & 2 & 4 \\ -3 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix}$ and $B=\begin{bmatrix} 2 & 0 & 0 \\ 1 & -4 & 3 \\ -1 & 3 & 2 \end{bmatrix}$ are given. Perform the matrix operations.

a. $3A$ **b.** $-B$ **c.** $3A-B$

Solution **a.** $3A=3\cdot\begin{bmatrix} 1 & 2 & 4 \\ -3 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix}=\begin{bmatrix} 3\cdot1 & 3\cdot2 & 3\cdot4 \\ 3\cdot(-3) & 3\cdot0 & 3\cdot(-1) \\ 3\cdot2 & 3\cdot1 & 3\cdot2 \end{bmatrix}=\begin{bmatrix} 3 & 6 & 12 \\ -9 & 0 & -3 \\ 6 & 3 & 6 \end{bmatrix}$

b. $-B=(-1)\begin{bmatrix} 2 & 0 & 0 \\ 1 & -4 & 3 \\ -1 & 3 & 2 \end{bmatrix}=\begin{bmatrix} -2 & 0 & 0 \\ -1 & 4 & -3 \\ 1 & -3 & -2 \end{bmatrix}$

c. $3A-B=3A+(-B)=\underbrace{\begin{bmatrix} 3 & 6 & 12 \\ -9 & 0 & -3 \\ 6 & 3 & 6 \end{bmatrix}}_{\text{from a}}+\underbrace{\begin{bmatrix} -2 & 0 & 0 \\ -1 & 4 & -3 \\ 1 & -3 & -2 \end{bmatrix}}_{\text{from b}}=\begin{bmatrix} 1 & 6 & 12 \\ -10 & 4 & -6 \\ 7 & 0 & 4 \end{bmatrix}$

EXAMPLE 9 Solve the matrix equation for x and y. $2\left(\begin{bmatrix} 3 & 2x \\ 6 & 4 \end{bmatrix}+\begin{bmatrix} -3 & 5 \\ -3y & 3 \end{bmatrix}\right)=\begin{bmatrix} 0 & 18 \\ 6 & 14 \end{bmatrix}$

Solution Simplify the left side of the equation:

$$2\left(\begin{bmatrix} 3 & 2x \\ 6 & 4 \end{bmatrix}+\begin{bmatrix} -3 & 5 \\ -3y & 3 \end{bmatrix}\right)=\begin{bmatrix} 0 & 18 \\ 6 & 14 \end{bmatrix}$$

$$2\begin{bmatrix} 3-3 & 2x+5 \\ 6-3y & 4+3 \end{bmatrix}=\begin{bmatrix} 0 & 18 \\ 6 & 14 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 4x+10 \\ 12-6y & 14 \end{bmatrix}=\begin{bmatrix} 0 & 18 \\ 6 & 14 \end{bmatrix}.$$

Equate corresponding entries and solve the two resulting equations:

$$\begin{aligned} 4x+10&=18 \\ 4x&=8 \\ x&=2 \end{aligned}\qquad\qquad \begin{aligned} 12-6y&=6 \\ 6y&=6 \\ y&=1. \end{aligned}$$

It is often useful to rewrite a matrix as a scalar multiple of a simpler matrix. For instance, in the following example the scalar $\frac{1}{2}$ has been factored out of the matrix:

$$\begin{bmatrix} \frac{1}{2} & -\frac{3}{2} \\ \frac{5}{2} & \frac{1}{2} \end{bmatrix}=\frac{1}{2}\begin{bmatrix} 1 & -3 \\ 5 & 1 \end{bmatrix}.$$

Check Yourself 2

Given the matrices $A = \begin{bmatrix} 6 & -1 & 4 \\ 2 & 4 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 & 2 \\ -1 & 5 & 0 \end{bmatrix}$, find

a. $-A$. **b.** $2A - 3B$. **c.** $5A + B$.

Answers

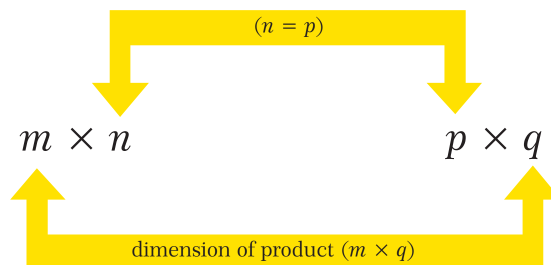
a. $\begin{bmatrix} -6 & 1 & -4 \\ -2 & -4 & -3 \end{bmatrix}$ **b.** $\begin{bmatrix} 9 & -14 & 2 \\ 7 & -7 & 6 \end{bmatrix}$ **c.** $\begin{bmatrix} 31 & -1 & 22 \\ 9 & 25 & 15 \end{bmatrix}$

3. Matrix Multiplication

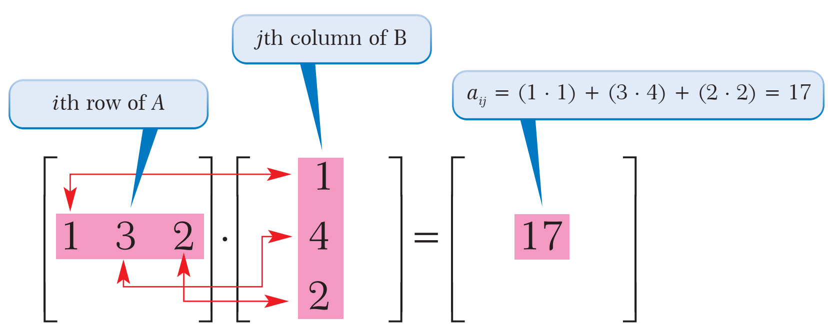
It is important to check the dimensions of two matrices before we start to multiply them. If matrix A has dimension $m \times n$ and matrix B has dimension $p \times q$, then the product AB only exists if $n = p$. Furthermore, the product will have dimension $m \times q$.

If A is an $m \times n$ matrix and B is an $n \times p$ matrix then the product AB is an $m \times p$ matrix.

$$\underbrace{\underset{m \times n}{A} \cdot \underset{n \times p}{B}}_{\text{equal: } n,\ n;\ \text{dimensions of } AB:\ m \times p} = \underset{m \times p}{AB}$$



We obtain each entry in the matrix AB (the product of A and B) from a row of A and a column of B as follows: multiply the entries in the ith row of A by the entries in the jth column of B and add the results to get a_{ij} in AB.



EXAMPLE 10 $A = \begin{bmatrix} 2 & 1 & -2 \\ 3 & 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 & 5 \\ -2 & 0 & 6 \\ 3 & 2 & 4 \end{bmatrix}$ are given. Find the products.

a. AB **b.** BA

Solution a. The product AB is defined because A has dimension 2×3 and B has dimension 3×3. Moreover, the product AB will have dimension 2×3. Look at the procedure for calculating the product:

1st row, 1st column $\begin{bmatrix} 2 & 1 & -2 \\ 3 & 1 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 & 5 \\ -2 & 0 & 6 \\ 3 & 2 & 4 \end{bmatrix} = \begin{bmatrix} -6 & & \\ & & \end{bmatrix}$

$2 \cdot 1 + 1 \cdot (-2) + (-2) \cdot 3 = -6$

1st row, 2nd column $\begin{bmatrix} 2 & 1 & -2 \\ 3 & 1 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 & 5 \\ -2 & 0 & 6 \\ 3 & 2 & 4 \end{bmatrix} = \begin{bmatrix} -6 & -6 & \\ & & \end{bmatrix}$

$2 \cdot (-1) + 1 \cdot 0 + (-2) \cdot 2 = -6$

1st row, 3rd column $\begin{bmatrix} 2 & 1 & -2 \\ 3 & 1 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 & 5 \\ -2 & 0 & 6 \\ 3 & 2 & 4 \end{bmatrix} = \begin{bmatrix} -6 & -6 & 8 \\ & & \end{bmatrix}$

$2 \cdot 5 + 1 \cdot 6 + (-2) \cdot 4 = 8$

2nd row, 1st column $\begin{bmatrix} 2 & 1 & -2 \\ 3 & 1 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 & 5 \\ -2 & 0 & 6 \\ 3 & 2 & 4 \end{bmatrix} = \begin{bmatrix} -6 & -6 & 8 \\ 16 & & \end{bmatrix}$

$3 \cdot 1 + 1 \cdot (-2) + 5 \cdot 3 = 16$

2nd row, 2nd column $\begin{bmatrix} 2 & 1 & -2 \\ 3 & 1 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 & 5 \\ -2 & 0 & 6 \\ 3 & 2 & 4 \end{bmatrix} = \begin{bmatrix} -6 & -6 & 8 \\ 16 & 7 & \end{bmatrix}$

$3 \cdot (-1) + 1 \cdot 0 + 5 \cdot 2 = 7$

2nd row, 3rd column $\begin{bmatrix} 2 & 1 & -2 \\ 3 & 1 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 & 5 \\ -2 & 0 & 6 \\ 3 & 2 & 4 \end{bmatrix} = \begin{bmatrix} -6 & -6 & 8 \\ 16 & 7 & 41 \end{bmatrix}$

$3 \cdot 5 + 1 \cdot 6 + 5 \cdot 4 = 41$

So the product is $\begin{bmatrix} 2 & 1 & -2 \\ 3 & 1 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 & 5 \\ -2 & 0 & 6 \\ 3 & 2 & 4 \end{bmatrix} = \begin{bmatrix} -6 & -6 & 8 \\ 16 & 7 & 41 \end{bmatrix}$.

b. Since the dimensions of B and A are 2×2 and 3×2 respectively, the product BA is not defined. This shows us that matrix multiplication is not always commutative: $AB \neq BA$.

Check Yourself 3

Find the products.

a. $\begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & -2 \end{bmatrix}\begin{bmatrix} -2 & 4 & 2 \\ 1 & 0 & 0 \\ -1 & 1 & -1 \end{bmatrix}$ b. $\begin{bmatrix} 3 & 4 \\ -2 & 5 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ c. $\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$

Answers

a. $\begin{bmatrix} -5 & 7 & -1 \\ -3 & 6 & 6 \end{bmatrix}$ b. $\begin{bmatrix} 3 & 4 \\ -2 & 5 \end{bmatrix}$ c. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

EXAMPLE 11 Given the matrices $A = \begin{bmatrix} 0 & 3 \\ -2 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 7 \\ 2 & 4 \end{bmatrix}$, show that $AB \neq BA$.

Solution Using the definition of the product of two matrices, we get

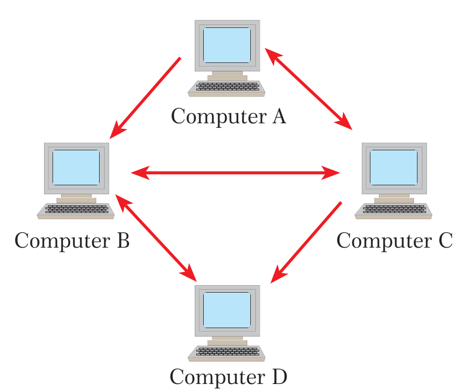
$$AB = \begin{bmatrix} 0 & 3 \\ -2 & 6 \end{bmatrix}\begin{bmatrix} -1 & 7 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 0\cdot(-1)+3\cdot 2 & 0\cdot 7+3\cdot 4 \\ (-2)\cdot(-1)+6\cdot 2 & (-2)\cdot 7+6\cdot 4 \end{bmatrix} = \begin{bmatrix} 6 & 12 \\ 14 & 10 \end{bmatrix}$$

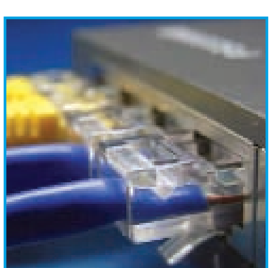
$$BA = \begin{bmatrix} -1 & 7 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} 0 & 3 \\ -2 & 6 \end{bmatrix} = \begin{bmatrix} (-1)\cdot 0+7\cdot(-2) & (-1)\cdot 3+7\cdot 6 \\ 2\cdot 0+4\cdot(-2) & 2\cdot 3+4\cdot 6 \end{bmatrix} = \begin{bmatrix} -14 & 39 \\ -8 & 30 \end{bmatrix}.$$

Thus $AB \neq BA$.

Matrix multiplication is not in general commutative: $AB \neq BA$.

EXAMPLE 12 The figure shows the computer network from Example 19. Find the matrix that represents the number of ways that data can be sent from one computer to another *through exactly one computer*.





Solution This example shows how the product of two matrices can be useful. The product of matrix X with itself (or the square of matrix X) gives us the number of ways that data can be sent from one computer to another through exactly one computer.

$$X^2 = X \cdot X = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 \\ 0 & 2 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{array}{cc} & \text{(to)} \\ & \begin{array}{cccc} A & B & C & D \end{array} \\ \text{(from)} \begin{array}{c} A \\ B \\ C \\ D \end{array} & \begin{bmatrix} 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 \\ 0 & 2 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{array}$$

The 1 in the first row, first column of X^2 means that Computer A can send data to itself in one way (via Computer C). Likewise, the 2 in the first row, fourth column means that Computer A can send data to Computer D in two ways (via either Computer B or Computer C). Notice that, for example, B cannot send data to C through any other computer, so the corresponding matrix entry is 0.

EXAMPLE 13 The following table shows the probabilities of a taxi ride ending at each of three destinations for taxis traveling among three sections of a city. For example, the probability of picking up a rider southside and dropping him off downtown is 30%.

Pickup	Destination		
	Northside	Downtown	Southside
Northside	50%	20%	30%
Downtown	10%	40%	50%
Southside	30%	30%	40%

What is the probability of starting downtown and being downtown again after two taxi rides?

Solution Let us represent the information we are given in a matrix:

$$\begin{array}{cc} & \begin{array}{ccc} N & D & S \end{array} \\ \begin{array}{c} N \\ D \\ S \end{array} & \begin{bmatrix} 0.5 & 0.2 & 0.3 \\ 0.1 & 0.4 & 0.5 \\ 0.3 & 0.3 & 0.4 \end{bmatrix} \end{array} = P.$$

Then $P^2 = P \cdot P = \begin{bmatrix} 0.5 & 0.2 & 0.3 \\ 0.1 & 0.4 & 0.5 \\ 0.3 & 0.3 & 0.4 \end{bmatrix}\begin{bmatrix} 0.5 & 0.2 & 0.3 \\ 0.1 & 0.4 & 0.5 \\ 0.3 & 0.3 & 0.4 \end{bmatrix} = \begin{bmatrix} 0.36 & 0.27 & 0.37 \\ 0.24 & 0.33 & 0.43 \\ 0.3 & 0.3 & 0.4 \end{bmatrix}$

The second column of the product matrix P^2 shows the probability of ending up downtown after two trips, given the original starting point of either N, D or S. The Downtown to Downtown probability is in the middle: 0.33, or 33%.

Remember!
If E_1 and E_2 are independent events then the probability of events E_1 and E_2 both happening is $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$.

PROPERTIES OF MATRIX MULTIPLICATION

If A, B, C are any three matrices whose products are defined and k is any real number, then the following properties hold.

1. $A(BC) = (AB)C$ (**associative**)
2. $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$ (**distinctive**)
3. $k(AB) = (kA)B = A(kB)$

Remark

1. In general, $AB \neq BA$.
2. If $AB = 0$, we cannot conclude (in general) that either $A = 0$ or $B = 0$.
3. If $AB = AC$ then it is not true in general that $B = C$. In other words, the cancellation laws do not hold for matrix multiplication.

We can verify 2 and 3 as follows:

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix},\ B = \begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix} \text{ but } AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Also, for $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix}$ we have

$$A \cdot B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ and } A \cdot C = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ but } B \neq C.$$

4. If A is a square matrix and $n \in \mathbb{N}$ then $A^0 = \mathrm{I}$, $A^1 = A$, $A^2 = AA$, $A^3 = AA^2$, ..., $A^n = AA^{n-1}$.
5. $\mathrm{I}^n = \mathrm{I}$
6. $A\mathrm{I} = \mathrm{I}A$

$A\mathrm{I} = A$

$\mathrm{I} \cdot \mathrm{I} = \mathrm{I}$

EXAMPLE 14 Given $X = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}$ and $Y = \begin{bmatrix} 0 & 1 \\ 2 & -3 \end{bmatrix}$, find XY, YX, X^2 and Y^2.

Solution

$$XY = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 2 & -3 \end{bmatrix} = \begin{bmatrix} 2\cdot 0 + 0\cdot 2 & 2\cdot 1 + 0\cdot(-3) \\ 1\cdot 0 + (-1)\cdot 2 & 1\cdot 1 + (-1)\cdot(-3) \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ -2 & 4 \end{bmatrix}$$

$$YX = \begin{bmatrix} 0 & 1 \\ 2 & -3 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0\cdot 2 + 1\cdot 1 & 0\cdot 0 + 1\cdot(-1) \\ 2\cdot 2 + (-3)\cdot 1 & 2\cdot 0 + (-3)\cdot(-1) \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix}$$

$$X^2 = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2\cdot 2 + 0\cdot 1 & 2\cdot 0 + 0\cdot(-1) \\ 1\cdot 2 + (-1)\cdot 1 & 1\cdot 0 + (-1)\cdot(-1) \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 1 & 1 \end{bmatrix}$$

$$Y^2 = \begin{bmatrix} 0 & 1 \\ 2 & -3 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 2 & -3 \end{bmatrix} = \begin{bmatrix} 0\cdot 0 + 1\cdot 2 & 0\cdot 1 + 1\cdot(-3) \\ 2\cdot 0 + (-3)\cdot 2 & 2\cdot 1 + (-3)\cdot(-3) \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -6 & 11 \end{bmatrix}$$

EXAMPLE 15 $A = \begin{bmatrix} 0 & 3 \\ 2 & 0 \end{bmatrix}$ is given. Find A^{206}.

Solution First we calculate A^2:

> **Remember!**
> 1. $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is the identity matrix.
> 2. $I^n = I$

$$A^2 = AA = \begin{bmatrix} 0 & 3 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} 0 & 3 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix} = 6\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 6I.$$

Now $A^{206} = (A^2)^{103}$, and substituting $A^2 = 6I$ gives us $(6I)^{103} = 6^{103} \cdot I^{103}$.

Finally, since $I^{103} = I$, we get $6^{103} \cdot I = \begin{bmatrix} 6^{103} & 0 \\ 0 & 6^{103} \end{bmatrix}$. This is the required matrix.

EXAMPLE 16 Let $A = \begin{bmatrix} 3 & 2 \\ 0 & -3 \end{bmatrix}$. Find A^{1986}.

Solution First we calculate A^2:

$$A^2 = AA = \begin{bmatrix} 3 & 2 \\ 0 & -3 \end{bmatrix}\begin{bmatrix} 3 & 2 \\ 0 & -3 \end{bmatrix} = \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix} = 9\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 9I.$$

Also, $A^{1986} = (A^2)^{993}$, and substituting $A^2 = 9I$ gives us $(9I)^{993} = 9^{993} \cdot I^{993}$.

Since $I^{993} = I$, we get $9^{993} \cdot I = \begin{bmatrix} 9^{993} & 0 \\ 0 & 9^{993} \end{bmatrix}$.

EXAMPLE 17 Solve $\begin{bmatrix} -4 & x \\ -x & 4 \end{bmatrix}^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$.

Solution First expand the left-hand side:

$$\begin{bmatrix} -4 & x \\ -x & 4 \end{bmatrix}^2 = \begin{bmatrix} -4 & x \\ -x & 4 \end{bmatrix}\begin{bmatrix} -4 & x \\ -x & 4 \end{bmatrix} = \begin{bmatrix} 16-x^2 & 0 \\ 0 & 16-x^2 \end{bmatrix}$$

Now equate corresponding entries:

$$16 - x^2 = -1$$
$$x^2 = 17$$
$$x = \pm\sqrt{17}.$$ This is the solution.

Check Yourself 4

1. Calculate the products.

a. $\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 3 & -6 \\ 5 & 7 \end{bmatrix}$ b. $\begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ -1 & 1 & -1 \end{bmatrix}\begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$

2. Given $\begin{bmatrix} 2 & 0 & 3 \\ 1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$, find A^2 and A^3.

3. Given the function $f(x) = x^2 - 5x + 2\mathrm{I}$ where I is the identity matrix, calculate $f(A)$ for $A = \begin{bmatrix} 2 & 0 \\ 4 & 5 \end{bmatrix}$.

Remember!

$p \cdot \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} pa & pb \\ pc & pd \end{bmatrix}$

Answers

1. a. $\begin{bmatrix} 14 & -11 \\ 7 & 20 \end{bmatrix}$ b. $\begin{bmatrix} 2 \\ 6 \\ -2 \end{bmatrix}$

2. $A^2 = \begin{bmatrix} 4 & 6 & 6 \\ 2 & 2 & 3 \\ 2 & 0 & 2 \end{bmatrix}$, $A^3 = \begin{bmatrix} 14 & 12 & 18 \\ 6 & 6 & 8 \\ 4 & 4 & 6 \end{bmatrix}$

3. $\begin{bmatrix} -4 & 0 \\ 8 & 2 \end{bmatrix}$

F. THE INVERSE OF A MATRIX

1. Definition

So far we have looked at some of the similarities between the algebra of real numbers and the algebra of matrices. In this section we will extend the algebra of matrices to include the solution of equations.

To begin, consider the real number equation $ax = b$. To solve this equation for x, we can multiply both sides of the equation by

$a^{-1} = \dfrac{1}{a}$ (provided that $a \neq 0$).

$$\begin{aligned} ax &= b \\ (a^{-1} \cdot a) \cdot x &= a^{-1} \cdot b \\ 1 \cdot x &= a^{-1} \cdot b \\ x &= a^{-1} \cdot b. \end{aligned}$$

The number a^{-1} is called the **multiplicative inverse** of a, because, $a^{-1}a = 1$ (the identity element for real number multiplication). We can define the multiplicative inverse of a matrix in a similar way.

Definition

inverse matrix, invertible matrix, singular matrix

Let A and B be two square matrices with dimension $n \times n$. Then B is called the **inverse of A** if

$$AB = BA = I_n,$$

where I_n is the identity matrix of order n. We write A^{-1} to mean the inverse of a matrix A.

A matrix which has an inverse is called an **invertible matrix**. A matrix which does not have an inverse is called a **noninvertible** (or **singular**) **matrix**.

$A^{-1} \neq \frac{1}{A}$

Only square matrices have invenses: non-square matrices do not have inverses. To see this, notice that if A is of dimension $m \times n$ and B is of dimension $n \times m$ (where $n \neq m$), then the products AB and BA are of different dimensions and therefore could not be equal to each other.

EXAMPLE 18 Given $A = \begin{bmatrix} -1 & 2 \\ -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix}$, show that A and B are inverses of one another.

Solution First calculate the products:

$$AB = \begin{bmatrix} -1 & 2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -1+2 & 2-2 \\ -1+1 & 2-1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} -1 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1+2 & 2-2 \\ -1+1 & 2-1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Both product matrices are the 2×2 identity matrix for multiplication, so $AB = BA = I_2$ and therefore A and B are inverses of one another.

EXAMPLE 19 Find the inverse of the matrix $A = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}$.

Solution To find the inverse of A, we need to solve the matrix equation $AX = I$ for X.

$$\begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Multiplying A and X gives us

$$\begin{bmatrix} a+4c & b+4d \\ -a-3c & -b-3d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Equating the corresponding entries gives

$$\begin{aligned} a + 4c &= 1 \\ -a - 3c &= 0 \end{aligned} \quad \text{and} \quad \begin{aligned} b + 4d &= 0 \\ -b - 3d &= 1. \end{aligned}$$

Solving these two systems, we find that $a = -3$, $b = -3$, $c = 1$ and $d = 1$.

So the inverse of A is

$$X = A^{-1} = \begin{bmatrix} -3 & -3 \\ 1 & 1 \end{bmatrix}.$$

As an exercise, check this result using matrix multiplication.

The following rule provides a simple way of calculating the inverse of a 2×2 matrix when it exists.

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} x & y \\ z & t \end{bmatrix}$ be two matrices which satisfy $A \cdot A^{-1} = \text{I}$.

Multiplying these two matrices gives us

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} x & y \\ z & t \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} ax+bz & ay+bt \\ cx+dz & cy+dt \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

So $\begin{matrix} ax + bz = 1 \\ cx + dz = 0 \end{matrix}$, $\begin{matrix} ay + bt = 0 \\ cy + dt = 1. \end{matrix}$

Solving these two systems gives us

$$x = \frac{d}{ad-bc}, \ y = \frac{-b}{ad-bc}, \ z = \frac{-c}{ad-bc}, \ t = \frac{a}{ad-bc}, \text{ i.e.}$$

$$A^{-1} = \begin{bmatrix} \dfrac{d}{ad-bc} & \dfrac{-b}{ad-bc} \\ \dfrac{-c}{ad-bc} & \dfrac{a}{ad-bc} \end{bmatrix}.$$ Factor out $\dfrac{1}{ad-bc}$ $(ad - bc \neq 0)$ to get

$$A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$ This is the form of the inverse of $[A]_{2\times2}$.

Rule

inverse of a 2×2 matrix

A matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible if and only if $ad - bc \neq 0$. If the inverse exists, it is given by

$$A^{-1} = \frac{1}{ad-bc} \cdot \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

EXAMPLE 20 Find A^{-1} if $A = \begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix}$.

Solution Using the rule for the inverse of a 2×2 matrix, we get

$AA^{-1} = \text{I} = A^{-1}A$

$$A^{-1} = \frac{1}{1\cdot3-5\cdot2} \cdot \begin{bmatrix} 3 & -5 \\ -2 & 1 \end{bmatrix} = \frac{1}{-7} \cdot \begin{bmatrix} 3 & -5 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -\dfrac{3}{7} & \dfrac{5}{7} \\ \dfrac{2}{7} & -\dfrac{1}{7} \end{bmatrix}.$$

EXAMPLE 21 For which value of x does the matrix $A=\begin{bmatrix}3x+1 & 6\\ 4 & 2\end{bmatrix}$ have no inverse?

Solution By the theorem for the inverse of a 2×2 matrix, A is singular when $ad-bc=0$. So

$$(3x+1)\cdot 2-(4\cdot 6)=0$$
$$6x+2-24=0$$
$$6x-22=0$$
$$x=\frac{11}{3}.$$

Remember!
A singular matrix is a matrix with no inverse.

EXAMPLE 22 $A=\begin{bmatrix}-3 & m\\ 2-m & 3\end{bmatrix}$ is a matrix such that $A=A^{-1}$. Find A.

Solution Since $A=A^{-1}$, $AA^{-1}=AA=\mathrm{I}$.

So $A\cdot A=\begin{bmatrix}-3 & m\\ 2-m & 3\end{bmatrix}\begin{bmatrix}-3 & m\\ 2-m & 3\end{bmatrix}=\begin{bmatrix}9+2m-m^2 & -3m+3m\\ -6+3m+6-3m & 2m-m^2+9\end{bmatrix}$

$=\begin{bmatrix}-m^2+2m+9 & 0\\ 0 & -m^2+2m+9\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, which gives us

$-m^2+2m+9=1$, i.e, $m^2-2m-8=0$.

Solving the quadratic equation for m gives us $m_1=4$ or $m_2=-2$.

If $m_1=4$, $A_1=\begin{bmatrix}-3 & 4\\ -2 & 3\end{bmatrix}$. If $m_2=-2$, $A_2=\begin{bmatrix}-3 & -2\\ 4 & 3\end{bmatrix}$. So there are two possibilities for the matrix A.

Check Yourself 11

1. Find A^{-1} if $\begin{bmatrix}4 & 6\\ 1 & 2\end{bmatrix}$.

2. Let $A=\begin{bmatrix}a & \frac{1}{3}\\ \frac{1}{12} & b\end{bmatrix}$. Find a if $A=A^{-1}$.

3. For which value of x does the matrix $A=\begin{bmatrix}2 & -4\\ -3 & \log x\end{bmatrix}$ have no inverse?

Answers

1. $\begin{bmatrix}1 & -3\\ -\frac{1}{2} & 2\end{bmatrix}$ 2. $\pm\frac{\sqrt{35}}{6}$ 3. 10^6

CRYPTOGRAPHY

If you have ever tried to send a secret message to someone, you have probably used cryptography. Cryptography is the study and use of special codes to send and receive messages. In the past, people used codes to communicate secret messages such as information about battleships and soldiers in a war. Today, cryptography helps us to send messages around the world in digital form, and also keeps private information about people and companies safe.

There are many different kinds of code. One kind involves the use of matrices and their inverse. For example, imagine that you have a matrix B that contains a message you want to send in code. To encode the message, create a second invertible matrix A so that the product AB exists, and then calculate AB. Make sure that the receiver of your message knows the inverse matrix A^{-1}, then you can send your message as AB. The receiver of the message then calculates the product of A^{-1} and your message. Indeed, we have

$$A^{-1}(AB) = B,$$

which is the original message.

The following table shows a simple code for sending messages. Each letter of the alphabet corresponds to a number: for example, the letter R corresponds to 17 and N corresponds to 13. The space between two words corresponds to the number 26.

A	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*I*	*J*	*K*	*L*	*M*	*N*	*O*
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
P	*Q*	*R*	*S*	*T*	*U*	*V*	*W*	*X*	*Y*	*Z*	–	?	.	,
15	16	17	18	16	20	21	22	23	24	25	26	27	28	29

Follow the steps to encode the word ESCAPE:

1) ESCAPE corresponds to the number sequence 4 18 2 0 15 4. We can write this as the 2 x 3 matrix B:

$$B = \begin{bmatrix} 4 & 18 & 2 \\ 0 & 15 & 4 \end{bmatrix}.$$

2) Create an invertible matrix A such as $A = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$ to encode the message, then calculate the product AB:

$$AB = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 4 & 18 & 2 \\ 0 & 15 & 4 \end{bmatrix} = \begin{bmatrix} 12 & 129 & 26 \\ 4 & 48 & 10 \end{bmatrix}.$$

3) The encrypted message is 12 129 26 4 48 10.

To decode the message, repeat the steps but use the inverse of A. The inverse of our encoding matrix is

$$A^{-1} = \frac{1}{6-5}\begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}.$$

Write the encrypted message as the 2 x 3 matrix $C = \begin{bmatrix} 12 & 129 & 26 \\ 4 & 48 & 10 \end{bmatrix}$.

Finally, multiply C by A^{-1} to get the original message:

$$A^{-1}C = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 12 & 129 & 26 \\ 4 & 48 & 10 \end{bmatrix} = \begin{bmatrix} 4 & 18 & 2 \\ 0 & 15 & 4 \end{bmatrix}.$$

The product gives the decoded message 4 18 2 0 15 4.

This product gives us the message 4 18 2 0 15 4, which corresponds to ESCAPE.

Notice that to make this process easy, both A and A^{-1} should contain only integers. We can use our knowledge of matrices to generate A: if A is a matrix with determinant ±1 and only integer entries, then its inverse will also contain only integers.

Try encoding other short messages with matrices. Can you find any disadvantages of this encoding technique?

EXERCISES 2.1

D. Operations on Matrices

1. Calculate $A + B$, $A - B$, $2A$ and $2A - B$ for each pair of matrices.

   a. $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -3 & -2 \\ 4 & 2 \end{bmatrix}$

   b. $A = \begin{bmatrix} 6 & -1 \\ 2 & 4 \\ -3 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 \\ -1 & 5 \\ 1 & 10 \end{bmatrix}$

   c. $A = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$, $B = \begin{bmatrix} -4 \\ 6 \\ 2 \end{bmatrix}$

2. Find c_{21} and c_{13} if $C = 3A - 2B$,

   $A = \begin{bmatrix} 5 & 4 & 4 \\ -3 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -7 \\ 0 & -5 & 1 \end{bmatrix}$.

3. Find c_{23} and c_{32} if $C = 2A + 5B$,

   $A = \begin{bmatrix} 4 & 11 & -9 \\ 0 & 3 & 2 \\ -3 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -7 \\ -4 & 6 & 11 \\ -6 & 4 & 9 \end{bmatrix}$.

4. Solve the matrix equation for a, b and c.

   $$4\begin{bmatrix} a & b \\ c & -1 \end{bmatrix} = 2\begin{bmatrix} b & c \\ -a & 1 \end{bmatrix} + 2\begin{bmatrix} 4 & a \\ 5 & -a \end{bmatrix}$$

5. Find k, m and n if

   $$\begin{bmatrix} n & k \\ 1 & -3 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 2 & -1 \end{bmatrix} + 2\begin{bmatrix} 1 & n \\ m & k \end{bmatrix}.$$

6. Solve the matrix equations for a, b, c and d.

   a. $\begin{bmatrix} a-b & 2b+c \\ c-2b & a+d \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 5 & 5 \end{bmatrix}$

   b. $\begin{bmatrix} a+b & b+c \\ a-c & b-a \end{bmatrix} = \begin{bmatrix} 0 & 3 \\ -3 & 2 \end{bmatrix}$

7. Calculate AB in each case.

   a. $A = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 3 & -3 \end{bmatrix}$

   b. $A = \begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}$

   c. $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 \\ -3 & 1 \end{bmatrix}$

   d. $A = \begin{bmatrix} 1 & -1 & 7 \\ 2 & -1 & 8 \\ 3 & 1 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 1 & -3 & 2 \end{bmatrix}$

   e. $A = \begin{bmatrix} 3 & 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$

8. Calculate AB in each case.

   a. $A = \begin{bmatrix} 2 & 1 \\ -2 & 4 \\ 1 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 & 0 \\ 4 & 0 & 2 \\ 8 & -1 & 7 \end{bmatrix}$

   b. $A = \begin{bmatrix} 0 & -1 & 0 \\ 4 & 0 & 2 \\ 8 & -1 & 7 \end{bmatrix}$, $B = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$

   c. $A = \begin{bmatrix} -1 & 3 \\ 4 & -5 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 0 & 7 \end{bmatrix}$

   d. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & -2 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

e. $A=\begin{bmatrix}0&0&-5\\0&0&-3\\0&0&7\end{bmatrix}$, $B=\begin{bmatrix}6&-11&4\\8&16&4\\0&0&0\end{bmatrix}$

f. $A=\begin{bmatrix}6\\-2\\1\\6\end{bmatrix}$, $B=[10\quad 12]$

g. $A=\begin{bmatrix}1&0&3&-2\\6&13&8&-17\end{bmatrix}$, $B=\begin{bmatrix}1&6\\4&2\end{bmatrix}$

9. Compute the given power of A for the matrix

$A=\begin{bmatrix}1&0\\0&-1\end{bmatrix}$.

a. A^{29}

b. A^{30}

10. Solve each matrix equation for A.

a. $\begin{bmatrix}1&2\\3&5\end{bmatrix}A=\begin{bmatrix}1&0\\0&1\end{bmatrix}$

b. $\begin{bmatrix}2&-1\\3&-2\end{bmatrix}A=\begin{bmatrix}1&0\\0&1\end{bmatrix}$

11. Verify that $AB = BA$ for the matrices

$A=\begin{bmatrix}\cos\alpha&-\sin\alpha\\\sin\alpha&\cos\alpha\end{bmatrix}$ and $B=\begin{bmatrix}\cos\beta&-\sin\beta\\\sin\beta&\cos\beta\end{bmatrix}$.

B. The Inverse of a Matrix

12. Show that B is the inverse of A in each case.

a. $A=\begin{bmatrix}1&2\\3&4\end{bmatrix}$, $B=\begin{bmatrix}-2&1\\\frac{3}{2}&-\frac{1}{2}\end{bmatrix}$

b. $A=\begin{bmatrix}2&-17&11\\-1&11&-7\\0&3&-2\end{bmatrix}$, $B=\begin{bmatrix}1&1&2\\2&4&-3\\3&6&-5\end{bmatrix}$

c. $A=\begin{bmatrix}2&0&0\\0&-3&0\\0&0&4\end{bmatrix}$, $B=\begin{bmatrix}\frac{1}{2}&0&0\\0&-\frac{1}{3}&0\\0&0&\frac{1}{4}\end{bmatrix}$

d. $A=\begin{bmatrix}0&1&0\\1&1&2\\0&-2&-1\end{bmatrix}$, $B=\begin{bmatrix}3&1&2\\1&0&0\\-2&0&-1\end{bmatrix}$

13. Find the inverse of each matrix, if it exists.

a. $\begin{bmatrix}1&2\\3&7\end{bmatrix}$

b. $\begin{bmatrix}1&-2\\2&-3\end{bmatrix}$

c. $\begin{bmatrix}2&4\\4&8\end{bmatrix}$

d. $\begin{bmatrix}11&1\\-1&0\end{bmatrix}$

e. $\begin{bmatrix}1&1&1\\3&5&4\\3&6&5\end{bmatrix}$

f. $\begin{bmatrix}1&2&2\\3&7&9\\-1&-4&-7\end{bmatrix}$

14. $A=\begin{bmatrix}x&-\frac{1}{2}\\y&\frac{1}{5}\end{bmatrix}$ and $A = A^{-1}$. Find $x + y$.

15. $A=\begin{bmatrix}x&2\\y&-2\end{bmatrix}$ and $A^{-1}\cdot A = A^2$. Find $x\cdot y$.

16. $A=\begin{bmatrix}1&3&5\\3&0&7\\1&3&a-9\end{bmatrix}$ is not invertible. Find a.

2 DETERMINANTS

A. THE DETERMINANT OF A MATRIX

Every square matrix can be assigned a real number which is called the **determinant** of the matrix. This number is useful when we are calculating the inverse of a matrix or solving a system of linear equations. We write **det** A, $|A|$, D or Δ to mean the determinant of a square matrix A.

Only square matrices have determinants.

1. Determinants of 1×1 and 2×2 Matrices

Definition

determinant of a 1×1 matrix

The determinant of a 1×1 matrix $[a]$ is $\boldsymbol{a}$.

$[A]$ is a matrix. $|A|$ is a determinant.

For example, $\det [6] = |6| = 6$, $\det [-5] = |5| = -5$ and $\det \begin{bmatrix} \frac{1}{3} \end{bmatrix} = \begin{vmatrix} \frac{1}{3} \end{vmatrix} = \frac{1}{3}$.

Definition

determinant of a 2×2 matrix

The determinant of a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $\boldsymbol{ad - bc}$.

We can remember this rule visually like this:

$$\overset{+}{}\begin{vmatrix} a & b \\ c & d \end{vmatrix}\overset{-}{} = ad - bc.$$

Look some examples of the determinants of 2×2 matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \ \det A = |A| = (1 \cdot 4) - (2 \cdot 3) = -2$$

$$B = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}, \ \det B = |B| = (2 \cdot 7) - (3 \cdot 5) = -1$$

$$C = \begin{bmatrix} 5 & -3 \\ 6 & -2 \end{bmatrix}, \ \det C = |C| = [5 \cdot (-2)] - [(-3) \cdot 6] = 8.$$

Notice that the determinant of a matrix can be either positive or negative.

EXAMPLE 23 Evaluate the determinant of each matrix.

a. $\begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix}$ b. $\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$ c. $\begin{bmatrix} 2-k & -3 \\ k & 3 \end{bmatrix}$

Solution a. $\begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix} = (1 \cdot 7) - (2 \cdot 4) = -1$ b. $\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} = (a \cdot a) - (0 \cdot 0) = a^2$

c. $\begin{bmatrix} 2-k & -3 \\ k & 3 \end{bmatrix} = (2-k) \cdot 3 - (-3) \cdot k = 6 - 3k + 3k = 6$

EXAMPLE 24 Evaluate the determinant of the matrix $\begin{bmatrix} 99876 & 99877 \\ 99874 & 99875 \end{bmatrix}$.

Solution Let $x = 99874$, then we can write

$$\begin{vmatrix} 99876 & 99877 \\ 99874 & 99875 \end{vmatrix} = \begin{vmatrix} x+2 & x+3 \\ x & x+1 \end{vmatrix}$$
$$= (x + 2)(x + 1) - x(x + 3)$$
$$= x^2 + 3x + 2 - x^2 - 3x$$
$$= 2.$$

EXAMPLE 25 Evaluate the determinant of each matrix.

a. $\begin{bmatrix} 15 & 18 \\ 18 & 15 \end{bmatrix}$ b. $\begin{bmatrix} 2006 & 2002 \\ 2005 & 1999 \end{bmatrix}$

Solution a. $\begin{vmatrix} 15 & 18 \\ 18 & 15 \end{vmatrix} = (15 \cdot 15) - (18 \cdot 18) = 15^2 - 18^2$

$$= (15 - 18) \cdot (15 + 18) = (-3) \cdot 33 = -99$$

b. Let $a = 1999$, then $\begin{vmatrix} 2006 & 2002 \\ 2005 & 1999 \end{vmatrix} = \begin{vmatrix} a+7 & a+3 \\ a+6 & a \end{vmatrix}$

$$= (a + 7)a - (a + 3)(a + 6)$$
$$= a^2 + 7a - (a^2 + 9a + 18)$$
$$= -2a - 18$$
$$= -2(a + 9)$$
$$= -2 \cdot 2008$$
$$= -4016.$$

2. Minors, Cofactors and Cofactor Matrices

Definition

minor

Let A be a square matrix. The **minor** of an entry a_{ij}, denoted by $\boldsymbol{M_{ij}}$, is the determinant of the square matrix formed when we delete the ith row and jth column from A.

An $n \times n$ matrix has n^2 minors.

For example, to find the minor of a_{24} in the matrix $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 2 & 2 & 4 \\ 4 & 5 & 6 & 4 \\ 7 & 8 & 9 & 4 \end{bmatrix}$, we first cross out the row and column that pass through entry a_{24} (i.e. delete the second row and the fourth column of the matrix). The determinant of the new, smaller matrix is the minor M_{24}.

matrix A	cross out all entries sharing a row or column with entry a_{24}	minor M_{24}
$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 2 & 2 & 4 \\ 4 & 5 & 6 & 4 \\ 7 & 8 & 9 & 4 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 3 & \cancel{4} \\ \cancel{2} & \cancel{2} & \cancel{2} & \cancel{4} \\ 4 & 5 & 6 & \cancel{4} \\ 7 & 8 & 9 & \cancel{4} \end{bmatrix}$	$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 0$

The determinant (therefore a minor) is a real number.

EXAMPLE 26 Find all the minors of the matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & -1 & 3 \\ 0 & 1 & 4 \end{bmatrix}$.

Solution The minors of the given matrix are M_{11}, M_{12}, M_{13}, M_{21}, M_{22}, M_{23}, M_{31}, M_{32} and M_{33}.

To find the minors, we eliminate each corresponding row and column and then calculate the determinant of the remaining entries.

$$M_{11} = \begin{bmatrix} \cancel{2} & \cancel{0} & \cancel{1} \\ \cancel{2} & -1 & 3 \\ \cancel{0} & 1 & 4 \end{bmatrix} = \begin{vmatrix} -1 & 3 \\ 1 & 4 \end{vmatrix} = (-1)\cdot 4 - 1\cdot 3 = -7 \qquad M_{12} = \begin{bmatrix} \cancel{2} & \cancel{0} & \cancel{1} \\ 2 & \cancel{-1} & 3 \\ 0 & \cancel{1} & 4 \end{bmatrix} = \begin{vmatrix} 2 & 3 \\ 0 & 4 \end{vmatrix} = 2\cdot 4 - 0\cdot 3 = 8$$

$$M_{13} = \begin{bmatrix} \cancel{2} & \cancel{0} & \cancel{1} \\ 2 & -1 & \cancel{3} \\ 0 & 1 & \cancel{4} \end{bmatrix} = \begin{vmatrix} 2 & -1 \\ 0 & 1 \end{vmatrix} = 2\cdot 1 - 0\cdot(-1) = 2 \qquad M_{21} = \begin{bmatrix} \cancel{2} & 0 & 1 \\ \cancel{2} & \cancel{-1} & \cancel{3} \\ \cancel{0} & 1 & 4 \end{bmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & 4 \end{vmatrix} = 0\cdot 4 - 1\cdot 1 = -1$$

$$M_{22} = \begin{bmatrix} 2 & 0 & 1 \\ 2 & -1 & 3 \\ 0 & 1 & 4 \end{bmatrix} = \begin{vmatrix} 2 & 1 \\ 0 & 4 \end{vmatrix} = 2 \cdot 4 - 0 \cdot 1 = 8 \qquad M_{23} = \begin{bmatrix} 2 & 0 & 1 \\ 2 & -1 & 3 \\ 0 & 1 & 4 \end{bmatrix} = \begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix} = 2 \cdot 1 - 0 \cdot 0 = 2$$

$$M_{31} = \begin{bmatrix} 2 & 0 & 1 \\ 2 & -1 & 3 \\ 0 & 1 & 4 \end{bmatrix} = \begin{vmatrix} 0 & 1 \\ -1 & 3 \end{vmatrix} = 0 \cdot 3 - (-1) \cdot 1 = 1 \qquad M_{32} = \begin{bmatrix} 2 & 0 & 1 \\ 2 & -1 & 3 \\ 0 & 1 & 4 \end{bmatrix} = \begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix} = 2 \cdot 3 - 2 \cdot 1 = 4$$

$$M_{33} = \begin{bmatrix} 2 & 0 & 1 \\ 2 & -1 & 3 \\ 0 & 1 & 4 \end{bmatrix} = \begin{vmatrix} 2 & 0 \\ 2 & -1 \end{vmatrix} = 2 \cdot (-1) - 2 \cdot 0 = -2$$

Definition

cofactor

A **cofactor** C_{ij} is a real number defined in terms of the minor M_{ij} as

$$\boldsymbol{C_{ij} = (-1)^{i+j} \cdot M_{ij}}.$$

Notice that the sign of the cofactor C_{ij} depends on i, j and the minor M_{ij}. If the power of (–1) is an even number then we leave the minor alone (multiply by 1). However, if the power of (–1) is an odd number then we multiply the minor by (–1).

We can summarize this idea in the following sign matrix. We can see that $C_{11} = M_{11}$, $C_{12} = -M_{12}$, $C_{13} = M_{13}$, and so on.

$$\begin{bmatrix} + & - & + & - & + & \cdots \\ - & + & - & + & - & \cdots \\ + & - & + & - & + & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

Definition

cofactor matrix

Let A be an $n \times n$ matrix and let C_{ij} be the cofactor of a_{ij}. Then the **cofactor matrix** of A is

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}.$$

EXAMPLE 27 Find the cofactors of the matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & -1 & 3 \\ 0 & 1 & 4 \end{bmatrix}$ from Example 4 and write the cofactor matrix of A.

Solution The minors of the matrix A (from Example 4) are

$$M_{11} = -7,\ M_{12} = 8,\ M_{13} = 2,$$
$$M_{21} = -1,\ M_{22} = 8,\ M_{23} = 2,$$
$$M_{31} = 1,\ M_{32} = 4,\ M_{33} = -2.$$

Using the formula $C_{ij} = (-1)^{i+j} \cdot M_{ij}$, we get

$$C_{11} = (-1)^{1+1} \cdot M_{11} = (-1)^2 \cdot (-7) = -7$$
$$C_{12} = (-1)^{1+2} \cdot M_{12} = (-1)^3 \cdot 8 = -8$$
$$C_{13} = (-1)^{1+3} \cdot M_{13} = (-1)^4 \cdot 2 = 2$$
$$C_{21} = (-1)^{2+1} \cdot M_{21} = (-1)^3 \cdot (-1) = 1$$
$$C_{22} = (-1)^{2+2} \cdot M_{22} = (-1)^4 \cdot 8 = 8$$
$$C_{23} = (-1)^{2+3} \cdot M_{23} = (-1)^5 \cdot 2 = -2$$
$$C_{31} = (-1)^{3+1} \cdot M_{31} = (-1)^4 \cdot 1 = 1$$
$$C_{32} = (-1)^{3+2} \cdot M_{32} = (-1)^5 \cdot 4 = -2$$
$$C_{33} = (-1)^{3+3} \cdot M_{33} = (-1)^6 \cdot (-2) = -2.$$

So the cofactor matrix of A is $\begin{bmatrix} -7 & -8 & 2 \\ 1 & 8 & -2 \\ 1 & -4 & -2 \end{bmatrix}$.

We could also write the minors in a matrix:

$$\begin{bmatrix} -7 & 8 & 2 \\ -1 & 8 & 2 \\ 1 & 4 & -2 \end{bmatrix}$$

and apply the signs

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

to get the same result.

3. The Determinant of a 3×3 Matrix

We have seen how to find the determinant of a 2 × 2 matrix. In this section we will look at two different methods for finding the determinant of a 3×3 matrix: first by using a process called **cofactor expansion**, and second by using the **rule of Sarrus**.

a. Finding the determinant with cofactors

Definition

cofactor expansion

We can calculate the determinant of a matrix $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ by multiplying the entries in any row or column by the cofactors of their minors. The **cofactor expansion** along the ith row is

$$|A| = (a_{i1} \cdot C_{i1}) + (a_{i2} \cdot C_{i2}) + (a_{i3} \cdot C_{i3}).$$

Similarly, the cofactor expansion of the jth column is

$$|A| = (a_{1j} \cdot C_{1j}) + (a_{2j} \cdot C_{2j}) + (a_{3j} \cdot C_{3j}).$$

EXAMPLE 28 Find the determinant of the matrix $A=\begin{bmatrix}-1 & 2 & 4\\ 2 & 1 & 3\\ 3 & 2 & 1\end{bmatrix}$ by cofactor expansion

a. along the first row.

b. along the second column.

Solution a. By the definition we have just seen,

For a 3×3 determinant, the sign matrix is $\begin{bmatrix}+ & - & +\\ - & + & -\\ + & - & +\end{bmatrix}$.

$$\begin{vmatrix}-1 & 2 & 4\\ 2 & 1 & 3\\ 3 & 2 & 1\end{vmatrix} = (-1)\cdot(-1)^{1+1}\begin{vmatrix}1 & 3\\ 2 & 1\end{vmatrix} + 2\cdot(-1)^{1+2}\begin{vmatrix}2 & 3\\ 3 & 1\end{vmatrix} + 4\cdot(-1)^{1+3}\begin{vmatrix}2 & 1\\ 3 & 2\end{vmatrix}$$

$$= -(1-6) - 2(2-9) + 4(4-3)$$

$$= 5 + 14 + 4 = 23.$$

b. Again by the definition,

$$\begin{vmatrix}-1 & 2 & 4\\ 2 & 1 & 3\\ 3 & 2 & 1\end{vmatrix} = 2(-1)^{1+2}\begin{vmatrix}2 & 3\\ 3 & 1\end{vmatrix} + 1(-1)^{2+2}\begin{vmatrix}-1 & 4\\ 3 & 1\end{vmatrix} + 2(-1)^{3+2}\begin{vmatrix}-1 & 4\\ 2 & 3\end{vmatrix}$$

$$= -2(2-9) + (-1-12) - 2(-3-8)$$

$$= 14 - 13 + 22 = 23.$$

Notice that the determinant is the same in each case. Indeed, it doesn't matter which row or column we choose for cofactor expansion: the determinant of a matrix will always be the same. In general, it is best to choose the row or column with the most zero entries, as this makes the calculations easier.

EXAMPLE 29 Evaluate the determinant of the matrix $A=\begin{bmatrix}2 & 4 & 0\\ 1 & 4 & 3\\ -2 & 3 & 5\end{bmatrix}$.

Solution Let us expand the cofactors along the first row, since it contains a zero.

$$\begin{vmatrix}2 & 4 & 0\\ 1 & 4 & 3\\ -2 & 3 & 5\end{vmatrix} = 2\cdot(-1)^{1+1}\begin{vmatrix}4 & 3\\ 3 & 5\end{vmatrix} + 4\cdot(-1)^{1+2}\begin{vmatrix}1 & 3\\ -2 & 5\end{vmatrix} + 0\cdot(-1)^{1+3}\begin{vmatrix}1 & 4\\ -2 & 3\end{vmatrix}$$

$$= 2(20-9) - 4(5+6) + 0$$

$$= 22 - 44 = -22$$

EXAMPLE 30 Solve for x: $\begin{vmatrix} x & 1 & x \\ 2 & 3 & 4 \\ x & 5 & x \end{vmatrix} = 16.$

Solution Let us expand the cofactors along the first row.

$$\begin{aligned}
\begin{vmatrix} x & 1 & x \\ 2 & 3 & 4 \\ x & 5 & x \end{vmatrix} &= x \cdot \begin{vmatrix} 3 & 4 \\ 5 & x \end{vmatrix} - \begin{vmatrix} 2 & 4 \\ x & x \end{vmatrix} + x \cdot \begin{vmatrix} 2 & 3 \\ x & 5 \end{vmatrix} \\
&= x(3x-20)-(2x-4x)+x(10-3x) \\
&= 3x^2-20x+2x+4x+10x-3x^2 \\
16 &= -8x \\
x &= -2
\end{aligned}$$

Check Yourself 5

1. Find the minors for the entries –3, 1 and –8 in the matrix $\begin{bmatrix} -3 & 4 & 2 \\ 6 & 3 & 1 \\ 4 & -1 & -8 \end{bmatrix}$.

2. Find the cofactors for the entries –2, 0 and 7 in the matrix $\begin{bmatrix} 1 & -2 & 4 \\ 3 & 0 & -1 \\ 5 & 7 & 2 \end{bmatrix}$.

3. Find the determinant of each matrix.

   a. $\begin{bmatrix} 1 & 10 \\ 5 & 8 \end{bmatrix}$ b. $\begin{bmatrix} 19 & 20 \\ 20 & 19 \end{bmatrix}$ c. $\begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & 3 \\ 2 & 4 & 1 \end{bmatrix}$ d. $\begin{bmatrix} -1 & 0 & 1 \\ 1 & 2 & 3 \\ 4 & 5 & 0 \end{bmatrix}$

Answers

1. $M_{11} = -23$, $M_{23} = -7$, $M_{33} = -29$ 2. $C_{12} = -11$, $C_{22} = -18$, $C_{32} = 6$

3. a. –42 b. –39 c. 7 d. 12

b. The rule of Sarrus

We can also find the determinant of a 3×3 matrix as shown in the figure below. First we copy the first two rows or columns, then we add the products of one set of diagonals and subtract the products of the other diagonals from the result.

$$\det A = \left[\begin{array}{ccc|cc} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \\ - & - & - & + \;\; + & + \end{array}\right] \quad \text{or} \quad \det A = \begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & - \\ a_{21} & a_{22} & a_{23} & - \\ a_{31} & a_{32} & a_{33} & - \\ a_{11} & a_{12} & a_{13} & + \\ a_{21} & a_{22} & a_{23} & + \\ & & & + \end{array}$$

We can define this rule more formally as follows:

Definition **rule of Sarrus**

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$. Then det A is defined as

$$\begin{aligned} \det A &= a_{11}a_{22}a_{33} + a_{21}a_{32}a_{13} + a_{31}a_{12}a_{23} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12} \\ &= \boldsymbol{a_{11}a_{22}a_{33} + a_{21}a_{32}a_{13} + a_{31}a_{12}a_{23} - (a_{31}a_{22}a_{13} + a_{32}a_{23}a_{11} + a_{33}a_{21}a_{12})}. \end{aligned}$$

EXAMPLE 31 Use the rule of Sarrus to evaluate the determinant of the matrix $A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 5 & -1 \\ 3 & -2 & 3 \end{bmatrix}$.

Solution Let us write the first column and the second column of A after the third column:

$$A = \left[\begin{array}{ccc} 1 & 3 & 4 \\ 2 & 5 & -1 \\ 3 & -2 & 3 \end{array}\right] \begin{array}{cc} 1 & 3 \\ 2 & 5 \\ 3 & -2 \end{array}$$

$$\begin{array}{ccc} - & \qquad & + \\ - & & + \\ - & & + \end{array}$$

So by the rule of Sarrus we have

$$\begin{aligned} \det A &= [(1\cdot 5\cdot 3) + (3\cdot(-1)\cdot 3) + (4\cdot 2\cdot(-2))] - [(4\cdot 5\cdot 3) + (1\cdot(-1)\cdot(-2)) + (3\cdot 2\cdot 3)] \\ &= (15 - 9 - 16) - (60 + 2 + 18) \\ &= -10 - 80 \\ &= -90. \end{aligned}$$

We can check this result by using cofactor expansion. Let us expand the cofactors along the first row:

$$\begin{vmatrix} 1 & 3 & 4 \\ 2 & 5 & -1 \\ 3 & -2 & 3 \end{vmatrix} = 1\cdot\begin{vmatrix} 5 & -1 \\ -2 & 3 \end{vmatrix} - 3\begin{vmatrix} 2 & -1 \\ 3 & 3 \end{vmatrix} + 4\begin{vmatrix} 2 & 5 \\ 3 & -2 \end{vmatrix}$$

$$\begin{aligned} &= 1\cdot(15-2) - 3(6+3) + 4(-4-15) \\ &= 13 - 27 - 76 \\ &= -90. \end{aligned}$$

As we can see, the determinant is the same in each case.

EXAMPLE 32 Use the rule of Sarrus to evaluate the determinant of the matrix $A = \begin{bmatrix} 1 & i & i+1 \\ 0 & 1 & i-1 \\ 0 & i & i \end{bmatrix}$, where $i^2 = -1$.

Solution Use the rule of Sarrus.

$$
\begin{array}{c}
A = \begin{bmatrix} 1 & i & i+1 \\ 0 & 1 & i-1 \\ 0 & i & i \end{bmatrix} \\
\begin{array}{ccccc} - & 1 & i & i+1 & + \\ - & 0 & i & i & + \\ - & & & & + \end{array}
\end{array}
$$

$$
\begin{aligned}
\det A &= (i + 0 + 0) - (0 + i^2 - i + 0) \\
&= i - (-1 - i) \\
&= i + 1 + i \\
&= 2i + 1.
\end{aligned}
$$

Check Yourself 6

1. Find the determinant of each matrix by using the rule of Sarrus.

a. $\begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 1 \\ 0 & 2 & 1 \end{bmatrix}$ b. $\begin{bmatrix} 1 & -2 & 3 \\ -1 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$

Answers

1. a. 1 b. –7

EXAMPLE 33 Find the determinant of the matrix $A = \begin{bmatrix} \frac{1}{2} & \frac{1}{6} & 1 \\ -\frac{2}{3} & \frac{1}{3} & 0 \\ \frac{1}{4} & -\frac{1}{4} & 0 \end{bmatrix}$.

Solution Since the third column has two zeros, we can expand the determinant along the third column. However, the fractions in the matrix will make the calculations complicated. Therefore, before expanding let us see if we can get rid of the fractions inside the matrix.

Notice that $\frac{1}{2} = \frac{3}{6}$. We can write

$$
\begin{vmatrix} \frac{1}{2} & \frac{1}{6} & 1 \\ -\frac{2}{3} & \frac{1}{3} & 0 \\ \frac{1}{4} & -\frac{1}{4} & 0 \end{vmatrix} = \begin{vmatrix} \frac{3}{6} & \frac{1}{6} & \frac{6}{6} \\ -\frac{2}{3} & \frac{1}{3} & 0 \\ \frac{1}{4} & -\frac{1}{4} & 0 \end{vmatrix} = \begin{vmatrix} \frac{1}{6}\cdot 3 & \frac{1}{6}\cdot 1 & \frac{1}{6}\cdot 6 \\ \frac{1}{3}\cdot(-2) & \frac{1}{3} & 0 \\ \frac{1}{4} & -\frac{1}{4} & 0 \end{vmatrix}.
$$

So there is a common factor of $\frac{1}{6}$ in the first row. Likewise, we can factor $\frac{1}{3}$ out of the second row and $\frac{1}{4}$ out of the third row. As a result, we have

$$\det A = \begin{vmatrix} \frac{1}{6}\cdot 3 & \frac{1}{6}\cdot 1 & \frac{1}{6}\cdot 6 \\ \frac{1}{3}\cdot(-2) & \frac{1}{3} & 0 \\ \frac{1}{4} & -\frac{1}{4} & 0 \end{vmatrix} = \frac{1}{6}\cdot\frac{1}{3}\cdot\frac{1}{4}\cdot\begin{vmatrix} 3 & 1 & 6 \\ -2 & 1 & 0 \\ 1 & -1 & 0 \end{vmatrix}.$$

Now the calculations are easier. Expanding along the third column we get

$$\frac{1}{6}\cdot\frac{1}{3}\cdot\frac{1}{4}\cdot\begin{vmatrix} 3 & 1 & 6 \\ -2 & 1 & 0 \\ 1 & -1 & 0 \end{vmatrix} = \frac{1}{72}\cdot 6\cdot\begin{vmatrix} -2 & 1 \\ 1 & -1 \end{vmatrix} = \frac{1}{12}.$$ This is the determinant.

Property

Let A be an $n \times n$ matrix. If each element in one row (or column) of its determinant is multiplied by a real number k and the resulting elements are then added to the corresponding elements of another row (or column), then the resulting determinant is the same as the original one.

EXAMPLE 34 Evaluate the determinant of the matrix $A = \begin{bmatrix} 1 & 4 & 1 \\ 2 & -1 & 0 \\ 0 & 18 & 4 \end{bmatrix}$.

Solution Add –2 times the first row to the second row:

$$\det A = \begin{vmatrix} 1 & 4 & 1 \\ 2 & -1 & 0 \\ 0 & 18 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 4 & 1 \\ 0 & -9 & -2 \\ 0 & 18 & 4 \end{vmatrix}$$

By the property we have just seen, this new determinant is equal to det A.

Notice that the third row is a multiple of the second row, i.e. $[0 \quad 18 \quad 4] = -2[0 \quad -9 \quad -2]$. So $|A| = 0$.

EXAMPLE 35 Prove that $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (b-a)(c-a)(c-b)$.

Solution Subtract the first row from the second row and the third row:

$$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = \begin{vmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{vmatrix}.$$

Expand the determinant by the first column:

$$\begin{vmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{vmatrix} = \begin{vmatrix} b-a & b^2-a^2 \\ c-a & c^2-a^2 \end{vmatrix} = \begin{vmatrix} b-a & (b-a)(b+a) \\ c-a & (c-a)(c+a) \end{vmatrix}.$$

Factor out $(b-a)$ and $(c-a)$:

$$= (b-a)(c-a)\begin{vmatrix} 1 & (b+a) \\ 1 & (c+a) \end{vmatrix}$$

$= (b-a)(c-a)(c-b)$, as required.

EXAMPLE 36 Find the determinant of $A = \begin{bmatrix} 2 & -3 & 10 \\ 1 & -2 & -2 \\ 0 & 1 & -3 \end{bmatrix}$.

Solution Using the properties we have seen, we can rewrite $|A|$ in triangular form as follows:

$$\begin{vmatrix} 2 & -3 & 10 \\ 1 & -2 & -2 \\ 0 & 1 & -3 \end{vmatrix} = -\begin{vmatrix} 1 & 2 & -2 \\ 2 & -3 & 10 \\ 0 & 1 & -3 \end{vmatrix}$$

Interchange the first two rows.

$$= -\begin{vmatrix} 1 & 2 & -2 \\ 0 & -7 & 14 \\ 0 & 1 & -3 \end{vmatrix}$$

Add –2 times the first row to the second row to get a new second row.

$$= 7\begin{vmatrix} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 0 & 1 & -3 \end{vmatrix}$$

Factor –7 out of the second row.

$$= 7\begin{vmatrix} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & -1 \end{vmatrix}.$$

Add –1 times the second row to the third row to get a new third row.

Now, because the final determinant is the determinant of a triangular matrix, we can conclude that the determinant is $|A| = 7 \cdot 1 \cdot 1 \cdot (-1) = -7$.

Property

Let A and B be two square matrices of the same size, then the determinant of their matrix product is the product of their determinants, i.e.

$$\det(AB) = \det A \cdot \det B.$$

EXAMPLE 37 Compute det (A^2) if det $A = 5$.

Solution det (A^2) = det (AA) = (det A) · (det A) = 5 · 5 = 25.

EXAMPLE 38 Verify the theorem det (AB) = det A · det B for the matrices $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix}$.

Solution $A \cdot B = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 3 & -5 \end{bmatrix}$

$\det(A \cdot B) = \begin{vmatrix} 2 & 0 \\ 3 & -5 \end{vmatrix} = -10.$ (1)

Also, $\det A = \begin{vmatrix} 1 & 2 \\ -1 & 3 \end{vmatrix} = 3-(-2) = 5$ and $\det B = \begin{vmatrix} 0 & 2 \\ 1 & -1 \end{vmatrix} = -2$, so

det A · det B = 5 · (–2) = –10. (2)

Combining (1) and (2) gives us det (AB) = det A · det B, as required.

Theorem

Let A be an $n \times n$ matrix. If the elements of a row or column are expressed as binomials, the determinant can be written as the sum of two determinants as follows:

$$\begin{vmatrix} a+k & b+l & c+m \\ d & e & f \\ g & h & i \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} + \begin{vmatrix} k & l & m \\ d & e & f \\ g & h & i \end{vmatrix}.$$

EXAMPLE 39 Evaluate the following determinants.

a. $\begin{vmatrix} 165 & 187 \\ 167 & 188 \end{vmatrix}$

b. $\begin{vmatrix} 2004 & 2005 & 2006 \\ 2003 & 2004 & 2005 \\ 2002 & 2003 & 2004 \end{vmatrix}$

Solution **a.** $\begin{vmatrix} 165 & 187 \\ 167 & 188 \end{vmatrix} = \begin{vmatrix} 165 & 187 \\ 165+2 & 187+1 \end{vmatrix} = \begin{vmatrix} 165 & 187 \\ 165 & 187 \end{vmatrix} + \begin{vmatrix} 165 & 187 \\ 2 & 1 \end{vmatrix}$

Since the rows are the same, $\begin{vmatrix} 165 & 187 \\ 165 & 187 \end{vmatrix} = 0.$

So $\begin{vmatrix} 165 & 187 \\ 167 & 188 \end{vmatrix} = \begin{vmatrix} 165 & 187 \\ 2 & 1 \end{vmatrix} = 165 \cdot 1 - 187 \cdot 2 = -209.$

b. $\begin{vmatrix} 2004 & 2005 & 2006 \\ 2003 & 2004 & 2005 \\ 2002 & 2003 & 2004 \end{vmatrix} = \begin{vmatrix} 2003+1 & 2004+1 & 2005+1 \\ 2003 & 2004 & 2005 \\ 2002 & 2003 & 2004 \end{vmatrix}$

$$= \underbrace{\begin{vmatrix} 2003 & 2004 & 2005 \\ 2003 & 2004 & 2005 \\ 2002 & 2003 & 2004 \end{vmatrix}}_{\substack{\text{since the first row and the} \\ \text{second row are the same,} \\ \text{the determinant is zero}}} + \begin{vmatrix} 1 & 1 & 1 \\ 2003 & 2004 & 2005 \\ 2002 & 2003 & 2004 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 2002+1 & 2003+1 & 2004+1 \\ 2002 & 2003 & 2004 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 2002 & 2003 & 2004 \\ 2002 & 2003 & 2004 \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 2002 & 2003 & 2004 \end{vmatrix}$$

$$= 0 + 0 = 0.$$

Check Yourself 7

1. Which property of determinants is illustrated by each equation?

a. $\begin{vmatrix} 1 & 2 \\ -2 & -4 \end{vmatrix} = 0$

b. $\begin{vmatrix} 2 & 3 & 5 \\ 0 & 0 & 0 \\ 4 & 2 & -1 \end{vmatrix} = 0$

c. $\begin{vmatrix} 1 & -2 & -3 \\ 7 & 2 & 0 \\ 1 & -2 & -3 \end{vmatrix} = 0$

d. $\begin{vmatrix} 1 & 3 & 2 \\ 2 & 5 & 4 \\ 0 & 2 & 1 \end{vmatrix} = -\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 0 & 1 & 2 \end{vmatrix}$

e. $\begin{vmatrix} 2 & 5 & 0 \\ 1 & 7 & 2 \\ 3 & 2 & 1 \end{vmatrix} = -\begin{vmatrix} 3 & 2 & 1 \\ 1 & 7 & 2 \\ 2 & 5 & 0 \end{vmatrix}$

f. $\begin{vmatrix} 4 & 12 \\ -3 & 2 \end{vmatrix} = 4\begin{vmatrix} 1 & 3 \\ -3 & 2 \end{vmatrix}$

g. $\begin{vmatrix} 3 & 6 & 0 \\ 15 & 10 & 20 \\ 1 & 0 & 0 \end{vmatrix} = 15 \cdot \begin{vmatrix} 1 & 2 & 0 \\ 3 & 2 & 4 \\ 1 & 0 & 0 \end{vmatrix}$

h. $\begin{vmatrix} 2 & 5 & 0 \\ 1 & 7 & 2 \\ 3 & 2 & 1 \end{vmatrix} = -\begin{vmatrix} 3 & 2 & 1 \\ 1 & 7 & 2 \\ 2 & 5 & 0 \end{vmatrix}$

i. $\begin{vmatrix} 3 & -2 \\ 9 & 5 \end{vmatrix} = \begin{vmatrix} 3 & -2 \\ 0 & 11 \end{vmatrix}$

j. $\begin{vmatrix} 0 & 5 & 2 \\ 2 & -1 & 3 \\ 7 & 5 & 0 \end{vmatrix} = -\begin{vmatrix} 0 & 5 & 2 \\ -2 & 1 & -3 \\ 7 & 5 & 0 \end{vmatrix}$

2. Compute $\det(A^2)$ if $\det A = -3$.

3. $A = \begin{bmatrix} 1 & 3 \\ 4 & -2 \end{bmatrix}$ and $\det(AB) = 28$ are given. Find $\det B$.

Answers

2. 9 3. –2

DATA MATRICES

If you visit any large supermarket today, the cashier will calculate your grocery bill using a bar code reader. The reader reads a set of vertical lines called a **bar code**, which is printed on grocery packaging to uniquely identify each product. Linear bar codes like this are used in many modern applications: they are printed on boxes, product labels, magazines and books, and help companies to track their stock as it is manufactured, transported and sold.

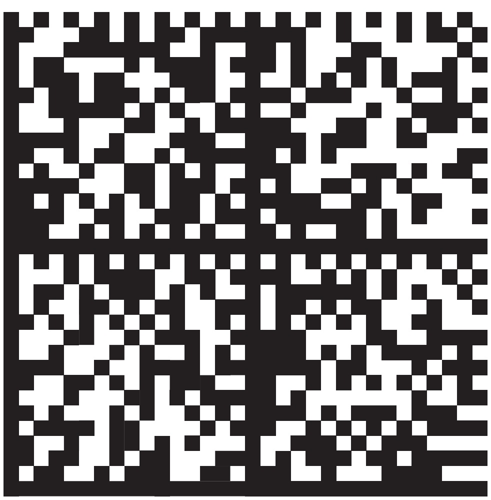
However, linear bar codes (also called *UPC bar codes*) have some limitations. In particular, the vertical lines can only be read accurately if they are printed in high contrast, which requires good quality printing and means that the bar code cannot be very small. This makes linear bar codes unsuitable for some things, such as identifying small electronic components.

A **data matrix** is a different kind of bar code. Instead of using vertical lines, a data matrix stores information as a set of black and white squares (called *cells*) which are arranged in a square or rectangle. Data matrices can be printed in low contrast and can also be very small, which makes them appropriate for marking and tracing small parts. Data matrices are widely used in the automotive, aerospace, electronics, semiconductor and pharmaceutical industries.

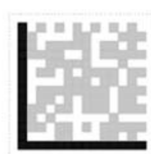
A data matrix has four main parts:

- an *L*-shaped solid border around two sides,
- an *L*-shaped dotted border along the other two sides,
- the data storage area inside the symbol, and
- a 'quiet zone' around the whole symbol, which must not contain any writing or marks.

The special coding used to create the data matrix is free for anyone to use, which means that anyone can create and read data matrix codes without paying a license.



Data matrix



Solid border

Broken border

Data storage area

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Linear bar code

Advantages of Data Matrices

Data matrices have the following advantages over traditional linear bar codes:

- They encode information digitally (as a set of 0 and 1 digits).
- They can be very small and printed with low contrast, so they can be printed directly on small parts.
- They offer high *information density*, which means that a small matrix can carry a lot of data.
- They can be printed at different sizes.
- They have good built-in error correction, which means that a data matrix can still be read even if up to 20% of it is missing or damaged.
- They are read by video cameras instead of a laser beam, which means that they can be read in any orientation (e.g. upside down).

Storage Capacity

A data matrix is square or sometimes rectangular, and is made up of an arrangement of between 8 x 8 and 144 x 144 black and white cells. A matrix can store up to 3116 numbers, or 2335 alphanumeric characters (i.e. letters and numbers). In computing terms, a data matrix can store anything between a few bytes and two kilobytes of data. This is much more efficient than a UPC linear bar code, which can encode only 12 numbers plus one 'check digit' for error correction.

Activities

- Search the Internet for online barcode generators. Try making a data matrix containing your name and the name of your school (select 'data matrix' as your symbology or bar code type). How big is each matrix?

  Now try making a linear bar code with the same information (select 'UPC' as your data symbology or bar code type). Can you do it?
- How could you write a data matrix as a numerical matrix like the ones you are studying in this book?
- How many distinct pieces of information can be encoded if the area inside the border of a data matrix is

  a) a 1 x 1 matrix?

  b) a 2 x 2 matrix?

  c) a 5 x 5 matrix?
- What size of square matrix would you need in order to encode every letter of the English alphabet, if each matrix had to contain one letter?

EXERCISES 2.2

A. The Determinant of a Matrix

1. Evaluate the determinants.

a. $\begin{vmatrix} 1 & 2 \\ 5 & 8 \end{vmatrix}$ b. $\begin{vmatrix} 7 & 0 \\ 11 & 6 \end{vmatrix}$ c. $\begin{vmatrix} 13 & 12 \\ 12 & 13 \end{vmatrix}$

d. $\begin{vmatrix} 101 & 100 \\ 100 & 101 \end{vmatrix}$ e. $\begin{vmatrix} a & a+1 \\ a-1 & a \end{vmatrix}$ f. $\begin{vmatrix} x-2 & x \\ x & x-2 \end{vmatrix}$

2. Evaluate the determinants.

a. $\begin{vmatrix} 25 & 26 \\ 27 & 28 \end{vmatrix}$ b. $\begin{vmatrix} 105 & 107 \\ 109 & 111 \end{vmatrix}$

c. $\begin{vmatrix} 1113 & 1117 \\ 1111 & 1115 \end{vmatrix}$ d. $\begin{vmatrix} \cos 25° & -\sin 25° \\ \sin 25° & \cos 25° \end{vmatrix}$

e. $\begin{vmatrix} \cos 15° & \sin 15° \\ \sin 15° & \cos 15° \end{vmatrix}$ f. $\begin{vmatrix} 2007-a & 2008-a \\ 2009-a & 2010-a \end{vmatrix}$

3. Given the matrix $A = \begin{bmatrix} 4 & 11 & 9 \\ 0 & 3 & 2 \\ 3 & 1 & 1 \end{bmatrix}$, compute the minors M_{12}, M_{23} and M_{31}.

4. Given the matrix $A = \begin{bmatrix} -1 & 2 & 7 \\ 3 & 0 & -2 \\ 4 & 11 & 1 \end{bmatrix}$, compute the cofactors C_{12}, C_{23} and C_{31}.

5. Evaluate the determinants.

a. $\begin{vmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \\ 3 & 0 & 5 \end{vmatrix}$ b. $\begin{vmatrix} 3 & -2 & 1 \\ 4 & 3 & 0 \\ 5 & 1 & 0 \end{vmatrix}$

c. $\begin{vmatrix} 2 & -1 & 3 \\ 1 & 0 & 2 \\ 1 & 2 & 1 \end{vmatrix}$ d. $\begin{vmatrix} 15 & 12 & 10 \\ 0 & 0 & 1 \\ 4 & 5 & 6 \end{vmatrix}$

e. $\begin{vmatrix} 15 & 15 & 10 \\ 2 & 2 & 4 \\ 1 & 1 & 6 \end{vmatrix}$ f. $\begin{vmatrix} 2 & -1 & 3 \\ 1 & 2 & -1 \\ 3 & -4 & 7 \end{vmatrix}$

7. Find x in each case.

a. $\begin{vmatrix} 3 & 4 & 3 \\ -2 & x & 4 \\ x & -1 & x \end{vmatrix} = 30$ b. $\begin{vmatrix} x & 1 & x \\ x & 1 & 1 \\ 1 & 1 & x \end{vmatrix} = 0$

c. $\begin{vmatrix} 2 & 2 & 0 \\ 2 & x & 0 \\ x & 2 & 6 \end{vmatrix} = 0$ d. $\begin{vmatrix} 1 & 2 & x-1 \\ x & 2x & x \\ 0 & x & 1 \end{vmatrix} = 0$

CHAPTER REVIEW TEST 2A

1. The following matrix shows the results of a recent poll.

$$\begin{array}{l} \\ \textit{Proposition 1} \\ \textit{Proposition 2} \\ \textit{Proposition 3} \end{array} \begin{array}{c} \begin{array}{cc} \textit{For} & \textit{Against} \end{array} \\ \begin{bmatrix} 1553 & 771 \\ 689 & 1633 \\ 2088 & 229 \end{bmatrix} \end{array}$$

Given these results, which conclusion is <u>false</u>?

A) There were 771 votes against Proposition 1.

B) More people voted against Proposition 1 than voted for Proposition 2.

C) Proposition 2 has little chance of passing.

D) More people voted for Proposition 1 than for Proposition 3.

E) There were 229 votes against Proposition 3.

2. $\begin{bmatrix} 4 & -1 \\ 7 & 2 \end{bmatrix} + \begin{bmatrix} -6 & 5 \\ -3 & -1 \end{bmatrix} = ?$

A) $\begin{bmatrix} -2 & 1 \\ 4 & 1 \end{bmatrix}$ B) $\begin{bmatrix} 4 & -2 \\ 4 & 1 \end{bmatrix}$ C) $\begin{bmatrix} -2 & 4 \\ 1 & 4 \end{bmatrix}$

D) $\begin{bmatrix} 4 & 1 \\ -1 & 4 \end{bmatrix}$ E) $\begin{bmatrix} -2 & 4 \\ 4 & 1 \end{bmatrix}$

3. If $A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$, what is $(2 \cdot a_{12}) + (3 \cdot a_{22})$?

A) –4 B) –2 C) 3 D) 5 E) 7

4. If $A = \begin{bmatrix} -2 & 1 & -1 \\ 1 & 0 & 2 \end{bmatrix}$, what is $a_{23} - (2 \cdot a_{12}) + a_{11}$?

A) 5 B) 3 C) 1 D) –2 E) –3

5. $A = \begin{bmatrix} a & 0 \\ 0 & 1-b \end{bmatrix}$ is a zero matrix. What is $a + b$?

A) –1 B) 0 C) 1 D) 2 E) 4

6. $A = \begin{bmatrix} 1 & 2 \\ -3 & a \end{bmatrix}$ and $B = \begin{bmatrix} b & a \\ -3 & c \end{bmatrix}$ are equal. Find $a + b + c$.

A) –1 B) 1 C) 2 D) 4 E) 5

7. If $\begin{bmatrix} 1 & -2 \\ 0 & b \end{bmatrix} + \begin{bmatrix} a & 1 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 3 & 3 \end{bmatrix}$, what is $a + b$?

A) 5 B) 1 C) 3 D) 4 E) 0

8. What is $[\sin x \quad \cos x] \cdot \begin{bmatrix} \sin x \\ \cos x \end{bmatrix}$?

A) $[\cos x \quad \sin x]$ B) $\begin{bmatrix} \sin x \\ \cos x \end{bmatrix}$ C) $[1]$

D) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ E) $[-1]$

9. What is $\begin{bmatrix} 1 & 5 & 6 \\ 2 & -4 & 0 \end{bmatrix} \cdot \begin{bmatrix} 6 & 4 \\ -2 & 0 \\ 8 & 0 \end{bmatrix}$?

A) $\begin{bmatrix} 44 & 4 \\ 20 & 8 \end{bmatrix}$ B) $\begin{bmatrix} 4 & 44 \\ 20 & 8 \end{bmatrix}$ C) $\begin{bmatrix} 44 & 4 \\ 8 & 20 \end{bmatrix}$

D) $\begin{bmatrix} 20 & 8 \\ 4 & 44 \end{bmatrix}$ E) $\begin{bmatrix} 4 & 20 \\ 44 & 4 \end{bmatrix}$

10. If $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, what is A^{207}?

A) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ B) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ C) $\begin{bmatrix} 1 & 207 \\ 0 & 1 \end{bmatrix}$

D) $\begin{bmatrix} 1 & 1 \\ 0 & 207 \end{bmatrix}$ E) $\begin{bmatrix} 1 & 0 \\ 0 & 207 \end{bmatrix}$

11. What is the inverse of $\begin{bmatrix} 5 & -3 \\ -2 & 1 \end{bmatrix}$?

A) $\begin{bmatrix} -5 & -3 \\ 2 & -1 \end{bmatrix}$ B) $\begin{bmatrix} -1 & -3 \\ -2 & -5 \end{bmatrix}$ C) $\begin{bmatrix} -1 & 3 \\ 2 & -5 \end{bmatrix}$

D) $\begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$ E) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

12. If $[1 \;\; 2 \;\; a \;\; 5] \cdot \begin{bmatrix} a \\ 2 \\ 3 \\ 4 \end{bmatrix} = [0]$, what is a?

A) –6 B) –4 C) 3 D) 4 E) 5

13. If $\begin{bmatrix} 3 & a \\ 2 & a+1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ x \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, what is a?

A) –3 B) 1 C) 2 D) –1 E) 3

14. $A = \begin{bmatrix} x & 1 \\ y & 1 \end{bmatrix}$ and $A^{-1} \cdot A = A^2$. What is $y - x$?

A) –5 B) –4 C) –3 D) –2 E) –1

15. Which matrix has no inverse?

A) $\begin{bmatrix} 6 & 0 \\ 0 & 5 \end{bmatrix}$ B) $\begin{bmatrix} 4 & 6 \\ -6 & -9 \end{bmatrix}$ C) $\begin{bmatrix} -2 & 4 \\ 3 & 6 \end{bmatrix}$

D) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ E) $\begin{bmatrix} -4 & 4 \\ 2 & 1 \end{bmatrix}$

16. Find $\begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} + 2 \cdot \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}$.

A) $\begin{bmatrix} -2 & 1 \\ 4 & 1 \end{bmatrix}$ B) $\begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$ C) $\begin{bmatrix} -2 & 4 \\ 1 & 4 \end{bmatrix}$

D) $\begin{bmatrix} 4 & 1 \\ -1 & 4 \end{bmatrix}$ E) $\begin{bmatrix} 4 & -1 \\ 3 & 0 \end{bmatrix}$

17. If $\begin{bmatrix} 4 & 2 & 1 & 0 \\ -5 & 2 & 0 & 8 \\ 2 & -3 & 8 & 1 \end{bmatrix}$, what is $a_{11} \cdot a_{23} - a_{13}$?

A) 32 B) 24 C) –1 D) 0 E) –2

18. What are the values of x and y in this matrix equation?

$$2x \cdot \begin{bmatrix} -2 & -1 \\ -10 & 5 \end{bmatrix} = \begin{bmatrix} -16 & -8 \\ y & 40 \end{bmatrix}$$

A) 4, –80 B) –4, –80 C) 4, 80

D) 8, –80 E) 2, –40

19. $A = \begin{bmatrix} a & 0 \\ b & 2-c \end{bmatrix}$ is an identity matrix. Find $a + b + c$.

A) –1 B) 0 C) 1 D) 2 E) 4

20. If $A = \begin{bmatrix} 1 & 1 \\ 4 & 2 \end{bmatrix}$ and $f(x) = x^2 + x - 2$, what is $f(A)$?

A) $\begin{bmatrix} 1 & 4 \\ 4 & 5 \end{bmatrix}$ B) $\begin{bmatrix} 4 & -2 \\ 4 & 1 \end{bmatrix}$ C) $\begin{bmatrix} 2 & 4 \\ 1 & 5 \end{bmatrix}$

D) $\begin{bmatrix} 4 & 4 \\ 16 & 8 \end{bmatrix}$ E) $\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$

21. If $A = \begin{bmatrix} 1 & 2 \\ -1 & a \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} \frac{3}{5} & \frac{-2}{5} \\ \frac{1}{5} & \frac{1}{5} \end{bmatrix}$, what is a?

A) 5 B) 1 C) 3 D) 4 E) 0

1. A triangle with vertices $(-4, 1)$, $(2, 3)$ and $(x, -1)$ has an area of 11 square units. What is x if x is a positive number?

A) 11 B) 9 C) 6 D) 2 E) 1

2. $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ is given. Find $\det(A^{-1})$.

A) 25 B) 5 C) 1 D) $\frac{1}{5}$ E) $\frac{1}{25}$

3. Solve $\begin{vmatrix} x & -1 \\ 1 & x^2 \end{vmatrix} = \begin{vmatrix} x^2 & -1 \\ x+1 & 1 \end{vmatrix}$.

A) 3 B) 2 C) 1 D) 0 E) -1

4. Evaluate the determinant $\begin{vmatrix} \log_2 6 & 1 \\ \ln e & \log_{36} 16 \end{vmatrix}$.

A) 0 B) 1 C) 2 D) 3 E) 4

5. $A = \begin{bmatrix} \cos 2x & \cos x \\ \sin 2x & \sin x \end{bmatrix}$ is given. What is $\dfrac{|A|}{\cos x}$?

A) $\tan x$ B) $\cot x$ C) 1 D) $\sin x$ E) $\sin 2x$

6. A is a 3×3 matrix with $|A| = 2$. Find $\det(2A)$.

A) 32 B) 30 C) 24 D) 16 E) 8

7. Given $\begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 11 \end{bmatrix}$, find z.

A) -2 B) $-\frac{1}{2}$ C) $-\frac{1}{3}$ D) 3 E) 4

8. Evaluate $\begin{vmatrix} a^2 & -a \\ b^2 & b \end{vmatrix}$ if a and b are the roots of the equation $x^2 + x - 2 = 0$.

A) -4 B) -2 C) 1 D) 2 E) 4

9. $f(x)=\begin{vmatrix} 2\sin x & 1+\cos x \\ 1-\cos x & \sin x \end{vmatrix}$. Find $f(\frac{\pi}{3})$.

A) $\frac{1}{4}$ B) $\frac{3}{4}$ C) 1 D) 2 E) 4

10. Given $\begin{vmatrix} e^x & e^2 \\ e^{-2} & e^x \end{vmatrix} = 24$, find x.

A) –5 B) 3 C) 2 D) ln 2 E) ln 5

11. Find the sum of the entries in the first column of the adjoint of the matrix $A=\begin{bmatrix} 1 & 2 & -1 \\ 2 & 3 & 0 \\ 1 & -1 & 2 \end{bmatrix}$.

A) –5 B) –4 C) –3 D) 2 E) 4

12. Given $A=\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$ and $B^{-1}=\begin{bmatrix} 1 & -2 \\ 3 & 2 \end{bmatrix}$, find $\det(A^{-1}B)$.

A) $\frac{1}{80}$ B) $\frac{1}{40}$ C) 2 D) $\frac{12}{5}$ E) $\frac{14}{5}$

13. If $\begin{bmatrix} 2x & x & x \\ -3x & -2x & -x \\ x & x & x \end{bmatrix} = -27$, what is x?

A) 3 B) 1 C) –1 D) –3 E) –4

14. $A=\begin{bmatrix} b+c & c & b \\ c & c+a & a \\ b & a & a+b \end{bmatrix}$. Simplify $\dfrac{|A|}{a\cdot b\cdot c}$.

A) 4 B) 1 C) $a + b$ D) $a + b + c$ E) –1

15. Find the determinant of the matrix $\begin{bmatrix} 1 & x & y+z \\ 1 & y & x+z \\ 1 & z & x+y \end{bmatrix}$.

A) $x \cdot y \cdot z$ B) 0 C) –1 D) $x + y + z$ E) 1

16. Which one of the following <u>cannot</u> be a value of x if the inverse of the matrix $\begin{bmatrix} 1 & x & -1 \\ 2 & -1 & 3 \\ 1 & -2 & 1 \end{bmatrix}$ exists?

A) –1 B) –2 C) –5 D) –8 E) –11

CHAPTER 3

MATHEMATICAL INDUCTION

1 INTRODUCTION TO MATHEMATICAL INDUCTION

Many interesting and important results in mathematics have been discovered by first observing patterns in some specific cases and then making generalisations from the observations, that is, by applying inductive reasoning. For example, it is easy to form a conjecture about the sum of the first k odd positive integers.

$$\begin{aligned} &1 &&= 1 \\ &1 + 3 &&= 4 \\ &1 + 3 + 5 &&= 9 \\ &1 + 3 + 5 + 7 &&= 16 \\ &1 + 3 + 5 + 7 + 9 &&= 25 \\ &\vdots \end{aligned}$$

The numbers 1, 4, 16, and 25 are the squares of the first five positive integers, and so it appears that the sum of the first k odd positive integers is exactly k^2. Since the kth odd positive integer may be written as (2k – 1), the conjecture may also be expressed as an equation:

$$1 + 3 + 5 + \dots + (2k - 1) = k^2$$

Although this particular formula is valid, it is important for you to see that recognising a pattern and then simply jumping to the conclusion that the pattern must be true for all values of n is not a logically valid method of proof. There are many examples in which a pattern appears to be developing for small values of n and then at some points the pattern fails. One of the most famous cases of this was the conjecture by the French mathematician Pierre de Fermat (1601 - 1655), who speculated that all numbers of the form

$$F_n = 2^{2^n} + 1, n = 0,1,2, \dots$$

are prime. For $n = 0, 1, 2, 3,$ and 4, the conjecture is true.

The fifth Fermat number ($F_5 = 4294967297$) is so great that it was difficult for Fermat to determine whether it was prime or not. However, another well-known mathematician, Leonhard Euler (1707 - 1783) , later found the factorisation

$$F_5 = 4294967297 = 641(6700417)$$

which proved that F_5 is not prime, and therefore Fermat's conjecture was false.

Just because a rule, pattern, or formula seems to work for several values of n, you cannot simply decide that it is valid for all values of n without goind through a legitimate proof.

A. MATHEMATICAL INDUCTION

Mathematical induction is based on simple characteristics of the natural numbers which are stated in the following theorem.

Theorem

Induction Theorem

If ID is a subset of natural numbers so that

1. $1 \in ID$ and
2. $k \in ID$ implies $k + 1 \in ID$, then $ID = IN^+$

Proof

On the contrary, that $ID \neq IN^+$, then there exists at least one element, say $m \in IN^+$ such that $m \notin ID$. So this implies that $(m - 1) \notin ID$, and if $(m - 1) \notin ID$, then $(m - 1) - 1 = (m - 2) \notin ID$. By this way, we can easily see that,

$$(m - 3), (m - 4), \ldots , 3, 2, 1$$

will not be an element of ID. But by induction theorem, $1 \in ID$, so we conclude the this is a contradiction. Therefore, $ID = IN^+$.

Axiom

Let IN+ be the non-empty set with

1. $1 \in IN^+$.
2. For each $n \in IN^+$, $\exists n^* \in IN+$ called successor of n.
3. For each $n \in IN^+$, we have $n^* \neq 1$.
4. If $m, n \in IN^+$ and $m^* = n^*$, then $m = n$.
5. An subset K of IN^+ having the properties
   a. $1 \in K$
   b. $k^* \in K$ whenever $k \in K$ is equal to IN^+

In order to prove a proposition by mathematical induction, it is enough to satisfy the fifth Peano Axiom. These axioms will investigated in advanced algebra.

Induction theorem is useful particularly for proving propositions involving the positive integers. To formulate the following principle of mathematical induction, it is convenient to let P_n be a proposition where n is the positive integer.

Conclusion

Principle of Mathematical Induction

Let $a \in IN^+$ and P_n be a proposition involving the positive integer $n \geq a$. If,

1. the proposition is true for $n = 0$, that is P_a is true, and
2. for every positive integer k, the truth of P_k implies the truth of P_{k+1}, then P_n must be true for all positive integers n.

The set $N_a = \{n \mid n \geq a\}$ where $a \in IN^+$ is called the truth set of P_n. Note that, after verifying that P_a is true you must assume that P_k is true for some $k \in N^+$. This assumption is called "Induction hypothesis".

A well - known illustration used to explain why the principle of mathematical induction works is the unending domino line. Proving that "P_1 is true" is like knocking over the first domina. Proving that "whenever P_k is true then P_{k+1} is also true" is likehaving the dominos arranged so that each one would knock down the next one as it fell. That means, you could knock them all down simply by pushing the first one. Mathematical induction works in the same way. Since P_1 is true, then P_2 must be true. Since P_2 is true, P_3 must be true. Since P_3 is true, hhen P_4 must also be true, and so on. This suggests that you can prove a statement Pn to be true for $n \geq k$ by showing that P_k implies P_{k+1}.

EXAMPLE 1 Prove by mathematical induction that the proposition

$$P_n : 1+2+3+...+n = \frac{n(n+1)}{2}$$

is true for all $n \in IN^+$.

Solution

1. For $n = 1$, P_1 is true since $1 = \frac{1\cdot(1+1}{2}$.
2. For $n = k$, assume that

$$p_k : 1 + 2 + 3 + ... + k = \frac{k(k+1)}{2}.$$

holds. We must show that, for $n = k + 1$,

$$P_{k+1} : 1 + 2 + 3 + ... + (k + 1) = \frac{(k+1)(k+2)}{2}.$$

Now, adding "$k + 1$" to the each side of the equality

$$P_k : 1 + 2 + 3 + ... + k = \frac{k(k+1)}{2}.$$

we obtain

$$\begin{aligned} 1+2+3+...+k+(k+1) &= \frac{k(k+1)}{2}+(k+1) \\ &= \frac{k^2+3k+2}{2} \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

Thus, P_{k+1} is true whenever P_k is true. Hence.

$$P_n : 1+2+3+...+n = \frac{n(n+1)}{2}$$

is true $\forall n \in IN^+$.

EXAMPLE 2 Prove by the mathematical induction that

a. $2 + 3 + 6 + ... + 2n = n(n + 1)$

b. $1 + 3 + 5 + ... + (2n - 1) = n^2$

are true for all $n \in IN^+$.

Solution You may work out for the proof by yourself.

EXAMPLE 3 Use mathematical induction to prove that

$$P_n : 1^2 + 2^2 + 3^2 + ... + n^2 = \frac{n(n+1)(2n+1)}{6}$$

for all $n \in IN^+$.

Solution

1. When $n = 1$, the formula is valid, because $P_1 : 1^2 = \frac{1 \cdot 2 \cdot 3}{6}$.
2. Assume that

$$P_k : 1^2 + 2^2 + 3^2 + ... + k^2 = \frac{k(k+1)(2k+1)}{6}$$

holds.

We must show that

$$P_{k+1} : 1^2 + 2^2 + 3^2 + ... + (k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}.$$

Now, adding $(k + 1)^2$ to each side of the induction hppothesis, we obtain

$$\begin{aligned} 1^2 + 2^2 + 3^2 + ... + (k+1)^2 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\ &= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \\ &= \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} \\ &= \frac{(k+1)(2k^2 + 7k + 6)}{6} \\ &= \frac{(k+1)(k+2)(2k+3)}{6}. \end{aligned}$$

Thus, P_{k+1} holds when P_k is true. Hence P_n is true $\forall n \in IN^+$.

EXAMPLE 4 Prove by mathematical induction that

$$P_n : 1^3 + 2^3 + 3^3 + \ldots + n^3 = \frac{n^2(n+1)^2}{4}$$

for all $n \in IN^+$.

Solution You may work out for the proof by yourself.

EXAMPLE 5 Prove by mathematical induction that, for all positive natural numbers

$$P_n : \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \ldots + \frac{1}{n\cdot(n+1)} = \frac{n}{n+1}.$$

Solution
1. For $n = 1$, $P_1 : \frac{1}{1\cdot 2} = \frac{1}{1+1}$ is true.
2. For $n = k$, assume

   $$P_k : \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \ldots + \frac{1}{k\cdot(k+1)} = \frac{k}{k+1}.$$

   Now, we will show that

   $$P_{k+1} : \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \ldots + \frac{1}{(k+1)\cdot(k+2)} = \frac{k+1}{k+2}.$$

   By hypothesis, we have

   $$P_{k+1} : \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \ldots + \frac{1}{(k+1)\cdot(k+2)} = \frac{k+1}{k+2}.$$

   By adding $\frac{1}{(k+1)\cdot(k+2)}$ to each side of this of this equality, we get

   $$\frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \ldots + \frac{1}{k\cdot(k+1)} + \frac{1}{(k+1)\cdot(k+2)} = \frac{k}{k+1} + \frac{1}{(k+1)\cdot(k+2)}.$$

   That is, we have

   $$\frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \ldots + \frac{1}{(k+1)\cdot(k+2)} = \frac{k^2+2k+1}{(k+1)\cdot(k+2)}$$

   $$= \frac{(k+1)^2}{(k+1)\cdot(k+2)}$$

   $$= \frac{(k+1)}{k+2}.$$

   Therefore, P_{k+1} is true. Hence, $\forall n \in IN^+$:

   $$P_n : \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \ldots + \frac{1}{n\cdot(n+1)} = \frac{n}{n+1}$$

EXAMPLE 6 Prove by mathematical induction that $n \in IN^+$

$$P_n: 1 + r + r^2 + \ldots + r^{n-1} = \frac{1-r^n}{1-r}, \quad (r \neq 1).$$

Solution

1. For $n = 1$, $P_1: 1 = \frac{1-r^1}{1-r}$, $(r \neq 1)$ is obviosly true.
2. Assume that for $n = k$,

$$P_k: 1 + r + r^2 + \ldots + r^{k-1} = \frac{1-r^k}{1-r}, \quad (r \neq 1)$$

holds. We will show that P_n is also true for $n = k + 1$. Now, by assumption

$$1 + r + r^2 + \ldots + r^{k-1} = \frac{1-r^k}{1-r}, \quad (r \neq 1)$$

Adding r^k to each side of the equality above, gives

$$1 + r + r^2 + \ldots + r^{k-1} = \frac{1-r^k}{1-r} + r^k, \quad (r \neq 1)$$

$$\Rightarrow 1 + r + r^2 + \ldots + r^{k-1} = \frac{1 - \cancel{r^k} + \cancel{r^k} - r^{k+1}}{1-r}, \quad (r \neq 1)$$

$$= \frac{1-r^{k+1}}{1-r}, \quad (r \neq 1)$$

That is, P_n is true for $n = k + 1$. Hence, $\forall n \in IN^+$:

$$1 + r + r^2 + \ldots + r^{k-1} = \frac{1-r^k}{1-r}, \quad (r \neq 1)$$

EXAMPLE 7 Prove by the mathematical induction that the sum of cubes of three consecutive natural numbers is divisible by 9.

Solution Let n, $n + 1$ and $n + 2$ be three consecutive natural numbers $(n \in IN^+)$. The given proposition in mathematical form is

$$P_n: 9 \mid n^3 + (n+1)^3 + (n+2)^3.$$

1. For n = 1 , P1 : 9 | 13 + 23 + 33 . P1 is clearly true, since 9 | 36.
2. Assume that P_k is true, i.e.,

$k^3 + (k + 1)^3 + (k + 2)^3$

is divisible by 9. We will show that P_k implies $P_k + 1$, i.e., we'll prove that P_{k+1} must also be true,

$9 \mid (k + 1)^3 + (k + 2)^3 + (k + 3)^3$

Now, by the assumption, we have

$k^3 + (k + 1)^3 + (k + 2)^3 = 9t \quad (t \in IN^+)$.

Adding $-k^3 + (k + 3)^3$ to the both sides of the equality gives

$$k^3 + (k + 1)^3 + (k + 2)^3 - k^3 + (k + 3)^3 = 9t - k^3 + (k + 3)^3$$

$$\begin{aligned} \Rightarrow (k + 1)^3 + (k + 2)^3 + (k + 3)3 &= 9t - k^3 + k^3 + 9k^2 + 27k + 27 \\ &= 9t + 9k^2 + 27k + 27 \\ &= 9(t + k^2 + 3k + 3). \end{aligned}$$

Since $9(t + k^2 + 3k + 3)$ is divisible by 9, we may easily conclude that $P_k + 1$ is also true. Hence, by 1° and 2°

$$P_n : \; 9 \mid n^3 + (n + 1)^3 + (n + 2)^3$$

is true for all $n \in IN^+$.

EXAMPLE 8 Use mathematical induction to prove that for all $n \in IN^+$ the number "$3^{2n} - 1$" is of the form $8k$, $k \in Z$.

Solution The proposition is $Pn : 8 \mid 3^{2n} - 1$.

1. For $n = 1$,

   $P_1 : 8 \mid 3^{2 \cdot 1} - 1$ is clearly true.

2. Assume that $P_k : 8 \mid 3^{2 \cdot k} - 1$ is true. Now, we will prove, for $n = k + 1$,

   $P_{k+1} : 8 \mid 32(k + 1) - 1$.

By hypothesis, $3^{2k} - 1 = 8t (t \in Z)$. Multiplying both sides of this equality by 3^2, we obtain

$$\begin{aligned} 3^2 \cdot (3^{2k} - k) = 3^2 \cdot 8t &\Rightarrow 3^{2k+2} - 3^2 = 72t \\ &\Rightarrow 3^{2(k+1)} - 9 = 72t, \\ &\Rightarrow 3^{2(k+1)} - 1 - 8 = 72t \\ &\Rightarrow 3^{2(k+1)} - 1 = 72\,\mathrm{t} + 8, \\ &\Rightarrow 3^{2(k+1)} - 1 = 8(9t + 1). \end{aligned}$$

It can easily be seen that $3^{2(k+1)} - 1$ is also of the form $8k$, $k \in Z$. i.e., P_n is al true for $n = k + 1$. Therefore, $\forall n \in IN^+$.

$$8 \mid 3^{2n} - 1.$$

EXAMPLE 9 Find the truth set N_a, $a \in IN^+$ of $P_n : 2^n < n!$ and prove it by mathematical induction.

Solution

1. $P_1 : 2^1 < 1! \Rightarrow 2 <$ is not true.

   $P_2 : 2^2 < 2! \Rightarrow 4 < 2$ is not true.

   $P_3 : 3^2 < 3! \Rightarrow 8 < 6$ is not ture.

   $P_4 : 2^4 < 4! \Rightarrow 16 < 24$ is true. So the truth set is

   $$N_4 \{4, 5, 6 ...\}$$

2. Assuming that P_k is true, i.e., $P_k : 2k < k!$, we have to prove that the truth of P_{k+1} follows from that of P_k. In other words, we will show that

   $$P_{k+1} : 2^{k+1} < (k + 1)!$$

   is true. It is clear that for $k > 3$, $2 < k + 1$. Also, by hypothesis $2k < k!$. Multiplying these inequalities side by side, we get

   $$2^k \cdot 2 < k! \cdot (k + 1) \Rightarrow 2^{k+1} < (k + 1)!$$

   We conclude that P_n is also true for $n = k + 1$, hence

   $$P_n : 2^n < n!C$$

   is true for all $n \in N_4$.

EXAMPLE 10 Prove "Bernoulli Inequality" by mathematical induction.

"Bernoulli Inequality"

$\forall n \in IN^+$ and for $1 + x \geq 0$, $(1 + x)^n \geq 1 + nx$.

Solution

1. For $n = 1$, $(1 + x)^1 \geq 1 + 1 \cdot x$ and also for $n = 2$, $(1 + x)^2 \geq 1 + 2 \cdot 2$

   Since

   $1 + 2x + x^2 \geq 1 + 2x \Rightarrow x^2 \geq 0$

   is always true.

2. Assuming for $n = k$, $Pk : (1 + x)^k \geq 1 + kx$ holds, we will verify that

   $P_{k+1} : (1 + x)^{k+1} \geq 1 + (k + 1)x$

   also holds. By hypothesis, we have

   $(1 + x)^k \geq 1 + kx$.

   Multiplying both sides of this inequality by $1 + x$ gives.

   $(1 + x)^k (1 + x) \geq (1 + kx)(1 + x) \Rightarrow (1 + x)^{k+1} \geq 1 + x + kx + kx^2$

   $\Rightarrow (1 + x)^{k+1} \geq 1 + (k + 1)x + kx^2$.

Recall by the transitive property, $\forall a, b, c \in IR$ if $a \leq$ and $b \leq c$, the $a \leq c$.

By using this this property, since

$(1 + x)^{k+1} \geq 1 + (k + 1)x + kx^2$

and, $kx^2 \geq 0$ for $k \in IN^+$,

$1 + (k + 1)x + kx^2 \geq 1 + (k + 1)x$,

we can get

$(1 + x)^{k+1} \geq 1 + (k + 1)x$.

That means, P_{k+1} is also true. Hence, Bernoulli Inequality is true for all $n \in IN^+$.

BIOGRAPHICAL SKETCH

BERNOULLI, JAKOP or JACQUES (1654 - 1705)

Swiss mathematician, born in Basel. He was the first of the famous Bernoulli family to achieve distinction in mathematics. He developed the newly invented calculus into a powerful mathematical tool, applying it to the solution of a variety of problems. In 1682 he opened a seminary in Basel for mathematics and experimental physics. He was appointed professor of mathematics at the University of Basel in 1687. In his Ars conjectandi he laid the foundations of the theory of probability; Bernoulli's Theorem, given there, is of great importance where the number of "trials" is large. He and his brother Johann were the first to be elected foreign associates of the Paris Academy of Sciences.

EXAMPLE 11 Prove that hep proposition

$$P_n : 1^2 + 2^2 + 3^2 + \ldots + (n-1)^2 < \frac{n^2}{3}$$

holds for all $n \in IN^+$.

Solution

1. For $n = 1$, $P_1 : (1 - 1)2 < \frac{1^3}{3}$ is true.
2. Induction hypothesis:

Assume P_k is true; i.e.,

$P_k : 1^2 + 2^2 + 3^2 + \ldots + (k - 1)^2 < \frac{k^3}{3}$

holds. Then, we will prove that P_{k+1} is also true. By adding k^2 to both sides of the inequality

$1^2 + 2^2 + 3^2 + \ldots + (k - 1)^2 < \frac{k^3}{3}$,

we obtain

$1^2 + 2^2 + 3^2 + \ldots + (k - 1)^2 + k^2 < \frac{k^3}{3} + k^2 \quad$ (I)

Now, it is clear that for every $k \in IN^+$

$$k^3 + 3k^2 < k^3 + 3k^2 + 3k + 1 \Rightarrow k^3 + 3k^2 < (k+1)^3,$$

$$\Rightarrow \frac{k^3+3k^2}{3}\frac{(k+1)^3}{3},$$

$$\Rightarrow \frac{k^3}{3}+k^2 < \frac{(k+1)^3}{3} \qquad \text{(II)}$$

By using the transitive property, from the inequalities I and II, we get

$$1^2+2^2+3^2+\ldots+k^2 < \frac{(k+1)^2}{3}.$$

Thus, P_{k+1} is also true. Hence, we can conclude that

$$1^2+2^2+3^2+\ldots+(n-1)^2 < \frac{n^3}{3}.$$

EXAMPLE 12 Use mathematical induction to prove that

$$P_n : IN^+ \xrightarrow{f} IN^+, \quad f(1)=1, \; f(n)=f(n-1)n=n!.$$

Solution

1. P_1 is true, since $f(2-1) \cdot 2 = 2!$.
2. First, we assume for $n = k$,

   $P_k : IN^+ \xrightarrow{f} IN^+, f(1) = 1, f(k) = f(k-1)k = k!.$

   holds. Yet, we must show P_{k+1} holds either. By induction hypothesis, we know

   $f(k) = f(k-1)k = k!$

   Multiplying both sides of this equality by $k + 1$ gives

   $f(k-1) \cdot k\,(k+1) = k! \cdot (k+1) = f(k)\,(k+1) = k+1)!.$

   That is

   $P_{k+1} : IN^+ \xrightarrow{f} IN^+, f(1) = 1, f(k+1) = f(k)\,(k+1) = (k+1)!$

   Hence, P_n is true $\forall n \in IN^+$.

EXAMPLE 13 For $n \geq 3$, $P_n : 2^n > 2n + 1$. Prove that P_n is true for all $n \in IN^+$.

Solution

1. For $n = 3$, $P_3 : 2^3 > 2 \cdot 3 + 1$ is clearly true.
2. Assuming

   $P_k : 2^k > 2k + 1$

   is true, we will verify that P_{k+1} is also true. Now, for $k \geq 3$ it is clear that $2k > 2$ and by hypothesis, we have

   $2^k > 2k + 1.$

   Adding these two inequalities side by side yields.

   $2^k + 2^k > 2k + 3 \Rightarrow 2^{k+1} > 2(k + 1) + 1.$

   Therefore, P_{k+1} is also true. Hence,

   $\forall n \geq 3,\ 2^n > 2n + 1.$

Theorem

De Moivre's Theorem

If $z = r(\cos\theta + i\sin\theta)$ is a complex number and n is a positive integer, then

$z^n = [r(\cos\theta + i\sin\theta)]^n = r^n(\cos n\theta + \mathrm{i}\sin n\theta).$

EXAMPLE 14 Prove "De Moivre's Theorem" by mathematical induction.

Solution

1. For $n = 1$, $[(r\cos\theta + i\sin\theta)]^1 = r(\cos(1\theta) + i\sin(1\theta)$ is true.
2. Assuming that the formula

   $[r(\cos\theta + i\sin\theta)]^k = r^k(\cos k\theta + i\sin k\theta)$

   is true, we must show that the formula

   $[r(\cos\theta + i\sin\theta)]^{k+1} = r^{k+1}[\cos(k+1)\theta + i\sin(k+1)\theta)$

   also holds. Now, by induction hypothesis, we have

   $[r(\cos\theta + i\sin\theta)^k = r^k(\cos k\theta + i\sin k\theta).$

   If we multiply each side of this equality by $r(\cos\theta + i\sin\theta)$, we obtain

   $$[r(\cos\theta + i\sin\theta)]^k[r(\cos\theta + i\sin\theta)] = r^k(\cos k\theta + i\sin k\theta)[r(\cos\theta + i\sin\theta)],$$
   $$[r(\cos\theta + i\sin\theta)]^{k+1} = r^k + 1[\cos(k\theta + \theta) + i\sin(k\theta + \theta)]$$
   $$= r^{k+1}[\cos(k+1)\theta + i\sin(k+1)\theta].$$

   Hence, for all positive integer n,

   $[r(\cos\theta + i\sin\theta)]n = r^n(\cos n\theta + i\sin n\theta).$

EXAMPLE 15 Prove by mathematical induction that $\forall n \in IN^{+}, n! \leq n^{n-1}$.

Solution

1. For $n = 1$, $P_1 : 1! \leq 1^{1-1}$, for $n = 2$, $P_2 : 2! \leq n^{2-1}$ and for $n = 3$, $P_3 : 3! \leq 3^{3-1}$. Thus P_1, P_2 and P_3 are all true.
2. Assume that, for $n = k$,

   $P_k : k! \leq k^{k-1}$

   is true.

   Now, we will show that

   $P_{k+1} : (k + 1)! \leq (k + 1)k$

   also holds. We know, by induction hypothesis, that $k! \leq k^{k-1}$. Multiplying both sides of this inequality by $k + 1$ yields.

   $k! \cdot (k + 1) \leq k^{k-1} \cdot (k + 1) = (k + 1)! \leq k^{k} + k^{k-1}$ (I).

   By using binomial expansion

   $$(k+1)^{k} = \binom{k}{0}k^{k} + \binom{k}{1}k^{k-1} + \binom{k}{2}k^{k-2} + \ldots + 1,$$

   we can obtain the following inequality

   $(k + 1)! \leq (k + 1)^{k}$.

   Thus, Pn is true for $n = k + 1$. Hence, we can conclude that

   $\forall n \in IN^{+}, n! \leq n^{n-1}$.

Theorem

Binomial Theorem

If a and b are real numbers and n is a positive integer, then

$$(a+b)^{n} = \binom{n}{0} + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^{2} + \ldots + \binom{n}{n-1}ab^{n-1} + \binom{n}{n}b^{n}$$

where $\binom{n}{k}$ is the binominal coefficient $\binom{n}{k} = \dfrac{n!}{k!(n-k)!}$.

EXAMPLE 16 Prove “Binomial Theorem” by mathematical induction.

Solution

1. For $n = 1$, P_1 : $(a + b)^{1} = \binom{1}{0}a^{1} + \binom{1}{1}b^{1}$. Since both coefficients equal to 1, the equation reduces to $(a + b)^{1} = a + b$.
2. Next, suppose that the formula holds for $n = k$, i.e.,

   $$P_k : (a+b)^{k} = \binom{k}{0}a^{k} + \binom{k}{1}a^{k-1}b + \ldots + \binom{k}{k-1}ab^{k-1} + \binom{k}{k}b^{k}.$$

Now we use the property to show that

$$P_{k+1}:(a+b)^{k+1}=\binom{k+1}{0}a^{k+1}+\binom{k+1}{1}a^{k}b+\ldots+\binom{k+1}{k+1}b^{k+1}.$$

is true. To do this, multiply both sides of the formula for $n = k$ by $a + b$ to obtain

$$(a+b)^{k+1}=\binom{k}{0}a^{k+1}+\binom{k}{1}a^{k}b+\binom{k}{2}a^{k-1}b^{2}+\ldots+\binom{k}{k}ab^{k}$$

$$+\binom{k}{0}a^{k}b+\binom{k}{1}a^{k-1}b^{2}+\ldots+\binom{k}{k-1}ab^{k}+\binom{k}{k}b^{k+1}.$$

Adding like terms, such as $\binom{k}{k}a^{k}b$ and $\binom{k}{0}a^{k}b$, ak b , using the property

$$\binom{k}{r-1}+\binom{k}{r}=\binom{k+1}{r}$$

gives

$$(a+b)^{k+1})=\binom{k+1}{0}a^{k+1}+\binom{k+1}{1}a^{k}b+\ldots+\binom{k+1}{k+1}b^{k+1}.$$

Hence, P_n is true $\forall n \in N^{+}$.

EXAMPLE 17 Prove that $\forall n \in N^{+}$.

$$\cos\alpha\cdot\cos 2\alpha\cdot\cos 4\alpha\cdot\ldots\cdot\cos 2\alpha=\frac{\sin 2^{n+1}\alpha}{2^{n+1}\sin\alpha}.$$

Solution 1. For $n = 1$, $P_1 : \cos\alpha\cdot\cos 2a=\dfrac{\sin 2^{1+1}\alpha}{2^{1+1}\sin\alpha}$,

$$\Rightarrow \cos\alpha\cdot\cos 2\alpha=\frac{\sin 4\alpha}{4\sin\alpha},$$

$$\Rightarrow 4\cdot\sin\alpha\cdot\cos\alpha\cdot\cos 2\alpha=\sin 4\alpha,$$

$$\Rightarrow 2\cdot\underbrace{2\cdot\sin\alpha\cdot\cos\alpha}_{\sin 2\alpha}\cdot\cos 2\alpha=\sin 4\alpha,$$

$$\Rightarrow 2\cdot\sin 2\alpha\cdot\cos 2\alpha=\sin 4\alpha.$$

So, P_1 is true.

2. Assume that

$$P_k : \cos\alpha \cdot \cos 2\alpha \ldots \cos 2^k\alpha = \frac{\sin 2^{k+1}\alpha}{2^{k+1}\cdot\sin\alpha}$$

is true, we will prove P_k implies $P_k + 1$. Multiplying both sides of the equation P_k by $\cos 2^{k+1}\alpha$ yields.

$$\cos\alpha\cdot\cos 2\alpha\cdot\ldots\cdot\cos 2^k\alpha\cdot\cos 2^{k+1}\alpha = \frac{\sin 2^{k+1}\alpha\cdot\cos 2^{k+1}\alpha}{2^{k+1}\sin\alpha},$$

$$= \frac{\frac{1}{2}\sin 2\cdot 2^{k+1}\alpha}{2^{k+1}\sin\alpha},$$

$$= \frac{\sin 2^{(k+1)+1}\alpha}{2^{(k+1)+1}\sin\alpha}.$$

Hence, P_{k+1} holds; that is, P_n is true $\forall n \in IN^+$.

EXAMPLE 18 Prove that for all $n \in IN^+$

$$\underbrace{\sqrt{2+\sqrt{2+\sqrt{2+\ldots+\sqrt{2+\sqrt{2}}}}}}_{n-\text{times}} = 2\cos\frac{\pi}{2^{n+1}}.$$

Solution 1. For $n = 1$, $P_1 : \sqrt{2} = 2\cos\frac{\pi}{2^{1+1}} \Rightarrow \frac{\sqrt{2}}{2} = \cos\frac{\pi}{4}$ is true.

2. Assuming for $n = k$,

$$P_k : \underbrace{\sqrt{2+\sqrt{2+\sqrt{2+\ldots+\sqrt{2+\sqrt{2}}}}}}_{k-\text{ times}} = 2\cos\frac{\pi}{2^{k+1}}$$

is true, we will show that P_{k+1} also holds. Now, by adding “2” to both sides of the equality given above, we get

$$2+\underbrace{\sqrt{2+\sqrt{2+\sqrt{2+\ldots+\sqrt{2+\sqrt{2}}}}}}_{k-\text{ times}} = 2+2\cos\frac{\pi}{2^{k+1}}.$$

Taking the square root of the both sides yields

$$\underbrace{\sqrt{2+\sqrt{2+\sqrt{2+\ldots+\sqrt{2+\sqrt{2}}}}}}_{k+1-\text{ times}} = \sqrt{(1+\cos\frac{\pi}{2^{k+1}})},$$

Since $\cos 2\pi = 1$

$$\sqrt{2}\sqrt{(\cos 2\pi+\cos\frac{\pi}{2^{k+1}})}.$$

Recall from “Algebra 3” that

$$\cos x + \cos y = 2 \cdot \cos(\frac{x+y}{2}) \cdot \cos(\frac{x-y}{2}).$$

By using this conversion formula, we obtain

$$\underbrace{\sqrt{2+\sqrt{2+\ldots+\sqrt{2}}}}_{k+1} = \sqrt{2}\sqrt{2\cos(\frac{2\pi+\frac{\pi}{2^{k+1}}}{2})\cos(\frac{2\pi-\frac{\pi}{2^{k+1}}}{2})}$$

$$= \sqrt{2}\sqrt{2}\sqrt{\cos(\pi+\frac{\pi}{2^{k+2}})\cos(\pi-\frac{\pi}{2^{k+2}})}$$

$$= 2\sqrt{(-\cos\frac{\pi}{2^{k+2}})(-\cos\frac{\pi}{2^{k+2}})}$$

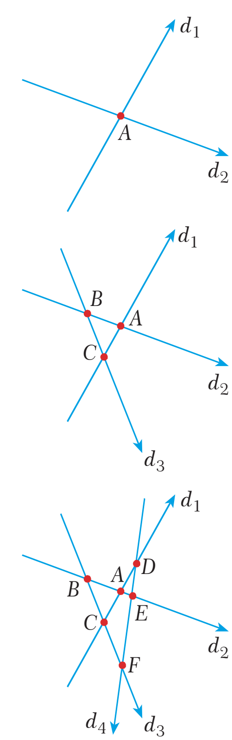
$$= 2\sqrt{\cos^2\frac{\pi}{2^{k+2}}} = 2\cos\frac{\pi}{2^{k+2}}$$

$$= 2\cos\frac{\pi}{2^{(k+1)+1}}.$$

Hence,

$$P_n : \underbrace{\sqrt{2+\sqrt{2+\sqrt{2+\ldots+\sqrt{2+\sqrt{2}}}}}}_{n-\text{times}} = 2\cos\frac{\pi}{2^{n+1}} \text{ is true } \forall n \in IN^{+}.$$

EXAMPLE 19 Prove by using the Gausian method that n lines, no two are parallel, intersect at most $C(n, 2)$ points in a plane.



Solution Two lines d_1 an d_2 which are not parallel to each other in tersect at most one point, say A.

The line d_3 which is not parallel to any one of the lines d_1 and d_2 intersects these lines at most two points, say B and C.

The line d_4 which is not parallel to any of the lines d_1, d_2 and d_3 intersects these lines at most three points, say D, E and F.

Continuing in this way, for n e IN, the line dn intersects the lines d_1, d_2, ... d_{n-1} at most $(n - 1)$ points.

If we add the number of all the points formed by the intersection of the lines given above, then we obtain

$$1+2+3+\ldots+(n-1) = \frac{(n-1)n}{2} = C(n,\ 2)$$

intersection points in a plane.

EXAMPLE 20 Prove by mathematical induction that by joining n points, no three are collinear, we can from $C(n, 2)$ different lines.

Solution

1. For $n = 2$, $C(2, 2) = 1$ is true, since there is only one line passing through two points.
2. Assume that $n = k$ points, no three are collinear, we can form $C(k, 2)$ different lines. Now we will show that for $n = k + 1$ points $C(k + 1, 2)$ different lines can be formed.

   Adding one more point forms k extra lines. So, we will have

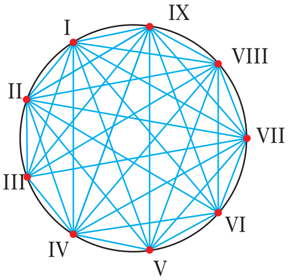
   $$C(k+2)+k=\frac{k(k-1)}{2}+k=\frac{k(k-1)+2k}{2}=\frac{k(k+1)}{2}=C(k+1,\ 2)$$

   different lines. Hence, we can form

   $$C(n,\ 2)=\frac{n(n-1)}{2}$$

   different lines by joining n points, no three are collinear.

EXAMPLE 21 Prove by mathematical induction that $C(n, 3)$ triangles can be drawn by using the vertices chosen from n points lie on a circle.



Solution

1. For $n = 3$, C(3, 3) = 1 is true since only one triangle can be drawn using three points
2. Assume that $C(k, 3)$ triangles can be formed using k points lie on a circle.Yet we will show,

   $C(k + 1, 3)$

   triangles can be drawn using $k + 1$ points lying on a circle. By previous problem, we know that

   $C(n , 2)$

   different lines can be formed by joining n points, no three are collinear. So adding one more point forms $C(k, 2)$ extra triangles. That is

   $C(k, 3) + C(k, 2) = C(k + 1, 3)$

   triangles can be formed using $k + 1$ points lying on a circle. Hence,

   $$C(n,\ 3)=\frac{n(n-1)(n-2}{6}$$

   triangles can be drawn by using n points lie on a circle. Investigate the Figure 1.2 drawn for $n = 9$ points.

EXERCISES 3.1

1. Use mathematical induction to prove the following equalities for every positive integer n.

   a. $2+5+8+11+\ldots+3n-1=\dfrac{n(3n+1)}{2}$

   b. $3+6+9+\ldots+3n=\dfrac{3n(n+1)}{2}$

   c. $1+4+7+10+\ldots+3n-2=\dfrac{n(3n-1)}{2}$

   d. $1+7+13+19+\ldots+6n-5=n(3n-2)$

   e. $1^2+3^2+5^2+7^2+\ldots+(2n-1)^2=\dfrac{n(4n^2-1)}{3}$

   f. $1^3+3^3+5^3+7^3+\ldots+(2n-1)^3=n^2(2n^2-1)$

   g. $1^2-2^2+3^2-4^2+\ldots+(-1)^{n-1}n^2=(-1)^{n-1}\dfrac{n(n+1)}{2}$

   h. $2^1\cdot 2+2^2\cdot 3+2^3\cdot 4+\ldots+2^n(n+1)=n\cdot 2^{n+1}$

   i. $1\cdot 2+2\cdot 3+\ldots+n(n+1)=\dfrac{n(n+1)(n+2)}{3}$

   j. $1\cdot 2\cdot 3+2\cdot 3\cdot 4+\ldots+n(n+1)(n+2)$
   $=\dfrac{n(n+1)(n+2)(n+3)}{4}$

   k. $1\cdot 2\ldots k+\ldots+n(n+1)\ldots(n+k-1)$
   $=\dfrac{n(n+1)(n+2)\ldots(n+k)}{(k+1)}$

   l. $1+3+3^2+\ldots+3^{n-1}=\dfrac{3^n-1}{2}$

   m. $2+5+13+\ldots+(2^{n-1}+3^{n-1})=2^n-1+\dfrac{3^n-1}{2}$

   n. $\dfrac{1}{2}+\dfrac{1}{2^2}+\dfrac{1}{2^3}+\ldots+\dfrac{1}{2^n}=1-\dfrac{1}{2^n}$

   o. $\dfrac{1}{1\cdot 3}+\dfrac{1}{3\cdot 5}+\dfrac{1}{5\cdot 7}+\ldots+\dfrac{1}{(2n-1)(2n+1)}=\dfrac{n}{2n+1}$

   o. $\dfrac{1}{1\cdot 3}+\dfrac{1}{3\cdot 5}+\dfrac{1}{5\cdot 7}\ldots+\dfrac{1}{(2n-1)(2n+1)}=\dfrac{n}{2n+1}$

   p. $\dfrac{1}{1\cdot 4}+\dfrac{1}{4\cdot 7}+\ldots+\dfrac{1}{(3n-2)(3n+1)}=\dfrac{n}{3n+1}$

   r. $\dfrac{5}{1\cdot 2\cdot 3}+\dfrac{6}{2\cdot 3\cdot 4}+\ldots+\dfrac{n+4}{n(n+)(n+2}$
   $=\dfrac{n(3n+7)}{2(n+1)(n+2)}$

   s. $\dfrac{1^2}{1\cdot 3}+\dfrac{2^2}{3\cdot 5}+\dfrac{3^2}{5\cdot 7}+\ldots+\dfrac{n^2}{(2n-1)(2n+1)}=\dfrac{n(n+1)}{2(2n+1)}$

   t. $1\cdot 1!+2\cdot 2!+3\cdot 3!+n\cdot n!=(n+1)!-1$

   u. $0+\dfrac{1}{2!}+\dfrac{2}{3!}+\dfrac{3}{4!}+\dfrac{n-1}{n!}=1-\dfrac{1}{n!}$

2. Find the truth set of the following propositions, and use mathematical induction to prove them.

   a. $2^n\ge n^2$

   b. $3^n\ge 1+4n$

   c. $n^3>n^2+3$

   d. $\dfrac{1}{n!}<\dfrac{1}{2^n}$

3. Use mathematical induction to prove the following inequalities for the indicated positive integer values of n.

   a. $2^n\ge 1+n,\ \forall n\in\mathbb{N}^+$

   b. $2^n>2n+1,\ \forall n\in\mathbb{N}^+$

   c. $3^n\ge n2^n,\ \forall n\in\mathbb{N}^+$

   d. $4n>n^2,\ \forall n\in\mathbb{N}^+$

   e. $5^n\in 1+4n,\ \forall n\in\mathbb{N}^+$

   f. $(2n)!\ge 2n\,(n!)^2,\ \forall n\in\mathbb{N}^+$

   g. $n!\ge n^{n-1},\ \forall n\in\mathbb{N}^+$

h. For $x_1, x_2, x_3 \dots x_n \in \mathbb{R}$ and $\forall n \in \mathbb{N}^+$

$$|x_1 + x_2 + \dots + x_n| \le |x_1| + |x_2| + \dots + |x_n|$$

i. $2! \cdot 4! \cdot 6! \dots (2n)! \ge [(n+1)!]^n$, $\forall n \in \mathbb{N}^+$

j. $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\dots+\frac{1}{2^n}<1, \quad \forall n \in \mathbb{N}^+$

k. $\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\dots+\frac{1}{3^n}<1, \quad \forall n \in \mathbb{N}^+$

4. Use mathematical induction to prove the following expressions for every positive ineger n.

a. $5 \mid 6^n - 1$

b. $6 \mid 7n - 1$

c. $2 \mid 2^{2n} + 2$

d. $3 \mid 2^{2n} + 2$

e. $3 \mid n^3 - n$

f. $7 \mid 13^{2n} + 6$

g. $11 \mid 2^{4n} - 5^n$

h. $11 \mid 12^n + 10$

i. $3 \mid 4^n - 1$

j. $5 \mid 5^n - 3^n$

k. $(a + b) \mid a^{2n} - b^{2n}$

l. $27 \mid (2^{5n+1} + 5^{n+2})$

m. $6 \mid (5n^3 + 13n - 24)$

n. $5 \mid 17^{4n+1} + 3 \cdot 9^{2n}$

o. $7 \mid 3^{2n+2} - 2^{n+1}$

p. $133 \mid 11^{n+2} + 12^{2n+1}$

r. $(x + y) \mid x^{2n-1} + y^{2n-1}$

s. $6 \mid n(2n + 1)(7n + 1)$

5. Prove that $\forall n \in \mathbb{N}^+$

$$\sin\theta+\sin 2\theta+\dots+\sin(n\theta) = \frac{\cos\frac{\theta}{2} - \cos(n+\frac{1}{2})\theta}{2\sin\frac{\theta}{2}}.$$

6. Prove that $\forall n \in \mathbb{N}^+$

$$\cos\theta+\cos 2\theta+\dots+\cos(n\theta) = \frac{\sin\frac{n\theta}{2}\cdot\cos(\frac{(n+1)}{2}\theta)}{\sin\frac{\theta}{2}}.$$

7. Prove that $\forall n \in \mathbb{N}^+$

$$\cos\theta+\cos 3\theta+\dots+\cos(2n-1)\theta = \frac{\sin 2n\theta}{2\sin\theta}.$$

8. Prove that $\forall n \in \mathbb{N}^+$

$$\log(\frac{2}{1})+2\log(\frac{3}{2})+\dots+n\log(\frac{n+1}{n}) = \log(\frac{(n+1)^n}{n!}).$$

9. Prove that $\forall n \in \mathbb{N}^+$

$$x+(x+y)+\dots+[x+(n-1)y] = \frac{n}{2}[2x+(n-1)y].$$

10. Prove that $\forall n \in \mathbb{N}^+$ and $a, x, y \in \mathbb{R}^+$

$f(x \cdot y) = f(x) \cdot f(y) \Rightarrow f(a^n) = [f(a)]^n$.

11. Prove that $\forall n \in \mathbb{N}^+$ and $a, x, y \in \mathbb{R}^+$

$f(x \cdot y) = f(x) + f(y) \Rightarrow f(a^n) = n \cdot f(a)$.

12. Prove that a polygon with n-sides ($n > 3$) has $\frac{n(n-3)}{2}$ diagonals.

13. If $x_1 > 0, x_2 > 0, \dots, x_n > 0$, then prove that $\forall n \in \mathbb{N}^+$

In $(x_1 x_2 x_3 \dots x_n)$ = In x_1 + In x_2 + ... + In x_n .

2 SUMMATION AND MULTIPLICATION NOTATION

Addition is as fundamental in advanced mathematics as it is in arithmetic. There is a story often told about the German mathematician Karl Friedrich Gauss. When he was in third grade, his class misbehaved and the teacher gave the following problem as punishment.

"Add the integers from 1 to 100."

It is said that Gauss solved the problem in almost no time at all. His method was of course different. In this section we will study sums of terms of various sequences, learn a special notation for sums and also for products of their terms.

A. SUMMATION NOTATION

A convenient notation for the sum of the terms of a given function defined as

$$f : \mathrm{Z} \to \mathbb{R}, f(k) = a_k$$

is summation notation or sigma notation, which involves the Greek capital letter sigma (Σ).

Definition

Let : $\mathrm{Z} \in \mathbb{R}$, $f(k) = a_k$ and $r, n \in \mathrm{Z}$ provided that $r \leq n$, then

$$\sum_{k=r}^{n} a_k = a_r + a_{r+1} + a_{r+2} + \ldots + a_n$$

is the sum of $(n - r + 1)$ terms of the function f from r to n where k is the index of summation, r is the lower bound and n is the upper bound.

Note that, $\sum\limits_{k=r}^{n} a_k$ can also be denoted by $\sum\limits_{r \leq k \leq n} a_k$ or $\sum\limits_{k \in [r,\, n]} a_k$ and is read as "the sum of a_k from $k = r$ to $k = n$".

Study the examples given below.

$$a_5 + a_6 + a_7 + \ldots + a_{18} = \sum_{k=5}^{18} a_k,$$

$$a_{-3} + a_{-2} + a_{-1} + \ldots + a_7 = \sum_{k=-3}^{7} a_k,$$

$$1^2 + 2^2 + 32 + \ldots + n^2 = \sum_{k=1}^{n} k^2,$$

$$1 + 3 + 5 + \ldots + 2n - 1 = \sum_{k=1}^{n} (2k - 1),$$

$$\sum_{k=3}^{10} (2k) = 6 + 8 + 10 + 12 + 14 + 16 + 18 + 20,$$

$$\sum_{k=-1}^{n} k = -1 + 0 + 1 + 2 + 3 + \ldots + n.$$

1. Summation Formulas

FORMULA

a. $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ b. $\sum_{k=1}^{n} (2k-1) = n^2$ c. $\sum_{k=1}^{n} (2k) = n(n+1)$

We will prove the first and the third formula, you may work out for the proof of second case by yourself.

Proof 1. Let $P = \sum_{k=1}^{n} k = 1+2+...+(n-1)+n.$

Adding $n + (n-1) + ... + 1$ to the expression given above, which also equal to P, gives

$$2P = \underbrace{(n+1)+(n+1)+...+(n+1)+(n+1)}_{n-\text{times}}$$

$$\Rightarrow 2P = n \cdot (n+1),$$

$$\Rightarrow P = \frac{n \cdot (n+1)}{2}.$$

3. Let $T = \sum_{k=1}^{n} 2k = 2+4+6+...+2n \qquad = 2(1+2+3+...+n$

By substituting the “Formula 1” into the above expression, we get

$$T = 2 \cdot \sum_{k=1}^{n} k = 2 \cdot \frac{k(k+1)}{2} = k(k+1)$$

FORMULA

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Proof In order to derive this formula we will use the expansion

$$(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

It is clear that

$$(1 + 3)^3 = 1^3 + 3 \cdot 1^2 \cdot 1 + 3 \cdot 1 \cdot 1^2 + 1^3,$$

$$(2 + 1)^3 = 2^3 + 3 \cdot 2^2 \cdot 1 + 3 \cdot 2 \cdot 1^2 + 1^3,$$

$$(3 + 1)^3 = 3^3 + 3 \cdot 3^2 \cdot 1 + 3 \cdot 3 \cdot 1^2 + 1^3,$$

....

....

$$(n + 1)^3 = n^3 + 3 \cdot n^2 \cdot 1 + 3 \cdot n \cdot 1^2 + 1^3.$$

Adding all of these equalities side by side yields

$(n+1)^3 = 1^3 + 3 \cdot (1^2 + 2^2 + 3^2 + \ldots + n^2) + 3 \cdot (1 + 2 + 3 + \ldots + n) + n \cdot 13$ .... (I)

Since

$$\sum_{k=1}^{n} k^2 = 1^2 + 2^2 + 3^2 + \ldots + n^2 \text{ and } \sum_{k=1}^{n} k = 1 + 2 + 3 + \ldots + n,$$

We can write the expression (I) as

$$n^3 + 3n^2 + 3n + 1 = 1 + 3 \cdot \sum_{k=1}^{n} k^2 + 3 \cdot \sum_{k=1}^{n} k + n,$$

$$\Rightarrow n^3 + 3n^2 + 2n = 3\sum_{k=1}^{n} k^2 + 3\frac{n \cdot (n+1)}{2}.$$

If we rearrange this equality, we will obtain

$$6\sum_{k=1}^{n} k^2 = 2n^3 + 6n^2 + 4n - 3n^2 - 3n = 2n^3 + 3n^2 + n$$

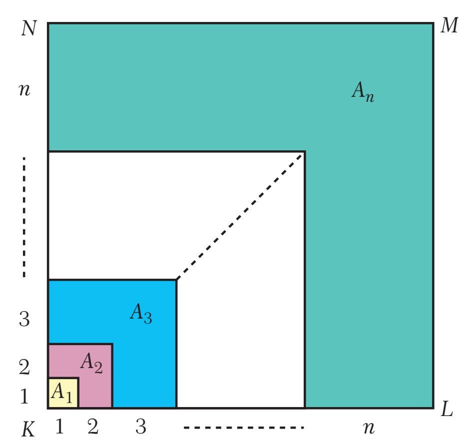
$$= n(2n^2 + 3n + 1) = n(n+1)(2n+1).$$

Hence, $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$.

FORMULA

$$\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

Proof We can prove Formula 1.3 by calculating the area of the square with sidelngth $1 + 2 + \ldots + n$ units. In Figure 1.1, the letters $A_1, A_2, A_3, \ldots, A_n$ denote the areas painted by the same color.



$A(KLMN) = A_1 + A_2 + A_3 + \ldots + A_n$

$= (1 + 2 + 3 + \ldots + n)^2$

$$\Rightarrow 1 + 8 + 27 + \ldots + n^3 = \left(\sum_{k=1}^{n} k\right)^2$$

$$\Rightarrow 1^3 + 2^3 + 3^3 + \ldots + n^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

FORMULA

$$\sum_{k=1}^{n} r^{k-1} = \frac{1-r^n}{1-r}, \quad r \neq 1$$

Proof Let $T = \sum_{k=1}^{n} r^{k-1} = 1 + r + r^2 + \ldots + r^{n-1}$ ... (I)

Multiplying both sides of this equality by r, we obtain

$r \cdot T = r + r^2 + r^3 + \ldots + r^n = 1$ ... (II)

Now, if we subtract the equality (II) from the equality (I) side by side, we will get

$$T - rT = 1 - r^n \Rightarrow T(1-r) = 1 - r^n \Rightarrow = \frac{1-r^n}{1-r}, \quad (r \neq 1).$$

FORMULA

$$\sum_{k=1}^{n} \frac{1}{k(k+1)} = \frac{n}{n+1}$$

Proof $\sum_{k=1}^{n} \frac{1}{k \cdot (k+1)} = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \ldots + \frac{1}{n \cdot (n+1)}.$

Since $\frac{1}{k \cdot (k+1)} = \frac{1}{k} - \frac{1}{k+1}$, we obtain

$$\sum_{k=1}^{n} \frac{1}{k \cdot (k+1)} = \sum_{k=1}^{n} \left(\frac{1}{k} - \frac{1}{k+1}\right)$$

$$= \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \ldots + \frac{1}{n} - \frac{1}{n+1}$$

$$= 1 - \frac{1}{n+1} = \frac{n}{n+1}.$$

2. Properties of Summation Notation

Properties 1

Let $c \in \mathbb{R}$ and $r \in \mathbb{Z}$,

$$\sum_{k=1}^{n} c = n \cdot c \quad \text{and} \quad \sum_{k=r}^{n} c = (n - r + 1)c.$$

Proof It is obvious that $\sum_{k=1}^{n} c = \underbrace{c + c + c + \ldots + c}_{n \text{ times}} = n \cdot c$ and $\sum_{k=1}^{n} c = \underbrace{c + c + c + \ldots + c}_{(n-r+1)-\text{ times}} = (n - r + 1) \cdot c.$

EXAMPLE 22 Evaluate the given expressions.

a. $\sum_{k=1}^{17} 3$ b. $\sum_{p=1}^{50} (-2)$ c. $\sum_{t=4}^{12} 5$ d. $\sum_{n=-7}^{13} (\frac{1}{8})$

Solution Here, we will solve only "*a*" and "c". Others are left as an exercise to the student.

a. $\sum_{k=1}^{17} 3 = 3 \cdot 17 = 51$ c. $\sum_{t=4}^{12} 5 = 5 \cdot (12 - 4 + 1) = 45$

Properties 2

For $c \in \mathbb{R}$, $\sum_{k=1}^{n} c \cdot a_k = c \cdot \sum_{k=1}^{n} a_k$.

Proof It is obvious that $\sum_{k=1}^{n} c \cdot a_k = ca_1 + ca_2 + ca_3 + \ldots + ca_n$

$$= c \cdot (a_1 + a_2 + a_3 + \ldots + a_n)$$

$$= c \cdot \sum_{k=1}^{n} a_k.$$

EXAMPLE 23 Evaluate the given expressions.

a. $\sum_{k=1}^{11} 2k$ b. $\sum_{k=1}^{10} 3k^2$ c. $\sum_{p=1}^{6} (-2p^3)$

Solution a. $\sum_{k=1}^{11} 2k = 2\sum_{k=1}^{11} k = 2 \cdot \frac{11 \cdot 12}{2} = 132.$

b. and c. are left as an exercise for the student.

Properties 3

For $1 < p < n$ and $p \in \mathbb{Z}$, $\sum_{k=1}^{n} a_k = \sum_{k=1}^{p-1} a_k + \sum_{k=p}^{n} a_k$.

Proof It is obvious that $\sum_{k=1}^{n} a_k = a_1 + a_2 + \ldots + a_{p-1} + a_p + a_{p+1} + \ldots + a_n,$

$$= \sum_{k=1}^{p-1} a_k + \sum_{k=p}^{n} a_k.$$

EXAMPLE 24 Evaluate the given expressions.

a. $\sum_{k=24}^{47} \frac{1}{k(k+1)}$ b. $\sum_{k=10}^{20} 2^{k-1}$

Solution a. $\sum_{k=24}^{47} \frac{1}{k \cdot (k+1)} = \sum_{k=1}^{47} \frac{1}{k \cdot (k+1)} - \sum_{k=1}^{23} \frac{1}{k \cdot (k+1)}$

$$= \frac{47}{48} - \frac{23}{24}$$

$$= \frac{47}{48} - \frac{46}{48} = \frac{1}{48}.$$

b. $\sum_{k=10}^{20} 2^{k-1} = \sum_{k=10}^{20} 2^{k-1} - \sum_{k=1}^{9} 2^{k-1}$

$$= \frac{1-2^{20}}{1-2} - \frac{1-2^{9}}{1-2} = 2^{20} - 2^{9}.$$

Properties 4

$\sum$ notation has distributive property over addition and subtraction, i.e.,

$$\sum_{k=1}^{n} (a_k \pm b_k) = \sum_{k=1}^{n} a_k \pm \sum_{k=1}^{n} b_k.$$

Proof It is obvious that $\sum_{k=1}^{n} (a_k \pm b_k) = (a_1 \pm b_1) + (a_2 \pm b_2) + \ldots + (a_n \pm b_n)$

$$= (a_1 + a_2 + \ldots a_n) \pm (b_1 + b_2 + \ldots + b_n)$$

$$= \sum_{k=1}^{n} a_k \pm \sum_{k=1}^{n} b_k.$$

EXAMPLE 25 Evaluate the given expressions.

a. $\sum_{k=1}^{5} (k^3 + 2k^2)$ b. $\sum_{k=1}^{6} (2k-1)^2$

Solution a. $\sum_{k=1}^{5} (k^3 + 2k^2) = \sum_{k=1}^{5} k^3 + \sum_{k=1}^{5} 2k^2$

$$= \sum_{k=1}^{5} k^3 + 2\sum_{k=1}^{5} k^2 = (\frac{5 \cdot 6}{2})^2 + 2 \cdot \frac{5 \cdot 6 \cdot 11}{6} = 335.$$

b. It left as an exercise for the student.

Note

$\sum$ notation does not have distributive property over multiplication and division, i.e.,

$$\sum_{k=1}^{n}(a_k \cdot b_k) \neq \sum_{k=1}^{n} a_k \cdot \sum_{k=1}^{n} b_k \text{ and } \sum_{k=1}^{n}(a_k : b_k) \neq \sum_{k=1}^{n} a_k : \sum_{k=1}^{n} b_k.$$

Properties 5

Let $r, p \in \mathbb{Z}$ then $\sum_{k=p}^{n} a_k = \sum_{k=p-r}^{n-r} a_{k+r} = \sum_{k=p+r}^{n+r} a_{k-r}$.

You may work out for the proof as an exercise.

EXAMPLE 26 Evaluate.

a. $\sum_{k=3}^{7}(k-1)^2$ b. $\sum_{k=-1}^{8}(2k+4)$ c. $\sum_{k=0}^{3}(3k^2-2)$

Solution

a. $$\sum_{k=3}^{7}(k-1)^2 = \sum_{k=3-2}^{7-2}(k+2-1)^2 = \sum_{k=1}^{5}(k+1)^2$$

$$= \sum_{k=1}^{5}(k^2+2k+1) = \frac{5 \cdot 6 \cdot 11}{6} + 2 \cdot \frac{5 \cdot 6}{2} + 5 = 90.$$

b. $$\sum_{k=-1}^{8}(2k+4) = \sum_{k=-1+2}^{8+2} 2(k-2)+4 = \sum_{k=1}^{10}(2k-4+4)$$

$$= \sum_{k=1}^{10} 2k = 2 \cdot \frac{10 \cdot 11}{2} = 110.$$

c. It is left as an exercise for the student.

Properties 6 Commutativity of $\sum$

$$\sum_{k=1}^{n}\sum_{r=1}^{m} a_{kr} = \sum_{r=1}^{m}\left(\sum_{k=1}^{n} a_{kr}\right)$$

Proof

$$\sum_{k=1}^{n}\sum_{r=1}^{m} a_{kr} = \sum_{k=1}^{n}(a_{k1}+a_{k2}+a_{k3}+\ldots+a_{km})$$

$$= (a_{11}+a_{12}+\ldots+a_{1m})+(a_{21}+a_{22}+\ldots+a_{2m})+\ldots+(a_{n1}+a_{n2}+\ldots+a_{nm})$$

$$= \sum_{r=1}^{m} a_{1r} + \sum_{r=1}^{m} a_{2r} + \ldots + \sum_{r=1}^{m} a_{nr}$$

$$= \sum_{r=1}^{m}(a_{1r}+a_{2r}+\ldots+a_{nr}) = \sum_{r=1}^{m}\left(\sum_{k=1}^{n} a_{kr}\right).$$

EXAMPLE 27 Evaluate the following expressions.

a. $\sum_{k=1}^{5}\sum_{r=1}^{3}(k^2-2r)$ b. $\sum_{a=1}^{4}\sum_{b=0}^{3}(a+b)$

Solution a. $$\sum_{k=1}^{5}\sum_{r=1}^{3}(k^2-2r)=\sum_{k=1}^{5}(3k^2-2\cdot\frac{3\cdot 4}{2})$$

$$=\sum_{k=1}^{5}(3k^2-12)=\sum_{k=1}^{5}3k^2-\sum_{k=1}^{5}12$$

$$=3\cdot\frac{5\cdot 6\cdot 11}{6}-12\cdot 5=165-60=105.$$

Also, $$\sum_{r=1}^{3}\sum_{k=1}^{5}(k^2-2r)=\sum_{r=1}^{3}(\frac{5\cdot 6\cdot 11}{6}-5\cdot 2r)$$

$$=55\cdot 3-10\frac{3\cdot 4}{2}=165-60=105.$$

b. The solution is left as an exercise for the student.

EXAMPLE 28 Find gof (2) if $f(x)=\frac{1}{14}\sum_{k=-1}^{x}\sum_{m=1}^{x+2}(2k+m)$ and $g(x)=\sum_{p=1}^{x+1}\sum_{r=0}^{x+3}(p+r)$.

Solution $$f(2)=\frac{1}{14}\sum_{k=-1}^{x}\sum_{m=1}^{x+2}(2k+m)$$

$$=\frac{1}{14}\sum_{k=-1}^{2}(8k+\frac{4\cdot 5}{2})=\frac{1}{14}\sum_{k=-1}^{2}(8k+10)$$

$$=\frac{1}{14}\sum_{k=-1}^{2}[8(k-2)+10]=\frac{1}{14}\sum_{k=1}^{4}(8k-6)$$

$$=\frac{1}{14}(8\cdot\frac{4\cdot 5}{2}-6\cdot 4)=\frac{1}{14}\cdot 56=4.$$

Since $g \text{ o } f(2)=g(f(2))$, we need to evaluate $g(4)$.

$$g(4)=\sum_{p=1}^{5}\sum_{r=0}^{7}(p+r)$$

$$=\sum_{p=1}^{5}\sum_{r=1}^{8}(p+r-1)$$

$$=\sum_{p=1}^{5}(8\cdot p+\frac{8\cdot 9}{2}-8)$$

$$=\sum_{p=1}^{5}(8p+28)=8\cdot\frac{5\cdot 6}{2}+28\cdot 5$$

$$=260.$$

B. MULTIPLICATIN NOTATION

In the preceding section, we have discussed the sum of the terms of a function given as:

$$f : Z \to \mathbb{R}, f(k) = a_k$$

Remember that, we used $\sum$ (sigma notation) in order to denote the sum of the terms of the function f. Now, we will introduce a new notation, $\prod$ (pi), used for representing the product of the terms of the above function.

Definition

Let $f : Z \to \mathbb{R}, f(k) = a_k$ and $r, n \in Z$ provided that $r \leq n$, then

$$\prod_{k=r}^{n} a_k = a_{r+1} \cdot a_{r+2} \ldots a_n$$

is the product of $(n - r + 1)$ terms of the function f from r to n where k is the index of multiplication, r is the lower bound and n is the upper bound.

$\prod_{k=r}^{n} a_k$ can also be denoted by $\prod_{r \leq k \leq n}^{n} a_k$ or $\prod_{k \in [r, n]} a_k$ and is read as "product of a_k's from $k = r$ to $k = n$"

Study the following examples.

$$a_3 \cdot a_4 \cdot a_5 \cdot \ldots \cdot a_{40} = \prod_{k=3}^{40} a_k,$$

$$a_{-9} \cdot a_{-8} \cdot a_{-7} \cdot \ldots \cdot a_0 = \prod_{k=-9}^{0} a_k,$$

$$1 \cdot 2 \cdot 3 \cdot \ldots \cdot n = \prod_{k=1}^{n} k,$$

$$\prod_{k=1}^{20} k^3 = 1^3 \cdot 2^3 \cdot 3^3 \cdot 4^3 \cdot \ldots \cdot \frac{n}{n+1},$$

$$\prod_{k=1}^{n} \frac{k}{k+1} = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \ldots \cdot \frac{n}{n+1},$$

$$\prod_{k=1}^{100} 3^k = 3^1 \cdot 3^2 \cdot 3^3 \cdot \ldots \cdot 3^{100}.$$

1. Multiplication Formulas

FORMULA

a. $\prod_{k=1}^{n} k = n!$

b. $\prod_{k=p}^{n} k = \frac{n!}{(p-1)!}$

Proof 1. By definition, $\prod_{k=1}^{n} k = 1 \cdot 2 \cdot 3 \cdot \ldots \cdot n,$

$$= n!$$

2. Obviously, $\prod_{k=p}^{n} k = p \cdot (p+1) \cdot (p+2) \cdot \ldots \cdot n$

$$= \frac{1 \cdot 2 \cdot 3 \cdot \ldots \cdot (p-1) \cdot p \cdot (p+1) \cdot \ldots \cdot n}{1 \cdot 2 \cdot 3 \cdot \ldots \cdot (p-1)}$$

$$= \frac{n!}{(p-1)!}.$$

FORMULA

$$\prod_{k=p}^{n} \log_k (k+1) = \log_p (n+1).$$

Proof Here, we shall use the properties of logarithm.

$$\prod_{k=p}^{n} \log_k (k+1) = \log_p (p+1) \cdot \log_{p+1} (p+2) \cdot \ldots \cdot (\log_n (n+1)$$

$$= \frac{\log(p+1)}{\log p} \cdot \frac{\log(p+2)}{\log(p+1)} \cdot \ldots \cdot \frac{\log(n+1)}{\log n}$$

$$= \frac{\log(n+1)}{\log p} = \log_p (n+1).$$

FORMULA

$$\prod_{k=p}^{n} \frac{k+1}{k} = \frac{n+1}{p}.$$

Proof It is obvious that, $\prod_{k=p}^{n} \frac{k+1}{k} = \frac{n+1}{p} \cdot \frac{p+2}{p+1} \ldots \frac{n+1}{n} = \frac{n+1}{p}.$

2. Properties of Multiplication Notation

Properties 7

For $c \in \mathbb{R}$ and $r \in ¢$, $\prod_{k=1}^{n} c = c^n$ and $\prod_{k=r}^{n} c = c^{n-r+1}$.

Proof Obviously, $\prod_{k=1}^{n} c = \underbrace{c \cdot c \cdot c \cdot \ldots \cdot c}_{n-\text{ times}} = c^n$ and $\prod_{k=r}^{n} c = \underbrace{c \cdot c \cdot c \cdot \ldots \cdot c}_{(n-r+1)-\text{ times}} = c^{n-r+1}$.

EXAMPLE 29 Evaluate the given expressions.

a. $\prod_{k=1}^{4} 3$ b. $\prod_{p=1}^{9} (-3)$ c. $\prod_{k=3}^{6} 2$ d. $\prod_{k=-5}^{7} \frac{1}{6}$

Solution a. $\prod_{k=1}^{4} 3 = 3^4 = 81.$ c. $\prod_{k=3}^{6} 2 = 2^{6-3+1} = 2^4 = 16.$

The solutions of b. and d. are left as an exercise for the student.

Properties 8

For $c \in \mathbb{R}$, $\prod_{k=1}^{n} c \cdot a_k = c^n \cdot \prod_{k=1}^{n} a_k.$

Proof Obviously, $\prod_{k=1}^{n} c \cdot a_k = c \cdot a_1 \cdot c \cdot a_2 \cdot c \cdot a_3 \cdot \ldots \cdot c \cdot a_n$

$$= \underbrace{c \cdot c \cdot c \cdot \ldots \cdot c}_{n-\text{ times}} \cdot a_1 \cdot a_2 \cdot_3 \cdot \ldots \cdot a_n = c^n \cdot c^n \cdot \prod_{k=1}^{n} a_k.$$

EXAMPLE 30 Evaluate the given expressions.

a. $\prod_{k=1}^{3} 4k$ b. $\prod_{p=1}^{5} 2p^2$ c. $\prod_{t=1}^{3} (-\frac{1}{t})$

Solution a. $\prod_{k=1}^{3} 4k = 4^3 \prod_{k=1}^{3} k = 4^3 \cdot (1 \cdot 2 \cdot 3 = 384.$

b. and c. are left as an exercise for the student.

Properties 9

For $1 < p < n$, $p \in \mathbb{Z}$, $\prod_{k=1}^{n} a_k = \prod_{k=1}^{p-1} a_k \cdot \prod_{k=p}^{n} a_k.$

Proof $\prod_{k=1}^{n} a_k = a_1 \cdot a_2 \cdot \ldots \cdot a_{p-1} \cdot a_p \cdot \ldots \cdot a_n = \prod_{k=1}^{p-1} a_k \cdot \prod_{k=p}^{n} a_k.$

EXAMPLE 31 Evaluate the given expressions.

a. $\sum_{k=4}^{7} k$ b. $\prod_{p=10}^{17}(1-\frac{1}{p})$

Solution a. By property, 3, we have $\sum_{k=1}^{7} k = \prod_{k=1}^{3} k \cdot \prod_{k=4}^{7} k$.

Dividing both sides of this equality by $\prod_{k=1}^{3} k$, gives

$$\prod_{k=4}^{7} k = \frac{\prod_{k=4}^{7} k}{\prod_{k=1}^{3} k} = \frac{1\cdot 2\cdot 3\cdot 4\cdot 5\cdot 6\cdot 7}{1\cdot 2\cdot 3} = \frac{7!}{3!}.$$

b. The solution is left as an exercise for the student.

Properties 10

$\prod$ notation has distributive property over multiplication and divison.

1. $\prod_{k=1}^{n}(a_k \cdot b_k) = \prod_{k=}^{n} a_k \cdot \prod_{k=1}^{n} b_k,$

2. $\prod_{k=1}^{n}(a_k : b_k) = \prod_{k=1}^{n} a_k : \prod_{k=1}^{n} b_k, \quad b_k \neq 0.$

Proof

1. $$\prod_{k=1}^{n}(a_k \cdot b_k) = a_1 \cdot b_1 \cdot a_2 \cdot b_2 \cdot a_3 \cdot b_3 \cdot \ldots \cdot a_n \cdot b_n$$
$$= (a_1 \cdot a_2 \cdot a_3 \cdot \ldots \cdot a_n)\cdot(b_1 \cdot b_2 \cdot b_3 \cdot \ldots \cdot b_n) = \prod_{k=}^{n} a_k \cdot \prod_{k=1}^{n} b_k.$$

2. $$\prod_{k=1}^{n}\frac{a_k}{b_k} = \frac{a_1}{b_1}\cdot\frac{a_2}{b_2}\cdot\frac{a_3}{b_3}\cdot\ldots\cdot\frac{a_n}{b_n}$$
$$= \frac{a_1 \cdot a_2 \cdot a_3 \cdot \ldots \cdot a_n}{b_1 \cdot b_2 \cdot b_3 \cdot \ldots \cdot b_n} = \prod_{k=1}^{n} a_k : \prod_{k=1}^{n} b_k, \quad b_k \neq 0.$$

EXAMPLE 32 Evaluate the given expressions.

a. $\prod_{k=1}^{3} 2^k \cdot k^2$ b. $\prod_{k=1}^{3}(\frac{k}{2^k})$

Solution a. $$\prod_{k=1}^{3} 2^k \cdot k^2 = \prod_{k=1}^{3} 2^k \cdot \prod_{k=1}^{3} k^2$$
$$= (2^1 \cdot 2^2 \cdot 2^3)\cdot(1^2 \cdot 2^2 \cdot 3^2) = 2^8 \cdot 9.$$

b. This part is left as an exercise for the student.

Note

$\prod$ notation does not have distributive property over addition and subtraction i.e.,

$$\prod_{k=1}^{n}(a_k \pm b_k) \neq \prod_{k=1}^{n}a_k \pm \prod_{k=1}^{n}b_k.$$

Properties 11

Rule for Changing Boundaries

For $r, p \in \mathbb{Z}$, $\prod_{k=2}^{n}a_k = \prod_{k=r-p}^{n-p}a_{k+p} = \prod_{k=r+p}^{n+p}a_{k-p}$.

You may work out for the proof by yourself.

EXAMPLE 33 Evaluate the given expressions.

a. $\prod_{k=3}^{10}\log_{k+1}k$. b. $\prod_{k=2}^{7}(1-\frac{1}{k})$ c. $\prod_{t=0}^{10}p$

Solution a.
$$\prod_{k=3}^{10}\log_{k+1}k = \prod_{k=3-2}^{10-2}\log_{k+2+1}(k+2) = \prod_{k=1}^{8}\log_{k+3}(k+2)$$
$$= \log_4 3 \cdot \log_5 4 \cdot \log_6 5 \cdot \ldots \cdot \log_{11}10$$
$$= \frac{\log 3}{\cancel{\log 4}} \cdot \frac{\cancel{\log 4}}{\cancel{\log 5}} \cdot \frac{\cancel{\log 5}}{\cancel{\log 6}} \cdot \ldots \cdot \frac{\cancel{\log 10}}{\log 11} = \log_{11}3.$$

b.
$$\prod_{k=2}^{7}(1-\frac{1}{k}) = \prod_{k=2-1}^{7-1}(1-\frac{1}{k+1}) = \prod_{k=1}^{6}\frac{k}{k+1}$$
$$= \frac{1}{\cancel{2}} \cdot \frac{\cancel{2}}{\cancel{3}} \cdot \frac{\cancel{3}}{\cancel{4}} \cdot \frac{\cancel{4}}{\cancel{5}} \cdot \frac{\cancel{5}}{\cancel{6}} \cdot \frac{\cancel{6}}{7} = \frac{1}{7}.$$

c. are left as an exercise for the student.

Properties 12

Commutative Property of $\prod$

$$\prod_{k=1}^{n}(\prod_{r=1}^{m}a_{kr}) = \prod_{r=1}^{m}(\prod_{k=1}^{n}a_{kr})$$

Proof

$$\prod_{k=1}^{n}(\prod_{r=1}^{m}a_{kr}) = \prod_{k=1}^{n}(a_{k1} \cdot a_{k2} \cdot a_{k3} \cdot \ldots \cdot a_{km})$$
$$= (a_{11} \cdot a_{12} \cdot \ldots \cdot a_{1m}) \cdot (a_{21} \cdot a_{22} \cdot \ldots \cdot a_{2m}) \cdot \ldots \cdot (a_{n1} \cdot a_{n2} \cdot \ldots \cdot a_{nm})$$
$$= \prod_{r=1}^{m}a_{1r} \cdot \ldots \cdot \prod_{r=1}^{m}a_{2r} \cdot \ldots \cdot \prod_{r=1}^{m}a_{nr}$$
$$= \prod_{r=1}^{m}(a_{1r} \cdot a_{2r} \cdot \ldots \cdot a_{nr})$$
$$= \prod_{r=1}^{m}(\prod_{k=1}^{n}a_{kr}).$$

EXAMPLE 34 Evaluate the given expressions.

a. $\prod_{k=1}^{4}(\prod_{r=1}^{3}\frac{2^r}{3^k})$ b. $\prod_{t=1}^{3}\prod_{p=1}^{2}(p\cdot t)$

Solution a. $\prod_{k=1}^{4}(\prod_{r=1}^{3}\frac{2^r}{3^k}) = \prod_{k=1}^{4}\frac{2\cdot 2^2\cdot 2^3}{3^{3k}} = \prod_{k=1}^{4}\frac{2^6}{3^{3k}}$

$$= \frac{2^{4\cdot 6}}{3^3\cdot 3^6\cdot 3^9\cdot 3^{12}} = \frac{2^{24}}{3^{30}}.$$

b. The solution is left as an exercise for the student.

Properties 13

For $r \in \mathbb{R}$, $\prod_{k=1}^{n} r^{a_k} = r^{\sum_{k=1}^{n} a_k}$

Proof Obviously, $\prod_{k=1}^{n} r^{a_k} = r^{a_1}\cdot r^{a_2}\cdot r^{a_3}\cdot \ldots \cdot r^{a_n}$

$$= r^{a_1+a_2+a_3+\ldots+a_n} = r^{\sum_{k=1}^{n} a_k}.$$

EXAMPLE 35 Evaluate the given expressions.

a. $\prod_{k=0}^{10} 3^{(k-2)^2}$ b. $\log_{64}(\prod_{k=1}^{10} 2^{3^{k-1}})$

Solution a. $\prod_{k=0}^{10} 3^{(k-2)^2} = 3^{\sum_{k=0}^{10}(k-2)^2} = 3^{\sum_{k=0}^{10}(k^2-4k+4)} = 3^{\frac{10\cdot 11\cdot 21}{6}-4\frac{10\cdot 11}{4}+4\cdot 11} = 3^{385-220+44} = 3^{209}.$

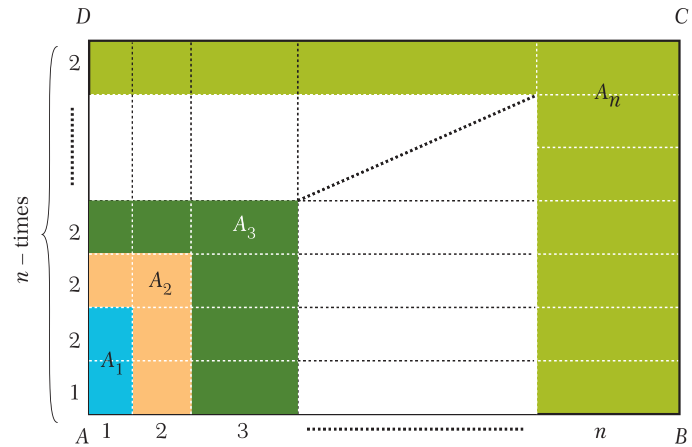
b. $\log_{64}(\prod_{k=1}^{10} 2^{3^{k-1}}) = \log_{64} 2^{\sum_{k=1}^{10} 3^{k-1}} = (\sum_{k=1}^{10} 3^{k-1})\log_{64} 2 = \frac{3^{10}-1}{2}\cdot\frac{\log 2}{6\log 2} = \frac{3^{10}-1}{12}.$

EXAMPLE 36 If $\sum_{k=1}^{n} 2k = -n + \prod_{k=1}^{5} k$ then find the value of n.

Solution The solution is left as an exercise for the student.

EXAMPLE 37 Prove that $1^2 + 2^2 + 3^2 + ... + n^2 = \dfrac{n(n+1)(2n+1)}{6}$.

Solution Remember that we have proven this formula before. But now, we will prove it by calculating the area of the rectangle having the sides $1 + 2 + 3 + ... + n$ and $2n + 1$ units. In Figure 1.2, the letters A_1, A_2, A_3, ..., A_n denote the areas painted by the same color.



$A(ABCD) = A_1 + A_2 + A_3 + . . . + A_n$

$\Rightarrow (1 + 2 + 3 + . . . + n)\ (2n + 1) = 3 + 12 + 27 + . . . + 3n^2$

$\Rightarrow \dfrac{n(n+1)(2n+1)}{2} = 3(1 + 4 + 9 + ... + n^2 = 3\sum_{k=1}^{n} k^2$

Hence $\sum_{k=1}^{n} k^2 = \dfrac{n(n+1)(2n+1)}{6}$.

EXAMPLE 38 Prove that $1^2 + 2^2 + 3^2 + ... + n^3 = (\dfrac{n(n+1)}{2})^2$.

Solution By the Binomial theorem that

$(x +)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$.

Using this expression yields

$(1 + 1)^4 = 1^4 + 4 \cdot 1^3 \cdot 1 + 6 \cdot 1^2 \cdot 1^2 + 4 \cdot 1 \cdot 1^3 + 1^4$

$(2 + 1)^4 = 2^4 + 4 \cdot 2^3 \cdot 1 + 6 \cdot 2^2 \cdot 1^2 + 4 \cdot 2 \cdot 1^3 + 1^4$

$(3 + 1)^4 = 3^4 + 4 \cdot 3^3 \cdot 1 + 6 \cdot 3^2 \cdot 1^2 + 4 \cdot 3 \cdot 1^3 + 1^4$

....

....

$(n + 1)^4 = n^4 + 4 \cdot n^3 \cdot 1 + 6 \cdot n^2 \cdot 1^2 + 4 \cdot n \cdot 13 + 14$

Adding these equalities side by side gives

$$(n+1)^4 = 1^4 + 4(1^3 + 2^3 + \ldots + n^3) + 6(1^2 + 2^2 + \ldots + n^2) + 4(1 + 2 + \ldots + n) + n.$$

$$n^4 + 4n^3 + 6n^2 + 4n + 1 = 1 + 4\sum_{k=1}^{n} k^3 + 6\sum_{k=1}^{n} k^2 + 4\sum_{k=1}^{n} k + n$$

Since

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6} \quad \text{and} \quad \sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$

we get

$$n^4 + 4n^3 + 6n^2 + 4n = 4\sum_{k=1}^{n} k^3 + 6\frac{n(n+1)(2n+1)}{6} + 4\frac{n(n+1)}{2} + n$$

By rearranging this equality, we will have

$$4\sum_{k=1}^{n} k^3 = n^4 + 2n^3 + n^2 = n^2 \cdot (n^2 + 2n + 1)$$

$$= n^2 \cdot (n+1)^2.$$

Hence, $\displaystyle\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}$

$$= \left(\frac{n(n+1)}{2}\right)^2.$$

EXAMPLE 39 Find the sum $-11 - 7 - 3 + 1 + 5 + \ldots + 63$.

Solution The solution is left as an exercise for the student.

EXAMPLE 40 Prove that $1 + 4 + 7 + \ldots + 3n - 2 = \dfrac{3n^2 - n}{2}$ by using summation notation .

Solution $\displaystyle 1 + 4 + 7 + \ldots + 3n - 2 = \sum_{k=1}^{n}(3k - 2)$

$$= \sum_{k=1}^{n} 3k - \sum_{k=1}^{n} 2$$

$$= 3 \cdot \frac{n(n+1)}{2} - 2n$$

$$= \frac{3n^2 + 3n - 4n}{2}$$

$$= \frac{3n^2 - n}{2}.$$

EXAMPLE 41 Find the sum $\sum_{k=1}^{n}(4k^3-6k^2+2k)$.

Solution $\sum_{k=1}^{n}(4k^3-6k^2+2k)=4\cdot(\frac{n(n+1)}{2})^2-6\cdot\frac{n(n+1)(2n+1)}{6}+2\cdot\frac{n(n+1)}{2}$

$$= n^2(n+1)^2-n(n+1)(2n+1)+n(n+1)$$
$$= n(n+1)[n(n+1)-(2n+1)+1]$$
$$= n(n+1)(n^2+n-2n-1+1)$$
$$= n(n+1)(n^2-n)$$
$$= n^4-n^2.$$

EXAMPLE 42 Prove that $1\cdot2+2\cdot3+3\cdot4+...+n\cdot(n+1)=\frac{n(n+1)\cdot(n+2)}{3}$.

Solution $1\cdot2+2\cdot3+3\cdot4+...+n\cdot(n+1)=\sum_{k=1}^{n}k\cdot(k+1)=\sum_{k=1}^{n}(k^2+k)$

$$=\frac{n(n+1)\cdot(2n+1)}{6}+\frac{n(n+1)}{2}$$
$$=(\frac{n(n+1)}{2})\cdot(\frac{2n+1}{3}+1)$$
$$=\frac{n(n+1)}{2}\cdot\frac{2n+4}{3}$$
$$=\frac{n(n+1)\cdot(n+2)}{3}.$$

EXAMPLE 43 Evaluate $(\sum_{k=1}^{10}2\cdot\sum_{k=3}^{11}(-3)):\sum_{k=-2}^{9}6.$

Solution Since $\sum_{k=p}^{n}(c)=c(n+p+1)$ where $n-p+1$ is the number of terms, $k\in[p,n]$, we get

$$(\sum_{k=1}^{10}2\cdot\sum_{k=3}^{11}(-3)):\sum_{k=-2}^{9}6=((2\cdot10)\cdot(-3\cdot9)):(6\cdot12)$$
$$=-\frac{20\cdot27}{72}=-\frac{15}{2}.$$

EXAMPLE 44 Prove that $1^4+2^4+3^4+\ldots+n^4=\dfrac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$ by using summation notation.

Solution The solution is left as an exercise for the student.

EXAMPLE 45 Find the sum $\sum\limits_{k=3}^{10}(k^2-4k)$.

Solution

$$\sum_{k=3}^{10}(k^2-4k)=\sum_{k=3-2}^{10-2}((k+2)^2-4(k+2))$$
$$=\sum_{k=1}^{8}(k^2+\cancel{4k}+4-\cancel{4k}-8)=\sum_{k=1}^{8}(k^2-4)$$
$$=\frac{8\cdot 9\cdot 17}{6}-4\cdot 8$$
$$=20-32=172.$$

EXAMPLE 46 Evaluate $\sum\limits_{k=0}^{4}(2k^2-3k-7)$.

Solution

$$\sum_{k=0}^{4}(2k^2-3k-7)=\sum_{K=0+1}^{4+1}[2(K-1)^2+3(K-1)-7]$$
$$=\sum_{K=1}^{5}(2k^2-k-8)$$
$$=2\cdot\frac{5\cdot 6\cdot 11}{6}-\frac{5\cdot 6}{2}-5\cdot 8$$
$$=110-15-40=55.$$

EXAMPLE 47 Evaluate $\sum\limits_{k=4}^{15}2^{2k}$.

Solution

$$\sum_{k=4}^{15}2^{2k}=\sum_{k=4}^{15}4^k=\sum_{k=4-3}^{15-3}4^{k+3}=\sum_{k=1}^{12}4^{k-1}\cdot 4^4$$
$$=\frac{1-4^{12}}{1-4}\cdot 4^4=\frac{4^{16}-4^4}{3}=\frac{2^{32}-2^8}{3}.$$

EXAMPLE 48 Evaluate $\sum_{k=1}^{20}(-1)^n(\frac{1}{2})^{n+1}$.

Solution The solution is left as an exercise for the student.

EXAMPLE 49 Find the sum $\sum_{k=6}^{71}\frac{1}{k(k+1)}$.

Solution **First way:**

Using the "Formula 1.5", we get

$$\sum_{k=6}^{71}\frac{1}{k(k+1)}=\sum_{k=1}^{71}\frac{1}{k(k+1)}-\sum_{k=1}^{5}\frac{1}{k(k+1)}$$
$$=\frac{71}{72}-\frac{5}{6}=\frac{11}{72}.$$

Second way:

Since $\frac{1}{k(k+1)}=\frac{1}{k}-\frac{1}{k+1}$, we obtain

$$\sum_{k=6}^{71}\frac{1}{k(k+1)}=\sum_{k=6}^{71}(\frac{1}{k}-\frac{1}{k+1})$$
$$=\frac{1}{6}-\frac{1}{7}+\frac{1}{7}-\frac{1}{8}+\ldots+\frac{1}{71}-\frac{1}{72}$$
$$=\frac{1}{6}-\frac{1}{72}=\frac{11}{72}.$$

EXAMPLE 50 Evaluate $\sum_{k=1}^{40}(\sqrt{2k+1}-\sqrt{2k-1})$.

Solution
$$\sum_{k=1}^{40}(\sqrt{2k+1}-\sqrt{2k-1})=\sqrt{3}-\sqrt{1}+\sqrt{5}-\sqrt{3}+\sqrt{7}-\sqrt{5}+\ldots+\sqrt{81}-\sqrt{79}$$
$$=\sqrt{81}-\sqrt{1}$$
$$=9-1=8.$$

EXAMPLE 51 Evaluate $\sum_{k=1}^{4}(\sum_{p=1}^{4}(2k+3p-5))$.

Solution
$$\sum_{k=1}^{4}(\sum_{p=1}^{4}(2k+3p-5))=\sum_{k=1}^{4}(2k\cdot 4+3\cdot\frac{4\cdot 5}{2}-5\cdot 4)=\sum_{k=1}^{4}(8k+10)$$
$$=8\cdot\frac{4\cdot 5}{2}+10\cdot 4=80+40=120.$$

EXAMPLE 52 Evaluate $\sum_{k=1}^{30} \frac{2}{4k^2-1}$.

Solution Since $\frac{2}{4k^2-1} = (\frac{1}{2k-1} - \frac{1}{2k+1})$, we get

$$\sum_{k=1}^{30} \frac{2}{4k^2-1} = \sum_{k=1}^{30} (\frac{1}{2k-1} - \frac{1}{22k+1})$$

$$= (1 - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \ldots + \frac{1}{59} - \frac{1}{61})$$

$$= (1 - \frac{1}{61}) = \frac{60}{61}.$$

EXAMPLE 53 Evaluate $\sum_{k=0}^{20} \frac{2}{k^2+3k+2}$.

Solution $$\sum_{k=0}^{20} \frac{2}{k^2+3k+2} = 2\sum_{k=0}^{20} \frac{1}{(k+1)(k+2)} = 2\sum_{k=0}^{20} (\frac{1}{k+1} - \frac{1}{k+2})$$

$$= 2 \cdot (1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \ldots + \frac{1}{21} - \frac{1}{22})$$

$$= 2 \cdot (1 - \frac{1}{22}) = \frac{21}{11}.$$

EXAMPLE 54 Evaluate $\sum_{a=0}^{3} \sum_{b=0}^{4} (b-a+3)$.

Solution $$\sum_{a=0}^{3} \sum_{b=0}^{4} (b-a+3) = \sum_{a=0}^{3} (\frac{4 \cdot 5}{2} - a \cdot (4+1) + 3 \cdot (4+1))$$

$$= \sum_{a=0}^{3} (25-5a)$$

$$= 25 \cdot (3+1) - 5 \cdot \frac{3 \cdot 4}{2}$$

$$= 100 - 30 = 70.$$

EXAMPLE 55 Let x_1 and x_2 be the roots of the quadratic equation $x^2 - 6x + 8 = 0$ provided that $x_1 < x_2$. If $f(x) = 4x - 1$ find the sum $\sum_{i=1}^{2} x_i f(x_i)$.

Solution Since the roots of the equation $x^2 - 6x + 8$ are $x_1 = 2$ and $x_2 = 4$, we have

$$\sum_{i=1}^{2} x_i f(x_i) = x_1 f(x_1) + x_2 f(x_2) = f(2) + 4 \cdot f(4)$$

$$= 2 \cdot (4 \cdot 2 - 1) + 4 \cdot (4 \cdot 4 - 1) = 14 + 60 = 74.$$

EXAMPLE 56 If $\sum_{k=0}^{5}(2k^2-3k-5)=5a$, fint the sum $\sum_{t=1}^{a}t^3$.

Solution
$$\sum_{k=0}^{5}(2k^2-3k-5)=2\cdot\frac{5\cdot 6\cdot 11}{6}-3\cdot\frac{5\cdot 6}{2}-5\cdot 6$$
$$=110-45-30$$
$$=35.$$
That is, $5a = 35 \Rightarrow a = 7$. Hence,
$$\sum_{t=1}^{7}t^3=(\frac{7\cdot 8}{2})^2=28^2=784.$$

EXAMPLE 57 If $\sum_{k=0}^{5}(2k^2-3k$ and $g(x)=\sum_{p=1}^{x}p^2$, then find $f \text{ o } g(5)$

Solution Since $f \text{ o } g(5) = f(g(5))$, we obatain
$$f(g(5))=f(\sum_{p=1}^{5}p^2)=f(\frac{5\cdot 6\cdot 11}{6})=f(55)=\sum_{k=1}^{55}k=\frac{55\cdot 56}{2}=1540.$$

EXAMPLE 58 Prove that $\sum_{p=1}^{n}p\cdot p!=(n+1)!-1$.

Solution
$$\sum_{p=1}^{n}p\cdot p!==\sum_{p=1}^{n}(p+1-1)\cdot p!$$
$$=\sum_{p=1}^{n}((p+1)\cdot p!-p!)=\sum_{p=1}^{n}((p+1)!-p!)$$
$$= 2!-1!+3!-2!+4!-3!+\ldots+(n+1)!-n!$$
$$=(n+1)!-1.$$

EXAMPLE 59 Prove that $\sum_{k=1}^{n}\frac{k}{(k+1)!}=1-\frac{1}{(n+1)!}$.

Solution
$$\sum_{k=1}^{n}\frac{k}{(k+1)!}=\sum_{k=1}^{n}\frac{k+1-1}{(k+1)!}$$
$$=\sum_{k=1}^{n}(\frac{k+1}{(k+1)!}-\frac{1}{(k+1)!})$$
$$=\sum_{k=1}^{n}(\frac{1}{k!}-\frac{1}{(k+1)!})$$
$$=\frac{1}{1!}-\frac{1}{2!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}+\ldots+\frac{1}{n!}-\frac{1}{(n+1)!}$$
$$=1-\frac{1}{(n+1)!}$$

EXAMPLE 60 Evaluate $\prod_{k=1}^{99}(1-\frac{1}{k+1})$.

Solution $\prod_{k=1}^{99}(1-\frac{1}{k+1}) = \prod_{k=1}^{99}\frac{k}{k+1}$

$$= \frac{99!}{100!} = \frac{99!}{99!\cdot 100} = \frac{1}{100}.$$

EXAMPLE 61 Evaluate $\prod_{k=5}^{624}\log_k(k+1)$.

Solution $\prod_{k=5}^{624}\log_k(k+1) = \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \ldots \log_{624} 625$

$$= \log_5 625 = \log_5 5^4 = 4.$$

EXAMPLE 62 Prove that $\prod_{k=2}^{n}(1-\frac{1}{k^2}) = \frac{n+1}{2n}$.

Solution $\prod_{k=2}^{n}(1-\frac{1}{k^2}) = \prod_{k=2}^{n}\frac{k^2-1}{k^2}$

$$= \prod_{k=2}^{n}\frac{(k-1)(k+1)}{k\cdot k} = (\prod_{k=2}^{n}\frac{k-1}{k})\cdot(\prod_{k=2}^{n}\frac{k+1}{k})$$

$$= (\frac{1}{2}\cdot\frac{2}{3}\cdot\frac{3}{4}\cdot\ldots\cdot\frac{n-1}{n})\cdot(\frac{3}{2}\cdot\frac{4}{3}\cdot\frac{5}{4}\cdot\ldots\cdot\frac{n+1}{n})$$

$$= \frac{1}{n}\cdot\frac{n+1}{2} = \frac{n+1}{2n}.$$

EXAMPLE 63 Find the value of x, if $\prod_{t=1}^{9}(\frac{3}{4})^{2t} = (\frac{16}{9})^{2x+5}$.

Solution By using "Property 7", we get

$$\prod_{t=1}^{9}(\frac{3}{4})^{2t} = (\frac{16}{9})^{2x+5} \Rightarrow (\frac{3}{4})^{\sum_{t=1}^{9}2t} = (\frac{16}{9})^{2x+5} \Rightarrow (\frac{3}{4})^{9\cdot 10} = (\frac{4}{3})^{4x+10}$$

$$\Rightarrow (\frac{3}{4})^{90} = (\frac{3}{4})^{-4x-10} \Rightarrow 90 = -4x-10.$$

This gives $x = -25$.

EXAMPLE 64 Evaluate $\prod_{k=1}^{5}\frac{k^2+3k+2}{k^2+4k}$.

Solution $$\prod_{k=1}^{5}\frac{k^2+3k+2}{k^2+4k}=\prod_{k=1}^{5}\frac{(k+1)(k+2)}{k(k+4)}=\prod_{k=1}^{5}\frac{k+1}{k}\cdot\prod_{k=1}^{5}\frac{k+2}{k+4}$$
$$=(\frac{2}{1}\cdot\frac{3}{2}\cdot\frac{4}{3}\cdot\frac{5}{4}\cdot\frac{6}{5})\cdot(\frac{3}{5}\cdot\frac{4}{6}\cdot\frac{6}{8}\cdot\frac{7}{9})=6\cdot(\frac{1}{6})=1.$$

EXAMPLE 65 Find the value of x, if $\prod_{t=2}^{x+1}4^t=8^{3x-3}$.

Solution $$\prod_{t=2}^{x+1}4^t=8^{3x-3}\Rightarrow\prod_{t=1}^{x}4^{t+1}=8^{3x-3}\Rightarrow 4^{\sum_{t=1}^{x}(t+1)}=8^{3x-3}$$
$$\Rightarrow 4^{\frac{x\cdot(x+1)}{2}+x}=8^{3x-3}\Rightarrow 2^{x(x+1)+2x}=2^{9x-9}$$
$$\Rightarrow x^2+3x=9x-9\Rightarrow x^2+3x-9x+9=0$$
$$\Rightarrow x^2-6x+9=0\Rightarrow (x-3)^2=0$$

Hence, we conclude that $x=3$.

EXAMPLE 66 Evaluate $\prod_{k=1}^{4}(\prod_{p=1}^{3}k\cdot p)$.

Solution $$\prod_{k=1}^{4}(\prod_{p=1}^{3}k\cdot p)=\prod_{k=1}^{4}(k^3\cdot 3!)$$
$$=\prod_{k=1}^{4}6\cdot k^3=6^4\cdot\prod_{k=1}^{4}k^3=6^4\cdot(4!)^3=2^{13}\cdot 3^7.$$

EXAMPLE 67 Evaluate $\prod_{i=1}^{3}\prod_{j=1}^{4}(\frac{i}{j})$.

Solution $$\prod_{i=1}^{3}\prod_{j=1}^{4}(\frac{i}{j})=\prod_{i=1}^{3}(\frac{i^4}{4!})=\prod_{i=1}^{3}\frac{i^4}{24}$$
$$=\frac{(3!)^4}{24^3}=\frac{3}{32}.$$

EXAMPLE 68 Evaluate $\prod_{a=1}^{10}\prod_{b=1}^{2}(a+b)$.

Solution $$\prod_{a=1}^{10}(\prod_{b=1}^{2}(a+b))=\prod_{a=1}^{10}(a+1)\cdot(a+2)=\sum_{a=1}^{10}(a^2+3a+2)$$
$$=\frac{10\cdot 11\cdot 21}{6}+3\cdot\frac{10\cdot 11}{2}+2\cdot 10$$
$$=385+165+20=570.$$

EXAMPLE 69 Evaluate $\prod_{k=1}^{2}(\sum_{n=1}^{100}(-1)^n \cdot n)$.

Solution $$\prod_{k=1}^{2}(\sum_{n=1}^{100}(-1)^n \cdot n) = \prod_{k=1}^{2}(\underbrace{-1+2}_{1}-\underbrace{3+4}_{1}-\underbrace{5+6}_{1}-\ldots-\underbrace{99+100}_{1})$$

$$= \prod_{k=1}^{2}\underbrace{(1+1+1+\ldots+1)}_{50 \text{ times}} = \prod_{k=1}^{2}50 = 50^2 = 2500.$$

EXAMPLE 70 Evaluate $\prod_{n=2}^{15}\sum_{k=1}^{n}(\log_n \frac{k+1}{k})$.

Solution $$\prod_{n=2}^{15}(\sum_{k=1}^{n}(\log_n \frac{k+1}{k}) = \prod_{n=2}^{15}(\log_n \frac{2}{1}+\log_n \frac{3}{2}+\log_n \frac{4}{3}+\ldots+\log_n \frac{n+1}{n})$$

$$= \prod_{n=2}^{15}(\log_n \frac{2}{1}\cdot\frac{3}{2}\cdot\frac{4}{3}\cdot\ldots\cdot\frac{n+1}{n})$$

$$= \prod_{n=2}^{15}\log_n(n+1)$$

$$= \log_2 3\cdot\log_3 4\cdot\log_4 5\cdot\ldots\cdot\log_{15}16$$

$$= \log_2 16$$

$$= 4.$$

EXAMPLE 71 If $f(x)=\prod_{k=1}^{x+2}\sum_{m=1}^{x}\frac{m}{6}$ and $g(x)=\frac{1}{12}\sum_{t=1}^{x}\prod_{n=1}^{x-1}n$, find $f \circ g(4)$

Solution Since fog (4) = *f*(g (4)), we obtain

$$fog(4) = f(\frac{1}{12}\sum_{t=1}^{4}\prod_{n=1}^{3}n)$$

$$= f(\frac{1}{12}\sum_{t=1}^{4}6)$$

$$= f(\frac{24}{12}) = f(2)$$

$$f(2) = \prod_{k=1}^{4}\sum_{m=1}^{2}\frac{m}{6}$$

$$= \prod_{k=1}^{4}\frac{3}{6} = \prod_{k=1}^{4}\frac{1}{2}$$

$$= (\frac{1}{2})^4 = \frac{1}{16}.$$

EXERCISES 3.2

A. Summation Notation

1. Evaluate the indicated sums below.

a. $\sum_{k=1}^{12} 3$ b. $\sum_{k=0}^{10} (-2)$ c. $\sum_{p=3}^{22} 7$

d. $\sum_{m=2}^{8} (-6)$ e. $\sum_{t=-2}^{3} 10$ f. $\sum_{r=-1}^{5} e$

g. $\sum_{k=0}^{2} \pi$ h. $\sum_{k=1}^{4} (n+1)$ i. $\sum_{n=-2}^{2} (k-1)$

2. Evaluate the indicated sums below.

a. $\sum_{k=1}^{50} k$ b. $\sum_{k=3}^{40} (2k)$

c. $\sum_{a=0}^{17} (2a+1)$ d. $\sum_{k=0}^{7} k^2$

e. $\sum_{k=0}^{12} (k^2-k+1)$ f. $\sum_{t=1}^{9} (3t-2)^2$

g. $\sum_{p=-7}^{1} (2p+16)^2$ h. $\sum_{p=3}^{7} (3^p+1)$

i. $\sum_{k=n}^{2n} (2k+1)$ j. $\sum_{m=1}^{20} (3k) + \sum_{k=0}^{10} (2k-1)$

k. $\sum_{k=0}^{3} (k-1)\cdot k!$ l. $\sum_{k=3}^{11} \log a^k$

g $\sum_{k=1}^{90} \sin^2 k°$ h. $\sum_{k=1}^{1000} (i)^k (i^2=-1)$

i. $\sum_{k=1}^{9} \frac{2}{3k^2+3k}$ j. $\sum_{p=3}^{50} \frac{1}{p^2-3p+2}$

k. $\sum_{k=2}^{50} \frac{k}{k^3-k^2}$ l. $\sum_{k=4}^{n+2} \frac{k-3}{k-2} - \sum_{i=-2}^{n-2} \frac{i+2}{i+3}$

m. $\sum_{k=2}^{n} (\sqrt{k-1}-\sqrt{k+1}$ n. $\sum_{k=1}^{89} (\sqrt{\sin k°}-\sqrt{\cos k°}$

o. $\sum_{k=1}^{100} \log\frac{1}{k+1}$ p. $\sum_{p=2}^{79} \log_3(1+\frac{1}{p+1})$

q. $\sum_{k=1}^{100} k\cdot k!$ r. $\sum_{k=1}^{50} \frac{k}{(k+1)!}$

s. $\sum_{k=-4}^{4} k^3$ t. $\sum_{r=-5}^{5} (k^2+k)$

u. $\sum_{k=-13}^{13} (2k^3+3k^2+k)$ v. $\sum_{x=1}^{13} 2^x - \sum_{x=1}^{13} 2^{x+1}$

w. $\sum_{k=-1}^{7} (k^2+2k) + \sum_{k=-1}^{7} (2k+4)$

x. $\sum_{k=2}^{8} (k^2-3k^2) + \sum_{k=2}^{8} (3k-1)$

3. Evaluate the indicated sums below.

a. $\sum_{k=1}^{21} (-1)^k a^2$ b. $\sum_{k=-2}^{3} (-1)^{k-1} k^2$

c. $\sum_{p=0}^{7} (-1)^p (2p-1)$ d. $\sum_{k=1}^{30} (-1)^n \sin\frac{n\pi}{2}$

e. $\sum_{k=180}^{359} \cos k°$ f. $\sum_{k=87}^{267} \sin(k+3)°$

4. Evaluate the indicated sums below.

a. $\sum_{m=1}^{7}\sum_{n=0}^{5} (2m+\frac{n}{3})$ b. $\sum_{k=1}^{5}\sum_{n=0}^{4} (n^2+kn-3)$

c. $\sum_{k=0}^{4}\sum_{m=1}^{k} (2^m-2)$ d. $\sum_{a=-1}^{1}\sum_{b=2}^{5} (3a-ab)$

e. $\sum_{k=1}^{3}\sum_{n=1}^{k}\sum_{m=1}^{n} \frac{t}{2}$ f. $\sum_{p=1}^{2}\sum_{r=1}^{k}\sum_{t=1}^{r} p$

5. Prove that $\sum_{k=0}^{n} \frac{1}{\sin 2^n a} = \cot \frac{a}{2} - \cot 2^n a$.

6. Prove the following expressions.

a. $\sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}$ b. $\sum_{k=0}^{n} (k+1)\binom{n}{k} = (n+2)2^{n-1}$

c. $\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$ d. $\sum_{k=0}^{n} \binom{r+k}{k} = \binom{r+n+1}{n}$

7. Evaluate the indicated sums below.

a. $\sum_{k=0}^{6} \binom{7}{k}$ b. $\sum_{k=0}^{60} \binom{5+k}{k}$

c. $\sum_{k=0}^{n} x^k \binom{n}{k}$ d. $\sum_{k=5}^{10} 2^k \binom{10}{k}$

e. $\sum_{k=0}^{10} \binom{10}{k}^2$ f. $\log_2 (\sum_{k=1}^{64} k\binom{64}{k})$

8. If $\sum_{k=1}^{6} (2a_k + 3) = 58$, find the sum $\sum_{k=1}^{6} 3a_k$.

9. If $\sum_{k=0}^{n} (2k+3) = 63$, find n.

10. If $\sum_{k=1}^{n} \log 3^{(k+1)} = 9 \cdot \log 27$, find n.

11. If $1 + \sum_{k=1}^{20} 2^k = 2^{4a+1}$, find a.

12. If $\sum_{k=2}^{n} \binom{9}{k-1} = 2^n - 2$, find n.

13. If $f(x) \sum_{k=0}^{4n} x^k$, find $f(i)$ where $i^2 = -1$.

14. Let $f: \mathbb{N}^+ \to \mathbb{N}^+$, $f(x) = \sum_{k=1}^{x} 2k+1$. Find the value of $f^{-1}(8)$

15. If $f(x) = x + 1$, $x_1 = 1$, $x_2 = 2$ and $\sum_{i=1}^{3} x_i^{\,2} f(x_i) = 50$, find $f(i)$ where $i^2 = -1$.

16. Let x_1 and x_2 are the roots of the quadratic equation $x^2 + 2x - 3 = 0$. If $f: \mathbb{N}^+ \to \mathbb{N}^+$, $f(x) = 2x + 3$ then evaluate $\sum_{i=1}^{2} x_1 f(x_i)$.

17. If $\sum_{k=1}^{n} 2a_k = 2^{2n-1}$, find $a_4 + a_5$.

18. If $a_1 = 1$ and $a_n = a_{n-1} + 2$, find the sum $\sum_{k=1}^{30} a_k$.

19. If $\sum_{k=1}^{6} (2a_k - 3) = 2\sum_{k=3}^{7} a_{k-2}$, find a_6.

20. If $\sum_{k=1}^{n} (6k+2) = \frac{an^2 + bn + c}{2}$ find the value of $a + b + c$.

21. If $\sum_{k=1}^{2}(kx+3y)=12$ and $\sum_{p=1}^{3}(px-y)=3$, find the value of $x-2y$.

22. Let x_1 and x_2 be the roots of the quadratic equation $x^2 + (2a + b)x + a - b = 0$. Find the value of a $(a \cdot b)$, if $\sum_{k=1}^{2}2x_k=2$ and $\sum_{k=1}^{2}{x_k}^2=5$

23. If $A\sum_{k=1}^{15}(4k^2+2k)$ find the sum $2 \cdot 5 + 4 \cdot 7 + \ldots + 30 \cdot 33$. in terms of A.

24. If $\sum_{k=1}^{n}\frac{1}{k}=24$ and $\sum_{k=n+1}^{2n}\frac{1}{k}=B$, then find $\sum_{k=1}^{n}\frac{1}{2k-1}$ in terms of A and B.

25. If $f(x) = x - 3$ and $g(x) = x^2$, then find the sum $\sum_{x=1}^{5}(gof)(x)$.

26. If $f(x) = 2x + 3$ and $g(x) = 3x - 2$ find the sum $\sum_{x=1}^{5}(fog)^{-1}(x)$.

27. If $x_1 = 2$, $x_2 = 4$ and $f(x) = 2^x$, $g(y) = 2^{\log_4 y}$ then find the sum $\sum_{i=1}^{2}(gof)(x_i)$.

28. Let $f : \mathbb{N}^+ \to \mathbb{R}$, $f(x) = \log(1-\frac{1}{x})$ and $g : N^+ \to \mathbb{R}$, $g(x) = \log(1+\frac{1}{x})$ then find the sum $\sum_{k=3}^{50}[f(k)+g(k-1)]$.

29. If $\sum_{k=1}^{3}\sum_{m=-x}^{x}(m^3+k)=30$, find x.

30. If $\sum_{k=a}^{a+3}\sum_{k=3}^{a+3}(2k-4)=168$, find a.

31. If $x_1 = 2$, $x_n = x_{n-1}^2 - 1$ and $f(x) = x + 1$ then find the sum $\sum_{j=1}^{3}\sum_{i=1}^{3}(x_j-x_i)f(x_i)$.

32. If $f(x)\sum_{k=1}^{x}\sum_{m=1}^{k}(g(m)-m^2)$ and $g(x)=\sum_{p=1}^{x+1}\sum_{t=-1}^{p}2$, find f(2).

B. Multiplication Notation

33. Evaluate the given expressions below.

a. $\prod_{k=1}^{4} 5$ b. $\prod_{k=0}^{5} (-1)$ c. $\prod_{t=3}^{7} (\frac{1}{3})$

d. $\prod_{k=-2}^{4} (\frac{3}{4})$ e. $\prod_{r=-1}^{3} 2$ f. $\prod_{k=0}^{4} (\frac{-3\pi}{2})$

g. $\prod_{p=2}^{7} e$ h. $\prod_{k=1}^{5} (n+2)$ i. $\prod_{n=3}^{7} (2k-1)$

34. Evaluate the given expressions below.

a. $\prod_{k=1}^{41} k$ b. $\prod_{r=3}^{7} (3r-6)$

c. $\prod_{t=0}^{7} 2^t$ d. $\prod_{k=-2}^{6} 3^{2t+1}$

e. $\prod_{k=1}^{5} k\cdot 2^k$ f. $\prod_{k=1}^{14} \frac{k}{3^k}$

g. $\prod_{m=-1}^{3} \log_a 2^{m+2}$ h. $\prod_{k=2}^{28} \log_k (k-1)$

i. $\prod_{k=1}^{50} k^3-9k$ j. $\sum_{p=3}^{54} 2^{\log(1+\frac{1}{p-1})}$

k. $\prod_{r=3}^{21} (1+\frac{1}{r+2})$ l. $\log \prod_{k=1}^{50} 3^{k\cdot k!}$

m. $\prod_{k=2}^{6} k\cdot 2^{(\frac{k+2}{k})}$ n. $\log_2 \prod_{k=1}^{5} 4^{k^2}$

o. $\prod_{k=2}^{6} k\cdot 2^{(\frac{k+2}{k})}$ p. $\prod_{k=1}^{250} (k^2-4k-12)$

q. $\prod_{k=1}^{75} \frac{k^2}{k+1}$ r. $\prod_{k=2}^{10} (1-\frac{4k-4}{k^2})$

s. $\prod_{k=-8}^{7} 3^{k^3+2k}$ t. $\prod_{k=1}^{89} \tan k$

u. $\prod_{x=1}^{3} (x+1)\cdot \prod_{x=0}^{3} (x^2-x+1)$

x. $\log \prod_{a=1}^{20} (1+\frac{1}{a}) + \log \prod_{a=1}^{20} (1-\frac{1}{a})$

w. $\prod_{k=0}^{50} (k^2+3k+2) : \prod_{k=1}^{50} (k^2+2k)$

35. Evaluate the given expressions below.

a. $\prod_{k=-3}^{3} \prod_{n=1}^{k} (-1)$ b. $\prod_{r=2}^{4} \prod_{n=1}^{5} (nr)$

c. $\prod_{k=1}^{180} \prod_{k=0}^{180} (2)^{\sin k\cdot \cos k}$ d. $\prod_{j=1}^{4} \prod_{i=1}^{3} \frac{i^2}{j}$

36. If $\prod_{k=1}^{x} 3^k = 81^{x+1}$, then find the value of x.

37. If $\prod_{p=1}^{2x+1} 2^{\log_4 (1-\frac{1}{p+1})} = \frac{\sqrt{3}}{6}$, then find the value of x.

38. If $\prod_{k=2}^{x} \log_k (k+1) = \log_2 (2x-5)$, find x.

39. If $\prod_{k=1}^{n} 2^{(k^3-1)} = 128$, find n.

40. Let $\prod_{k=1}^{n} (k^3-7k^2+12k)$. Find the smallest value of n which makes A zero.

41. Evaluate $\prod_{k=1}^{6} (f^{-1}(k)+1)$, if $f(x) = 3x + 2$.

42. If $f(x) = x$, then find $\prod_{k=1}^{100} [f(i)]^k$ where $i^2 = -1$.

43. If $x_1 = 1$, $x_2 = 2$, $x_3 = 3$ and $f(x) = 2x - 1$ then evaluate $\prod_{i=1}^{3} \frac{x_i}{[f(x_i)]^2}$.

44. If $f(x) = \prod_{k=1}^{x} 2^k$ and $g(x) = \prod_{k=1}^{x+1} k$, find $(g \circ f)(3)$.

45. Evaluate the given expressions below.

a. $\sum_{r=1}^{5} \prod_{k=1}^{3} 2$

b. $\prod_{p=1}^{10} \sum_{k=1}^{7} \frac{k}{2}$

c. $\prod_{k=1}^{5} \sum_{r=1}^{10} \frac{r}{k}$

d. $\sum_{k=1}^{7} \prod_{k=1}^{5} (1 + \frac{1}{k})$

e. $\prod_{p=1}^{7} \sum_{k=1}^{10} (-1)^k (-2)^k$

f. $\prod_{k=1}^{2x} 2^{\sum_{k=3}^{5} (2k-4)}$

46. If $\prod_{k=1}^{3} 3^k = 9 \sum_{k=1}^{8} 2k + n$, then find n.

47. If $\prod_{k=1}^{n} x_k = 2n + 1$, then find $\prod_{k=1}^{4} {x_k}^2$

48. If $f(x) = \sum_{k=1}^{x} \frac{1}{\sqrt{k+1} + \sqrt{k}}$ and $g(x) = \prod_{p=1}^{f(x)} (1 + \frac{1}{p})$ find $g(63)$.

49. If $a = \sum_{k=2}^{7} (k-2)(k-3)(k-4)$ then find $\prod_{k=a}^{2a} \log_k (k+1)$.

50. If $f(x) = \sum_{k=1}^{2x} \prod_{k=1}^{x} 2^{k-1}$ and $g(x) = \prod_{k=1}^{x} \sum_{k=1}^{2x} (-1)^k k$, find $g \circ f\,(3)$.

51. If $f(x) = \prod_{k=1}^{x} 2^{k-1}$ and $x_1 = 2$, $x_2 = 4$, find the sum $\sum_{i=1}^{2} x_i f(x_i)$.

52. Let x_1 and x_2 be the roots of the quadratic equation $x^2 + (a + b)x + b - 2a = 0$. If $\sum_{i=1}^{2} x_i = 5$ and $\prod_{i=1}^{2} x_i = -6$ then find the value of the pair (a, b).

53. Evaluate $\sum_{k=1}^{n} \prod_{r=1}^{k+1} \sqrt[k+1]{g(k)} \cdot f(r)$, if $f : \mathbb{N}^+ \to \mathbb{R}$, $f(x) = \frac{1}{x}$ and $g : \mathbb{N}^+ \to \mathbb{N}^+$, $g(x) = x$.

54. Let x_1 and x_2 be the roots of the equation $x^2 - 5x - n = 0$. If $\sum_{k=1}^{2} \prod_{i=1}^{2} ((x_i^{k+1} + x_i^k)) = 0$ find the sum of the values which n can take.

CHAPTER REVIEW TEST 3A

1. If the sum of the first $3n$ positive integers is 150 more than the sum of the first n positive integers, then what is the sum of the first 4n positive integers.

A) 270 B) 300 C) 330 D) 360 E) 390

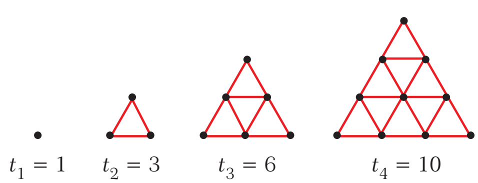
2. The sum of the cubes of the numbers from 1 to n is $K = 13 + 23 + 33 + \ldots + n^3$. How much will K increase if each of the numbers 1, 2, 3, ..., n is increased by 1?

A) n B) n^3 C) $n(n^2 + 3n)$ D) $n(n^2 + 3n + 3)$ E) $(\frac{n(n+1)}{2})^2$

3. What is the value of the sum $m^2 + (m + 1)^2 + (m + 2)^2 + \ldots + (2m)^2$ where m is a natural number?

A) $\frac{m(m+1)(14m+1)}{6}$ B) $\frac{m(m+1)(18m+1)}{6}$ C) $\frac{m(m+1)(2m+1)}{12}$ D) $\frac{2m(m+1)(12m+1)}{3}$ E) $\frac{m(m+1)(14m+3)}{3}$

4. Triangular numbers can be represented by dots that are arranged in the shape of an equilateral triangle, as shown below. The first four triangular numbers are given.



What is the 15'the triangular number?

A) 3 B) 5 C) –5 D) 10 E) 2

5. If $\sum_{k=1}^{n} k = M$ then what is the value of the sum $2n + (2n + 1) + (2n + 2) + \ldots + 3n$ in terms of M?

A) $M^2 + M$ B) $M + 1$ C) $3M$ D) $4M$ E) $5M$

6. If $\frac{1^3+3^3+5^3+\ldots+(2n-1)^3}{2^3+4^3+6^3+\ldots+(2n)^3} = \frac{199}{242}$ then what is the value of n?

A) 9 B) 10 C) 11 D) 12 E) 13

7. If $\sum_{k=15}^{14} k = \sum_{k=15}^{x} k$ then the value of x is

A) 28 B) 25 C) 23 D) 20 E) 19

8. What is the value of the sum $\sum_{k=1}^{32} k\binom{32}{k}$ where $\binom{n}{k} = \frac{n!}{(n-k)!k!}$?

A) 2^{33} B) 2^{34} C) 2^{35} D) 2^{36} E) 2^{37}

9. The consecutive odd integers are grouped as follows 1; (3, 5); (7, 9, 11); (13, 15, 17, 19); ... What is the sum of the numbers in the 10'th group?

A) 760 B) 897 C) 981 D) 1000 E) 1021

10. What is the value of the sum $\sum_{k=4}^{120} \frac{1}{k\sqrt{k+1}+(k+1)\sqrt{k}}$?

A) $\frac{7}{9}$ B) $\frac{9}{11}$ C) $\frac{11}{13}$ D) $\frac{9}{22}$ E) $\frac{11}{9}$

11. What is the value of the sum

$$\sum_{n=0}^{10}\binom{n+7}{n} \text{ where } \binom{n}{r}+\binom{n}{r+1}=\binom{n+1}{r+1}=?$$

A) $\binom{17}{6}$ B) $\binom{17}{8}$ C) $\binom{18}{7}$ D) $\binom{18}{8}$ E) $\binom{18}{9}$

12. Consider the triangular array formed by rotating the product in a standard multiplication table, that is,

				1				
			2		2			
		3		4		3		
	4		6		6		4	
5		8		9		8		5

....

What is the sum of the elements in the 18'th row?

A) 1020 B) 1060 C) 1100 D) 1140 E) 1180

13. If $A=\sum_{n=1}^{20}\frac{(-1)^{n+1}}{n}$ and $B=\sum_{n=11}^{20}\frac{1}{n}$ then the value of $A-B$ is

A) 0 B) $\frac{26}{41}$ C) $\frac{111}{101}$ D) $\frac{126}{107}$ E) $\frac{143}{132}$

14. What is the value of the product

$\sqrt{a}\sqrt[4]{a}\sqrt[8]{a}...\sqrt[1024]{a}$?

A) $a^{\frac{511}{512}}$ B) $a^{\frac{1023}{1024}}$ C) $a^{\frac{2047}{2048}}$ D) $a^{\frac{1024}{1023}}$ E) $a^{\frac{2047}{2047}}$

15. What is the value of the product

$(1-\frac{4}{4})(1-\frac{4}{9})(1-\frac{4}{25})...(1-\frac{4}{625})$?

A) $-\frac{17}{15}$ B) $-\frac{23}{25}$ C) $-\frac{27}{25}$ D) $-\frac{31}{29}$ E) $-\frac{37}{25}$

16. What is the value of the sum $\sum_{n=1}^{15}(2^n+\frac{1}{2^n})^2$?

A) $30+\frac{(2^{15}-1)(2^{16}+1)}{3\cdot 2^{15}}$

B) $30+\frac{(4^{15}-1)(4^{16}+1)}{3\cdot 4^{15}}$

C) $30+\frac{(4^{15}+1)(4^{16}-1)}{3\cdot 4^{15}}$

D) $30+\frac{(2^{15}+1)(2^{16}-1)}{3\cdot 2^{15}}$

E) $30+\frac{(4^{15}-1)(4^{16}-1)}{3\cdot 4^{15}}$

17. If $\sum_{n=1}^{x-1}\frac{x-n}{x}=3$ then the value of x is

A) 3 B) 4 C) 5 D) 6 E) 7

18. What is the value of the sum $\sum_{n=1}^{27}(2n+1)\cdot 3^n$?

A) 3^{28} B) 3^{29} C) 3^{30} D) 3^{31} E) 3^{32}

19. What is the value of the sum

26 + 37 + 50 + 65 + ... + 401?

A) 2796 B) 2818 C) 2848 D) 2856 E) 2872

20. Define $(n_a)!$ for n and a positive to be

$(n_a)!=\prod_{k=0}^{m}(n-ka)$ where m is the greatest integer for which $n > ma$. Then the quotient $\frac{(72_8)!}{(18_2)!}$ is equal to

A) 2^{10} B) 2^{12} C) 2^{16} D) 2^{18} E) 2^{20}

21. If $i = \sqrt{-1}$, then the sum $\sum_{n=0}^{40} i^n \cos(45+90n)°$ equals

A) $\frac{\sqrt{2}}{2}$ B) $-10i\sqrt{2}$ C) $\frac{21\sqrt{2}}{2}$ D) $\frac{\sqrt{2}}{2}(21-20i)$ E) $\frac{\sqrt{2}}{2}(21+20i)$

22. What is the value of the sum

1 + (1 + 2) + (1 + 2 + 3) + ... + (1 + 2 + 3 + ... + 20)?

A) 1750 B) 1540 C) 1190 D) 970 E) 875

23. If $x_{k+1} = x_k + \frac{1}{2}$ for k = 1, 2, ..., n – 1 and x_1 = 1 the value of the sum $\sum_{k=1}^{n} x_k$ equals

A) $\frac{n+1}{2}$ B) $\frac{n+3}{2}$ C) $\frac{n^2-1}{2}$ D) $\frac{n^2+n}{4}$ E) $\frac{n^2+3n}{4}$

24. The smallest value of n such that the procut $\prod_{k=1}^{n} 10^{\frac{k}{11}}$ exceeds 100,000 is

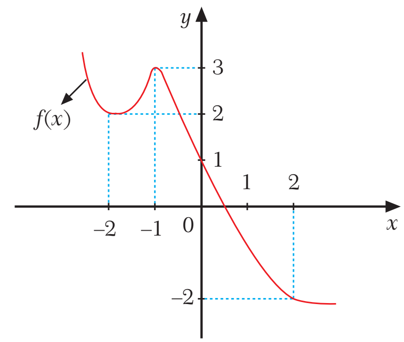
A) 7 B) 8 C) 9 D) 10 E) 11

25. The roots of the equation $x^3+4ax^2+bx+2-3a=0$ are x_1, x_2 and x_3. If $\sum_{n=1}^{3} x_n = \prod_{n=1}^{3}(2x_n)$ then the value of a is

A) $-\frac{4}{5}$ B) $\frac{2}{5}$ C) $\frac{4}{7}$ D) $\frac{4}{5}$ E) $\frac{5}{4}$

26. The graph of the function $f(x)$ is given below.

What is the value of the sum $\sum_{x=-2}^{2} x \cdot f(x)$?



A) –11 B) –10 C) –8 D) –6 E) –4

27. What is the value of the sum

$\sum_{n=1}^{24} \frac{1}{\sqrt{2n-1}+\sqrt{2n+1}}$?

A) 2 B) 3 C) 4 D) 5 E) 6

28. What is the value of the sum

$\sum_{n=1}^{45} \log(2^{2n-1}\tan(2n-1))$?

A) 1075 log 2 B) 1375 log 2 C) 1825 log 2 D) 2025 log 2 E) 2175 log 2

29. If x_1 = –3, x_2 = 1 and $f(x) = x^2 - 4$ then what is the value of the sum $\sum_{n=1}^{2}(x_n+4)\cdot f(x_n)$?

A) –15 B) –10 C) –5 D) 0 E) 5

30. If $\sum_{k=3n}^{5n} k^3 = an^4 + bn^3 + cn^2$ then what is the value of $a + b + c$?

A) 189 B) 200 C) 216 D) 225 E) 256

1. $a = 1 + 2 + 3 + \ldots + 23$

$b = 12 + 22 + 32 + \ldots + 232$

$c = 13 + 23 + 33 + \ldots + 233$ is given.

what is the value of the sum

$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \ldots + 23 \cdot 24 \cdot 25$

in terms of a, b, and c?

A) $a + 3b + 2c$ B) $a + 2b + 3c$

C) $3a + 2b + c$ D) $2a + 3b + c$

E) $3a + b + 2c$

2. What is the result of the product $\prod_{k=2}^{29} \log(\tan 3k°)$?

A) 0 B) 1 C) log (tan 87°)

D) $\log(\tan \dfrac{29!}{2 \cdot 3^{27}})$ E) $\dfrac{1}{\log(\tan 3°)}$

3. $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \ldots + n(n+1) = \dfrac{n(n+1)(n+2)}{3}$

$(\forall n \in N^+)$ is given.

Which one of the following is equal to the sum $(m + 1)(m + 2) + (m + 3) + \ldots (2m - 1)(2m)$?

A) $\dfrac{(m-1)m(7m+4)}{3}$ B) $\dfrac{m(m+1)(7m+4)}{3}$

C) $\dfrac{(m-1)(m+1)(2m+3)}{3}$ D) $\dfrac{(m-1)m(m+1)}{3}$

E) $\dfrac{m(m+1)(m+2)}{3}$

4. Which one of the following express the sum

$2 + 7 + 12 + \ldots + 102$?

A) $\sum_{n=1}^{21}(5n+2)$ B) $\sum_{n=1}^{21}(5n-3)$ C) $\sum_{n=1}^{102}(2n)$

D) $\sum_{n=1}^{102}(5n-3)$ E) $\sum_{n=1}^{21}(3n-1)$

5. Which one of the following express the sum

$1 \cdot 3 + 3 \cdot 7 + 5 \cdot 11 + \ldots + 21 \cdot 43$?

A) $\sum_{n=1}^{11}(8n^2 - 6n + 1)$ B) $\sum_{n=1}^{21}(4n^2 - 6n + 5)$

C) $\sum_{n=1}^{43}(3n^2 - 2n + 2)$ D) $\sum_{n=1}^{11}(8n^2 - 7n + 2)$

E) $\sum_{n=1}^{11}(2n^2 + n)$

6. $A = \sum_{k=1}^{2} k + \sum_{k=1}^{3} k + \sum_{k=1}^{4} k + \ldots + \sum_{k=1}^{11} k$

$B = \sum_{k=1}^{11} 2 + \sum_{k=1}^{11} 3 + \sum_{k=1}^{11} 4 + \ldots + \sum_{k=1}^{11} 11$ is given.

The value of $B - A$ is

A) 390 B) 410 C) 420 D) 430 E) 450

7. Which one of the following is equal to $\prod_{k=1}^{n}(4k^3)$?

A) $4^n \cdot (n^3)!$ B) $((4n)!)^3$ C) $64(n!)^3$

D) $2^{2n}(n!)^3$ E) $4 \cdot (n!)^3$

8. Which one of the following is equal $\sum_{k=2n}^{5n}(4k-1)$?

A) $42n^2 + 8n - 1$ B) $12n^2 - 11n - 1$

C) $42n^2 - 3n$ D) $42n^2 - 3n - 1$

E) $42n^2 + 11n - 1$

9. $f : \mathbb{N}^+ \Rightarrow \mathbb{N}^+$, $g : \mathbb{N}^+ \Rightarrow \mathbb{N}^+$

$f(x) = \prod_{k=1}^{x} 3^k$ and $g(x) = \sum_{k=1}^{x} k!$ are given.

What is the value of fog (4)

A) 3^{501} B) 3^{511} C) 3^{526} D) 3^{541} E) 3^{561}

10. What is the value of the sum $\sum_{n=2}^{12}\binom{n}{2}$?

A) 188 B) 202 C) 249 D) 286 E) 304

11. If $\frac{2\cdot4\cdot6\cdot\ldots\cdot2n}{1\cdot3\cdot5\cdot\ldots\cdot(2n-1)}\cdot\binom{2n}{n}=1024$

then the value of n is

A) 5 B) 7 C) 8 D) 9 E) 10

12. If $\sum_{k=1}^{13}(x+k)^2=ax^2+bx+c$, then which one of the following is equal to $4a - 2b + c$?

A) 476 B) 493 C) 507 D) 521 E) 546

13. What is the value of the product $\prod_{k=5}^{25}(1+\frac{7}{k^2-16})$?

A) $\frac{156}{24}$ B) $\frac{163}{24}$ C) $\frac{168}{25}$ D) $\frac{176}{29}$ E) $\frac{191}{29}$

14. If $P(x)=\prod_{k=2}^{14}(x-k)$ then what is the sum of the coefficients of the terms of $P(x)$ with even powers?

A) $53\cdot14!$ B) $-53\cdot13!$ C) $-56\cdot13!$

D) $56\cdot13!$ E) $-56\cdot14!$

15. What is the result of sum $\sum_{n=4}^{48}\frac{(-1)^n}{\sqrt{n+1}-\sqrt{n}}$?

A) 7 B) 8 C) 9 D) 10 E) 11

16. What is the value of the sum $\sum_{k=0}^{4n}((k+1)i^k)$ where $i = \sqrt{-1}$?

A) $2 + 2ni$ B) $2n + 1$ C) $n + 2i$

D) $2n + 1 - 2ni$ E) $2n^2 + 1 - 2ni$

17. If $f : \mathbb{N}^+ \to \mathbb{N}^+$, $\sum_{k=1}^{x}k$ and $g : \mathbb{N}^+ \to \mathbb{R}$

$g(x)=\sum_{k=2}^{x}\frac{f(k)}{f(k)-f(k-1)}$ then what is the value of $g(17)$?

A) $\frac{18!}{3\cdot2^{17}}$ B) $\frac{17!}{2^{18}}$ C) $\frac{17!}{2^{17}}$ D) $\frac{18!}{2^{18}}$ E) $\frac{18!}{2^{17}}$

18. What is the value of the product $\prod_{k=0}^{9}(3^{(2^k)}+1)$?

A) $3^{1024} - 1$ B) $3^{1024} + 1$ C) $\frac{3^{1024}-1}{2}$

D) $\frac{3^{1024}+1}{2}$ E) $\frac{3^{1024}+1}{4}$

19. What is the result of the sum

$\sum_{k=-10}^{10}(2k^3+5k^2+4k-1)$?

A) 3136 B) 3442 C) 3558

D) 3726 E) 3829

20. $\sum_{k=-8}^{n+9}(5k+2)=an^2+bn+c$ then what is the numerical value of $a - b + c$?

A) 8 B) 14 C) 21 D) 29 E) 34

21. What is the value of the product $\prod_{k=1}^{p-1}(pk^2-k^3)$?

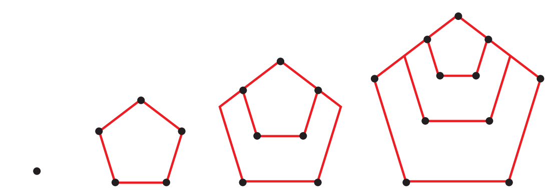
A) $[(p-1)^!]^2$ B) $[(p-1)!]^3$ C) $(p!)^3$

D) $(p!)^2$ E) $(p!)^3-[(p-1)!]^3$

22. What is the value of the sum $\sum_{k=0}^{25}\binom{8+k}{8}$?

A) $\binom{33}{9}$ B) $\binom{34}{8}$ C) $\binom{34}{8}$ D) $\binom{18}{8}$ E) $\binom{35}{9}$

23. Pentagonal numbers can be represented by dots that are arranged in the shape of a pentagon, as shown below. The first four pentagonal numbers are given. What is the tenth pentagonal number?



$t_1 = 1$ $t_2 = 5$ $t_3 = 12$ $t_4 = 22$

A) 128 B) 130 C) 145 D) 164 E) 182

24. What is the value of the product $\prod_{k=1}^{100}\left(1+\frac{(-1)^{k+1}}{k}\right)$?

A) 1 B) 1,1 C) 1,01 D) 1,11 E) 1,101

25. If the sum $1 + 2 + 3 + \ldots + n = M$ then what is the value of the sum

$n + (n + 3) + (n + 6) + (n + 9) + \ldots + 4n$ in terms of M?

A) 3M B) 4M C) 5M D) 6M E) 7M

26. $\sum_{x=1}^{n} f(x) = 3^n$ then what is the value of $f(5)$?

A) 27 B) 54 C) 81 D) 162 E) 243

27. If $m = 1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\ldots+\frac{1}{11}+\frac{1}{12}$ then what is the value of the sum $\sum_{k=0}^{11}\frac{4k^2+6k+5}{2k+2}$ in terms of m?

A) $\frac{176+3m}{2}$ B) $\frac{288+3m}{2}$ C) $\frac{144+3m}{2}$

D) $\frac{192+5m}{2}$ E) $\frac{224+5m}{4}$

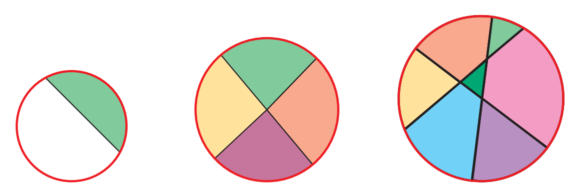
28. What is the value of the sum

$1 + (1 + 2) + (1 + 2 + 2^2) + \ldots + (1 + 2 + 2^2 + \ldots + 2^{n-1})$?

A) 2^n B) $2^n - n$ C) $2^{n+1} - n$

D) $2n + 1 - n - 2$ E) $n \cdot 2^n$

29. If n cords of a circle are drawn, then let t_n be the maximum number of non-overlaping regions that can be formed within the circle illustrated below.



1 chord 2 chords 3 chords

t_2 = 2 regions t_2 = 4 regions t_3 = 7 regions

What is the value of t_{10}?

A) 52 B) 54 C) 56 D) 58 E) 60

30. Which one of the following is equal to product $\prod_{r=1}^{n} P(n, r)$ where $P(n, r) = \frac{n!}{(n-r)!}$?

A) $\prod_{k=1}^{n} k^n$ B) $\prod_{k=1}^{n}(k+1)^k$ C) $\prod_{k=1}^{n} k!$

D) $\prod_{k=1}^{n} n^k$ E) $\prod_{k=1}^{n} k^k$

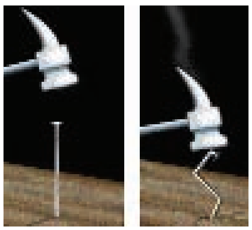
CHAPTER 4

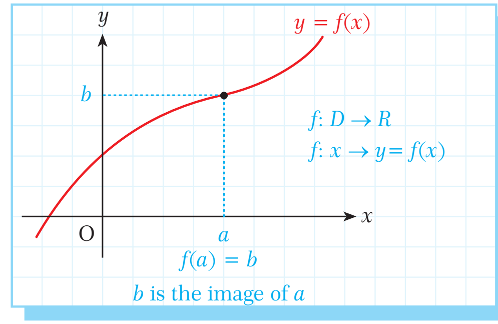
FUNCTIONS

1 TYPES OF FUNCTION

A **function** from a set D to a set R is a rule that maps each element of D to a single element of R.

$$f: \underset{\text{domain}}{\underset{\downarrow}{D}} \to \underset{\text{range}}{\underset{\downarrow}{R}}$$





For each x in D there exists a single element y in R such that $f(x) = y$. x is called the **variable** of f and $y = f(x)$ is called the **image** of x. The set of images of all the elements of D is called the **image set** of f.

Consider the function $f: A \to B$, $f(x) = x^2$. We can write this function in different ways: $f(x) = x^2$, $y = x^2$ or $f: x \to x^2$. All of these mean the same function. A is the domain and B is the range of f.

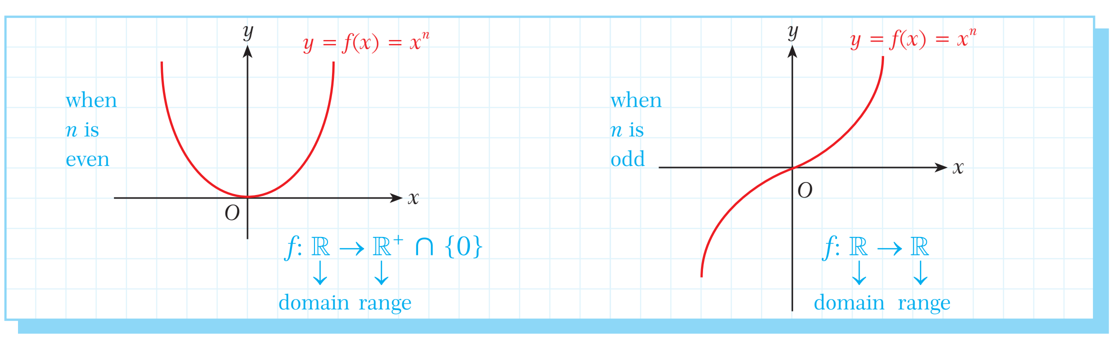
A. DOMAIN AND RANGE OF A FUNCTION

Unless stated otherwise, the **domain** of a function $f(x)$ is the largest set of real x-values for which $f(x)$ is defined. The **range** of the function is a set which includes at least all the images of the elements in its domain. The domain and range of many functions are subsets of the set of real numbers. The largest possible range of a function is $\mathbb{R}$.

Let us look at the domain and range of some common types of function. In these examples we will use the letter D to mean the domain of the function. The notation $D(f)$ means the domain of the function f.

Type of function	Form	Domain	Examples
polynomial function	$f(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_0$ $n \in \mathbb{Z}^+ \cup \{0\}$	$\mathbb{R}$	$f(x) = 2x+5$ $D(f)$: $\mathbb{R}$ $f(x) = 2x^2-3x+1$ $D(f)$: $\mathbb{R}$

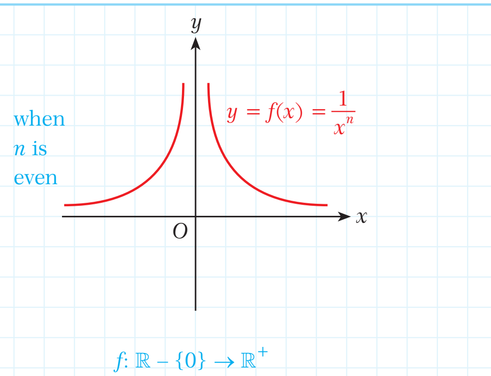
As stated in the table, the domain of any polynomial function f is $\mathbb{R}$. The range of the polynomial function depends on the function itself. For example, let us draw the graph of the function $y = f(x) = x^n$ and find its domain and range when n is odd or even.

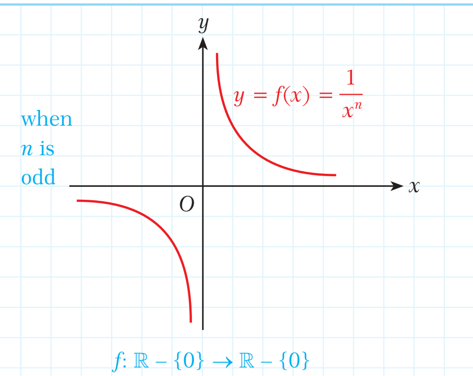


Type of function	Form	Domain	Examples
rational function	$f(x) = \dfrac{g(x)}{h(x)}$	$\mathbb{R} - \{x \mid h(x) = 0\}$	$f(x) = \dfrac{2x-3}{x+1}$ $D(f)$: $\mathbb{R} - \{-1\}$ $f(x) = \dfrac{x^2+5}{(x-5)(x+2)}$ $D(f)$: $\mathbb{R} - \{5, -2\}$

The value of the denominator in a rational expression cannot be zero, so any numbers which make the denominator zero must be excluded from the domain of a rational function.

As an example, let us look at the graph of the function $y = x^{-n} = \dfrac{1}{x^n}$.



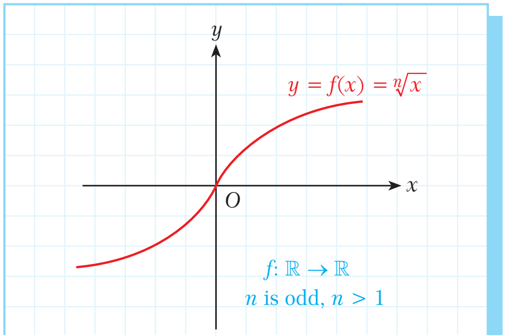


We can see that the domain of a rational function changes according to the function.

Type of function	Form	Domain	Example
radical function	$f(x) = \sqrt[n]{g(x)}$ n is an odd integer	$\mathbb{R}$	$f(x) = \sqrt[3]{x^2 + 5x}$ $D(f)$: $\mathbb{R}$

When the index of a radical expression is odd, the radicand can be negative, positive or zero. Therefore there is no restriction on the value of x and so the domain is $\mathbb{R}$.

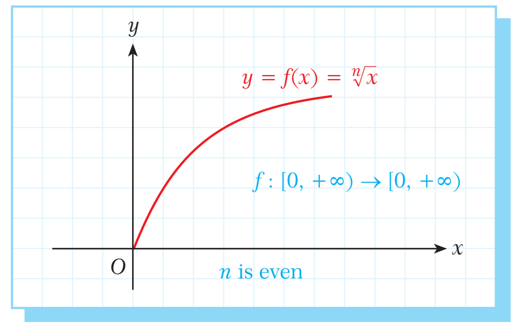
The figure at the right shows the graph of $y = \sqrt[n]{x}$ when n is odd and $n > 1$. We can see that the range of the function is the set of real numbers.



Type of function	Form	Domain	Examples
radical function	$f(x) = \sqrt[n]{g(x)}$ n is an even integer	$\mathbb{R} - \{x \mid g(x) < 0\}$	$f(x) = \sqrt{x^2 - 2}$ $\quad D(f)$: $\mathbb{R} - (-\sqrt{2}, \sqrt{2})$ $f(x) = \sqrt[4]{x^2 - 3x + 2}$ $\quad D(f)$: $\mathbb{R} - (1, 2)$

When the index of a radical expression is even, the radicand cannot be negative, so we must exclude any numbers which make the radicand negative from the domain.

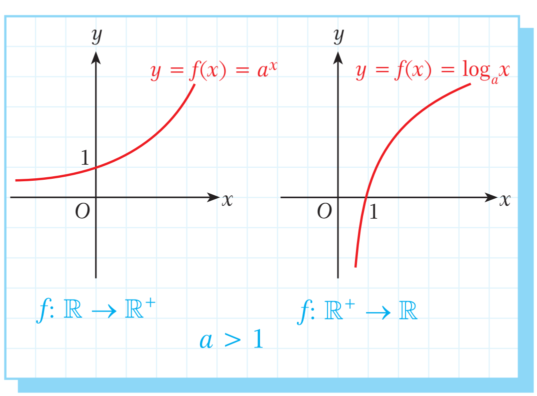
As we can see in the graph opposite, since the radicand is non-negative both the domain and range of $y = \sqrt[n]{x}$ are the set $\mathbb{R}^+ \cup \{0\}$.

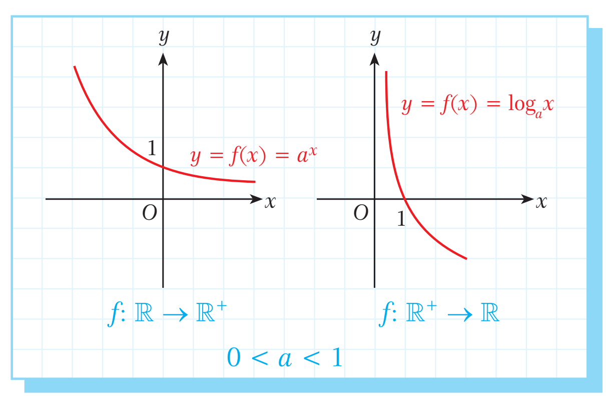


Type of function	Form	Domain	Examples
exponential function	$f(x) = a^x$ $(a \in \mathbb{R}^+ - \{1\})$	$\mathbb{R}$	$f(x) = 3^x$ $\quad D(f)$: $\mathbb{R}$ $f(x) = 5^{x^2+2x}$ $\quad D(f)$: $\mathbb{R}$
logarithmic function	$f(x) = \log_a g(x)$ $(a \in \mathbb{R}^+ - \{1\})$	$\mathbb{R} - \{x \mid g(x) \leq 0\}$	$f(x) = \log(x^2 - 4)$ $\quad D(f)$: $\mathbb{R} - [-2, 2]$ $f(x) = \ln(x^2 - 2x - 3)$ $\quad D(f)$: $\mathbb{R} - [-1, 3]$

An exponential function is defined for all real numbers but the logarithmic function is defined only for positive real numbers. Therefore we must exclude any numbers which make the function negative or zero from the domain of a logarithmic function.

Look at the following graphs of exponential and logarithmic functions:





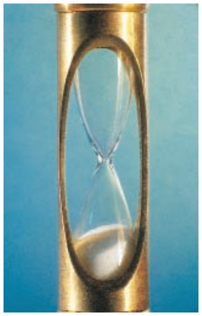
EXAMPLE 1 The following table shows some more examples of the domain and range of different functions. Write the missing values in the table.

Function	Domain	Range
$f(x) = x^2 + 2x + 1$	$\mathbb{R}$	$[0, +\infty)$
$f(x) = x^2 - 2x - 3$	$\mathbb{R}$	
$f(x) = x^3 - x^2 + x + 1$		$\mathbb{R}$
$f(x) = \dfrac{1}{x+1}$		$(-\infty, 0) \cup (0, +\infty)$
$f(x) = \dfrac{x^3 - 1}{x^2 - 5x + 6}$	$\mathbb{R} - \{2, 3\}$	
$f(x) = \sqrt{1 - x^2}$		$[0, 1]$
$f(x) = \log(x^2 + 5x)$	$(-\infty, -5) \cup (0, +\infty)$	

Solution The solution is left as an exercise for you.

EXAMPLE 2 Find the domain and range of the function $f(x) = \sqrt{x^2 + 5x + 6}$.

Solution We know that the radicand of a square root function cannot be negative. Let us look at the sign of the radicand $x^2 + 5x + 6$:



x		-3		-2	
$x^2 + 5x + 6$	$+$	0	$-$	0	$+$

The radicand is non-negative in the intervals $(-\infty, -3]$ and $[-2, +\infty)$.

Therefore the domain of the function is $(-\infty, -3] \cup [-2, +\infty)$.

As x increases, the value of y also increases without limit.

So the range is $[0, +\infty)$.

In conclusion, the domain of f is $(-\infty, -3] \cup [-2, +\infty)$ and the range is $[0, +\infty)$.

Note

If a function f is a sum or difference of different functions then the domain of f is the intersection of the domains of each function.

EXAMPLE 3 Find the domain of the function $f(x)=\dfrac{1}{\sqrt{x}}+\log(x^2+2x-8)$.

Solution We can think of this function as the sum of two functions: $g(x)=\dfrac{1}{\sqrt{x}}$ and $h(x) = \log(x^2 + 2x - 8)$.

$\dfrac{1}{\sqrt{x}}$ is defined when $0 < x < +\infty$.

The logarithmic function is defined in $\mathbb{R}^+$, so let us look at the sign of $x^2 + 2x - 8$:

x		-4		2	
$x^2 + 2x - 8$	$+$	0	$-$	0	$+$

We can see that $\log(x^2 + 2x - 8)$ is defined when $x < -4$ or $x > 2$.

The domain of f is the intersection of the domains of g and h, so $D(f) = (2, +\infty)$.

EXAMPLE 4 Find the domain of $f(x)=\sqrt{x+3}+\dfrac{5}{x^2-4}+\log_2(4x-3)$.

Solution Let us consider the domain of each separate function.

By solving the inequalities $\begin{cases} x+3\geq 0 \\ x^2-4\neq 0 \\ 4x-3>0 \end{cases}$, we get $\begin{cases} x\geq -3 \\ x\neq 2 \text{ and } x\neq -2 \\ x>\dfrac{3}{4}. \end{cases}$

So the domain of f is the intersection of these three intervals: $D(f)=\left(\dfrac{3}{4},\ +\infty\right)-\{2\}$.

EXAMPLE 5 Find the image set of each function over the given interval.

a. $f(x) = 3x + 6$, $x \in [0, +\infty)$

b. $f(x) = x^2 - 2x + 8$, $x \in [-1, 2]$

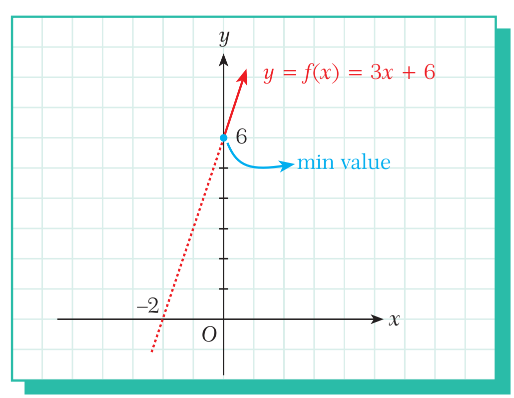
c. $f(x) = x^2 - 4x - 5$, $x \in [-1, 1]$

d. $f(x) = 2x - x^2$, $x \in [0, 3]$

Solution We can find each range by drawing the graph of the function over the given interval.

a. As we can see in the graph, $f(x)$ is an increasing function. The solid line shows the graph on the interval $x \in [0, +\infty)$. On this interval the minimum value of f is $f(0) = 6$, and the maximum value goes to infinity.

So the image set of f on this interval is $[6, +\infty)$.



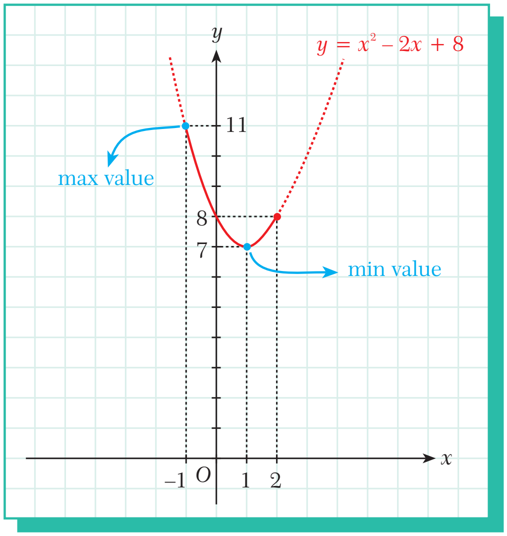
The image set I of a function $f: A \to B$ is the set of images of all the elements of A, so $I \subseteq B$.

b. The figure shows the graph of the function $y = f(x) = x^2 - 2x + 8$. The solid line shows the graph on the interval $x \in [-1, 2]$.

Since the extremum (vertex) of a function $f(x) = ax^2 + bx + c$ is $(-\frac{b}{2a}, f(-\frac{b}{2a}))$, the minimum value of the function on this interval is $f(-\frac{b}{2a}) = f(-\frac{-2}{2}) = f(1) = 7$.

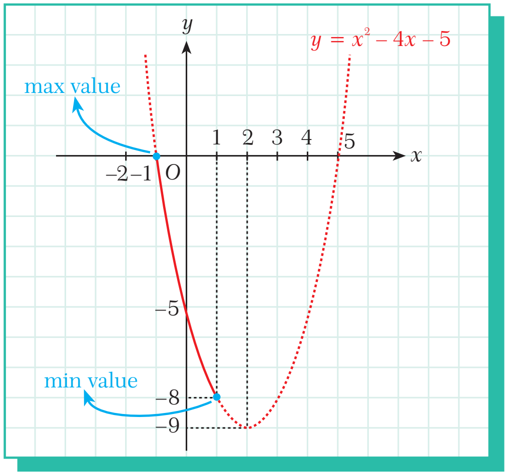
Its maximum value on this interval is $f(-1) = 11$.

So the image set of f on this interval is $[7, 11]$.



c. The figure shows the graph of $y = f(x) = x^2 - 4x - 5$. The solid line shows the graph on the interval $x \in [-1, 1]$.

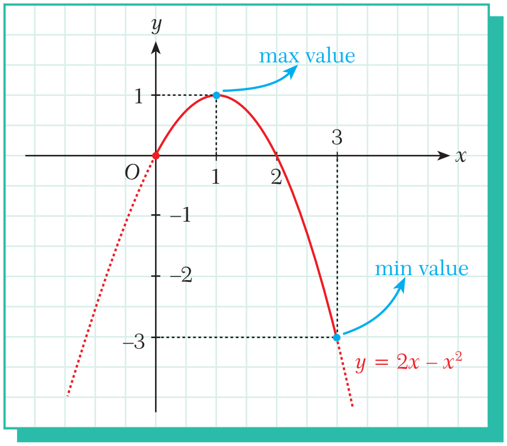
On this interval,
$\min f(x) = f(1) = -8$ and
$\max f(x) = f(-1) = 0$.
So the image set of f on this interval is $[-8, 0]$.



d. The solid line in the figure shows the graph of $y = f(x) = 2x - x^2$ on the interval $x \in [0, 3]$.

On this interval,
$\min f(x) = f(3) = -3$ and
$\max f(x) = f(1) = 1$.

So the image set of the function on this interval is $[-3, 1]$.



EXAMPLE 6 Find the range of each function for its largest domain.

a. $f(x) = 2\sin x - 3$ b. $f(x) = \sqrt{-x^2 + 4x + 5}$ c. $f(x) = \sqrt{5^x + 7}$ d. $f(x) = \dfrac{2}{3^x + 1}$

Solution a. The trigonometric function $f(x) = \sin x$ is defined from $\mathbb{R}$ to $[-1, 1]$.

For all $x \in \mathbb{R}$, $-1 \le \sin x \le 1$

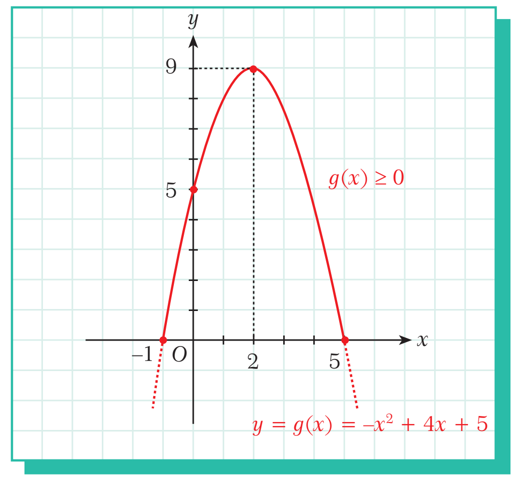
$-2 \le 2\sin x \le 2$

$-2 - 3 \le 2\sin x - 3 \le 2 - 3$

$-5 \le 2\sin x - 3 \le -1.$

Hence the range of $f(x) = 2\sin x - 3$ is $[-5, -1]$.

b. Let us define $g(x) = -x^2 + 4x + 5$ and plot its graph (shown opposite).



As we can see, the function is positive over the interval $x \in [-1, 5]$.

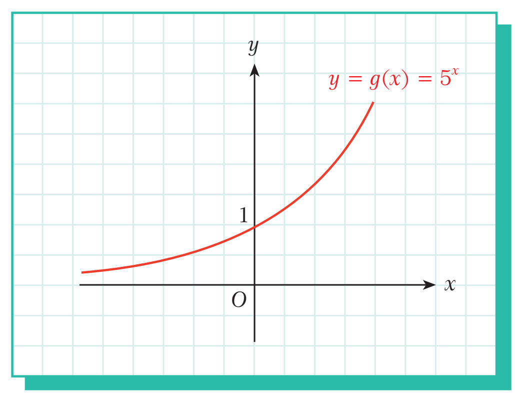
So $f(x) = \sqrt{-x^2 + 4x + 5}$ is defined on the interval $x \in [-1, 5]$. Also, on this interval the minimum value of $f(x)$ is 0 and the maximum value of $f(x)$ is $\sqrt{9}$, so we can write

$0 \le \sqrt{-x^2 + 4x + 5} \le \sqrt{9}$

$0 \le \sqrt{-x^2 + 4x + 5} \le 3$

$0 \le f(x) \le 3$. So the range of f is $[0, 3]$.

c. Since the exponential function $g(x) = 5^x$ is always positive, the domain of $f(x) = \sqrt{5^x + 7}$ is $\mathbb{R}$. As we can see in the graph opposite, the value of 5^x lies between 0 and ∞.



$0 < 5^x < \infty$

$7 < 5^x + 7 < \infty$

$\sqrt{7} < \sqrt{5^x + 7} < \infty$

So the range of f is $(\sqrt{7}, \infty)$.

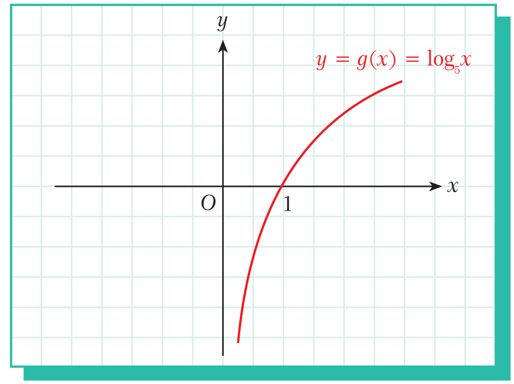
d. Let us begin by considering the range of 3^x: $0 < 3^x < \infty$.

This gives us $1 < 3^x + 1 < +\infty$, and when we take the reciprocal of each side, the inequality signs are reversed.

So we get $\dfrac{1}{1} > \dfrac{1}{3^x + 1} > 0$, i.e. $2 > \dfrac{2}{3^x + 1} > 0$.

So the range of f is $(0, 2)$.

EXAMPLE 7 Find the domain of $f(x) = \log_5(\log_5 x)$.



Solution The logarithmic function is defined for all positive real numbers, so the function $g(x) = \log_5 x$ is defined from $\mathbb{R}^+$ to $\mathbb{R}$. Now we must specify on which interval $g(x)$ is non-positive (i.e. negative or zero).

As we can see in the graph, $g(x) = \log_5 x$ is non-positive for $0 < x \leq 1$, so $(0, 1]$ must be excluded from the domain of f. Therefore, the domain of f is $\mathbb{R}^+ - (0, 1]$.

EXAMPLE 8 Find the domain of $f(x) = \sqrt{|x-1| - |x+2|}$.

Solution The radicand $|x - 1| - |x + 2|$ must be non-negative, i.e.

$|x - 1| - |x + 2| \geq 0$.

This gives $|x - 1| \geq |x + 2|$. Taking the square of both sides gives us $x^2 - 2x + 1 \geq x^2 + 4x + 4$

$-6x \geq 3$

$x \leq -\frac{1}{2}$. So the domain of the function is $(-\infty, -\frac{1}{2}]$.

EXAMPLE 9 Find the domain of $f(x) = \sqrt{3 - \sqrt{12 - x^2}}$.

Solution We have the radicands $12 - x^2$ and $3 - \sqrt{12 - x^2}$, and both of them must be non-negative:

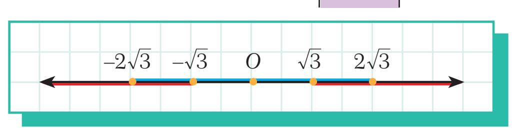
$12 - x^2 \geq 0$ and $3 - \sqrt{12 - x^2} \geq 0$

$$9 \geq 12 - x^2$$
$$x^2 - 3 \geq 0.$$

Let us solve each quadratic inequality by constructing its sign table:

x		$-2\sqrt{3}$		$2\sqrt{3}$	
$12 - x^2$	$-$	●	$+$	●	$-$

x		$-\sqrt{3}$		$\sqrt{3}$	
$x^2 - 3$	$+$	●	$-$	●	$+$



The domain of the combined function $f(x) = \sqrt{3 - \sqrt{12 - x^2}}$ is the intersection of these intervals: $D(f) = [-2\sqrt{3}, -\sqrt{3}] \cup [\sqrt{3}, 2\sqrt{3}]$.

EXAMPLE 10 Find the domain of $f(x)=\sqrt{\log\frac{5x-1}{x+2}}$.

Solution To find the domain we have to solve three inequalities:

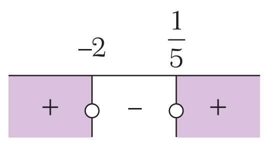
$x + 2 \neq 0,$ (1)

$\frac{5x-1}{x+2}>0$ (2)

and $\log\frac{5x-1}{x+2}\geq 0.$ (3)

(1) gives $x + 2 \neq 0$, i.e. $x \neq -2$.

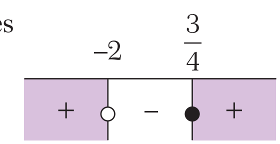
(2) gives $\frac{5x-1}{x+2}>0,$



i.e. $x < -2$ or $x > \frac{1}{5}$.

(3) gives $\log\frac{5x-1}{x+2}\geq 0,$ $\log\frac{5x-1}{x+2}\geq \log 1$ i.e. $\frac{5x-1}{x+2}\geq 1$. This gives

$\frac{5x-1}{x+2}-1\geq 0,$ $\frac{5x-1-x-2}{x+2}\geq 0,$ $\frac{4x-3}{x+2}\geq 0.$



The domain is the intersection of the three solution sets: $D(f) = (-\infty,-2)\cup[\frac{3}{4},+\infty)$.

EXAMPLE 11 Find the domain of $f(x) = \arccos(3x + 1)$.

Solution The trigonometric function $\cos x$ is defined from $\mathbb{R}$ to the closed interval $[-1, 1]$. Therefore its inverse $\arccos x$ will be defined from $[-1, 1]$ to $\mathbb{R}$. In other words,

$\cos x\colon \mathbb{R} \to [-1, 1],$

$\arccos x\colon [-1, 1] \to \mathbb{R}.$

This means that the value of $3x + 1$ must lie between -1 and 1:

$-1 \leq 3x + 1 \leq 1$

$-2 \leq 3x \leq 0$

$-\frac{2}{3} \leq x \leq 0.$

So the domain of the function $f(x) = \arccos(3x + 1)$ is $[-\frac{2}{3}, 0]$.

EXAMPLE 12 Find the domain and the range of $f(x) = \arcsin\sqrt{-x^2 + x}$.

Solution First let us find the domain of the function $g(x) = \sqrt{-x^2 + x}$.

The radicand must be non-negative, so

$-x^2 + x \geq 0$,

$-x(x - 1) \geq 0$.

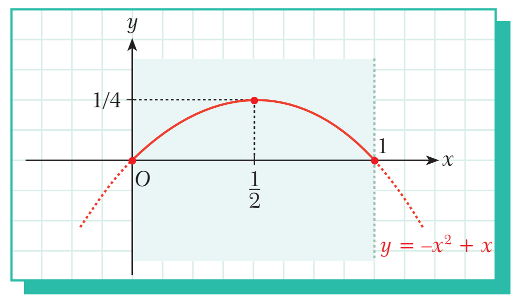
x		0		1	
$-x(x-1)$	$-$	●	$+$	●	$-$

This is true for $0 \leq x \leq 1$, so the domain of $g(x)$ is $[0, 1]$.

Now let us find the range of $g(x) = \sqrt{-x^2 + x}$.

Let x_1 and x_2 be the roots of the equation $-x^2 + x = 0$, i.e. $x_1 = 0$ and $x_2 = 1$.

For $x \in [0, 1]$ the expression $-x^2 + x$ reaches its maximum value at $x = \dfrac{x_1 + x_2}{2} = \dfrac{0+1}{2} = \dfrac{1}{2}$ which is $(\dfrac{1}{2}, \dfrac{1}{4})$. It reaches its minimum value at $x = 0$.



So $g(\dfrac{1}{2}) = \sqrt{-(\dfrac{1}{2})^2 + \dfrac{1}{2}} = \sqrt{-\dfrac{1}{4} + \dfrac{1}{2}} = \sqrt{\dfrac{1}{4}} = \dfrac{1}{2}$ is the maximum value of $g(x)$.

Similarly, $g(0) = g(1) = 0$ is the minimum value of $g(x)$.

Hence the range of $g(x) = \sqrt{-x^2 + x}$ is $[0, \dfrac{1}{2}]$.

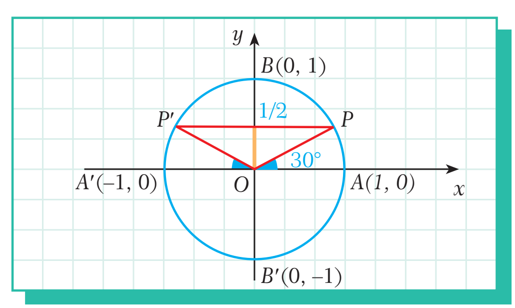
Since the function $\arcsin x$ is defined on the interval $[0, \dfrac{1}{2}]$, the domain of $f(x) = \arcsin\sqrt{-x^2 + x}$ is also the interval $[0, 1]$.

Now we can find the range of $f(x) = \arcsin\sqrt{-x^2 + x} = \arcsin(g(x))$.

We have found that $0 \leq \sqrt{-x^2 + x} \leq \dfrac{1}{2}$. So the range of the function is the set of angles whose sine value is between 0 and $\dfrac{1}{2}$. As we can see in the figure:

for $0° \leq \theta \leq 30°$, $\quad 0 \leq \sin\theta \leq \dfrac{1}{2}$ and

for $150° \leq \theta \leq \pi$, $\quad 0 \leq \sin\theta \leq \dfrac{1}{2}$.



So the range of $f(x) = \arcsin\sqrt{-x^2 + x}$ is the set $[0, \dfrac{\pi}{6}] \cup [\dfrac{5\pi}{6}, \pi]$, and we can write

$f : [0, 1] \to [0, \dfrac{\pi}{6}] \cup [\dfrac{5\pi}{6}, \pi]$.

Check Yourself 1

1. Find the domain of each function.

a. $f(x) = \dfrac{x}{x^2+2}$ b. $f(x) = \sqrt[3]{2x+1} + \sqrt{x^2-1}$ c. $f(x) = 2x - \sqrt[4]{\dfrac{x+1}{x-1}}$

d. $f(x) = \sqrt{2x-1-\sqrt{3x+1}}$ e. $f(x) = 3^{\frac{1}{x-2}}$ f. $f(x) = 2^{\log_3\left(\frac{x+2}{x^2-4}\right)}$

g. $f(x) = \sqrt{3-\log_2 x}$ h. $f(x) = \sqrt{\log_2\left(\dfrac{x+2}{2x+2}\right)}$ i. $f(x) = \log_{\frac{1}{2}}(\log_2(2x+1))$

j. $f(x) = \arccos(2x-3)$ k. $f(x) = \arcsin\left(\dfrac{2x+1}{x+1}\right)$ l. $f(x) = \dfrac{3x+1}{\log(x^2+x+1)-1}$

m. $f(x) = \dfrac{\sqrt{2-\cos x}}{2+\sin x}$ n. $f(x) = \sqrt{\tan x},\ \ x \in [0,\ \pi)$

2. Find the image of each function over the given interval.

a. $f(x) = 2x + 1,\ x \in [1, 5)$ b. $f(x) = x^2 - 4x - 5,\ x \in [1, 3)$

c. $f(x) = \cos x + \sin x,\ x \in [0, 2\pi)$ d. $f(x) = \log_3(x^2 - 2x - 8),\ x \in [6, 7]$

e. $f(x) = 2^{x+1},\ x \in [-1, 3)$ f. $f(x) = \sqrt{x^2+2},\ \ x \in (\sqrt{2}, \sqrt{7}]$

3. Find the range of each function for its largest domain.

a. $f(x) = \sqrt{-x^2+7x-12}$ b. $f(x) = -3\cos x + 1$ c. $f(x) = \dfrac{5}{2^x+3}$

Answers

1. a. $\mathbb{R}$ b. $\mathbb{R} - (-1, 1)$ c. $\mathbb{R} - (-1, 1]$ d. $[\frac{7}{4}, \infty)$ e. $\mathbb{R} - \{2\}$ f. $(2, \infty)$ g. $(0, 8]$ h. $(-1, 0]$

i. $(0, \infty)$ j. $[1, 2]$ k. $[-\frac{2}{3}, 0]$ l. $\mathbb{R} - \{\frac{-1+\sqrt{37}}{2}, \frac{-1-\sqrt{37}}{2}\}$ m. $\mathbb{R}$ n. $[0, \frac{\pi}{2})$

2. a. $[3, 11)$ b. $[-9, -8]$ c. $[-\sqrt{2}, \sqrt{2}]$ d. $[4\log_3 2,\ 3]$ e. $[1, 16)$ f. $(2, 3]$

3. a. $[0, \frac{1}{2}]$ b. $[-2, 4]$ c. $(0, \frac{5}{3})$

B. INVERSE OF A FUNCTION

Recall the definition of inverse function: if the function f: $D \to R$ is both a one-to-one function and an onto function then the function f^{-1}: $R \to D$ is called the **inverse** of f.

$$f(x) = y \iff f^{-1}(y) = x$$

A function f: $A \to B$ is a one-to-one function if for each $x_1 \neq x_2$ in A, $f(x_1) \neq f(x_2)$.

A function f: $A \to B$ is an onto function if for any $y \in B$ there exists an $x \in A$ such that $f(x) = y$.

To find the inverse of a given function $y = f(x)$ it is enough to find x in terms of the variable y.

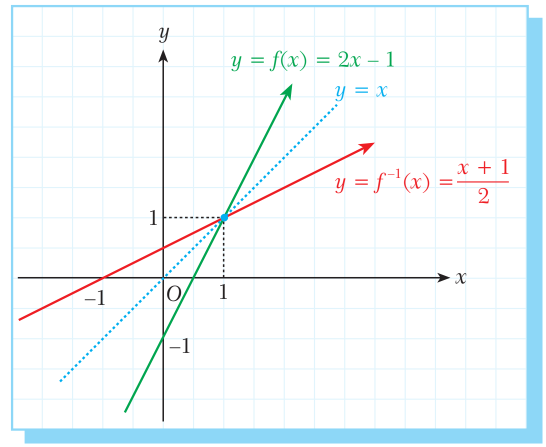
For example, let us find the inverse of the polynomial function f: $\mathbb{R} \to \mathbb{R}$, $f(x) = 2x - 1$:

Write $y = f(x)$: $y = 2x - 1$.

Express x in terms of y: $x = \dfrac{y+1}{2} = f^{-1}(y)$.

Finally, express the inverse function in terms of the variable x: $f^{-1}(x) = \dfrac{x+1}{2}$.

This is the inverse of $f(x) = 2x - 1$.



Recall that the graph of a function and the graph of its inverse are symmetric with respect to the line $y = x$.

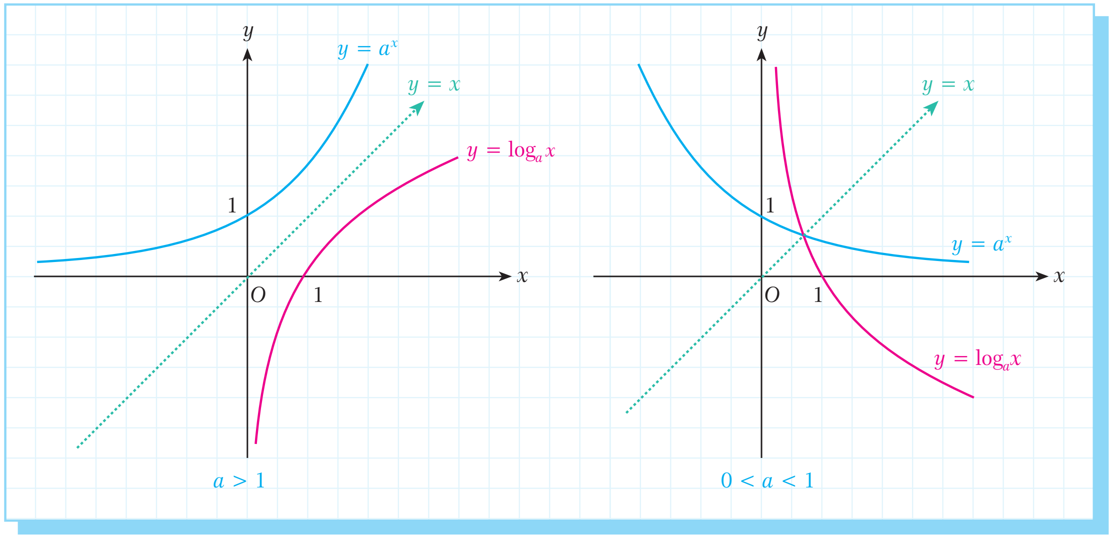
Let us recall the inverse of some common types of function:

Function	Form	Inverse
linear function	$f(x) = ax + b$	$f^{-1}(x) = \dfrac{x-b}{a}$
rational function	$f(x) = \dfrac{ax+b}{cx+d}$	$f^{-1}(x) = \dfrac{-dx+b}{cx-a}$

Remember that the exponential function and the logarithmic function are inverse of each other:

Function	Form	Inverse
exponential function	$f(x) = a^x$ ($a \in \mathbb{R}^+ - \{1\}$)	$f^{-1}(x) = \log_a x$
logarithmic function	$f(x) = \log_a x$ ($a \in \mathbb{R}^+ - \{1\}$)	$f^{-1}(x) = a^x$

As we can see in the figure below, the graphs of the exponential and logarithmic functions are symmetric with respect to the line $y = x$.



Look at some more examples of inverse functions:

Function	Inverse
$f\colon \mathbb{R} \to \mathbb{R},\quad f(x) = \dfrac{3x + 1}{2}$	$f^{-1}\colon \mathbb{R} \to \mathbb{R},\quad f^{-1}(x) = \dfrac{2x - 1}{3}$
$f\colon \mathbb{R} \to \mathbb{R},\quad f(x) = 2 - 3x$	$f^{-1}\colon \mathbb{R} \to \mathbb{R},\quad f^{-1}(x) = \dfrac{-x + 2}{3}$
$f\colon \mathbb{R} \to \mathbb{R}^+,\quad f(x) = 3^x$	$f^{-1}\colon \mathbb{R}^+ \to \mathbb{R},\quad f^{-1}(x) = \log_3 x$
$f\colon \mathbb{R} - \{1\} \to \mathbb{R} - \{2\},\quad f(x) = \dfrac{2x + 1}{x - 1}$	$f^{-1}\colon \mathbb{R} - \{2\} \to \mathbb{R} - \{1\},\quad f^{-1}(x) = \dfrac{x + 1}{x - 2}$

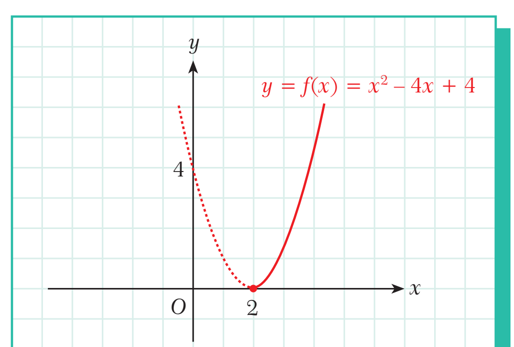
EXAMPLE 13 Find the inverse of the function $f\colon [2, \infty) \to [0, \infty),\ f(x) = x^2 - 4x + 4$.

Solution In the given domain and range, $f(x)$ is both one-to-one and onto, so its inverse is a function. We can find this inverse function as follows:

$$\begin{aligned} x^2 - 4x + 4 &= y \\ (x - 2)^2 &= y \\ x - 2 &= \sqrt{y} \\ x &= \sqrt{y} + 2 = f^{-1}(y). \end{aligned}$$

So the inverse is

$f^{-1}\colon [0, \infty) \to [2, \infty),\ f^{-1}(x) = \sqrt{x} + 2.$



EXAMPLE 14 Find the inverse of the function $f: \mathbb{R} \to (2, \infty)$, $f(x) = 10^{x-1} + 2$.

Solution

$$\begin{aligned} f(x) = 10^{x-1} + 2 &= y \\ 10^{x-1} &= y - 2 \\ x - 1 &= \log_{10}(y - 2) \\ x &= \log_{10}(y - 2) + 1 = f^{-1}(y) \end{aligned}$$

So the inverse is $f^{-1}: (2, \infty) \to \mathbb{R}$, $f^{-1}(x) = \log_{10}(x - 2) + 1$ or $f^{-1}(x) = \log(10x - 20)$.

EXAMPLE 15 Find the inverse of the function $f: (-4, \infty) \to \mathbb{R}$, $f(x) = -1 + 3\log_2(x + 4)$.

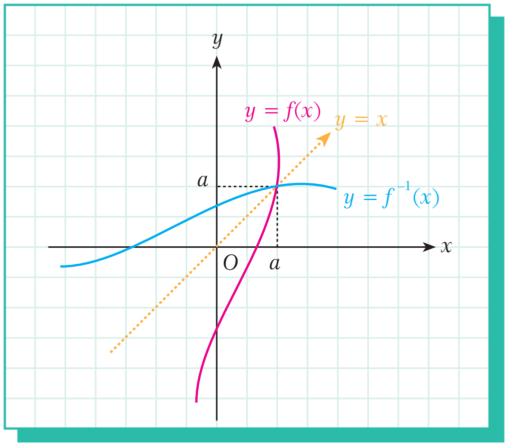
Solution

$$\begin{aligned} f(x) = -1 + 3\log_2(x + 4) &= y \\ \log_2(x + 4) &= \frac{y+1}{3} \\ x + 4 &= 2^{(\frac{y+1}{3})} \\ x &= 2^{(\frac{y+1}{3})} - 4 = f^{-1}(y) \end{aligned}$$

So the inverse is $f^{-1}: \mathbb{R} \to (-4, \infty)$, $f^{-1}(x) = 2^{(\frac{x+1}{3})} - 4$.

EXAMPLE 16 The function $f: \mathbb{R} \to \mathbb{R}$, $f(x) = x^3 - 3x^2 + 4x - 1$ is given. Find the real number a which satisfies the equation $f(a) = f^{-1}(a)$.

Solution To solve the problem we have to find the intersection of the graphs of f and f^{-1}. However, we know that the graph of a function and the graph of its inverse are symmetric with respect to the line $y = x$. In other words, the intersection of the two graphs will be on this line. At the intersection point, therefore, $y = f(x) = x$, and so $f(a) = a = f^{-1}(a)$, as shown at the right.



If $f(a) = a$ then

$$\begin{aligned} a^3 - 3a^2 + 4a - 1 &= a \\ a^3 - 3a^2 + 3a - 1 &= 0 \\ (a - 1)^3 &= 0 \\ a &= 1. \end{aligned}$$

Check Yourself 2

1. Find the inverse of each function.

   a. $f: \mathbb{R} \to \mathbb{R},\ f(x) = 3 - \dfrac{x}{2}$

   b. $f: \mathbb{R} - \{1\} \to \mathbb{R} - \left\{\dfrac{5}{2}\right\},\ f(x) = \dfrac{5x-2}{2x-2}$

   c. $f: \mathbb{R} \to \mathbb{R}^+, f(x) = 5^{2x-1}$

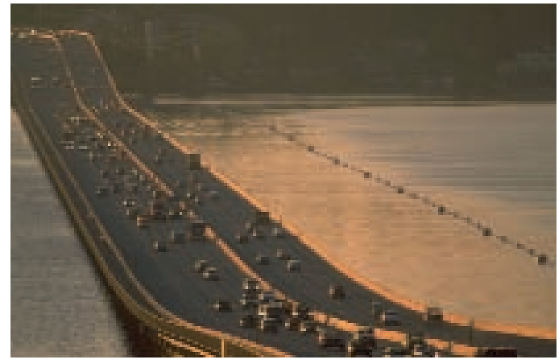
   d. $f: \mathbb{R}^+ \to \mathbb{R}, f(x) = \log_3(2x + 5)$

   e. $f: \mathbb{R} \to \mathbb{R}, f(x) = x^3 - 3x^2 + 3x$

   f. $f: \mathbb{R} \to (1, \infty), f(x) = 3^{x-1} + 1$

   g. $f: \mathbb{R} \to (1, \infty), f(x) = 1 + \log(3^x + 1)$

   h. $f: \mathbb{R} - \{1\} \to (1, \infty) - \{11\},\ f(x) = 10^{\frac{x+1}{x-1}} + 1$



2. For each function, find the real number a which satisfies the equation $f(a) = f^{-1}(a)$.

   a. $f(x) = 3x + 1$ b. $f(x) = 8x^3 - 12x^2 + 7x - 1$

Answers

1. a. $6 - 2x$ b. $\dfrac{2x-2}{2x-5}$ c. $\log_5 \sqrt{x} + \dfrac{1}{2}$ d. $\dfrac{3^x - 5}{2}$ e. $\sqrt[3]{x-1} + 1$ f. $\log_3(x - 1) + 1$

g. $\log_3(10^{x-1} - 1)$ h. $\dfrac{2}{\log \dfrac{x-1}{10}} + 1$ 2. a. $-\dfrac{1}{2}$ b. $\dfrac{1}{2}$

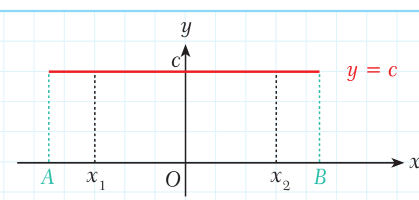
C. CONSTANT, INCREASING AND DECREASING FUNCTIONS

Let $f: D \to R$ be a function and let $I \subset D$ and $x_1, x_2 \in I$ and such that $x_1 < x_2$.

1. If $f(x_1) = f(x_2)$ for all $x_1, x_2 \in I$ then f is called a **constant function** on the interval I.

   We write a constant function as $f(x) = c$ $(c \in \mathbb{R})$.

   If f is a constant function, $x_1 < x_2 \Leftrightarrow f(x_1) = f(x_2) = c$.



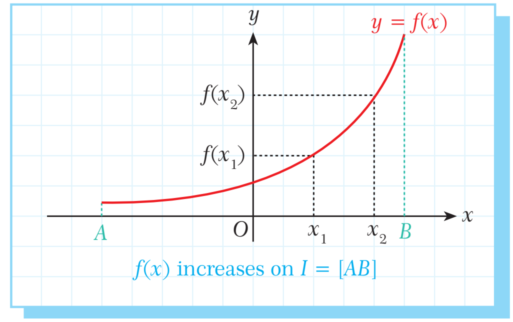
$f(x)$ is constant on $I = [AB]$

2. If $f(x_1) < f(x_2)$ for all $x_1, x_2 \in I$ then f is called an **increasing function** on the interval I.

If f is an increasing function,

$x_1 < x_2 \Leftrightarrow f(x_1) < f(x_2)$.

If $x_1 < x_2$ and $f(x_1) \leq f(x_2)$ then f is called a **non-decreasing function**.



$f(x)$ increases on $I = [AB]$

Note

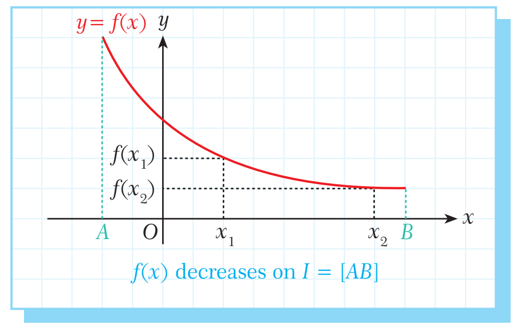
The increasing function $f: \mathbb{R} \to \mathbb{R}$, $f(x) = x$ is called the **identity function**.

3. If $f(x_1) > f(x_2)$ for all $x_1, x_2 \in I$ then f is called a **decreasing function** on the interval I.

If f is a decreasing function,

$x_1 < x_2 \Leftrightarrow f(x_1) > f(x_2)$.

If $x_1 < x_2$ and $f(x_1) \geq f(x_2)$ then f is called a **non-increasing function**.



$f(x)$ decreases on $I = [AB]$

EXAMPLE 17 Given that $f(x) = (5 - a)x^2 + (b + 2)x - 3$ is a constant function, find a and b.

Solution Since f is a constant function, the coefficients of x^2 and x must be zero:

$5 - a = 0$ and $b + 2 = 0$. So $a = 5$ and $b = -2$.

EXAMPLE 18 Show that

a. $f: [0, \infty) \to \mathbb{R}$, $f(x) = \sqrt{x} + x^3$ is an increasing function.

b. $f: \mathbb{R} - \{0\} \to \mathbb{R}$, $f(x) = \dfrac{1}{x}$ is a decreasing function.

Solution a. Let $x_1, x_2 \in [0, \infty)$ such that $x_1 < x_2$.

For all $x_1 < x_2$, $\sqrt{x_1} < \sqrt{x_2}$ and $x_1^3 < x_2^3$. So $\sqrt{x_1} + x_1^3 < \sqrt{x_2} + x_2^3$ and $f(x_1) < f(x_2)$.

So f is an increasing function on $[0, \infty)$.

b. First let $x_1, x_2 \in (0, \infty)$ such that $x_1 < x_2$. For all $x_1 < x_2$, $\dfrac{1}{x_1} > \dfrac{1}{x_2}$ and so $f(x_1) > f(x_2)$.

Now let $x_1, x_2 \in (-\infty, 0)$ such that $x_1 < x_2$. For all $x_1 < x_2$, $\dfrac{1}{x_1} > \dfrac{1}{x_2}$ and so $f(x_1) > f(x_2)$.

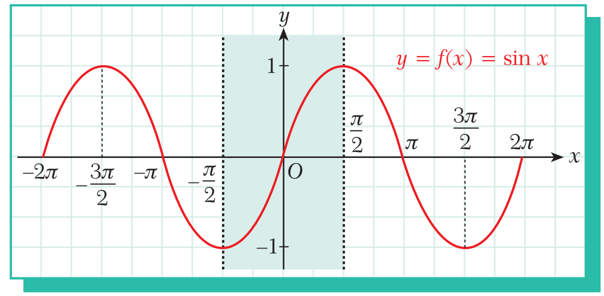
In both cases f is decreasing, so f is a decreasing function on $\mathbb{R} - \{0\}$.

EXAMPLE 19 Determine whether each function increases or decreases on the given interval.

a. $y = \sin x,\ x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ b. $y = 0.5^x - 3x,\ x \in \mathbb{R}$ c. $y = \ln x + x^2,\ x \in \mathbb{R}^+$

Solution a. $y = \sin x$ is a periodic function. Its graph is shown in the figure opposite.

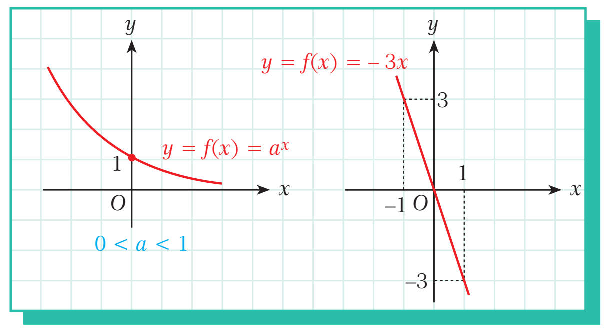
We can see that on the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$, $y = \sin x$ increases.



Let $t \in \mathbb{R}$. If $f(x + t) = f(x)$ then f is called a periodic function.

b. $y = \left(\frac{1}{2}\right)^x + (-3x)$

Recall that the exponential function $f(x) = a^x$ increases when $a > 1$ and decreases when $0 < a < 1$ as shown in the figure opposite.



The exponential function $\left(\frac{1}{2}\right)^x$ is therefore a decreasing function because $a = \frac{1}{2} < 1$.

We can easily see from the graph of $y = -3x$ that this is also a decreasing function. The sum of two decreasing functions is also a decreasing function, so f is a decreasing function.

The sum of two increasing functions is also an increasing function. The sum of two decreasing functions is also a decreasing function.

c. $y = \ln x + x^2$

$\ln x$ is an increasing function because $\ln x = \log_e x$ and $e \cong 2.71 > 1$.

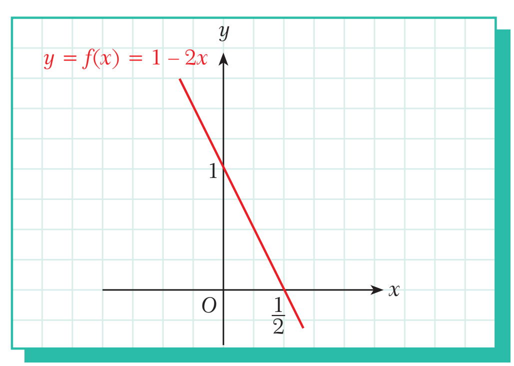
x^2 is also an increasing function in $\mathbb{R}^+$, so $y = \ln x + x^2$ is increasing function.

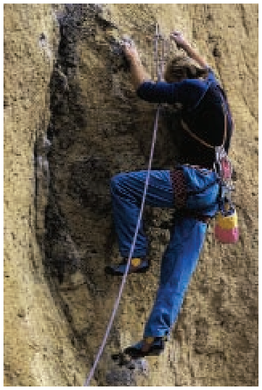
EXAMPLE 20 Find the interval(s) on which each function decreases and/or increases.

a. $y = 1 - 2x$ b. $y = x^2 - 3x$ c. $y = \frac{1}{x^2}$ d. $y = \cos x$

Solution We can draw a graph of each function to determine the intervals.

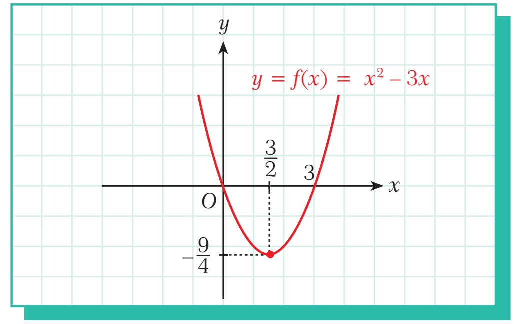
a. We can see from the graph that $f(x) = 1 - 2x$ decreases on $(-\infty, \infty)$.





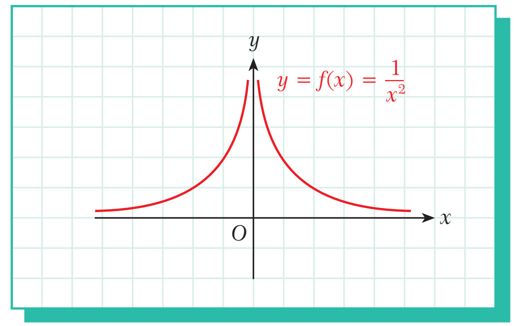
b. f decreases on the interval $(-\infty, \frac{3}{2}]$.

f increases on the interval $[\frac{3}{2}, +\infty)$.



c. f increases on the interval $(-\infty, 0)$.

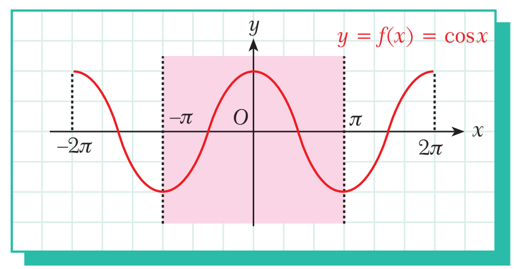
f decreases on the interval $(0, +\infty)$.



d. $f(x) = \cos x$ is a periodic function, so let us consider its value in the interval $[-\pi, \pi]$.

f increases on the interval $[-\pi, 0]$.

f decreases on the interval $[0, \pi]$.



EXAMPLE 21 Find the value $a + b$ if $f(x) = \dfrac{(a-2)x^2 + bx + 4}{3x+2}$ is a constant function.

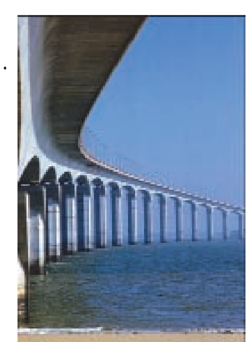
Solution Since f is constant, we can write $f(x) = \dfrac{(a-2)x^2 + bx + 4}{3x+2} = k \quad (k \in \mathbb{R})$.

This gives $(a - 2)x^2 + bx + 4 = 3kx + 2k$.

By the equality of polynomials, we can write

$$(a-2) = 0$$
$$b = 3k$$
$$4 = 2k, \text{ which gives}$$

$a = 2$, $k = 2$, $b = 6$. So $a + b = 8$.



EXAMPLE 22 The function $f(x)=\dfrac{(a-1)x^2+(b+5)x}{2x+1}$ is an identity function. Find a and b.

Solution Since f is an identity function, $f(x) = x$. So

$$\frac{(a-1)x^2+(b+5)x}{2x+1}=x, \text{ i.e.}$$

$$(a-1)x^2+(b+5)x=2x^2+x.$$

By the equality of polynomials, $a - 1 = 2$ and $b + 5 = 1$. So $a = 3$ and $b = -4$.

Check Yourself 3

1. $f(x)=\dfrac{3mx+1}{6x+5}$ is a constant function. Find m.
2. $f(x) = (m - 2)x + n + 5$ is an identity function. Find $m + n$.
3. Decide whether each function increases or decreases on the given interval.
   a. $f(x) = 2x + 1,\ x \in \mathbb{R}$
   b. $f(x) = 1 - x,\ x \in \mathbb{R}$
   c. $f(x) = -x^2 - 8x + 1,\ x \in (-4, \infty)$
   d. $f(x) = -x^2 - 2x - 1,\ x \in (-\infty, 1)$
   e. $f(x) = x^3 + 1,\ x \in \mathbb{R}$

Answers

1. $\dfrac{2}{5}$ 2. -2

D. EVEN AND ODD FUNCTIONS

Definition even function, odd function

Let $f: D \to R$ be a function.

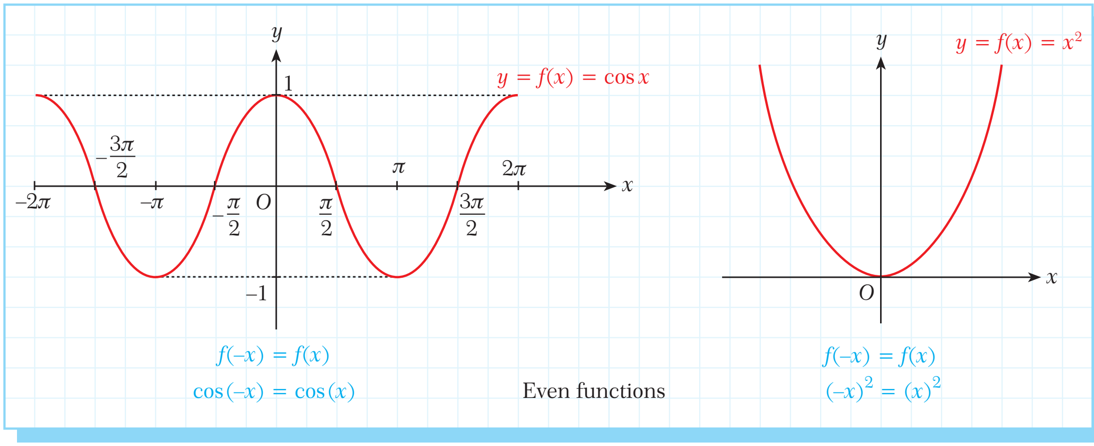
1. If $f(-x) = f(x)$ for all $x \in D$ then f is called an **even function**.
2. If $f(-x) = -f(x)$ for all $x \in D$ then f is called an **odd function**.

For example, the cosine function is an even function because $\cos(-x) = \cos x$.

Similarly, the sine, tangent and cotangent functions are odd functions because $\sin(-x) = -\sin x$, $\tan(-x) = -\tan x$ and $\cot(-x) = -\cot x$.

Not all functions are even or odd. For example, $f(x) = x + 1$ is neither even nor odd.

The graph of an even function is symmetric with respect to the y-axis.



The following rules help us to calculate the parity (even or odd) of the sum and product of even functions (E) and odd functions (O):

$E \pm E = E$

$O \pm O = O$

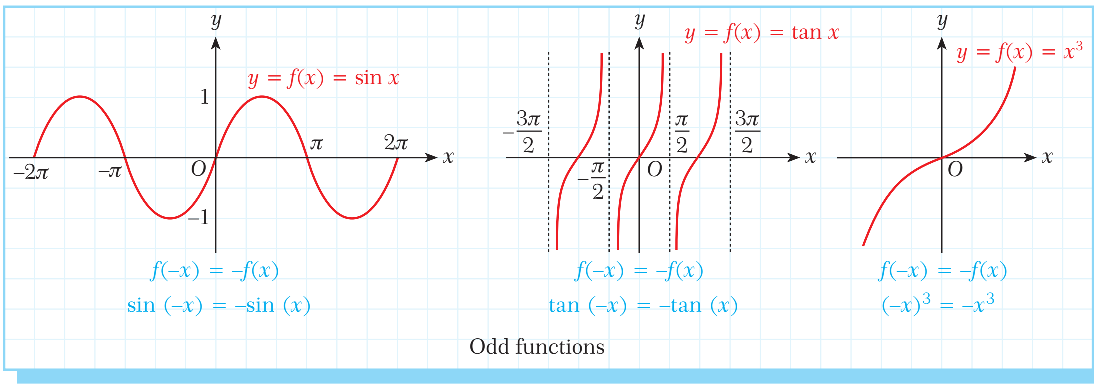
$E \pm O =$ neither E nor O

$E \cdot E = E$

$E \cdot O = O$

$O \cdot O = E$.

The graph of an odd function is symmetric with respect to the origin.



EXAMPLE 23 Determine whether each function is even, odd, or neither even nor odd.

a. $f(x) = x^2 + 3x + 2$ **b.** $f(x) = 7\tan x + x^3$ **c.** $f(x) = \dfrac{x^2 - 2}{x^6 + x^8}$ **d.** $f(x) = 3^x + 3^{-x}$

e. $f(x) = \sqrt{x^2 - 6x + 9} + \sqrt{x^2 + 6x + 9}$ **f.** $f(x) = |x - 3| + |x + 4|$ **g.** $f(x) = -3x^2 + 2|x| - 5$

Solution Let us find $f(-x)$ and compare it with $f(x)$ in each case.

a. $f(-x) = x^2 - 3x + 2$, so f is neither even nor odd.

b. $f(-x) = -7\tan x - x^3 = -(7\tan x + x^3) = -f(x)$, so f is odd.

c. $f(-x) = \dfrac{x^2 - 2}{x^6 + x^8} = f(x)$, so f is even.

d. $f(-x) = 3^{-x} + 3^x = 3^x + 3^{-x} = f(x)$, so f is even.

e. $f(-x) = \sqrt{x^2 + 6x + 9} + \sqrt{x^2 - 6x + 9} = f(x)$, so f is even.

f. $f(-x) = |-x - 3| + |-x + 4|$, so f is neither even nor odd.

g. $f(-x) = -3x^2 + 2|x| - 5 = f(x)$, so f is even.

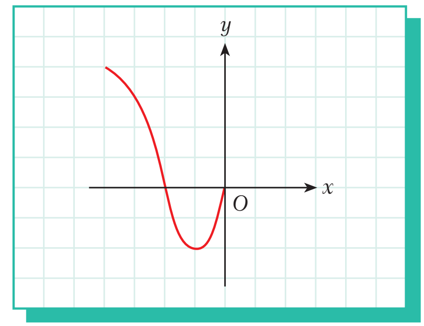
EXAMPLE 24 $f: \mathbb{R} \to \mathbb{R}$, $f(x)$ is an odd function such that $f(-2) = k + 5$ and $f(2) = 2k + 3$. Find k.

Solution Since f is an odd function, $f(-x) = -f(x)$ and $f(-2) = -f(2)$.

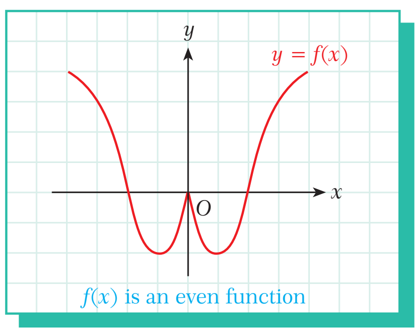
So
$$k + 5 = -(2k + 3)$$
$$k + 5 = -2k - 3$$
$$3k = -8$$
$$k = -\frac{8}{3}.$$

EXAMPLE 25 Complete the graph of the function if it is

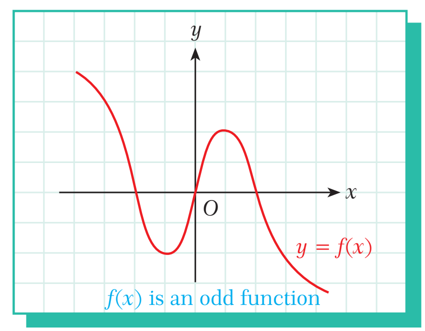
a. even. b. odd.



Solution a. The graph of an even function is symmetric with respect to the y-axis.



b. The graph of an odd function is symmetric with respect to the origin.



Check Yourself 4

1. Determine whether each function is even, odd, or neither even nor odd.

a. $f(x) = |x| + \cos x$ b. $f(x) = x^3 + \sin x$ c. $f(x) = x^4 + x^2 + 1$

d. $f(x) = \cos x^4 - x^3 \sin x$ e. $f(x) = \dfrac{x}{\cos(x^3)}$ f. $f(x) = 2^{\frac{\sin x + \tan x}{x^3}}$

2. f is an odd function and g is an even function. $f(-2) + g(1) = 8$ and $g(-1) + f(2) = 6$ are given. Find $f(-2)$ and $g(-1)$.

Answers

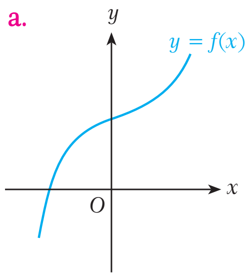
1. a. even b. odd c. even d. even e. odd f. even 2. $f(-2) = 1$, $g(-1) = 7$

EXERCISES 4.1

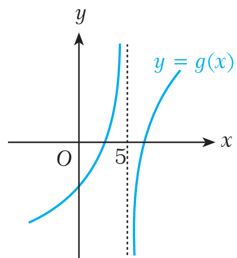
A. Domain and Range of a Function

1. State the domain and range of each function.

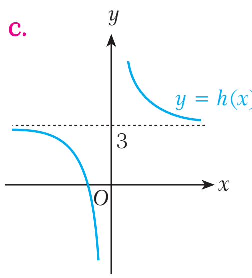
a.



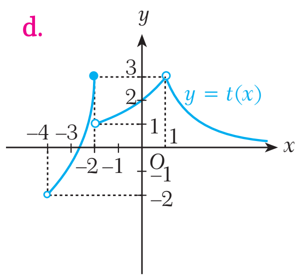
b.



c.



d.



2. Find the domain of each function.

a. $f(x)=\sqrt[5]{\dfrac{x+3}{x-3}}$

b. $f(x) = \log_3(x^2 + 5x + 6)$

c. $f(x)=5^{\sqrt{\frac{1}{x+1}}}$

d. $f(x)=\sqrt{9-|x^2-4|}$

e. $f(x)=\dfrac{\sqrt{x^2-3x-4}}{x^2-1}$

f. $f(x)=\sqrt{3x^2+1+\sqrt{\dfrac{2x}{x-5}}}$

g. $f(x) = \ln(x^2 - 7x + 10) + \sqrt{x^2-4}$

h. $f(x)=\sqrt{\dfrac{1-2\sin x}{2}},\ x\in[0,\ 2\pi]$

i. $f(x)=\arccos\dfrac{2-x}{x+1}$

j. $f(x) = \arcsin 2^x$

k. $f(x)=\sqrt{\log_{\frac{1}{2}}(\dfrac{x}{x^2-4})}$

l. $f(x) = \log_{2x-5}(x^2 - 3x - 10)$

m. $f(x) = \log(\log^2 x - 3\log x - 10)$

n. $f(x)=\sqrt{\cos^2 x-\cos x},\ \ x\in(0,\ 2\pi)$

3. Find the range of each function over the given interval.

a. $f(x) = 1 - 3x, \quad x \in [-2, 4)$

b. $f(x) = x^2 - 2x - 3, \quad x \in (2, 4]$

c. $f(x) = -x^2 + 4x + 5, \quad x \in [0, 1)$

d. $f(x)=\log_{\frac{1}{2}}(3x+1), \quad x \in (1, 5)$

e. $f(x) = 3^{2x+5}, \quad x \in (-1, 1)$

f. $f(x)=(\dfrac{1}{2})^{x+2}, \quad x \in [-2, 2)$

4. Find the range of each function for its largest domain.

a. $f(x)=\sqrt{-x^2+4}$

b. $f(x) = 1 - 2\sin x$

c. $f(x)=\dfrac{2}{1+7^x}$

d. $f(x)=\sqrt{-x^2-10x-9}$

B. Composite Function

5. Given $f(x) = \sqrt{x}$, $g(x) = x^2$, and $h(x) = x + 1$, write each function.

a. $g(h(f(x)))$ b. $f(h(g(x)))$

6. Write each function as a composite of elementary functions.

a. $f(x) = 5 - \dfrac{1}{\sqrt{x+1}}$

b. $f(x) = \log_3(\dfrac{1}{x^2+5})$

C. Inverse of a Function

7. Find the inverse of each function.

a. $f: \mathbb{R} \to \mathbb{R},\ f(x) = \dfrac{2-x}{5}$

b. $f: \mathbb{R} - \{3\} \to \mathbb{R} - \{2\},\ f(x) = \dfrac{-2x-3}{x+3}$

c. $f: \mathbb{R} \to \mathbb{R}^+$, $f(x) = 2 \cdot 3^{3x+4}$

d. $f: \mathbb{R}^+ \to \mathbb{R}$, $f(x) = 2 + \ln(5x - 4)$

e. $f: [1, +\infty) \to [1, +\infty)$, $f(x) = 4x^2 - 8x + 5$

8. For each function, find the real number a which satisfies the equation $f(a) = f^{-1}(a)$.

a. $f(x) = 5x - 2$

b. $f(x) = x^3 - 6x^2 + 13x - 8$

D. Constant, Increasing and Decreasing Functions

9. $f(x) = \dfrac{ax^2+2x+b}{3x^2+bx+2b}$ is a constant function. Find a and b.

10. Determine whether each function is increasing or decreasing on the given interval.

a. $f(x) = x^2 - 6x + 1$, $x \in (3, \infty)$

b. $f(x) = -x^2 + 4x - 3$, $x \in (-\infty, 2)$

c. $f(x) = -x^3 + 3$, $x \in \mathbb{R}$

d. $f(x) = \sin x$, $x \in (0, \dfrac{\pi}{2})$

11. The function $f(x) = 2x^2 - 7x - 15$ is given.

a. On which interval does the function decrease?

b. On which interval does the function increase?

E. Even and Odd Functions

12. Determine whether each function is even, odd, or neither even nor odd.

a. $f(x) = x^5 + x^3 + x$

b. $f(x) = \dfrac{\cos x + x^2}{3 + x^4}$

c. $f(x) = (\dfrac{x \cdot \tan x}{x^3 + \sin x})^5$

d. $f(x) = \sin(\tan(x^3 + x))$

e. $f(x) = e^{\frac{5x^3+x}{\cos x}}$

f. $f(x) = x^4 \cdot \sin x$

2 SPECIAL FUNCTIONS

A. PIECEWISE FUNCTION

Definition

piecewise function

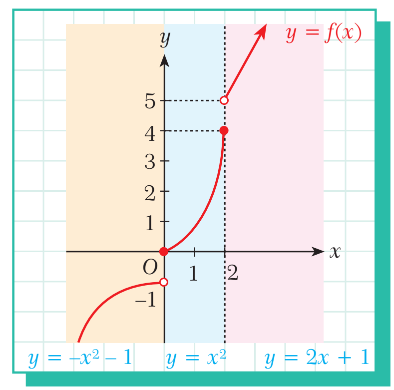
A function that is defined by different formulas on different intervals of its domain is called a **piecewise function**.

EXAMPLE 26 The piecewise function $f: \mathbb{R} \to \mathbb{R}$, $f(x) = \begin{cases} 2x+1 & \text{if } x > 2 \\ x^2 & \text{if } 0 \le x \le 2 \\ -x^2-1 & \text{if } x < 0 \end{cases}$ is given.

a. Draw the graph of f.

b. Find $f(-5) + f(2) + f(3)$.

Solution a. When $x > 2$, we draw the graph of $y = 2x + 1$, when $0 \le x \le 2$, we draw the graph of $y = x^2$ and when $x < 0$, we draw the graph of $y = -x^2 - 1$.



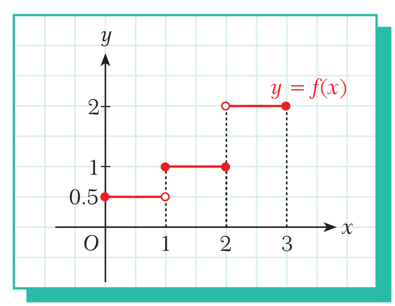
b. When $x = -5$, $f(x) = -x^2 - 1$. So $f(-5) = -(-5)^2 - 1 = -26$.

When $x = 2$, $f(x) = x^2$. So $f(2) = 2^2 = 4$.

When $x = 3$, $f(x) = 2x + 1$. So $f(3) = 2 \cdot 3 + 1 = 7$.

Hence, $f(-5) + f(2) + f(3) = -26 + 4 + 7 = -15$.

EXAMPLE 27 The domain of the function $f(x)$ shown in the figure is $[0, 3]$. Define $f(x)$ as a piecewise function.



Solution The graph consists of three line segments. Working from left to right:

the first line segment is valid for $0 \le x < 1$ and $f(x) = \frac{1}{2}$,

the second segment is valid for $1 \le x \le 2$ and $f(x) = 1$,

and the third segment is valid for $2 < x \le 3$ and $f(x) = 2$.

So the definition is

$$f(x) = \begin{cases} \frac{1}{2} & \text{if } 0 \le x < 1 \\ 1 & \text{if } 1 \le x \le 2 \\ 2 & \text{if } 2 < x \le 3. \end{cases}$$

EXAMPLE 28 Sketch the graph of the piecewise function $f: \mathbb{R} \to \mathbb{R}$,

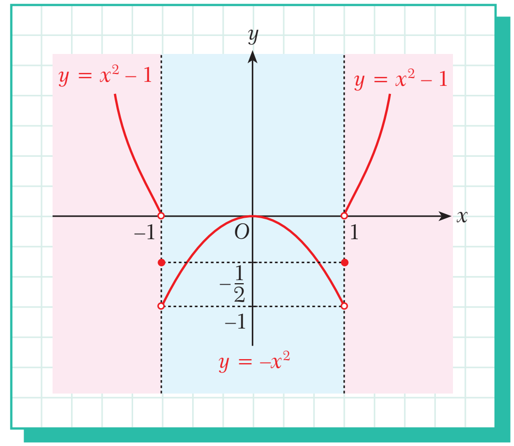
$$f(x) = \begin{cases} x^2 - 1 & \text{if } x < -1 \text{ or } x > 1 \\ -\dfrac{1}{2} & \text{if } x = -1 \text{ or } x = 1 \\ -x^2 & \text{if } -1 < x < 1. \end{cases}$$

Solution We draw the graph of $y = x^2 - 1$ for the interval $(-\infty, -1) \cup (1, \infty)$.

Since $f(-1) = -\dfrac{1}{2}$ and $f(1) = -\dfrac{1}{2}$,

we plot the single points $(-1, -\dfrac{1}{2})$ and $(1, -\dfrac{1}{2})$.

We draw the curve $y = -x^2$ for the interval $(-1, 1)$.



Check Yourself 5

1. The piecewise function $f: \mathbb{R} \to \mathbb{R}$, $f(x) = \begin{cases} x^2 & \text{if } x > 3 \\ 3x + 4 & \text{if } 0 \le x \le 3 \\ x^3 + 2 & \text{if } x < 0 \end{cases}$ is given.

   Calculate $f(f(f(-1)))$.

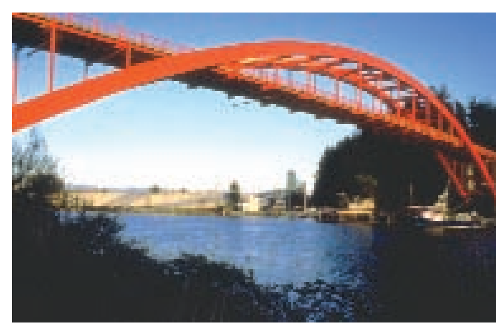
2. Sketch the graph of each piecewise function.

   a. $f: \mathbb{R} \to \mathbb{R}$, $f(x) = \begin{cases} x & \text{if } x \ge 1 \\ -x & \text{if } x < 1 \end{cases}$

   b. $f: \mathbb{R} \to \mathbb{R}$, $f(x) = \begin{cases} x^2 & \text{if } x > 1 \\ -x^2 & \text{if } x \le 1 \end{cases}$

   c. $f: \mathbb{R} \to \mathbb{R}$, $f(x) = \begin{cases} 2x - 1 & \text{if } x < 0 \\ 1 & \text{if } 0 \le x < 1 \\ 1 - x & \text{if } x \ge 1 \end{cases}$

   d. $f: \mathbb{R} \to \mathbb{R}$, $f(x) = \begin{cases} \ln x & \text{if } 0 < x \le e \\ e^x & \text{if } x \le 0 \text{ or } x > e \end{cases}$



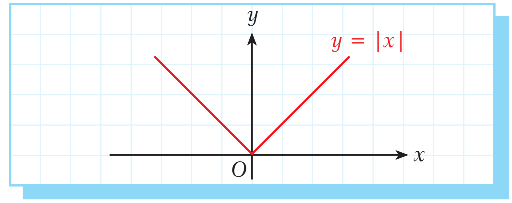
Answers

1. 49

B. ABSOLUTE VALUE FUNCTION

Recall that for any number x, the **absolute value** of x (written $|x|$) is the distance between x and the origin on a number line.

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$



Definition

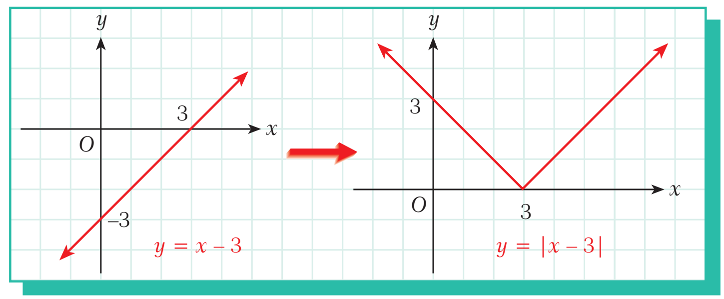
absolute value function

The **absolute value function** $|f(x)|$ is defined as

$$|f(x)| = \begin{cases} f(x) & \text{if } f(x) \geq 0 \\ -f(x) & \text{if } f(x) < 0. \end{cases}$$

EXAMPLE 29 Draw the graph $y = |x - 3|$.

Solution We begin by drawing the graph $y = x - 3$. We then draw the graph $y = |x - 3|$ by reflecting the negative part of the graph.



Note

When solving absolute value equations or inequalities or when drawing the graph of an absolute value function, begin by finding the intervals in which the value of the function is negative, positive or zero.

EXAMPLE 30 Draw the graph $y = |x^2 - 1|$.

Solution 1 Let us construct the sign table for $x^2 - 1$.

x		-1		1	
$x^2 - 1$	+	0	–	0	+

$x^2 - 1$ is positive for $x < -1$ or $x > 1$ and zero for $x = -1$ or $x = 1$.

For these values of x, $|x^2 - 1| = x^2 - 1$.

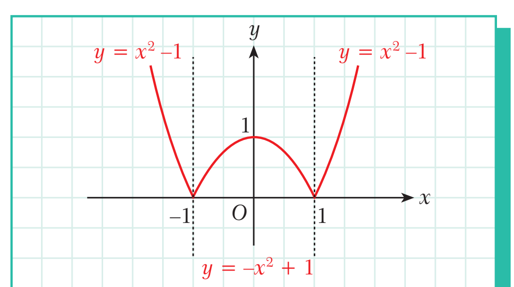
$x^2 - 1$ is negative for $-1 < x < 1$.

For these values of x,

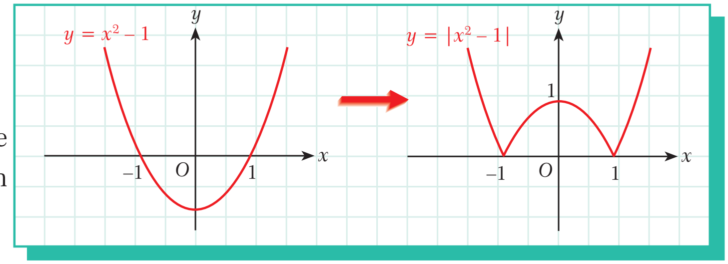
$|x^2 - 1| = -|x^2 - 1| = -x^2 + 1$.

$$\text{So } |x^2 - 1| = \begin{cases} x^2 - 1 & \text{if } x \leq -1 \text{ or } x \geq 1 \\ -x^2 + 1 & \text{if } -1 < x < 1. \end{cases}$$

Now we can draw the graph, shown opposite.



Solution 2 We could also graph $y = |x^2 - 1|$ using a different method: first we graph $y = x^2 - 1$ and then we reflect the negative y-values in the graph with respect to the x-axis.



EXAMPLE 31 Draw the graph of $f(x) = |x - 1| + |x + 2|$.

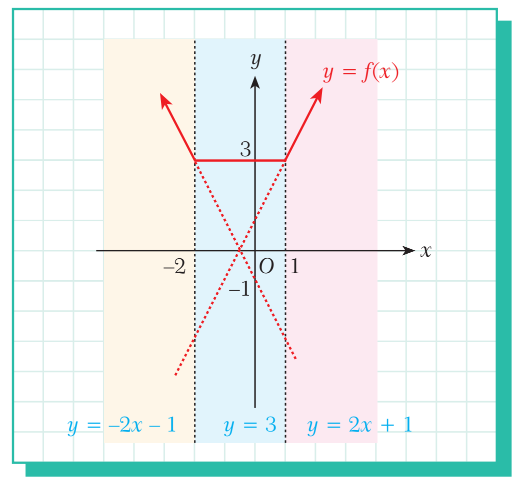
Solution First we construct a sign table.

x		-2		1	
$x - 1$	$-$		$-$	0	$+$
$x + 2$	$-$	0	$+$		$+$

When $x < -2$, $f(x) = -x + 1 - x - 2 = -2x - 1$.
When $-2 \leq x < 1$, $f(x) = -x + 1 + x + 2 = 3$.
When $x \geq 1$, $f(x) = x - 1 + x + 2 = 2x + 1$.
So $f(x)$ can be defined piecewise as

$$f(x) = \begin{cases} -2x-1 & \text{if} \quad x < -2 \\ 3 & \text{if} \quad -2 \leq x < 1 \\ 2x+1 & \text{if} \quad x \geq 1. \end{cases}$$

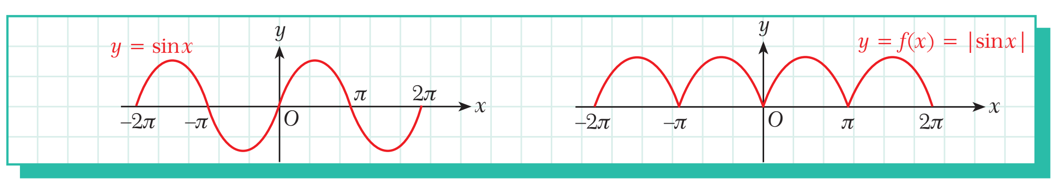
Now we can draw the graph, shown opposite.



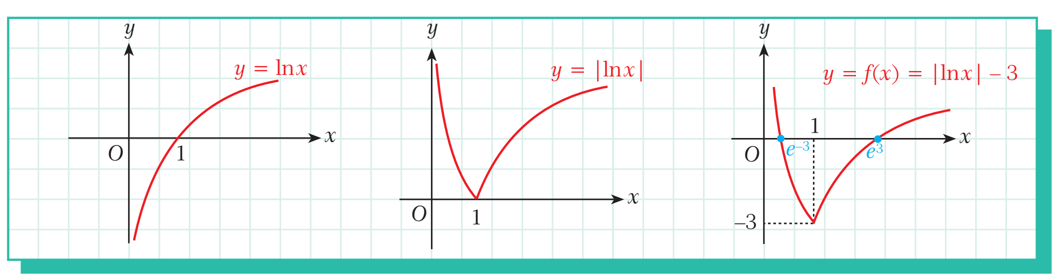
EXAMPLE 32 Draw the graph of each absolute value function.

a. $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = |\sin x|$ **b.** $f\colon \mathbb{R}^+ \to \mathbb{R}$, $f(x) = |\ln x| - 3$

Solution **a.**



b.



EXAMPLE 33

Draw the graph of $f(x) = -x \cdot |x + 2| + 3x$.

Solution Since the expression includes an absolute value, let us begin by defining the function in pieces:

x		-2	
$x + 2$	$-$	$\circ$	$+$
$\lvert x + 2\rvert$	$-x - 2$	$\circ$	$x + 2$

$$f(x) = \begin{cases} -x(-x-2)+3x & \text{if} \quad x < -2 \\ -x(x+2)+3x & \text{if} \quad x \geq -2 \end{cases}$$

$$f(x) = \begin{cases} x^2+5x & \text{if} \quad x < -2 \\ -x^2+x & \text{if} \quad x \geq -2. \end{cases}$$

To draw the graph precisely, let us find the x-intercepts by calculating the roots of the equations:

$x^2 + 5x = 0$ and $-x^2 + x = 0$

$x(x + 5) = 0$ $\qquad -x(x - 1) = 0$

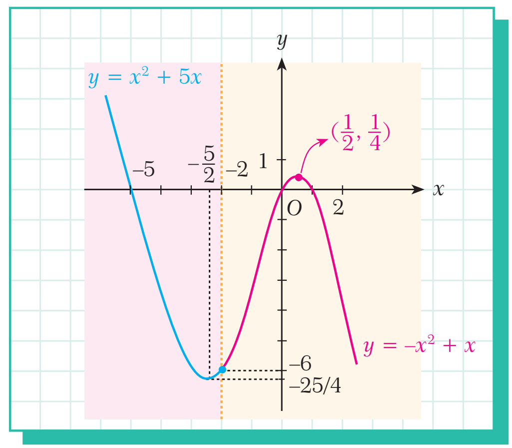
$x_1 = 0,\ x_2 = -5$ $\qquad x_3 = 0,\ x_4 = 1.$

At $x = -2$, $f(-2) = -(-2)^2 + (-2) = -6$ and also $(-2)^2 + 5 \cdot (-2) = -6$.

The vertex point of $y = x^2 + 5x$ is $(-\frac{5}{2}, -\frac{25}{4})$

and the vertex point of $y = -x^2 + x$ is $(\frac{1}{2}, \frac{1}{4})$.

Now we can draw the graph.



Check Yourself 6

Draw the graphs.

1. $y = |2x - 3|$
2. $y = |x^3|$
3. $y = |x^2 + 2x - 3|$
4. $y = |x - 5| + |x + 3|$
5. $y = |x - 5| \cdot x + 2x - 1$
6. $y = |2x + 1| + x - 3$

EXERCISES 4.2

A. Piecewise Function

1. Given

$$f(x)=\begin{cases} x-1 & \text{if} \quad x>2 \\ x & \text{if} \quad 0<x\le 2 \\ 4x & \text{if} \quad x\le 0 \end{cases}, \text{ find } \frac{f(0)+f(2)}{f(3)-f(1)}.$$

2. Draw the graph of each piecewise function.

a. $f(x)=\begin{cases} 1 & \text{if } x>1 \\ -2 & \text{if } x\le 1 \end{cases}$

b. $f(x)=\begin{cases} 2x+4 & \text{if } x>1 \\ -x & \text{if } x\le 1 \end{cases}$

c. $f(x)=\begin{cases} x^2+1 & \text{if } x>0 \\ x^2-1 & \text{if } x\le 0 \end{cases}$

d. $f(x)=\begin{cases} \cos x & \text{if } 0<x<\pi \\ \sin x & \text{if } \pi\le x<2\pi \end{cases}$

B. Absolute Value Function

3. Write each absolute value function as a piecewise function.

a. $f(x) = |x + 3|$
b. $f(x) = |x| + x$
c. $f(x) = |x^2 - x - 2|$
d. $f(x) = |x - 2| + |x - 3|$
e. $f(x) = |x + 1| + |x - 1|$
f. $f(x) = |x + 4| \cdot x + x^2 - 2x$

4. Draw the graphs.

a. $y = |-x|$
b. $y = |2 - 4x|$
c. $y = |x^2 - 1|$
d. $y = |x^2 - 4x - 5|$
e. $y = |\log x|$
f. $y = |\cos x|$
g. $y = x|x + 1| + 3$
h. $y = |5x + 4| + 2x - 1$

CHAPTER REVIEW TEST 4A

1. What is the domain of $f(x)=\dfrac{2x}{\log 2x}$?

A) $\mathbb{R}$ B) $(0, \infty)$ C) $(0, \infty)-\{\frac{1}{2}\}$

D) $(0, \infty)-\{2\}$ E) $(0, \frac{1}{2})$

2. What is the domain of $f(x)=\sqrt{|2x-1|-|3x+1|}$?

A) $[-2, 0]$ B) $(-2, 0)$ C) $[0, \infty)$

D) $(-\infty, -2]$ E) $\mathbb{R}$

3. What is the domain of $f(x)=\sqrt{\dfrac{2-x}{x+2}}$?

A) $[-4, -2)$ B) $[0, 2]-\{1\}$ C) $(-2, 2)$

D) $(-2, 2]$ E) $(-2, 0]$

4. What is the range of $f(x) = -x + 5$ for $x \in [-5, 5]$?

A) $[-5, 5]$ B) $[0, 5]$ C) $[-5, 10]$

D) $[5, 10]$ E) $[0, 10]$

5. What is the range of $f(x) = 2^{x+1}$ for $x \in (-1, 3)$?

A) $(2, 8)$ B) $(1, 16)$ C) $(7, 15)$

D) $(2, 16)$ E) $(0, 2)$

6. Find the inverse of f: $[1, \infty) \to [4, \infty)$, $f(x) = x^2 - 2x + 5$.

A) $\sqrt{x-4}$ B) $\sqrt{x-4}+1$ C) $\sqrt{x^2-2}+1$

D) $\sqrt{x-2}+1$ E) $\sqrt{x-4}-1$

7. Find the inverse of f: $\mathbb{R} \to (1, \infty)$, $f(x) = 3^{x-1} + 1$.

A) $\log_3 3x$ B) $\log_3(x-1)$ C) $\log_3(x-1)+2$

D) $\log_3(3x-3)$ E) $\log_3 x-1$

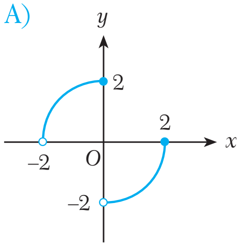
8. Which one of the following is an odd function?

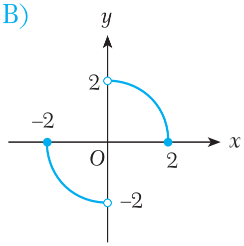
A) $f(x)=\dfrac{x^2}{|x|}$ B) $f(x)=\sqrt{5-x}+\sqrt{5+x}$

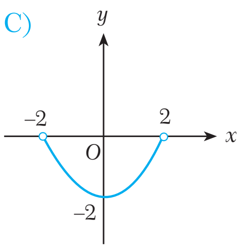
C) $f(x)=\dfrac{x^5}{x^3-x}$ D) $f(x) = x^3 + x$

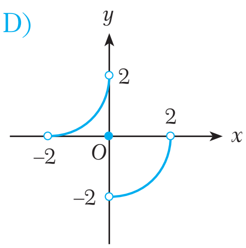
E) $f(x)=\sqrt{\dfrac{x^2}{x-1}}$

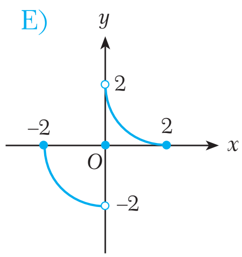
9. Which one of the following is the graph of an odd function?

A)


B)


C)


D)


E)


10. Given that $f(x)=\dfrac{-2x+1}{ax+3}$ is a constant function, find a.

A) –2 B) –3 C) –4 D) –6 E) –10

11. What is the domain of $f(x)=\log(\dfrac{2x}{x+5})$?

A) $\mathbb{R}$ B) $\mathbb{R}-(-5,\ 2)$ C) $\mathbb{R}-[2,\ 5]$

D) $\mathbb{R}-[-5,\ 0]$ E) $\mathbb{R}-(0,\ 2)$

12. What is the domain of $f(x)=\sqrt[5]{x^2-4x-12}$?

A) $(-6,\ 2)$ B) $(-2,\ 6)$ C) $\mathbb{R}-(-2,\ 6)$

D) $\mathbb{R}^+$ E) $\mathbb{R}$

13. What is the domain of $f(x)=\dfrac{2x+1}{\sqrt{4-x^2}}$?

A) $[-2,\ 2]$ B) $[0,\ -2]$ C) $(-2,\ 2)$

D) $\mathbb{R}-(-2,\ 2)$ E) $\mathbb{R}$

14. What is the range of $f(x)=\sqrt{-x^2+5x+36}$ for the largest domain of f?

A) $[0,\ \frac{9}{2}]$ B) $[0,\ \frac{11}{2}]$ C) $[0,\ \frac{13}{2}]$

D) $(0,\ \frac{15}{2})$ E) $[0,\ \frac{17}{2})$

15. What is the range of $f(x)=\log_{\frac{1}{2}}(3x+1)$ for the largest domain of f?

A) $(-\frac{1}{3},\ \infty)$ B) $(-1,\ 1)$ C) $(-\infty,\ \frac{1}{3})$

D) $\mathbb{R}$ E) $\mathbb{R}-\{-\frac{1}{3}\}$

16. Find the inverse of $f: \mathbb{R} \to \mathbb{R}$, $f(x) = x^3 - 3x^2 + 3x$.

A) $\sqrt[3]{x}-1$ B) $\sqrt[3]{x-1}+1$ C) $\sqrt[3]{x+1}-1$

D) $\sqrt{x}+1$ E) $\sqrt{x-1}+1$

17. Find the inverse of

$f: \mathbb{R} \to (1, \infty)$, $f(x) = 1 + \log(3^x + 1)$.

A) $\log_3(10^{x-1} - 1)$ B) $\log(x - 1)$

C) $\log_3(x-1)$ D) 10^{x-1}

E) $\log 3^{x-1}$

18. Which one of the following is an even function?

A) $f(x) = x^7 - x^3$ B) $f(x) = 2x$

C) $f(x) = x\sin x$ D) $f(x) = x\cos x$

E) $f(x) = \dfrac{\cos x}{x^3}$

19. Given that

$$f: \mathbb{R} - \{-1, 1\} \to \mathbb{R},\ f(x) = \frac{3x^2 - (a-2)x + 3}{1 - x^2}$$

is an even function, find the value of $a^2 \cdot f(a)$.

A) 5 B) 10 C) –15 D) –20 E) 25

20. Given $f: [-a, a] \to \mathbb{R}$, which of the following must be an odd function?

A) $f(x) \cdot f(-x)$ B) $\dfrac{f(x)+f(-x)}{2}$

C) $\dfrac{f(x)-f(-x)}{2}$ D) $f(x)+f(\dfrac{1}{x})$

E) $f(x)-f(\dfrac{1}{x})$

21. Which one of the following functions is equivalent to $f(x)=\begin{cases} 2x+2 & \text{if } x \geq -2 \\ -2 & \text{if } x < -2 \end{cases}$?

A) $g(x) = |x + 2|$ B) $g(x) = |x - 2|$

C) $g(x) = |x| + x + 2$ D) $g(x) = x - |x + 2|$

E) $g(x) = x + |x + 2|$

ANSWERS TO EXERCISES

EXERCISES 1.1

1. $B(0, 8)$, $B'(0, -8)$ 2. $A(12, 0)$, $A'(-12, 0)$ 3. $F(7, 0)$, $F'(-7, 0)$ 4. $A(6, 0)$, $A'(-6, 0)$, $B(0, 3)$, $B'(0, -3)$ 5. 18 cm
6. $e = \frac{4}{5}$; vertices: $B(0, 6)$, $B'(0, -6)$; major circle: $x^2 + y^2 = 100$; minor circle: $x^2 + y^2 = 36$; directrices: $x = \pm\frac{25}{2}$
7. major axis = 10, minor axis = 6 8. $A(\frac{15}{2}, 0)$, $A'(-\frac{15}{2}, 0)$, $B(0, \frac{9}{2})$, $B'(0, -\frac{9}{2})$; directrices: $x = \pm\frac{225}{24}$
9. minor axis = $2\sqrt{11}$ cm; foci: $F(0, 5)$, $F'(0, -5)$ 10. $BB' = 2\sqrt{106}$ cm; $e = \frac{5\sqrt{106}}{106}$
11. minor axis = $4\sqrt{21}$ cm; major circle: $x^2 + y^2 = 100$; minor circle: $x^2 + y^2 = 84$ 12. major axis $BB' = 18$; minor axis $AA' = 6\sqrt{5}$; d: $y = \pm\frac{27}{2}$ 13. $\frac{\sqrt{3}}{2}$ 14. $A(\frac{82}{5}, \frac{69}{5})$, $A'(-\frac{22}{5}, -\frac{9}{5})$, $B(\frac{66}{5}, -\frac{18}{5})$, $B'(-\frac{6}{5}, \frac{78}{5})$
15. $\frac{x^2}{121} + \frac{y^2}{49} = 1$ 16. $\frac{x^2}{25} + \frac{y^2}{64} = 1$ 17. major axis = 20, minor axis = 8 18. a. 20 b. 16 c. 0.6 d. 12
19. a. major axis = 0.5, minor axis = 0.4 b. $\frac{3}{5}$ 20. $\frac{x^2}{49} + \frac{y^2}{33} = 1$ 21. $\frac{x^2}{36} + \frac{y^2}{144} = 1$ 22. 54π 23. $4\sqrt{6}\pi$ 24. 40π
25. $(4\sqrt{5} - 5)\pi$ 26. 28π

EXERCISES 1.2

1. $e = \frac{5}{4}$; foci: $F(5, 0)$, $F'(-5, 0)$ 2. conjugate axis = $6\sqrt{3}$, $e = 2$ 3. major circle: $x^2 + y^2 = 25$; minor circle: $x^2 + y^2 = 9$; directrices: $x = \pm\frac{25\sqrt{34}}{34}$ 4. transverse axis = 4; conjugate axis = $8\sqrt{2}$; vertices: $A(2, 0)$, $A'(-2, 0)$ 5. conjugate axis = $4\sqrt{11}$ cm; distance between the foci = 24 cm; directrices: $x = \pm\frac{25}{3}$
6. asymptotes: $y = \pm\frac{\sqrt{5}}{2}x$; directrices: $x = \pm 4$ 7. a. $\frac{\sqrt{13}}{2}$ b. $y = \pm\frac{3}{2}x$ 8. transverse axis = 30 cm, conjugate axis = 72 cm 9. transverse axis = 24 cm; conjugate axis = 18 cm; $e = \frac{5}{4}$ 10. transverse axis = 48 cm, $e = \frac{25}{24}$
11. a. $F(0, 13)$, $F'(0, -13)$ b. $\frac{13}{12}$ c. $y = \pm\frac{144}{13}$ 12. $e = \frac{\sqrt{29}}{2}$; directrices: $y = \pm\frac{8\sqrt{29}}{29}$ 13. $\frac{x^2}{16} - \frac{y^2}{36} = 1$
14. transverse axis = 24, conjugate axis = 16 15. a. transverse axis = 16, conjugate axis = 8 b. $\frac{\sqrt{5}}{2}$ c. $d: x = \pm\frac{16\sqrt{5}}{5}$; asymptotes: $y = \pm\frac{1}{2}x$ 16. transverse axis = 14; conjugate axis = 6; distance between the foci = $2\sqrt{58}$

17. transverse axis = 2; conjugate axis = 8; $e = \sqrt{17}$ 18. a. transverse axis $= \frac{2}{9}$, conjugate axis $= \frac{2}{3}$

b. $y = \pm\frac{\sqrt{10}}{90}$ 19. distance between the foci $= 14\sqrt{2}$; angle $= 45°$ 20. a. transverse axis = 6, conjugate axis = 8 b. $\frac{5}{3}$

21. $\frac{x^2}{4} - \frac{y^2}{9} = 1$ 22. 18

EXERCISES 1.3

1. $x = -5$ 2. $F(-3, 0)$ 3. 12 4. $F(3, 0)$, $d: x = -3$ 5. a. $y^2 = 24x$ b. $x^2 = -8y$ c. $y^2 = -16x$ d. $x^2 = 32y$

6. a. $y^2 = -6x$ b. $x^2 = -\frac{8}{5}y$ c. $y^2 = 4\sqrt{3}x$ d. $x^2 = 3y$ 7. a. $F(\frac{3}{2}, 0)$, $d: x = -\frac{3}{2}$ b. $F(-\frac{1}{6}, 0)$, $d: x = \frac{1}{6}$

c. $F(\frac{3}{20}, 0)$, $d: x = -\frac{3}{20}$ d. $F(0, -9)$, $d: y = 9$ e. $F(0, -\frac{1}{4})$, $d: y = \frac{1}{4}$ f. $F(0, \frac{9}{8})$, $d: y = -\frac{9}{8}$

8. $\pm 4\sqrt{2}$ 9. $x^2 = \frac{9}{4}y$; $F(0, \frac{9}{16})$ 10. 12 11. 10

EXERCISES 2.1

1. a. $A + B = \begin{bmatrix} -2 & 0 \\ 6 & 3 \end{bmatrix}$ b. $A + B = \begin{bmatrix} 7 & 3 \\ 1 & 9 \\ -2 & 15 \end{bmatrix}$ c. $A + B = \begin{bmatrix} -1 \\ 8 \\ 1 \end{bmatrix}$ 2. $c_{21} = -9$, $c_{13} = 26$ 3. $c_{23} = 59$, $c_{32} = 56$

$A - B = \begin{bmatrix} 4 & 0 \\ -2 & -1 \end{bmatrix}$ $A - B = \begin{bmatrix} 5 & -5 \\ 3 & -1 \\ -4 & -5 \end{bmatrix}$ $A - B = \begin{bmatrix} 7 \\ -4 \\ -3 \end{bmatrix}$

$2A = \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}$ $2A = \begin{bmatrix} 12 & -2 \\ 4 & 6 \\ -6 & 10 \end{bmatrix}$ $2A = \begin{bmatrix} 6 \\ 4 \\ -2 \end{bmatrix}$

$2A - B = \begin{bmatrix} 5 & 6 \\ 0 & 0 \end{bmatrix}$ $2A - B = \begin{bmatrix} 11 & -6 \\ 5 & 1 \\ -7 & 0 \end{bmatrix}$ $2A - B = \begin{bmatrix} 10 \\ -2 \\ -4 \end{bmatrix}$

4. $a = 3$, $b = 2$, $c = 1$ 5. $k = -1$, $m = -\frac{1}{2}$, $n = -2$ 6. a. $a = 4$; $b = -1$; $c = 3$; $d = 1$ b. $a = -1$; $b = 1$; $c = 2$

7. a. $\begin{bmatrix} -3 & 3 \\ 12 & -12 \end{bmatrix}$ b. $\begin{bmatrix} 0 & -10 \\ 10 & 0 \end{bmatrix}$ c. $\begin{bmatrix} 4 & 2 \\ -2 & 4 \end{bmatrix}$ d. $\begin{bmatrix} 6 & -21 & 15 \\ 8 & -23 & 19 \\ 4 & 7 & 5 \end{bmatrix}$ e. $[12]$

8. a. undefined b. $\begin{bmatrix} 3 \\ 10 \\ 26 \end{bmatrix}$ c. $\begin{bmatrix} -1 & 19 \\ 4 & -27 \\ 0 & 14 \end{bmatrix}$ d. $\begin{bmatrix} 3 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & -10 \end{bmatrix}$ e. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ f. undefined g. undefined

9. a. $A^{29} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ b. $A^{30} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ 10. a. $\begin{bmatrix} -5 & 2 \\ 3 & 5 \end{bmatrix}$ b. $\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}$ 11. $AB = \begin{bmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) \\ \sin(\alpha+\beta) & \cos(\alpha+\beta) \end{bmatrix}$

$BA = \begin{bmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) \\ \sin(\alpha+\beta) & \cos(\alpha+\beta) \end{bmatrix}$

13. a. $\begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$ b. $\begin{bmatrix} -3 & 2 \\ -2 & 1 \end{bmatrix}$ c. no inverse d. $\begin{bmatrix} 0 & -1 \\ 1 & 11 \end{bmatrix}$ e. $\begin{bmatrix} 1 & 1 & -1 \\ -3 & 2 & -1 \\ 3 & -3 & 2 \end{bmatrix}$ f. $\begin{bmatrix} -13 & 6 & 4 \\ 12 & -5 & -3 \\ -5 & 2 & 1 \end{bmatrix}$

14. a. $-\dfrac{53}{25}$ 15. –3 16. 14

EXERCISES 2.2

1. a. –2 b. 42 c. 25 d. 201 e. 1 f. –4 2. a. –2 b. –8 c. 8 d. 1 e. $\dfrac{\sqrt{3}}{2}$ f. –2 3. $M_{12} = -6$, $M_{23} = -29$, $M_{31} = -5$ 4. $C_{12} = -11$, $C_{23} = 19$, $C_{31} = -4$ 5. a. –4 b. –11 c. –3 d. –27 e. 0 f. 0 6. a. 0 b. 0 7. a. $\dfrac{1}{2}$ b. 1 c. 2 d. $x = 0$, $x = 2$

EXERCISES 3.1

1. a. 36 b. –22 c. 140 d. –42 e. 60 f. $7e$ g. 3π h. $4n + 4$ i. $5k - 5$ 2. a. 1275 b. 1634 c. 324 d. 140 e. 585 f. 2061 g. 1140 h. 3272 i. $3n^2 + 4n + 1$ j. $60k + 99$ k. 13 l. $63\log a$ 3. a. $-a^2$ b. 3 c. –8 d. –1 e. –1 f. 0 g. $\dfrac{91}{2}$ h. 0 i. $\dfrac{3}{5}$ j. $\dfrac{48}{49}$ k. $\dfrac{49}{50}$ l. $-\dfrac{n}{n+1}$ m. $1 + \sqrt{2} - \sqrt{n} - \sqrt{n+1}$ n. 0 o. $-\log 101$ p. 3 q. $101! - 1$ r. $1-\dfrac{1}{51!}$ s. 0 t. 110 u. 4914 v. $1 - 2^{14}$ w. 285 x. 784 4. a. 371 b. 240 c. 32 d. 0 e. 5t f. $\dfrac{3}{2}k\cdot(k+1)$ 5. Hint : $\dfrac{1}{\sin a} = \cot\dfrac{a}{2} - \cot a$ 6. a. Hint : $\binom{n}{k} = \binom{n}{n-k}$ b. Hint : $\binom{n}{k} = \binom{n}{n-k}$ c. Hint : $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \ldots + \binom{n}{n}x^n$ d. Hint : $\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$ 7. a. 127 b. $\binom{66}{60}$ c. $(x + 1)^n$ d. $3^{10} - 3^4$ e. $\binom{20}{10}$ f. 69 8. 60 9. 6 10. 6 11. 5 12. 9 13. 1 14. 2 15. 3 16. 14 17. 240 18. 900

19. 9 **20.** 16 **21.** $-\frac{8}{5}$ **22.** –1 **23.** $A + 480$ **24.** $A + B$ **25.** 10 **26.** $\frac{10}{3}$ **27.** 6 **28.** 0 **29.** 2 **30.** 5

31. 118 **32.** 46 **33. a.** 625 **b.** 1 **c.** $\frac{1}{243}$ **d.** $(\frac{3}{4})^7$ **e.** 32 **f.** $(-\frac{3\pi}{2})^5$ **g.** e^6 **h.** $(n + 2)^5$ **i.** $(2k - 1)^5$

34. a. 41! **b.** 29160 **c.** 2^{28} **d.** 3^{45} **e.** $5!2^{15}$ **f.** $\frac{14!}{3^{105}}$ **g.** $5!(\log_a 2)^5$ **h.** 0 **i.** 0 **j.** 8 **k.** $\frac{24}{5}$ **l.** $(51! - 1) \cdot \log 3$

m. $720 \cdot 2^{80}$ **n.** 110 **o.** $2^{91/2}$ **p.** 0 **q.** $\frac{75!}{76}$ **r.** 0 **s.** 1 **t.** 1 **u.** 504 **v.** $\frac{21}{20}$ **w.** 102 **35. a.** 1 **b.** $(5!)^3 \cdot (4!)^5$

c. 1 **d.** $\frac{3^5}{2}$ **36.** 8 **37.** 5 **38.** 6 **39.** 2 **40.** 3 **41.** $\frac{7!}{3^6}$ **42.** –1 **43.** $\frac{2}{5}$ **44.** 65! **45. a.** 40 **b.** 14^{10}

c. $\frac{55^5}{5!}$ **d.** 42 **e.** $(2^{11} - 2)^7$ **f.** 12^{2x} **46.** $\frac{9}{8}$ **47.** 81 **48.** 7 **49.** $\log_{90}181$ **50.** 2^{24} **51.** 260 **52.** $(\frac{1}{3} - \frac{16}{3})$

53. $1 - \frac{1}{(k+1)!}$ **54.** 7

EXERCISES 4.1

1. a. $f: \mathbb{R} \to \mathbb{R}$ **b.** $g: \mathbb{R} - \{5\} \to \mathbb{R}$ **c.** $h: \mathbb{R} - \{0\} \to \mathbb{R} - \{3\}$ **d.** $t: (-4, \infty) - \{1\} \to (-2, 3]$ **2. a.** $\mathbb{R} - \{3\}$

b. $\mathbb{R} - [-3, -2]$ **c.** $(-1, \infty)$ **d.** $[-\sqrt{13}, \sqrt{13}]$ **e.** $\mathbb{R} - [-1, 4)$ **f.** $\mathbb{R} - (0, 5]$ **g.** $\mathbb{R} - (-2, 5]$ **h.** $[0, 2\pi] - (\frac{\pi}{6}, \frac{5\pi}{6})$

i. $[\frac{1}{2}, \infty)$ **j.** $(-\infty, 0]$ **k.** $(\frac{1-\sqrt{17}}{2}, 0) \cup (\frac{1+\sqrt{17}}{2}, \infty)$ **l.** $(5, \infty)$ **m.** $(0, 0.01) \cup (100000, \infty)$ **n.** $[\frac{\pi}{2}, \frac{3\pi}{2}]$ **3. a.** $[-11, 7]$

b. $(-3, 5]$ **c.** $[5, 8)$ **d.** $(-4, -2)$ **e.** $(27, 2187)$ **f.** $(\frac{1}{16}, 1]$ **4. a.** $[0, 2]$ **b.** $[-1, 3]$ **c.** $(0, 2)$ **d.** $[0, 4]$ **5. a.** $(\sqrt{x} + 1)^2$

b. $\sqrt{x^2 + 1}$ **6. a.** $g(x) = x + 1$, $h(x) = \sqrt{x}$, $t(x) = \frac{1}{x}$, $r(x) = 5 - x$, $f(x) = r(t(h(g(x))))$ **b.** $g(x) = x^2 + 5$,

$h(x) = \frac{1}{x}$, $t(x) = \log_3 x$, $f(x) = t(h(g(x)))$ **7. a.** $f^{-1}: \mathbb{R} \to \mathbb{R}, f^{-1}(x) = 2 - 5x$ **b.** $f^{-1}: \mathbb{R} - \{2\} \to \mathbb{R} - \{3\}, f^{-1}(x) = \frac{-3x-3}{x+2}$

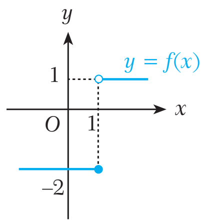
c. $f^{-1}: \mathbb{R}^+ \to \mathbb{R}, f^{-1}(x) = \log_3 \sqrt[3]{\frac{x}{162}}$ **d.** $f^{-1}: \mathbb{R} \to \mathbb{R}^+, f^{-1}(x) = \frac{e^{x-2} + 4}{5}$ **e.** $f^{-1}: [1, +\infty) \to [1, +\infty), f^{-1}(x) = \frac{\sqrt{x-1} + 2}{2}$

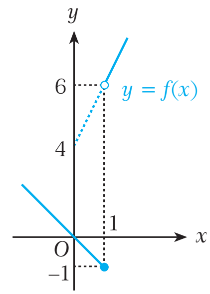
8. a. $\frac{1}{2}$ **b.** 2 **9.** $a = \frac{3}{2}, b = 4$ **10. a.** increasing **b.** increasing **c.** decreasing **d.** increasing **11. a.** $(-\infty, \frac{7}{4}]$

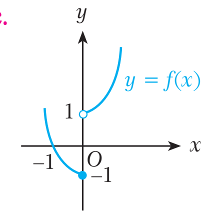
b. $[\frac{7}{4}, \infty)$ **12. a.** odd **b.** even **c.** odd **d.** odd **e.** neither even nor odd **f.** odd

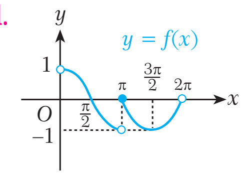
EXERCISES 4.2

1. 2

2. a.


b.


c.


d.


3. a. $f(x)=\begin{cases} x+3 & \text{if } x\geq -3 \\ -x-3 & \text{if } x<-3 \end{cases}$

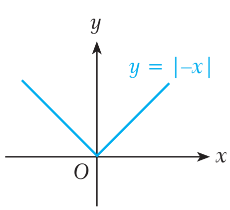
b. $f(x)=\begin{cases} 2x & \text{if } x\geq 0 \\ 0 & \text{if } x<0 \end{cases}$

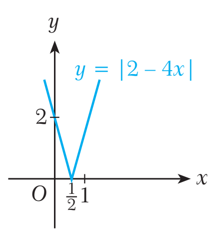
c. $f(x)=\begin{cases} x^2-x-2 & \text{if } x\notin(-1,2) \\ -x^2+x+2 & \text{if } x\in(-1,2) \end{cases}$

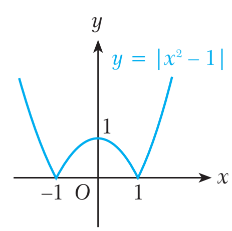
d. $f(x)=\begin{cases} -2x+5 & \text{if } x<2 \\ 1 & \text{if } 2\leq x<3 \\ 2x-5 & \text{if } x>3 \end{cases}$

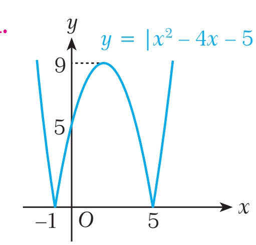
e. $f(x)=\begin{cases} -2x & \text{if } x<-1 \\ -2 & \text{if } -1\leq x<1 \\ 2x & \text{if } x\geq 1 \end{cases}$

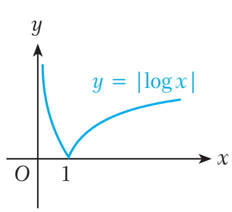
f. $f(x)=\begin{cases} 2x^2+2x & \text{if } x\geq -4 \\ -6x & \text{if } x<-4 \end{cases}$

4. a.


b.


c.


d.


e.


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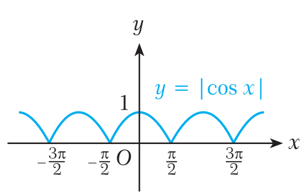
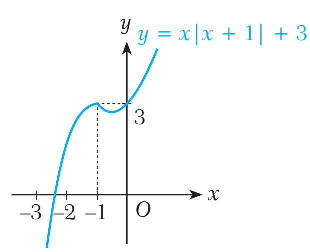
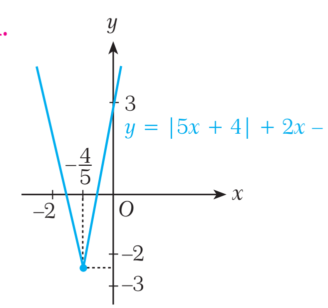


g.


h.


ANSWERS TO TESTS

TEST 1A

1.	D	9.	D	17.	B	25.	A
2.	C	10.	A	18.	E	26.	D
3.	E	11.	E	19.	A	27.	E
4.	E	12.	B	20.	D	28.	C
5.	A	13.	A	21.	E	29.	C
6.	C	14.	D	22.	D	30.	B
7.	B	15.	B	23.	B	31.	E
8.	D	16.	D	24.	A	32.	-

TEST 2A

1.	B	9.	A	17.	C
2.	E	10.	C	18.	A
3.	E	11.	B	19.	D
4.	D	12.	A	20.	D
5.	C	13.	D	21.	C
6.	E	14.	E	22.	-
7.	A	15.	B	23.	-
8.	C	16.	E	24.	-

TEST 2B

1.	E	9.	B
2.	D	10.	E
3.	E	11.	C
4.	B	12.	B
5.	A	13.	A
6.	D	14.	A
7.	E	15.	B
8.	D	16.	D

TEST 3A

1.	B	9.	D	17.	E	25.	C
2.	D	10.	D	18.	-	26.	A
3.	A	11.	D	19.	D	27.	B
4.	C	12.	-	20.	D	28.	D
5.	E	13.	A	21.	D	29.	B
6.	B	14.	B	22.	B	30.	C
7.	D	15.	C	23.	E		
8.	D	16.	B	24.	E		

TEST 3B

1.	D	9.	E	17.	E	25.	C
2.	A	10.	D	18.	C	26.	D
3.	A	11.	A	19.	E	27.	B
4.	B	12.	C	20.	E	28.	D
5.	A	13.	D	21.	B	29.	C
6.	D	14.	B	22.	-	30.	E
7.	D	15.	C	23.	C		
8.	E	16.	D	24.	A		

TEST 4A

1.	C	9.	B	17.	A
2.	A	10.	D	18.	C
3.	D	11.	D	19.	D
4.	E	12.	E	20.	C
5.	B	13.	C	21.	E
6.	B	14.	C	22.	-
7.	D	15.	D	23.	-
8.	D	16.	B	24.	-

GLOSSARY

A

absolute value: the non-negative difference of a number x and zero, written $|x|$. For example, $|8| = 8$ and $|-8| = 8$.

absolute value function: the function $|f(x)|$ defined as follows: if $f(x) \geq 0$ then $|f(x)| = f(x)$, and if $f(x) < 0$ then $|f(x)| = -f(x)$.

adjoint matrix: the transpose of a cofactor matrix.

antisymmetric matrix: If the entries on the main diagonal of a square matrix are all zero and the sums of the symmetric entries with respect to the main diagonal are zero then this matrix is called an antisymmetric matrix.

asymptote of a hyperbola: a line (or a curve) that a hyperbola gets closer and closer to but never touches.

axis of a parabola: the line which passes through the focus of a parabola perpendicular to its directrix.

axis of symmetry: a line that divides a figure into two symmetric parts.

C

canonical equation of an ellipse: the equation of an ellipse which is not rotated or translated and whose center is at the origin of the coordinate plane.

center of an ellipse: the point of intersection of the axes of an ellipse.

center of a hyperbola: the point of intersection of the transverse and conjugate axes of a hyperbola.

circle: the set of points in a plane whose distance from a fixed point in the plane is constant. A circle is a conic section whose eccentricity is zero.

circle of the directrix of an ellipse: a circle whose radius is the major axis of the ellipse and whose center is at the focus of the ellipse.

conic section: a geometric figure which results from a plane intersecting a pair of right circular cones positioned vertex to vertex. The standard conic sections are the circle, the ellipse, the parabola and the hyperbola.

conjugate axis of a hyperbola: the axis of a hyperbola which passes through the midpoint of the foci and which is perpendicular to the transverse axis.

coefficient: a constant multiplier of the variable(s) in an algebraic term. For example, the coefficient of $2xy$ is 2.

coefficient matrix: a matrix which is obtained from the coefficients of a system of linear equations. For example, the coefficient matrix for the system

$$\begin{aligned} 2x + y &= 1 \\ 3x - y &= 9 \end{aligned} \quad \text{is} \quad \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}.$$

communication matrix: a matrix which shows the possible paths of communication between different things, places or points.

consistent system of equations: a system of linear equations which has either one solution or infinitely many solutions.

column (of a matrix): a vertical array of numbers in a matrix.

cofactor: The cofactor of a matrix entry a_{ij} is

$C_{ij} = (-1)^{i+j} \cdot M_{ij}$, where M_{ij} is the minor of a_{ij}.

cofactor expansion: a way of calculating the determinant of a matrix by using some of its cofactors.

cofactor matrix: The cofactor matrix of a matrix A is the matrix which consists of the cofactors of all the entries of A.

Cramer's rule: a method for solving a system of n linear equations in n unknowns by using matrices.

composite function: a function which is formed by composing two or more elementary functions, for example: $f(g(x))$.

constant function: a function of the form $f(x) = c$, where c is constant.

continuous function: a function whose graph contains no breaks or gaps.

continuous function at a point: a function which is defined at a point and whose right-hand and left-hand limits at the point are equal to the image of the point.

continuous function on an interval: a function which is continuous at every point on an interval.

crucial point: a point at which we need to check the right-hand limit and the left-hand limit.

D

determinant: a real number which is associated with a square matrix. Every square matrix has one determinant.

decreasing function: A function $y = f(x)$ is a decreasing function on an interval I if y decreases as x increases on I. $f(x) = 6 - x$ is a decreasing function in $\mathbb{R}$.

directrices: the plural form of directrix.

directrix: a fixed line in a conic section which, along with the focus or foci, defines the section.

discriminant: the quantity $b^2 - 4ac$ in the quadratic equation $ax^2 + bx + c = 0$.

dimensions (of a matrix): the number of rows and columns in a given matrix (also called the *order* of the matrix). For example, the matrix $[1 \quad 2 \quad 3]$ has one row and three columns, so its dimensions are 1×3.

discontinuous function: a function whose graph contains breaks or gaps.

discontinuous function at a point: a function which is not continuous at a given point.

domain of a function: the largest set of real x-values for which a function f is defined.

E

eccentricity of an ellipse: the ratio of the distance between the foci of an ellipse to the length of its major axis. It is a number between 0 and 1.

eccentricity of a hyperbola: the ratio of the distance between the foci of a hyperbola to the length of its transverse axis. It is always greater than 1.

eccentricity of a parabola: the ratio of the distance between a point on a parabola and its focus to the distance between the same point and the directrix. It is always 1.

ellipse: the set of points in a plane whose distances from two fixed points in the plane have a constant sum.

elliptical: having the shape of an ellipse.

elementary matrix operation: one of three basic matrix operations: multiplying a row or column of a matrix by a scalar, adding a scalar multiple of another row or column to a given row or column, or transposing two rows or columns.

elimination method: a method for solving a system of linear equations in which equations are added together to eliminate one or more of the variables.

entry: one of the numbers in a matrix. We write a_{ij} to mean the entry in the ith row and jth column of a matrix A.

equal matrices: Two matrices are equal matrices if they have the same dimension and their corresponding entries are all equal. $A = B$ means that two matrices A and B are equal. Note that

$[1 \quad 2 \quad 3] \neq \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, since the dimensions are different.

ε-neighborhood of a number: for $\varepsilon > 0$, the open interval $(x_0 - \varepsilon, x_0 + \varepsilon)$ is the ε-neigborhood of the number x_0.

even function: a function $f: D \to R$ for which $f(-x) = f(x)$ for all $x \in D$. The graph of an even function is symmetric with respect to the y-axis.

exponential function: For $a \in \mathbb{R}^+ - \{1\}$, the function $f: \mathbb{R} \to \mathbb{R}^+$, $f(x) = a^x$ is called the exponential function.

Extreme Value Theorem: Let f be a function defined from $[a, b]$ into $\mathbb{R}$. If f is continuous on $[a, b]$ then it has a maximum and a minimum on the interval $[a, b]$.

F

foci: the plural form of focus.

focus: a fixed point in a conic section which, along with the directrix or directrices, defines the section.

floor function: the function $[\![f(x)]\!]$ defined as follows: if $f(x) \in \mathbb{Z}$ then $[\![f(x)]\!] = f(x)$, and if $f(x) \notin \mathbb{Z}$ then $[\![f(x)]\!]$ is the greatest integer which is smaller than $f(x)$.

function: a rule that maps each element of a set D (called the domain) to a single element of a set R (called the range).

H

hyperbola: the set of points in a plane whose distances from two fixed points in the plane have a constant difference.

homogeneous linear system: a system of linear equations in which the right-hand side of every equation is zero.

I

identity function: a function which is of the form $f(x) = x$.

identity matrix: a diagonal matrix in which every entry on the main diagonal (from top left to bottom right) is 1 and whose other elements are all 0. The $n \times n$ identity matrix is denoted by I_n.

imaginary axis of a hyperbola: another name for the conjugate axis.

image of a function: For a function $f: D \to R$, the image I (also called the image set) of f is the set of images of all the elements in D: $I \subseteq R$.

image of a point: For a function f and a point c, $f(c)$ is the image of c.

image set: see *image of a function*.

isosceles hyperbola: a hyperbola whose transverse and conjugate axes are congruent to each other.

inconsistent system of equations: a linear system which has no solution.

inverse matrix: A matrix B is the inverse of a matrix A if $AB = I$, the identity matrix. A^{-1} means the inverse of A.

invertible matrix: a matrix which has an inverse.

increasing function: A function $y = f(x)$ is an increasing function on an interval I if y increases as x increases on I. $f(x) = x + 6$ is an increasing function in $\mathbb{R}$.

indeterminate form: a limit of a function which is not defined or not bounded at a point.

infinite discontinuity: If at least one of the right-hand or left-hand limits of a function f at a point x_0 tends to infinity then f has infinite discontinuity at x_0.

Intermediate Value Theorem: Let f be a function which is continuous on the closed interval $[a, b]$ with $f(a) \neq f(b)$. Let k be any number between $f(a)$ and $f(b)$. Then there exists at least one real number $c \in (a, b)$ such that $f(c) = k$.

inverse function: The inverse of a function $f: A \to B$ is the function $f^{-1}: B \to A$ such that $f^{-1}(f(x)) = x$ for every $f(x) \in B$. The inverse of f can only exist if f is a one-to-one and onto function.

J

jump discontinuity: If the right-hand and left-hand limits of the function f are not equal at a point x_0 then f has jump discontinuity at x_0.

L

linear equation in n variables: an equation that can be written in the form $a_1x_1 + \ldots + a_nx_n = b$, where $a_1, \ldots, a_n$ and b are real numbers and $x_1, \ldots, x_n$ are variables.

left-hand limit: the limit of a function $f(x)$ at a point x_0 as x approaches x_0 from the left ($x < x_0$).

limit of a function: Informally, a function f has a limit L at a point x_0 if the value of $f(x)$ gets closer and closer to L as x approaches x_0.

linear function: a function which is of the form

$f(x) = ax + b$ for $a, b \in \mathbb{R}$.

logarithmic function: For $a \in \mathbb{R}^+ - \{1\}$, the function $f: \mathbb{R}^+ \to \mathbb{R}, f(x) = \log_a^x$ is called the logarithmic function.

M

main diagonal (of a matrix): the sequence of elements of a square matrix that run from the top left to the bottom right of the matrix.

matrix: an ordered rectangular array of elements set out in rows and columns. An $m \times n$ (pronounced 'm by n') matrix has m rows and n columns.

major axis of an ellipse: the longer axis of an ellipse, which contains the two focii.

major axis of a hyperbola: another name for the transverse axis.

major circle of an ellipse: the circle whose center is the center of the ellipse and whose diameter is the major axis of the ellipse.

major circle of a hyperbola: the circle whose center is at the center of the hyperbola and whose diameter is the transverse axis of the hyperbola

minor axis of an ellipse: the shorter axis of an ellipse.

minor circle of an ellipse: the circle whose center is the center of the ellipse and whose diameter is the minor axis of the ellipse

minor circle of a hyperbola: the circle whose center is at the center of the hyperbola and whose diameter is the conjugate axis of the hyperbola.

minor: If A is a square matrix then the minor of an entry a_{ij} is the determinant of the square matrix formed by deleting the ith row and jth column of A. M_{ij} means the minor of a_{ij}.

N

normal: a line which passes through a point of tangency perpendicular to the tangent line.

non-decreasing function: A function $y = f(x)$ is a non-decreasing function on an interval I if y does not decrease as x increases on I.

non-increasing function: A function $y = f(x)$ is a non-increasing function on an interval I if y does not increase as x increases on I.

O

orbit: a curved path in space that is followed by an object that is moving round a planet, a moon or the Sun.

origin: the intersection point of the x-axis and the y-axis in the coordinate plane. Its coordinates are (0, 0).

orthogonal: perpendicular.

order of a matrix: another name for the dimension of a square matrix.

odd function: a function $f: D \to R$ for which $f(-x) = -f(x)$ for all $x \in D$. The graph of an odd function is symmetric with respect to the origin.

one-sided limit: the right-hand limit or left-hand limit of a function at a point.

onto function: A function $f: A \to B$ is an onto function if for any $y \in B$ there exists an $x \in A$ such that $f(x) = y$.

one-to-one function: A function $f: A \to B$ is a one-to-one function if for each $x_1 \neq x_2$ in A, $f(x_1) \neq f(x_2)$.

P

parabola: a set of points in a plane that are equidistant from a fixed point and a fixed line.

parameter of a parabola: the length of the line segment between the focus and the directrix of a parabola.

parametric equations: a pair of equations which define a point on a curve in terms of a third variable, for example t.

point of tangency: the point at which a tangent to a curve touches the curve.

periodic function: a function f which satisfies $f(x + t) = f(x)$ for some $t \in \mathbb{R}$.

piecewise function: a function which is defined by different formulas in different intervals of its domain.

polynomial function: a function of the form $a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$, where n is a positive integer and $a_n \neq 0$.

R

radical function: a function which contains one or more radical expressions such square roots, cube roots, etc.

range (of a function): the set which includes at least all the images of the elements in the domain of a function f.

rational function: a function of the form $\dfrac{P(x)}{Q(x)}$ where $P(x)$ and $Q(x)$ are polynomials.

removable discontinuity: A function f has removable discontinuity at a point x_0 if the limit of the function f at x_0 exists but is not equal to $f(x_0)$.

right circular cone: a cone whose base is a circle and whose altitude passes through the center of this circle.

right-hand limit: the limit of a function $f(x)$ at a point x_0 as x approaches x_0 from the right ($x > x_0$).

row (of a matrix): a horizontal line of numbers in a matrix.

row echelon form: A system of linear equations is in row echelon form if the leading coefficient (i.e. the first non-zero coefficient from the left) of each equation is always strictly to the right of the leading coefficient of the equation above it.

S

slope of a line: the tangent of the angle between a line and the x-axis in the coordinate plane. In the line equation $y = mx + c$, m is the slope of the line.

standard equation of a parabola: the equation of a parabola in the form $y = ax^2 + bx + c$ or $x = ay^2 + by + c$.

scalar matrix: a diagonal matrix whose entries are all equal.

singular matrix: a square matrix whose determinant is equal to zero (and which therefore has no inverse).

square matrix: a matrix which has the same number of rows and columns.

substitution method: a method for solving a system of linear equations in which we substitute a variable in an equation with an equivalent expression from a different equation.

symmetric matrix: If all the entries in a square matrix are symmetric with respect to the main diagonal (i.e. $a_{ij} = a_{ji}$ for all possible i and j) then the matrix is called a symmetric matrix.

system of equations: a set of equations that need to be solved simultaneously, i.e. the solution must fit every equation in the set.

sign function: the function $\text{sgn}(f(x))$ defined as follows: if $f(x)$ is positive then $\text{sgn}(f(x))$ has value 1, if $f(x)$ is 0 then $\text{sgn}(f(x)) = 0$ and when $f(x)$ is negative, $\text{sgn}(f(x)) = -1$.

square root function: a function of the form $f(x) = \sqrt{g(x)}$.

T

tangent: a line in a plane which touches a curve in the same plane at a single point, called the point of tangency.

transverse axis of a hyperbola: the axis of a hyperbola which includes the foci.

triangular matrix: a square matrix which has all zero entries either above or below the main diagonal.

transpose of a matrix: the matrix which is obtained by interchanging the rows and columns of a given matrix. For example, the transpose of

$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $\begin{bmatrix} a & c \\ b & d \end{bmatrix}$.

trivial solution: the solution of a homogeneous linear system in which all variables are zero.

V

vertex: an intersection point of a conic section and one of its axes.

vertices: the plural form of vertex.

Z

zero matrix: a matrix whose entries are all zeros. We write 0 to mean a zero matrix.